

USO Administration Guide - 6.0.1-GA (AI)

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1. Preface - USO Administration Guide

This guide contains instructions on how to carry out administration, implementation and deployment tasks relating to the AccessMatrix Universal Sign-On (USO) system. The AccessMatrix SSO Portal, USO Trainer, and USO Client are components specific to USO:

- **AccessMatrix SSO Portal**

The AccessMatrix SSO Portal is a web application, installed together with other AccessMatrix components during USO installation.

- **USO Trainer**

The USO Trainer is an essential component of the AccessMatrix USO solution. It is used to "teach/train USO on how to sign-on to the application". The USO Trainer collects application information (e.g. login screen type, application path, etc), that in turn will be used by USO Client to do Single Sign-on (SSO) to the applications. The USO Application Training Guide section presents the different features of USO Trainer, as well as the detailed steps to utilize these features. Each application should go through the training steps before it becomes effective for SSO to the users.

The trainer can also be installed independently from other USO components. There is no manual configuration needed for USO Trainer.

- **USO Client** is a Win32 application, installed in the user's workstation and communicate with AccessMatrix SSO Portal during user login to USO Server. The USO Integration section contains instructions on how to integrate AccessMatrix Universal Sign-On (abbr. as USO) and its related component, i.e. USO Client.

Intended Reader

This document is intended for Security Administrators and i-Sprint's Professional Service personnel. The [USO Client User Guide](#), [USO Client Mobile App User Guide](#), and [AccessMatrix SSO Portal User Guide](#) can also be used as references for end users.

Document Scope

This document provides instructions on the following operations:

- **USO System**
 - Perform administration tasks in the USO system

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- **USO Client**
 - Access and use USO Client features
 - Access and use USO Client Mobile App features
 - **USO Trainer**
 - Perform USO application training using the USO Trainer
 - **USO Integration**
 - Integrate USO with the USO Client
 - **SSO Portal and SSF Portal**
 - Deploy and configure SSO Portal and Self-Service Portal
 - SSO Portal and Self-Service Portal Configuration
 - Access and use SSO Portal and Self-Service Portal features

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Universal Sign-On documentation set:

- [AccessMatrix Release Notes](#)
 - [AccessMatrix Common Administration Guide](#)
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2. Introduction to USO Administration

USO provides sign-on automation, enhancing the authentication process and enforcing consistent password management policies across applications. USO improves user productivity by simplifying access to all applications through a single authentication while enhancing security by integrating strong authentication mechanisms and proven security practices.

USO Components

USO is comprised of the following components:

- **USO Application**

This refers to user's business systems, applications or any resources meant to be enabled with AccessMatrix to provide for USO features.

- **USO Application Group**

This is a feature that allows the Administrator to group the USO applications to allow sharing of application login information. This is quite common in an organization when multiple users are granted a sharing login account to an application. Another usage is to allow sharing of login information among the defined applications. This does happen in organizations where users from the external store (i.e. Active Directory) are used as standard login account for login across some applications in an organization.

- **USO Client and Server Configurations**

USO specific configurable attributes (known as USO registry-based properties) are dynamically retrieved from the security registry during runtime. These attributes control the behaviour of the AccessMatrix SSO Portal and USO Client components. The USO registry-based properties can be configured via the Administration Console and grouped into 2 categories:

- **USO Client Configuration**

It consists of attributes that control the behaviour of USO Client, e.g. whether to launch a particular application during USO Client startup, whether to minimize the USO Client upon startup, etc.

- **USO Server Configuration**

It consists of attributes that control the behaviour of USO server particularly the AccessMatrix SSO Portal, e.g. the default authentication method for USO server login, whether to show the upgrading page upon the availability of a new version of USO Client, etc.

- **USO File-Based Properties**

Apart from the above-mentioned registry-based properties, there are other file-based properties. The main property file used by USO server is:

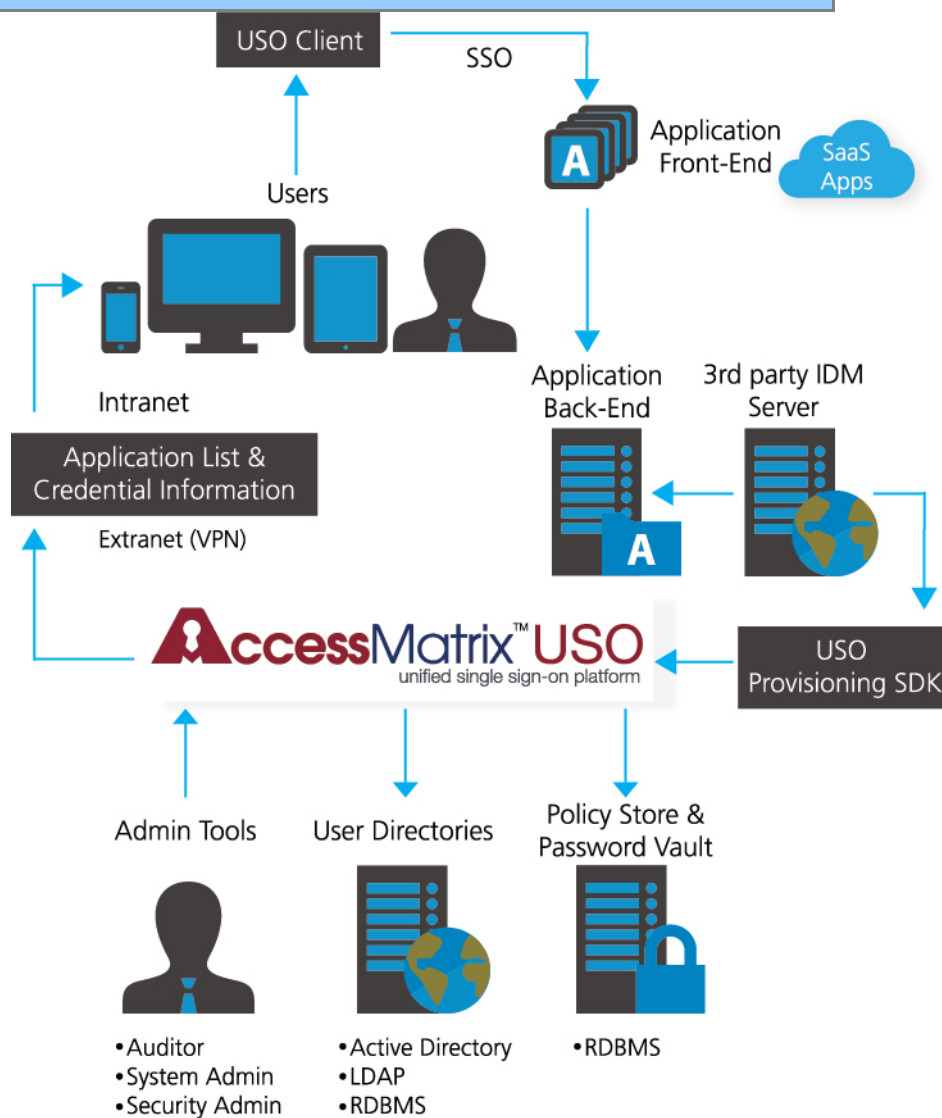
- **amlog4j2.properties**

This property file is used by the USO server for system logging purposes. It specifies the configuration of the log file, e.g. the log level, location, and format.

- **USO Self-Service Portal**

Self-Service Portal is a competitive feature provided by USO. It is governed by system-defined Self-Service Portal policy which determines the self-service functions to be made available to USO users upon successful USO server login.

Note: "Setup of Challenge Question" and "Reset of AD Password" are not supported from this version.



Supported Features

- Web-Based Self-Service Facility
- Sign-On to any application
- Application Login Profile Generator
- Policy Driven Access Management
- Flexible Integration Options

AccessMatrix License

USO FEATURE license must be enabled in order to use the USO functions. Otherwise, only **USO Server Configuration** will be displayed as it is required by other AccessMatrix products.

3. USO Admin Console

SSO Management

This section provides details on managing USO entitlements, USO applications, USO application groups and configurations for Single Sign-On.

Configure USO Application

Prior to creating a USO application, Administrator must ensure to train and export the application information into an Application Definition File (ADF). To find out more, read the [USO Application Training](#) section for detailed instructions on how to train an application.

To access this page, from the menu bar, go to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Application** on the left pane.

From this page, the following actions can be performed:

- **Create** USO application - Click the **Create** button to display the [Create USO Application](#) page. Save the changes.
 - **Search** for USO application - Enter a **USO Application Id**, **USO Application Name** and/or **Segment**, then click the **Search** button. Tick the **Recursive** checkbox to do a recursive search through all levels of segments.
 - **View** USO application - Click on a USO Application Id under **Search USO Application Result** to display the **View USO Application** page.
 - **Edit** USO application - Click the **Edit** button from the **View USO Application** page to display the [Edit USO Application](#) page and modify the attribute(s) to change. Save the changes.
 - **Delete** USO application - Click the **Delete** button from the **View USO Application** page. Click **OK** on the confirmation pop-up to delete the USO application.
 - **Import** USO application - Click the **Import** button from the **Create USO Application** page. Click the **Browse** button and select the ADF file of the application to import. Click the **Submit** button. Save the changes.
 - **Export to ADF** USO application - Click the **Export to ADF** button from the **View USO Application** page. Select whether to open or save the application to a location.
-

Create USO Application Attribute Descriptions

i **Note:** ' * ' denotes a mandatory attribute.

Attribute Name	Description
*USO Application Id	Identification assigned to the USO application.
*USO Application Name	The name of the USO application.
Description	Brief description for the USO application.
Version	The USO application revision. (uneditable)
Password Sharing Application Group	Refers to a USO application group where this USO application belong to. All USO applications with the same USO application group will share the same password. Note: USO application group must be created prior to assignment.
Status	Status of USO application.
Segment	Name of segment the USO application belong to.
Attributes Tab	
Required Authentication Level	
Required Authentication Level	The number of authentication level required to login to application. Note: This attribute is obsolete. Default: 1
Self-Service Portal	
Allow user to view password via SSF portal	If set to "true", user will be allowed to view password via SSF portal. Default: true
Allow user to enroll application via SSF portal	If set to "true", user will be allowed to enroll application via SSF portal. Default: true
Application Info	
Application path	Path of the application.
Working directory	Directory location of the application.
Argument	Application arguments.
Launch	
Citrix application path	Path of Citrix application.

Attribute Name	Description
Launch option	Launch options for an application.
Process external launch	To allow external launch of an application. Default: Yes
Offline accessibility	To allow offline access. Default: Yes
Allow launch app	To allow launching of application. Default: Yes
Manual	
Allow drag and drop	Determines whether to enable the drag and drop feature for all applications.
Password	
Allow password reset	To allow reset of password. Default: Yes
Use SID as password	To use SID as password. Default: No
Set SID to environment	To set SID to environment variable. Default: No
Change password mode	Refers to the change password mode of the application. Default: Manual
Password min length	Minimum password length. Default: 6
Password max length	Maximum password length. Default: 50
Password min uppercase	Minimum number of uppercase characters in a password. Default: 0
Password min lowercase	Minimum number of lowercase characters in a password. Default: 0
Password min numeric	Minimum number of numeric in a password. Default: 0
Password min special char	Minimum number of special characters in a password. Default: 0

Attribute Name	Description
Password max repeat char	Maximum number of repeated characters in a password. Default: 0
Password min age (days)	Minimum age of password. Default: 0
Password max expiry (days)	Maximum number of days before the password is expired. Default: 30
Customize special characters	Specifies the special characters that can be used in the password. Supported special characters: • "~" • "!" • "@" • "#" • "\$" • "%" • "^" • "&" • "*" • "(" • ")" • "-" • "+" Leaving this attribute blank means that all of the above supported special characters may be used.
Password mask	The value specified here defines the password format used when generating new passwords. Supported characters: • "A" - an uppercase letter • "a" - a lowercase letter • "n" - a numeric digit • "s" - a special character • "x" - a random character (i.e. any of the above types of characters) For example, specifying the value "AaAansnx" may generate the password of "StRe2@35".
Randomized by system	To allow the system to generate random password which will contain at least one numeric, lowercase and uppercase characters. Default: No

Attribute Name	Description
Include special characters	To include special characters. Default: No
Reauthentication	
Verify replay	To allow verify replay. Default: Disabled
Verify replay method	Defines the verify replay method.
Verify replay before launch	To allow verify replay before launching application. Default: No
Profile	
Number of multi pwd/login	To allow the application to authenticate more than one user in a sequence. Default: 0
Join multi userid	Default: 0
Join multi password	Default: 0
Emulator	
Emulator product name	Refers to the emulator's product name.
Emulator product version	Refers to the emulator's product version.
Emulator session's host	Refers to the emulator's session's host.
Sign Off Policy	
Sign off	Defines the sign off action. Default: Graceful
Sign off timeout	Defines the sign off timeout value. Default: 5
Others	
Integrity checking checksum	Specifies the integrity checking checksum value.
Enable integrity checking	To allow integrity checking. Default: No
Screen info for iOS	Defines the screen information. This is normally used for iOS.

Attribute Name	Description
ADF for Reverse Proxy	Contains the reverse proxy configuration.
Data version	Specifies the data version. Default: 2
Record last login time	Specifies the record last login time. Default: No
Debug mode for mobile	To allow debug mode for USO Client on mobile. Default: No

Edit USO Application Attribute Descriptions

Attribute Name	Description
Attributes Tab	
Refer to Create USO Application Attribute Descriptions for the list of attributes.	
User/Group Application Attributes Tab	
%argument	Defines the default application argument.
%pse.version	Defines the default PSE version.
%workDir	Defines the default working directory of the application.
%disableUserIdIfNotEmpty	Defines default behavior of user Id if it's not empty.
%rememberprofile	Defines the default remember profile value.
%APIName	Defines the default API name.
%dllPath	Defines the default DLL path.
%lastLoginTime	Defines the default last login time.
%prnapppath	Defines the default principal application path.
%lastprofile	Defines the default last profile value.

Adding Custom Attributes

1. From the **Edit USO Application** page, click the **Add Custom Attribute** button. The **Add New Attribute** popup dialog is displayed.

2. Specify the attribute name, data type and value and then click **Apply** to save the attribute.



Note: If the attribute value is to be stored in an encrypted form in the security registry, check the **Encryption** option. This action is not reversible once the entry is saved.

3. After adding the attribute, administrator is allowed to edit its value or delete it by clicking on the associated **Delete** button.
4. Click **OK** button.

List of USO Application Custom Attributes

Attribute	Description
%AppldAsOneInstance	To combine different process instances as one application instance. The valid value is either "Yes" or "No".
%URLasOneProcess	To combine all URL from different tabs into one instance if it contains the specified URL value.
%externDllProfileKey	Refers to the shared secret used to verify the DLL specified in _SwitchProfileByExternalDll function.
%user.min.length	Refers to the minimum length of user Id. If length of user Id entered in Set Application Information dialog is less than the minimum length, then USO Client will display "The User ID must have a minimum length of N", where N is the value of %user.min.length.
%user.max.length	Refers to the maximum length of user Id. If length of user Id entered in Set Application Information dialog is greater than the maximum length, then USO Client will display "The User ID must have a maximum length of N", where N is the value of %user.max.length.
%ocr.scanAll	Setting this attribute to "No" will turn off the attempt to scan screens when using OCR approach. The valid value is either "Yes" or "No".
%pwd.confirm.hide	Setting this attribute to "Yes" will hide the Confirm Password field. The valid value is either "Yes" or "No".
%allow.changeAppArg	If specified as "No", Application Argument cannot be configured using the Change App Path option in the USO Client.
%allow.changeHLLAPIPath	If specified as "No", HLLAPI Library Path cannot be configured using the Change App Path option in the USO Client.

Attribute	Description
	Note: This custom attribute is only applicable to USOCClient with versions 5.6.2.0029-GA-E27 and above.

Configure USO Application Group

To access this page, from the menu bar, go to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Application Group** on the left pane.

From this page, the following actions can be performed:

- **Create** USO application group - Click the **Create** button to display the [Create USO Application Group](#) page. Save the changes.
- **Search** for USO application group - Enter a **USO Application Group Id** and/or **USO Application Group Name** and click the **Search** button.
- **View** USO application group - Click on a USO Application Group Id under **Search USO Application Group Result** to display the **View USO Application Group** page.
- **Edit** USO application group - Click the **Edit** button from the **View USO Application Group** page and modify the attribute(s) to change. Save the changes.
- **Delete** USO application group - Click the **Delete** button from the **View USO Application Group** page. Click **OK** on the confirmation pop-up to delete the USO application group.

Create USO Application Group Attribute Descriptions

 Note: ' * ' denotes a mandatory attribute.	
Attribute Name	Description
*USO Application Group Id	Identification assigned to the USO application group.
*USO Application Group Name	The name of the USO application group.
Description	Brief description for the USO application group.
Version	The USO application group revision. (uneditable)
Attributes Tab	

Attribute Name	Description
This tab contains the custom attributes added by the Administrator. The Administrator can add, edit or delete any custom attributes.	
Password Sharing Application Tab	
This tab contains the list of USO applications with password sharing. Only the USO Application Id and USO Application Name attributes are displayed. Note: Adding or deleting of USO Application is not allowed.	

Administer USO Client Configuration

USO Client Configuration is specific to USO product. Thus, the **USO Client Configuration** menu option will only be shown provided Administrator has subscribed to the USO features for the AccessMatrix license.

i **Note:** Administrator may refer to [B. Admin Role and Admin Right Definitions](#) to find out the relevant admin role and admin rights required to perform administrative tasks. Administrator will not be able to edit the USO related configuration if there is a pending/rejected admin request pertaining to the same USO configuration entry.

To access this page, from the menu bar, go to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Client Configuration** on the left pane. The **View Configuration** page is displayed.

From this page, the following actions can be performed:

- **View** USO client configuration - This page will be displayed after clicking **USO Client Configuration** from the left pane. See [View Configuration](#) for the attribute descriptions.
- **Edit** USO client configuration - Click the **Edit** button from the **View Configuration** page and modify the attribute(s) to change. Save the changes.

i **Note:** Server restart is required for the changes to take effect.

View Configuration Attribute Descriptions

i **Note:** ' * ' denotes a mandatory attribute.

Attribute Name	Description
*Configuration Id	Identification assigned to the USO client configuration.
*Configuration Name	The name of the USO client configuration.
Description	Brief description for the USO client configuration.
Version	The USO client configuration revision. (uneditable)
Implementation	Implementation type. (uneditable)
Attributes Tab	
Auto Launch Application	[OPTIONAL] Specifies the application to be launched automatically by USO Client during startup. - The value can either be URL or application path without argument, e.g.: "http://www.xyz.com" or "C:/application.exe" - The value can also be application path with argument separated with a space character, e.g.: "C:\Program Files (x86)\Google\Chrome\Application\chrome.exe" "https://internet-banking.xyzs.com/Welcome" - The value can also be multiple applications separated by semi-colon (;) e.g.: "C:/application1.exe;c:\application2.exe;..." - The value can also be multiple applications with arguments separated by semi-colon (;) e.g.: "C:/application1.exe arg1 arg2; C:/application2.exe arg1 arg2;..." Note: The value of the entire application list must not exceed the maximum length of 250 characters.
Minimized	Specifies whether the USO Client will be minimized when the USO Server has been started. Default value: No Note: This attribute takes effect only when the SSO Portal is closed. For more details, refer to the Close SSO Portal When USO Client Active attribute.
Auto Login	Determines whether to automatically login to the USO Server after USO Client startup. Default value: No
Auto Login Local	Determines whether to automatically login to the local PSE after USO Client startup. Default value: No
Auto Switch To PSE When Server Unavailable	When running in online mode (logged-in and connected to USO server) or for the first time, setting this parameter to "Yes" will enable USO Client to keep running although the server is (or suddenly) not available by automatically switching to local PSE.

Attribute Name	Description
	Default value: No Note: The setting is only effective provided the following parameters are set as described: 1. EnablePSE must be set to "Yes". 2. PTDType must be set to either "1", "3" or "4". 3. The PAM used for USO server login must be set to "NTLM PAM Module".
Auto Switch To Server When Available	When running in off-line mode (logged-in to local PSE), setting this parameter to "Yes", will enable USO Client to automatically switch to online mode (i.e. to connect to server automatically) when the USO server connection is available. USO Client will do this switching in two ways: i) By using MinSessionCheckInterval setting, USO Client will do the connection polling based on this value; ii) Every time user wants to launch an application, USO Client will also try to check for the USO server connection. Default value: No Note: The setting is only effective provided the following parameters are set as described: 1. EnablePSE must be set to "Yes". 2. PTDType must be set to either "1", "3" or "4". 3. The PAM used for USO server login must be set to "NTLM PAM Module".
Enable Manual Drag And Drop	Determines whether to enable the drag and drop feature for all applications. • When set to "No", it will globally disable the manual drag and drop feature for all applications. • When set to "Yes", USO Client will allow manual drag and drop feature for applications that are configured for the feature. Default value: No
Force Screen Saver	Determines whether to force the user to set password protected screen saver. The valid values are: • "Yes" - USO Client will actively monitor for system idle and when time set under the ScreenSaverTimeout is reached, it will lock out the screen (WinNT, Win2K) or bring up the screen saver (Win98) • "OS" - USO Client will force the user to set the workstation's password protected screen saver. • "No" - USO Client will not force the user to set the password protected screen saver. • "OS_with_Lenovo_NPDirect" Default value: Yes
Screen Saver Timeout	The maximum allowable screen saver waiting period in millisecond (1 second = 1000 milliseconds). • If no value is specified or set to 0, AccessMatrix SSO Portal will only ensure that the password protected screensaver is set.

Attribute Name	Description
	This parameter is applicable when Force Screen Saver is set to "Yes" or "OS". Default value: 600000 (10 minutes)
Min Session Check Interval	To specify the minimum interval in minutes for USO Client to check the user's USO session validity. At the same time, it prevents idle timeout by refreshing the user's USO session when user's activity does not interact with the server. USO Client refreshes the login session whenever there is a computer activity (i.e. keyboard or mouse events) on USO Client and on any of USO applications. - If set to value of "0", USO Client will perform user's session validity checking every time before replaying application's login information submission. - If set to value of x minutes, USO Client will perform user's session validity checking before replaying application's login information submission AND if the last checking done is longer than x minutes. Default value: 5 Note: The MinSessionCheckInterval should be set shorter than idle timeout.
Idle Timeout Prompt Time	To specify the interval in seconds, of how soon before the session idle timeout that USO Client will show the session extension dialog box. - If set to "60", means USO Client will prompt the session extension dialog box 60 seconds before the session idle timeout. - If set to "0", USO Client will not show the session extension dialog box and it will turn off the auto refresh of server login session. - If set to a value $\geq$ idle timeout, the behavior will be the same as setting it to "0". Default value: 0
Kill Browser	Determines whether to close the relevant browser upon logging out from USO Client. The relevant browser is the browser that retrieves Session ID from USO Client. Default value: No
Logout When Browser Closed	Specifies whether USO Client will logout automatically when the browser instance (that initiated the login) is closed. Default value: No
Polling Time	Specifies the interval in milliseconds of which USO Client performs polling on the current foreground window. This option is useful to serve as a backup for applications that do not reliably send the update events. (For example: Java Application, Oracle Application, etc.) If the value is set to "0", the polling mechanism will be turned off. Default value: 1000 (1 second) Note: Do not specify value that is too low, as the machine CPU may not be able to handle it.

Attribute Name	Description
Disable Menu	[OPTIONAL] This parameter is used to disable the desired File menu options in the USO Client. This parameter is case-insensitive. The valid values: • "Login" - Disables access to "login" to server on main menu, toolbar, and tray icon. • "Local" - Disables access to login to "local" PSE on main menu, toolbar, and tray icon. • "Logout" - Disables access to "logout" on main menu, toolbar, and tray icon. • "Exit" - Disables access to "exit" on main menu, toolbar, and tray icon. • "Change Password" - Disables access to "change password" on main menu. • "Change PSE Password" - Disables access to "change psepassword" on main menu. • "About" - Disables access to "about" on main menu. Multiple values separated by a comma can be specified, e.g.: "Login,Change Password"
Autorun When Login to OS	Determines whether to automatically launch the USO Client after user logs in to the operating system. Default value: No
Reload Application List After Password Change	This parameter determines whether USO Client should reload application list from server after primary password is changed successfully. Default value: No
Dialog Timeout	This parameter is to specify the maximum time (in second) the User Verification window will appear before USO Client considered it as a cancel action. • This setting is only applicable for the Domain-Verification function during re-authentication. When set to "0", it means no timeout. Default value: 600
Log Level (obsolete)	The log level that determines how detailed USO Client will log the USO Client activities to the log file. The valid values, explains in increasing comprehensiveness of the log file are: • "0" – No log file will be written. • "1" – User level log, will only log error and informational events. • "99" – High level debugging log, will log all details including debug information. Default value: 1 Note: This attribute is obsolete. It is only applicable for USO Client version 5.0.8.0806 downward.

Attribute Name	Description
Enable PSE	This parameter enables/disables the creation of the user's Personal Security Environment (PSE) file, which stored application information locally. PSE file is encrypted using encryption method provided by PTD. The valid values for this parameter are: • "Yes" - A PSE file will be created where application login information will be stored locally. • "NoPwd" - A PSE file will be created that stores all application information and screen information, but without the application login information. • "No" - A PSE file will not be created; application login information will be discarded upon logout. Default value: NoPwd
PSE File Option	Specifies PSE file creation option and whether multiple copies of PSE files are allowed in same OS logon Id. Applicable when EnablePSE is "Yes" or "NoPwd". The valid values are: • "0" - means one copy of PSE file per OS logon Id, warning prompt will be shown when another USO user tries to login to same machine, the PSE file name is <UserHomeDir>\isprint\uso.pse. • "1" - means one copy of PSE file per OS logon Id, no warning prompt, but when another USO user tries to login, local PSE is automatically disabled for this user. • "2" - means one copy of PSE file per USO user Id, the PSE file name is <UserHomeDir>\isprint<userid>.pse. • "3" - means one copy of PSE file per USO user Id, the PSE file name is <UserHomeDir>\isprint<userid>.pse, no warning prompt when another USO user tries to login. Default value: 1
PTD Type	Determines the type of personal trusted device (PTD) used to encrypt the PSE. PTDType is applicable when EnablePSE is set to "Yes" or "NoPwd". The valid values are: • "0" - Software PTD, based on software encryption using a key derived from PSE password. • "1" - Auto PTD (MSCSP), based on MS Crypto environment, auto generated key. • "3" - Roaming PTD, similar to Auto PTD but with auto generated portable key. This option is only applicable when PSEFileOption is "0" or "1". • "4" - Server PTD, based on software encryption using a key derived from user's AM5 credential (applicable only for realm that contains "SystemPasswordBasic" login module). Choosing this PTD Type will require to define the Realm for PTDType=Server attribute.

Attribute Name	Description
	Note: If this option is selected, PSE login should always be available regardless of the PSE File Option value. Default value: 1
Download Mode	Determines how USO Client should download and/or cache the user's application login information. The valid values are described in increasing level of security: • "Download per login" - USO Client will download all the application information assigned to the client's workstation upon successfully login to the server and cache it until user logout. • "Download per application" - USO Client will download particular application information upon processing the application and cache it until user logout. • "Download per screen matched" - USO Client will download particular application information upon screen matching and immediately discard after used. Default value: Download per login Notes: A. Download Mode is applicable when EnablePSE is set to "No" or "NoPwd". B. When EnablePSE is set to "Yes", USO Client will always cache the login information.
Enable Hook	Specifies which hook to be enabled or disabled from USO Client. By default all hooks should be enabled, i.e. "EnableHook=GetMsg,CallWnd,Mouse,Key". However if specific requirement is needed, certain hook may be disabled. Note: If certain hook is disabled, some features that depend on the hook may not be functioning. For instance, if CallWnd hook is disabled: • Windows activate event trigger will not function. In such case, configure PollingTime so that USO Client can still perform screen matching. • Passing of UsoClientAppld environment variable will not work. This affects bind process Id applications that are launched thru wrapper process. If Mouse and Key hooks are disabled, the mouse and keyboard activities do not refresh the idle timeout (See ForceScreenSaver, MinSessionCheckInterval setting) This is an advanced setting for special customization usage. The default values are: "GetMsg", "CallWnd", "Mouse", "Key".
Enable Single Signoff	Specifies whether USO Client should perform sign-off on supported application upon USO logout. • This option works together with option at USO Application level. Thus, use USO Trainer to edit the ADF file to specify the option. Default value: No
Http Timeout	Specifies the timeout period (in milliseconds) for HTTP requests.

Attribute Name	Description
	• If value is > 0, that value will be the timeout period (in ms). • If value is < 0, the default timeout period will be used, i.e. 30000 ms. • If value is "0", there will be no timeout. Default value: 30000 (30 seconds)
Server List	Specifies the list of servers for the USO Client to connect to for failover purpose when the default ServerURL is not accessible. The list is in the format of comma-separated server IP addresses, e.g. "192.168.1.35,192.168.1.36, 192.168.1.37" OR by server names, e.g. "server1, server2, server3". By default, failover is disabled or turned off if no value is specified.
Audit Filter Wildcard Event Type	Specifies a wildcard representing audit event type to be filtered. By default, system does not filter any audit events if no value is specified. The filtered audit events will not be sent to server and instead will be logged into USO Client's log file locally. In addition to common wildcard characters * and ?, the wildcard pattern also supports regular expression " " to express "OR" operator. Example Usages: • auditFilter.filter.wildcard.eventType=* (it means to filter all audit events) • auditFilter.filter.wildcard.eventType=uso* (it means to filter audit events for eventType starting with 'uso') • auditFilter.filter.wildcard.eventType=usoAppProcessScreen usoAppLaunch (it means to filter audit events for usoAppProcessScreen and usoAppLaunch events)
Compress Application Info	Specifies whether the application information list sent from USO Server to USO Client upon USO login will be compressed. Default value: No
Allow Disable Application Temporarily	Specifies whether to allow the user to disable the application matching for the selected application throughout current logged-in session from the USO Client context menu temporarily. Default value: Yes
Allow Disable Application Permanently	Specifies whether to allow the user to disable the application matching for the selected application throughout current as well as subsequent logged-in sessions from the USO Client context menu permanently. Default value: Yes
Allow Changing App Path	Specifies whether to allow the user to change the application path from the USO Client context menu. Default value: Yes
Sort Application List By	Specifies the sorting criteria for the application list. The valid values are as follows:

Attribute Name	Description
	• "ID": means application list will be sorted by application ID • "Name": means application list will be sorted by application name • "Priority": means application list will be sorted by designated "%priority" attribute defined at application level . The default value is "Name". Notes: If application list sorting is by Priority, administrator needs to set the "%priority" attribute to a value of an integer number that is less than 100000 (caution: the value MUST be an integer number). If %priority attribute's values are same among the applications, system will sort the applications by application name. If the "%priority" attribute is not defined or its value is not set, system will sort them by application name. If there is a mix of values defined e.g. ID is specified while applications have "%priority" attribute set, system will first sort the applications based on the "%priority" attribute in a sequential manner, and then followed by application name for those that do not have the "%priority" attribute set.
Allow User To Manage Their Own Profile	Specifies whether users are allowed to manage their own application login profile. The default value is "Yes". Note: If you change the value from "Yes" to "No" while some users already have more than one application login profile defined for the application, those profiles will stay intact. In addition, the Set Active and Reset Profile menus will be disabled.
Default Language	Specifies the default language for USO Client. The valid values are: • "English" • "Simplified Chinese" • "Traditional Chinese" • "Japanese" • "Thai" The default value is "English". Note: For fresh or upgrade installation of USO Client, user is required to restart USO Client for the setting to take effect.
Window State Upon Switching To Offline Mode	Specifies the state of the window for USO Client upon switching from online to offline mode. The valid values are: • "Keep Current State" • "Restored" • "Minimized"

Attribute Name	Description
	The default value is "Keep Current State".
AutoShowLogin	Specifies whether to automatically show USO Client login screen upon session expiration. Default: No
Automatically login to UCM Gateway Server	Specifies whether to automatically login to UCM Gateway server. Default: No
Hide USO client	Enabling this will hide USO Client and won't allow the user to access the user interface. Default: No
Offline Audit	Specifies whether USO client should perform audit records during PSE mode. Note: If PSE File Option is set to "2" or "3", then offline audit cache file is always <userId>.oma. Default: No
Offline Audit Server	Refers to another AM5 server where the recorded audit during offline operation is saved. Must be set as "[IPAddress]:[Port]" format. For example: "192.168.1.63:8080"
Realm for PTDTType=Server	Specifies the realm that contains the SystemPasswordBasic login module. The credential will be used to derive key for encrypting PSE file. This is only applicable if PTD Type = Server. Note: Separate the items with comma if there are more realms applicable.
Merge Login Menu	Specifies whether Login to Server and PSE Login menu is being integrated. User will be logged in to server when connection is available, otherwise user will be switched to PSE Login. Default: No
Maximum PSE Login (in days)	Specifies the maximum time for user to login to PSE only. Default value is "0" days, which means PSE Login will never expire.
ClientCertificatePort	Specifies the user certificate port value. Note: This is only applicable if realm is 40153. Default: 8444
PSE File Custom Directory	Specifies the custom directory where to save the PSE File. If this value is not set, PSE File will be saved to the default directory which is USOCient application data folder. Notes:

Attribute Name	Description
	- Supports directory in shared folder on a file server, e.g. K:\%USERNAME% - Supports directory in UNC (Uniform Naming Convention) Path Format, e.g. file_server\shared_folder\%USERNAME% - If the path specified is inaccessible, then no PSE file will be created. Therefore, USO Client can only work using Online mode only. No error message will be displayed when USO client fails to locate the path.
Hide PSE File	Determines if the PSE file should be created or be hidden. If value is set to "Yes", PSE file will be hidden, otherwise PSE file will be created with default attribute. Default: No
Auto Launch Application On Log Out	[OPTIONAL] Determines if the application file should be automatically launched by USO Client upon logout. The value can be multiple applications separated by semi-colon (;) e.g.: "C:\Program Files\Demo\demo.exe";"C:\Program Files\Sample\sample.exe";...'
Auto Restart Time	This attribute will set USO Client to auto restart daily at a certain time (retaining user's session, application list and states), without interfering user's experience. This feature is necessary as USO Client connects internally with other applications. There are times that a defect in other applications will also affect USO Client subtly and slowly, therefore this auto restart ensures USO Client stability. If it is empty, or filled with wrong format, USO Client will revert to default time at 6 am. The value is a proper 24-hour time format, e.g. 21:30:00. Note: This auto restart time is applicable for both Server and PSE modes. Default: 06:00:00
CLP URL	This attribute refers to the target CLP URL. The purpose of this attribute is to let USO Client accept request of Session ID from the CLP.
Close SSO Portal When USO Client Active	If specified as "Yes", the SSO Portal will be closed automatically once the user successfully logs on to the USO client from CLP. Note: The USO client displays only USO applications (e.g. ESSO applications and native applications). Federated applications (e.g. SAML and OIDC), are accessible only through the SSO Portal and will not be available through the USO client when this attribute has been set to "Yes".
Bypass SSL Certificate Validation	If specified as "Yes", the USO Client will not validate the SSL certificate. This attribute is used if a firewall is preventing the SSL certificate from being validated and causing the USO Client to have significant slowdowns in loading applications.
Disable CSP Checking	If specified as "Yes", the USO Chrome Extension will not check for a website's Content-Security-Policy (CSP) headers. Default: Yes

Attribute Name	Description
	Note: This may cause error messages on the extension page for websites with a certain level of CSP configured.
Allow Changing Java Runtime	If specified as "Yes", users will be able to configure Java Runtime Settings in the USO Client.
Allow Changing SendKey DelayTime	If specified as "Yes", users will be able to configure Set SendKey DelayTime in the USO Client.
Allow Changing Open Browser	If specified as "Yes", users will be able to configure Set Open Browser in the USO Client.
Reauthentication Timeout	Specifies the reauthentication timeout window for the USO Client, in milliseconds. Once this window has been exceeded, subsequent reauthentication attempts from CLP will not be successful.
Reauthentication Browser Width Ratio	Specifies the reauthentication browser width ratio, in percentages. Default: 100
Reauthentication Launch Extra Edge Tab	Specifies if a new tab will be launched in Microsoft Edge if reauthentication is required when using the USO Client to perform SSO. Values: "Yes" The SSO Portal will be launched in a new tab, regardless if Microsoft Edge is open. "No" If Microsoft Edge is open, the SSO Portal will not be launched in a new tab. If Microsoft Edge is not already opened, the browser will be launched, prompting the user to log in. Note: This attribute takes effect only when the Close SSO Portal When USO Client Active attribute has been set to "Yes".

List of USO Client Configuration Custom Attributes

Attribute	Data Type	Description
DesktopIdleTimeout	Integer	To invalidate the current session after a certain period of idle time. The time specified is in milliseconds.
ShowTextInIcons	String	Setting this attribute to "Yes" will display the text in USO Client toolbar icons. The valid value is either "Yes" or "No".
SendRequestByThread	String	Setting to "Yes" will send HTTP connection for SSL with timeout of 30 seconds. Default: Yes
AllowChangingJavaRuntime	String	If specified as "Yes", users will be able to configure Java Runtime Settings in the USOCient. Note: This custom attribute is only applicable to USOCient with versions 5.6.2.0029-GA-E27 and above.
AllowChangingSendKey	String	If specified as "Yes", users will be able to configure Set SendKey DelayTime in the USOCient. Note: This custom attribute is only applicable to USOCient with versions 5.6.2.0029-GA-E27 and above.
AllowChangingOpenBrowser	String	If specified as "Yes", users will be able to configure Set Open Browser in the USOCient. Note: This custom attribute is only applicable to USOCient with versions 5.6.2.0029-GA-E27 and above.

Administer USO Server Configuration

USO Server Configuration is specific to USO product. Thus, the **USO Server Configuration** menu option will only be shown provided Administrator has subscribed to the USO features for the AccessMatrix license.

Administrator may refer to [A. Admin Role and Admin Right Definitions](#) to find out the relevant admin role and admin rights required to perform administrative tasks. Administrator will not be able to edit the USO related configuration if there is a pending/rejected admin request pertaining to the same USO configuration entry.

To access this page, from the menu bar, go to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Server Configuration** on the left pane. An information message: "Some settings in this menu are marked as Deprecated, which means that these settings will be phased out in the future." will be displayed. Click the **OK** button and the **View Configuration** page is displayed.

From this page, the following actions can be performed:

- **View** USO server configuration - This page will be displayed after clicking **USO Server Configuration** from the left pane. See [View Configuration](#) for the attribute descriptions.
- **Edit** USO server configuration - Click the **Edit** button from the **View Configuration** page and modify the attribute(s) to change. Save the changes.



Note: Server restart is required for the changes to take effect.

View Configuration Attribute Descriptions



Note: ' * ' denotes a mandatory attribute.

Attribute Name	Description
*Configuration Id	Identification assigned to the USO server configuration.
*Configuration Name	The name of the USO server configuration.
Description	Brief description for the USO server configuration.
Version	The USO server configuration revision. (uneditable)
Implementation	Implementation type. (uneditable)
Attributes Tab	
Authentication	
Default authentication for USO users	Determine the default authentication method used for authenticating USO users during USO server login. Default: 40151 (Default Password Basic)

Attribute Name	Description
Specify Default Authentication to CLP	Setting to "true" will bypass the selection of authentication realm in Common Login Page and instead will use the authentication realm specified in the Default authentication for USO users attribute. Default: Yes
ASA Cache	
ASA Cache System Policy	Service API caches some ASA settings, e.g. application attributes (including application screen training information). The cache can store up to a maximum number of applications (ASA_CACHE_SIZE, defaulted to 1000). It is refreshed automatically every polling time (ASA_CACHE_POLLING_SECS, defaults to 600 seconds). The cache item is kept up to maximum of expiration time (ASA_CACHE_EXPIRATION_SECS, defaulted to 3600 seconds). The expired cached item is removed and the next API calls to get the item will retrieve from server. When it is full, the next cache entry put will shift out the older entry based on item last access time. ASA SystemPolicy Cache when enabled will cache _session_idle_timeout, _session_timeout, _default_pam. Default: Yes
Flow Control	
FlowControl.Start	The default request Id for AccessMatrix SSO Portal request. Default: start.uso
FlowControl.USOClient	[DEPRECATED] The request Id for USO Client request. Default: usoclientrequest.uso&
Authentication	
Authentication.PAM.ChgPwdAllowedList	[DEPRECATED] The list of PAM Id that are allowed to change password. Default: 40151 40151 refers to Password Basic.
Authentication.SetLoginIDToCookie	[DEPRECATED] Whether to set logged-in user's Id to cookie. Default: No
Cookie	

Attribute Name	Description
Cookie.Domain	The domain of the Web Server on which the cookie will be stored. Only applications under the specified domain will be able to access the cookie. Note: This enables USO to work with UAM and USO Reverse Proxy.
Cookie.Path	The path of the cookie. Only applications under the specified path will be able to access the cookie. Default: /
Send Secure Cookies	If enabled, USOAgent will process secure cookies. Default: No Note: This attribute was previously named OnlySendCookieOnHTTPS .
Send HTTP-only Cookies	If enabled, USOAgent will process HTTP-only cookies. Warning: Setting this to "Yes" will disable USOCClient. Set to "Yes" only if USOCClient is not used. Default: No
Send SameSite Cookies	Determines which type of SameSite cookies USOAgent will process. Options: • "None" • "Strict" • "Lax" Default: None
Installation	
Installation.TrustedLocation	This is to specify the trusted download location from within which the user is allowed to download the USO Client upon successful server log in. It contains a list of IP address and subnet mask pairs. The default value is empty, which means the download location is not restricted. The value to be specified is in the following format (with no spacing in between): <IP Address 1>/<Subnet Mask 1>,<IP Address 2>/<Subnet Mask 2>,... For example: "192.168.1.44/255.255.255.0,168.128.1.1/255.255.0.0" In this example, those users logging in from workstations having IP addresses of 192.168.1.? and 168.128.1.? will be

Attribute Name	Description
	able to download and install USO Client from the Web browser upon successful server log in.
Installation.Version	Version number of the available USO Client, e.g.: "5,0,7,0713". The version number must be specified by using ',' as the separator. AccessMatrix SSO Portal will check the version defined here and the version of USO Client installed at client side. If the value is different, AccessMatrix SSO Portal will prompt the user to install the latest version of USO Client.
Show Upgrading Page Upon Availability of New Version of USOCClient	This parameter is to determine whether show the upgrading page upon availability of a new version of USO Client whenever user access to the AccessMatrix SSO Portal login page. Default: Yes
Show USOCClient Installer URL in application list	This parameter will provide option for users to download and install USOCClient. Default: Yes
Language	
Language.WebPage.Charset	[DEPRECATED] To be set as HTML meta header content-type on the entire output page served out by AccessMatrix SSO Portal. This header is useful when AccessMatrix SSO Portal is customized for another language locale, and it is used by Web Browser to display the page content. To decide which setting to be used, the basic guideline is, if the language involves only basic alphabets, then this parameter can be set to iso-8859-1, otherwise UTF-8 is recommended especially for handling double-byte characters. Default: UTF-8
Language.MsgFile.Encoding	[DEPRECATED] The encoding format of the message properties file. To decide which setting to be used, the basic guideline is, if the language involves only basic alphabets, then this parameter can be set to 8859_1, otherwise UTF-8 is recommended especially for handling double-byte characters. Default: 8859_1
Language.Locale.Language	[DEPRECATED] The default locale (language) that will be used by AccessMatrix SSO Portal when displaying UI and

Attribute Name	Description
	message. Default: en
Language.Locale.Country	[DEPRECATED] The default locale (Country) that will be used by AccessMatrix SSO Portal when displaying UI and message. For further reference, refer to: http://www.chemie.fu-berlin.de/diverse/doc/ISO_3166.html Default: US
Custom	
Custom.WaitTillUSOCompleteDownload	[DEPRECATED] Determines whether to wait for USO Client to completely download the application list before AccessMatrix SSO Portal display the application list. Default: No
Custom.LogoutWhenUnloadShowAppList	[DEPRECATED] Determines whether to logout from USO when unload show application list page. The following actions will log USO Client out if this parameter is set to 'Yes': • User clicks the back button on the browser. • User refreshes the landing page. • User presses the "BackSpace" key on keyboard. • User diverts the landing page to another URL. Default: No
AppListStyle	[DEPRECATED] Determines the style of USOAgent application list. The value is either "Tiles" or "List" Default: Tiles

List of USO Server Configuration Custom Attributes

Attribute	Description
AllowMultipleUSOEntitlment	Specifies whether to allow multiple USO entitlements. Setting to "Yes" will display a warning message and allows saving multiple USO entitlements. Setting to "No" will display a warning message and does not allow saving of multiple USO entitlements. Default: No

Configure USO Entitlement

USO entitlement can be done either on group or user level as described in following sections respectively.

Configure Group USO Entitlement

Configuration of USO entitlement for user group can be done during creation or modification of the group entry. Refer to the [AccessMatrix Common Administration Guide](#) for detailed instructions on creating or editing a group.

To configure the USO entitlement for a new group:

1. From the menu bar, go to **Operation Dashboard > User & Segment > Group** and click the **Create** button. The **Create Group page** is displayed.
2. Select the **USO Entitlements** tab and tick the respective **Entitlement** checkbox of the USO application(s) to assign to the group. Untick the checkbox to unassign.
3. Having assigned the entitlement of the USO application, the Administrator can decide whether to allow the user (member of the group) to use the application login information of either his/her own or the group under the **Password Sharing** column.
4. Provide the necessary information such as **Group Id, Group Name**, etc.
5. Save the changes.

Configure Group-Application Attributes

The **Edit Group Application Attributes** button will be displayed if "this group" is selected in **Password Sharing** column of the **USO Entitlements** tab. To configure the group's group-application attributes, perform the following steps:

1. Under the **USO Entitlements** tab, click the **Edit Group Application Attributes** button under Action column. The system will retrieve the list of group-application attributes pertaining to the selected USO application.
 2. Specify the values in the **Edit Group Applications Attributes** popup dialog, if necessary.
 3. Save the changes.
 4. If the save action is successful, system will display an Information window with this message:
"Your action has been executed."
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- Click **OK** to close the message window.

i Note: Under normal circumstances, administrator is not required to specify any value for the group-application attributes. Administrator is only required to do so provided the application needs to be initialized with certain initial values.

Configure User USO Entitlement

Configuration of USO entitlement for the user can be done during creation or modification of the user entry. Refer to [AccessMatrix Common Administration Guide](#) for detailed instructions on creating or editing a user.

To configure the USO entitlement for a new user:

- From the menu bar, go to **Operation Dashboard > User & Segment > User** and click the **Create** button. The **Create User** page is displayed.
- Select **USO Entitlements** tab and tick the respective **User Entitlement** and/or **Group Entitlement** checkboxes of the USO application(s) to assign to the user. Untick the checkbox to unassign.

Entitlement Condition	Description
User-Level User/Group Entitlement	If the user is assigned with a group membership where the group has been assigned with the same USO entitlement, then the Group Entitlement of the same USO application will be displayed in the USO Entitlements tab. Moreover, if a user is entitled with the same USO application on the User Entitlement level, then the user entitlement will take precedence over the group entitlement.
User-Level User-Group Password Sharing	If the user is assigned with both user and group entitlements for the same USO application, then the Administrator can decide where to allow the user to use the application login information of "his/her own" or of "this group" in the Password Sharing column.
User-Level Application Group Password Sharing	If the user is entitled to an USO application that is also a member of an USO Application Group, then the user will be using the application login information of the USO Application Group as shown in Password Sharing column.
User-Level User-Group cum Application Group Password Sharing	If the user is assigned with a group membership where the group has been assigned with the same USO entitlement and at the same time the USO application is also a member of a USO Application Group, then, the effective

Entitlement Condition	Description
	application login information will be the one shown by Group Entitlement in the Password Sharing column.

3. Having assigned the entitlement of the desired USO application, the Administrator can decide whether to enable/disable application matching for the desired USO application throughout subsequent USO logged-in sessions under the **Enabled** column.
4. By default, the option is checked, i.e. the application is enabled for application matching. To disable it, uncheck this option.

i Note: Assuming user is allowed to enable/disable the application matching for the desired application, after logging in to the USO server, user may enable/disable the application matching for desired application for current as well as subsequent USO sessions by checking/unchecking the relevant option under USO Client context menu (refer to [USO Client User Guide - Manipulating USO Client - Application Settings](#) for further details). Such changes will also be updated accordingly to the USO server for the particular application.

5. Save the changes.

Configure User-Application Attributes

Configuration of user-application attributes for the user can only be done after saving the changes made during creation or modification of the user entry or during viewing of the user entry.

Assuming user entry has been saved or is being viewed, configure the user's user application attributes by performing the following steps:

1. Under the **USO Entitlements** tab, click the **Edit User Application Attributes** button under the **Action** column. The system will retrieve the list of group-application attributes pertaining to the selected USO application.

i Note: Under normal circumstances, the Administrator is not required to specify any value for the user-application attributes. Administrator is only required to do so provided the application needs to be initialized with certain initial values.

2. Specify the values in the **Edit User Applications Attributes** popup dialog, if necessary.
3. Save the changes.

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4. If the save action is successful, system will display an **Information** window with this message:
"Your action has been executed."
 5. Click **OK** to close the message window.

Configure User's Group-Application Attributes

 **Notes:**

- Configuration of user's group-application attributes is only applicable when user is assigned with USO entitlement through **User Entitlement** while user uses application password sharing of a particular group.
- Configuration of group-application attributes for the user can only be done after saving the changes made during creation or modification of the user entry or during viewing of the user entry.

Assuming user entry has been saved or is being viewed, configure the user's group-application attributes by performing the following steps:

1. Under the **USO Entitlements** tab, click the **Edit Group Application Attributes** button under the **Action** column. The system will retrieve the list of group-application attributes pertaining to the selected USO application.

 **Note:** Under normal circumstances, the Administrator is not required to specify any value for the user's group-application attributes. Administrator is only required to do so provided the application needs to be initialized with certain initial values.

2. Specify the values in the **Edit User Applications Attributes** popup dialog, if necessary.
3. Save the changes.
4. If the save action is successful, system will display an **Information** window with this message:
"Your action has been executed."
5. Click **OK** to close the message window.

Configure USO Self-Service Portal

To enable self-service functions for USO, it is required to define the USO Self-Service Portal Policy first. By default, the system-defined Self-Service Portal policy is created upon successful installation of AccessMatrix. From the menu bar, go to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Self-Service Portal** on the left pane. The information message: "Some settings in this menu are marked as Deprecated, which means that these settings will be phased out in

the future." is displayed. Click the **OK** button. The **View USO Self-Service Portal Policy** page is displayed.

View USO Self-Service Portal Policy Attribute Descriptions

	Note: ' * ' denotes a mandatory attribute.	
Attribute Name		Description
*USO Self-Service Portal Policy Id		Identification assigned to the USO self-service portal policy.
*USO Self-Service Portal Policy Name		The name of the USO self-service portal policy.
Description		Brief description for the USO self-service portal policy.
Version		The USO self-service portal policy revision. (uneditable)
Attributes Tab		
Enable "Setup Challenge Question" function		[DEPRECATED] To enable the "Setup Challenge Question" function. Default: false
Enable "Reset AD Password" function		[DEPRECATED] To enable the "Reset AD Password" function. Default: false
Enable "USO Application Enrollment" function		To enable the "USO Application Enrollment" function. Default: true
Enable "Retrieve USO Application Password" function		To enable the "Retrieve USO Application Password" function. Default: true
Password display time (in seconds)		Specifies the password display time. Default: 30
Min number of challenge questions		[DEPRECATED] Minimum number of challenge questions (Question & Answer) to set up. Default: 1
Max number of challenge questions		[DEPRECATED] Maximum number of challenge questions (Question & Answer) that can be set up. Default: 3
Min number of responses		[DEPRECATED] Minimum number of responses required to be answered during identity verification. Default: 1

Attribute Name	Description
Min length of response	[DEPRECATED] Minimum length of response. Default: 2
Max length of response	[DEPRECATED] Maximum length of response. Default: 30
Case sensitive	[DEPRECATED] If set to "true", USO will check case sensitivity of the response to challenge question. Default: false
Max failure per challenge question	[DEPRECATED] Maximum failure count per challenge question. Default: 3
Max failure all challenge questions	[DEPRECATED] Maximum failure count for all challenge questions. Default: 5
Realm Id For Q&A KBA Login	[DEPRECATED] The realm Id for Q&A KBA Login.
*Reauthentication Login Module for USO Application Password Retrieval	Specifies the login module used for reauthentication before an application's password can be retrieved. The following login module implementations are supported: System Password Basic, i-Sprint SMS OTP Login, Ldap Login for Active Directory, OpenLDAP and FIDO2 Token Login. Default: SSFOTPLLogin
Enable "FIDO2 Registration" function	To enable the "FIDO2 Registration" function. Default: true
*FIDO2 Registration Login Module Id	Specifies the login module used when using FIDO2 authentication to log in to the AccessMatrix SSO Portal.
FIDO2 Attestation	To enable the use of FIDO2 attestation. Attestation is the process where a FIDO2 authenticator provides cryptographic proof about its model to the FIDO2 Relying Party (the server requesting access to online resources). Default: none

Enable Self-Service Functions for USO User

In order for the USO user to use the Self-Service functions, the Administrator is required to assign the relevant login account and admin role to the USO user. To do so, perform the steps below:

1. Navigate to **Operation Dashboard > User & Segment > User** and edit an existing USO user.
2. Select the **Login Accounts** tab and assign an authentication realm.
3. Select the **Admin Privilege** tab and assign "SelfServiceRole" as its admin role.

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4. Save the changes.

i **Note:** If the above user is assigned with System Password Basic as one of the login modules, administrator needs to reset user password for the login module so that user can login to USO server using Password Basic authentication. To do so, refer to [AccessMatrix Common Administration Guide](#) for detailed instructions on resetting user's password.

Refer to the [AccessMatrix SSO Portal Implementation](#) and [AccessMatrix SSO Portal User Guide](#) sections for detailed details on Self-Service Portal.

View USO Reports

USO Reports contain the records of all USO applications' login events, entitlements and application summary by segment.

Real Time Reports

Real Time Report contains the current list of USO applications and its entitlements.

Application Summary By Segment

This report provides information about the number of applications (ASA, WSA and USO) assigned per segment.

To navigate to **Application Summary By Segment Report** page, select **Report Dashboard > Enterprise SSO & Web SSO > Real Time > Application Summary By Segment**. Input the search criteria (**Segment Id**, **Segment Name**), then select the report format (i.e. HTML, PDF or CSV) and generate the report. See below for the attributes description of the generated report:

Attribute Name	Description
Segment Id	Unique Id of segment.
Segment Name	Name of segment.
Hierarchical Path	Path of parent segment.
Number of Application Security Agent	Number of Application Security Agents associated with the segment.
Number of Web Security Agent	Number of Web Security Agents associated with the segment.

Attribute Name	Description
Number of USO Application	Number of USO Applications associated with the segment.
Total Applications	Total number of applications associated with the segment.

For sample report, refer to [Application Summary By Segment Report Sample](#).

USO Entitlements

This report provides the list of applications entitled to the user or group.

To navigate to **User USO Entitlements Report** page, select **Report Dashboard > Enterprise SSO & Web SSO > Real Time > USO Entitlements**. Input the search criteria (**User Store, User** or **Group, User** or **Group Id, Application Id**), then select the report format (i.e. HTML, PDF or CSV) and generate the report. See below for the attributes description of the generated report:

Attribute Name	Description
User/Group Id	User or group's unique Id.
User/Group Name	User of group's name.
Description	User/group's description.
Account Status	User's account status.
Application Id	Unique Id of the USO application.
Application Name	Name of the USO application.
Entitlement	Specify whether it's a user or group entitlement.
Password Sharing	Specify whether the user uses password sharing of his/her own or of the group.
Profile	Display the profile(s) of the application.
User Id	Display the user Id of each profile.

For sample report, refer to [USO Entitlements Report Sample](#).

Historical Reports

USO Application Login

This report will display user's latest USO session if user has ever done USO Application login, otherwise, it will list latest USO session.

To navigate to USO Application Login Report page, select **Report Dashboard > Enterprise SSO & Web SSO > Historical > USO Application Login**. Input the search criteria (**User Store**, **Actor User Id** (Blank or * for all users), **Session End Time -- From**, and **Session End Time -- To**) then select the report format (i.e. HTML, PDF or CSV) and generate the report. See below for the attributes description of the generated report:

Attribute Name	Description
Session End Time	Specifies the session end time of user login session or USO application login.
User Id@User Store	Specifies the user Id with its user store.
User Name	Name of the user.
User Created Time	Creation time of the user.
Realm Id	The authentication realm assigned to the user.
Last Login	Last login time of the user.
USO App Login Time	Login time to the USO application. Note: The login time will only be displayed if Record last login time of the application is set to "Yes".

For sample report, refer to [USO Application Login Report Sample](#).

4. USO Mobile Administration

Customize USO Application and Application Group Icons

The tile icon of a USO application and USO application group can be changed and customized into a desired image. Following criteria should be met to customize the application icon:

- The image must be a PNG file.
- The image file name and the application Id or application group Id must be the same. For example, if USO application Id is "Gmail", then image filename should be **Gmail.png**)

Place the image file to this location:

- **Tomcat**

`<ACMATRIX_HOME>/assets/images/appicon`

(e.g. `/opt/acmatrix/tomcat/webapps/am5/assets/images/appicon`)



Notes:

- If the AccessMatrix server is yet to start, start AccessMatrix and log in to USO Mobile to see the new application icon.
- `<ACMATRIX_HOME>` is used to refer to the installation directory of AccessMatrix Server, e.g. `/opt/acmatrix/tomcat/webapps/am5`.

5. USO Client User Guide

The Universal Sign-On (USO) Client is an important component of the AccessMatrix USO package. Through the USO Client, a user can perform Single Sign-On (SSO) to various Web and non-Web applications. This section will provide detailed information on the USO Client, its features, and the functionality of each feature.

Installing the USO Client is seamless and does not require any source code modification or manual software installation. Upon successful authentication to the AccessMatrix Security Server, users will be presented with the list of applications they are authorized to access.

USO Client Unique Features

USO has many unique features and functions. A few notable ones are summarized below:

- **Wizard-Driven Installation and Auto-Detection Upgrade of Client Software**

USO Client is to be downloaded and installed following a wizard-driven installation onto user's workstation during first time sign-on to the USO Server via the web browser. For upgrade installation, USO Client is detected automatically for download and installation whenever a newer version is available during the subsequent login to the USO Server.

- **Single Sign-On without Any Source Code Modification**

After user has been successfully authenticated to the USO Server, USO Client allows the user to achieve SSO to various types of applications without any source code modification.

- **Personal Security Environment and Offline Mode**

This feature allows the users, to create a local secure storage also known as the Personal Security Environment (PSE). This enables the USO Client to function without having to connect directly to the USO Server: while USO Client is in offline mode. If PSE is enabled, every change made to the application during offline mode, will be synchronized to the server once the user is connected to the server.

- **State-Based**

This feature provides more flexibility in processing matched screens. Users will have a better control over the application since USO Client processes the matched screen according to the state defined for the application. State-based also provides a seamless control over the application.

- **Event-Driven**

USO Client will process the application based on events detected on the user's desktop environment. USO Client will monitor the events occurring and examine the application execution as well as the screen flows. If it detects a screen matching the pre-defined screen (i.e. screen trained via USO Trainer), it will pass in the information learnt to the screen input fields of the target application automatically.

- **USO Client Enabling/Disabling**

This feature allows the users to specify whether to allow or disallow the USO Client to fill in the login information automatically either globally for all applications or for the selected application only.

- **Multiple Login Profiles**

This feature allows the users to have multiple profiles for an application, which can be reset anytime.

- **Multi-language Support**

USO Client supports multiple languages (English, Simplified Chinese, Japanese, etc.). However, the selected language must also be supported by the operating system as well. The user is able to select the desired language dynamically.

- **Various Types of Applications Supported**

USO Client supports various types of applications such as Desktop, Web, Java, Host, and Console applications.

- **Availability of WinAttach Icon**

USO Client provides the WinAttach icon - that is attached to a matched screen. This icon provides shortcut to default as well as user-defined menus. Hence, user can perform certain functions to the application directly, instead of going through USO Client.

5.1. Installing the USO Client

System Requirements of USO Client

This section describes the requirements for the system in order to install and utilize USO Client.

Web Browser

To install USO Client, there are certain requirements to be adhered for the browser settings.

Chromium Browsers (Google Chrome and Microsoft Edge)

USO requires the USO Chrome Extension to support USO and Federated SSO applications with functionalities of USO Client UI on Google Chrome and Microsoft Edge (Chromium-based). Refer to [AccessMatrix SSO Portal User Guide - Install USO Chrome Extension](#) section for the installation steps.



Notes:

- After the installation of USO Chrome Extension, access the AccessMatrix SSO Portal page (e.g. *http://<SSOPortal_Host>:<port>/usoserver/start.uso*) and log in. If login is successful, the SSO Portal page will be displayed and USO Client functionalities on Google Chrome and Microsoft Edge (Chromium-based) are now supported.
- USO Chrome Extension supports chrome browser in Windows 10 (minimum).
- USO Chrome Extension prevents the content script from printing the sensitive information to the Chrome web console.

User Access Rights

To be able to install USO Client properly, user needs to have certain access control rights. The required access control rights are:

- The right to write to user's Windows registry.
 - For installation of Current User (a.k.a. Per Profile) edition, the right to create and write to the user's profile directory such as *C:\<User_Profile_Directory>\USOClient*.
 - For installation of All Users (a.k.a. Common Users) edition, the right to create and write to the program files directory such as *C:\Program Files\USOClient*.
-

Operating System

- USO Client must be installed on Windows OS machine, with Windows 10 as the minimum version.

Applications

There are certain types of applications that USO Client supports. For the list of supported and certified applications, refer to [USO Application Training - Types of Supported Applications](#) and [List of Certified Applications](#) sections.

For the list of unsupported applications, refer to [USO Application Training Guide - Known Unsupported Applications](#).

USO Client Installation & Startup

This section contains detailed explanation on different types of USO Client installation and starting up.

i **Note:** The USO Client can be falsely flagged as malware by antivirus software. For more information, refer to [False Positive Detection by Antivirus Scan](#).

Install USO Client

Fresh Installation of USO Client	For a fresh installation of USO Client, i.e. there is no previous or existing version of USO Client installed, the detailed installation steps are described below: 1. In order to login to the USO Server for the first time, obtain the following information from the administrator: ○ The URL of the USO Web UI, e.g. <i>http://localhost:8080/usoserver/start.uso</i> ○ The USO User ID and password. 2. Open a Web browser and enter the above-mentioned URL to access the USO Web UI page. The Common Login Page is displayed. 3. Select the Authentication Realm from the drop down box, e.g. "System Password Basic". Then, click Next button. 4. Enter the valid User ID and password and click on Next button to login to the USO Server. 5. For first time login, USO system will request the user to change the initial password.
----------------------------------	---

	6. Specify the current and new passwords and click the Change Password button. 7. Upon successful login, a page indicating that USO Client has yet to be installed is displayed. 8. Right-click on the USO Client Installer hyperlink and select Save Target As to save the installer to a local folder. 9. Specify the desired folder to save the USO Client Installer. 10. After the installer is successfully downloaded and saved, the download complete window appears. 11. Click on Open Folder and run the USO Client Installer from within the folder to install the USO Client.  Important Note: User must not click on Run to run the installer to avoid improper installation. 12. If the security warning window is prompted, click Run to continue. The USO Client setup wizard is displayed. Click the Install button to continue installation. 13. Upon successful installation, click on the Finish button to close the USO Client setup wizard. 14. As instructed on the USO Web UI web page, after USO Client installation is completed, click on the here hyperlink to continue with the login process. 15. Once user has logged in completely, USO Client user interface will be displayed. Notice that the USO Client icon will also appear in the Windows task bar. 16. User is now ready to perform SSO to the application.
Upgrade Installation of USO Client	For an upgrade installation of USO Client, i.e. there is already a previous or existing version of USO Client installed, the installation steps are described below in details:  Important Note: If the existing version of USO Client has been started or logged in, log out and exit from the USO Client before performing the following steps. 1. Open a Web browser and enter the USO Web UI URL. The Common Login Page is displayed. 2. Select the Authentication Realm from the drop down box, e.g. "System Password Basic". Then, click the Next button. 3. Enter a valid User ID and password and click Next to login to the USO Server. 4. Once login is successful, the user will be presented with a page informing that a new version of USO Client is available.  Note: At this point, user can choose not to upgrade the USO Client and proceed to login to USO Server. Otherwise, proceed to steps below to upgrade USO Client. 5. To upgrade the USO Client, follow the instructions given on the screen, i.e. right-click on the USO Client Installer hyperlink and select Save Target As to save the installer to a local folder.

	- Specify the desired folder to save the USO Client Installer. - After the installer is successfully downloaded and saved, the download complete window will appear. - Close the USO Web UI associated page. - From the download complete windows, click on Open Folder and run the USO Client Installer from within the folder to install the USO Client.  Important Note: User must not click on Run to run the installer to avoid improper installation. - If the security warning window is prompted, click Run to continue. USO Client setup wizard will be displayed. Click the Install button to continue installation. - Upon successful installation, click Finish to close the USO Client setup wizard. - At this point, USO Client upgrade is completed and user may proceed to login to USO server as usual.
(For System Administrators) Centralized Installation	For a centralized distribution of the USO Client in an enterprise setup, the system administrator may configure the Active Directory Group Policy so that USO Client can be automatically installed on the target user's computer. System administrators seeking to perform a centralized installation of the USO Client should refer to the steps outlined in AccessMatrix SSO Portal Implementation - SSO Portal with Windows Native Apps and Web Apps - USO Client Centralized Deployment.

Create USO Client Properties in Windows Registry

Upon installation, USO Client requires some properties to run immediately, e.g. ServerURL, AutorunWhenLoginOS, etc. To pre-create any of the registry entries before USO Client connects to USO server for the very first time, perform the steps described below:

- Create a file, e.g. **USOClient.reg**. The content should be the same format as the registry file, e.g.

```
Windows Registry Editor Version 5.00
[HKEY_CURRENT_USER\Software\isprint\usoagent]
"ServerURL"="http://localhost:8080/am5/start.uso?reqid=usoclientrequest.uso&"
```

- Place the created file in the same folder as the **USOClient.msi**.
- Run **USOClient.msi**. The properties added in the file will be created automatically in the registry.

Upgrade USO Client By Using Auto Update Feature

As a thick client, USO Client must be deployed into the user's computer, and there are cases where there's a need to redeploy the USO Client if there are some changes or fixes. In typical bank implementation, USO Client may need to be redeployed to thousands of computers. To overcome this problem, a feature that allows auto update of USO Client from the server is implemented.



Important Note: This auto update feature is only applicable to the **Per Profile** edition of USO Client as it allows the application that runs in the user security context to update USO Client without asking for the Administrator's permission.

To enable this feature, perform the steps below:

1. Create a *usoclient* folder in this location: <AxMx Root>\webapps\am5\WEB-INF\classes\.
2. Place the latest **usoclient.exe** to the previously created *usoclient* folder.
3. Restart AM server.
4. Login to USO Server via USO Web UI login page and USO Client will be started upon successful login to USO Server.
5. If the previously installed USO Client version is an old version, USO Client will automatically download the latest **usoclient.exe** and replace the old version. It will then store the latest **usoclient.exe** in the USO Client - Per Profile installation directory.
6. Logout and Login to USO Web UI again. USO Client will be started if login is successful.
7. Check the version of the USO Client. If auto update is successful, then the version of the USO Client should be the version of the **usoclient.exe** found in *usoclient* folder.

Start USO Client

Typically, user will login to USO Server via USO Web UI login page and USO Client will be started upon successful login to USO Server. USO Client will not be started automatically for a USO default installation.

However, user may choose to login to USO Server via USO Client directly. In this case, user may start the USO Client manually in order to login to USO Server.

To start USO Client manually, perform the following steps:

1. Go to **Start** and open **Run**.
2. For installation of Current User edition, type in following command and click **OK**:
`"C:\<User_Profile_Directory>\AppData\Roaming\USOClient\usoclient.exe"`
For installation of All Users edition, type in following command and click **OK**:
`"C:\Program Files\USOClient\usoclient.exe"`

-
3. USO Client will be started.

Notice that the USO Client icon will appear in the Windows task bar.

Alternatively, user may start the USO Client by clicking **Start > Programs > AccessMatrix > Universal Sign-On (USO) > USO Client**.

4. If USO Client is started-up in minimized mode, double-click on the USO Client icon from the Windows task bar to restore the USO Client. Alternatively, right-click the mouse button and choose **Restore** to restore the USO Client user interface.



Note: To automatically start USO Client whenever the user workstation is started-up successfully, set the **Autorun When Login** to OS attribute to "Yes" under **USO Client Configuration** via Administration Console.

Refer to [USO Client Configuration](#) for the detailed discussion about the USO Client Configuration.

False Positive Detection by Antivirus Scan

The USO Client can be falsely detected as malware by antivirus software.

While it is designed to provide convenience for user authentication and does not perform unauthorized actions such as those performed by viruses and malware, the techniques used to achieve SSO can be mistakenly identified as malicious behavior. These techniques include:

- Hooking into native applications
- Reading content within native applications
- Inputting data into native applications
- Modifying the Windows Registry

To conclude that a false positive has been made, the vast majority of scanning engines should not detect any malware. Regular scans of the USO Client are conducted using VirusTotal, which contains more than 50 scanning engines around the world. We validate that it has not been flagged as malware by any of the security vendors.

USO Client Uninstallation

It is advisable to exit USO Client and the USO Web UI associated web page, if they are running prior to uninstalling USO Client.

To uninstall the USO Client, proceed with the following steps:

-
1. On the Windows task bar, click **Start > Programs > AccessMatrix > Universal Sign-On (USO) > Uninstall USO Client.**

The USO Client Uninstallation Confirmation screen will be displayed.

2. Click **Yes** to proceed with the uninstallation and a screen will appear showing the status of the uninstallation process.
 3. The uninstallation is considered successful as long as no error is encountered during the uninstallation.
 4. Click **OK** on the **USO Client Uninstallation Complete** dialog.
-

5.2. USO Client User Interface

Main Menu

Upon logging in to USO Client, there are three menu items available: **File**, **Settings** and **Help**. The following sections discuss the usage of each menu item.

USO Client File Menu

The USO Client **File** menu provides the **Login to Server**, **Change USO Password**, **Login to Local PSE**, **Change PSE Password**, **Logout** and **Exit** sub-menu options.

Login to Server

This sub-menu will allow users to login to USO server.

Refer to [Performing USO Server Login \(Online Mode\)](#) for detailed information.

Change USO Password

This sub-menu lets the user change the USO password.

To change the USO password, perform the following:

1. Login to the USO Server if not already logged in.
2. Click on the **File** menu option and select the **Change USO Password** sub-menu option.
3. The **Change USO Password** screen is displayed.
4. Enter the old password and new password twice as indicated, and click **OK**.
5. System changes the USO password and displays the **USO change password successful** screen.
6. Click **OK** to return to the USO Client **Main Screen**.



Note: Users will not be able to change the USO password if they login to the local PSE only.

Login to Local PSE

This sub-menu will allow users to do PSE login.

Refer to [Performing Local PSE Login \(Offline Mode\)](#) for detailed information.

Change PSE Password

This sub-menu lets the user change the PSE password.

To change the PSE password, perform the following:

1. Login to the USO Server if not already logged in.
2. Click on the **File** menu option and select the **Change PSE Password** sub-menu option.
3. The **Change PSE Password** screen is displayed.
4. Enter the current password and the new password twice as indicated, and click **OK**.
5. The system changes the PSE password and displays the **PSE password change successful** screen.
6. Click **OK** to return to the USO Client **Main Screen**.

Logout

This sub-menu will trigger logout from the USO Server or local PSE assuming the user has already logged in to the USO Server or local PSE.

To logout from the USO Server or local PSE, user can proceed with the following steps:

1. Click on the **File** menu and then select the **Logout** sub-menu. Alternatively, if USO Client is minimized, user can right-click the USO Client icon on the icon bar, and select the **Logout** option.
2. The **logout confirmation** dialog box is displayed.
3. Click **Yes** to confirm to log out from the USO Server or local PSE.



Note: If user is logged in to the USO Server via a web browser, the user may click on the Logout hyperlink at the top right corner to logout from the USO Server.

4. System will log the user out and depending on the system setting, it may close the USO-related browser or redirect user back to USO **Login** page.



Note:

If the **Idle Timeout Prompt Time** attribute is set, the USO Client will alert the user of when the USO Client session will expire and prompt for user's action. For example, if the attribute is set to 5 minutes, USO Client will alert the user 5 minutes before the session timeout. Select **Yes** to continue

by refreshing the current session, or **No** to continue without refreshing the current session until timed-out and get terminated.

Refer to [View USO Client Configuration Attributes](#) for the detailed explanation on the **Idle Timeout Prompt Time** attribute.

Exit

This sub-menu will trigger exit from the USO Client.

To exit from the USO Client completely, perform the following steps:

1. Click the **File** menu option, and select the **Exit** sub-menu option. Alternatively, if USO Client is minimized, user can right-click the USO Client icon on the icon bar, and select the **Exit** option.
2. The **confirmation** dialog box for **Exit** is displayed.
3. Click **Yes** to confirm exit from the USO Client.

USO Client Settings Menu

The USO Client **Settings** menu provides the **Globally Disable USO Client**, **Enable Debug Log**, **Language** and **Java Runtime Settings** sub-menu options.

Globally Disable USO Client

This sub-menu will allow user to specify whether to allow USO Client to perform SSO for all applications in a global manner. Users can enable/disable USO Client anytime globally. By default, USO Client is enabled.

To switch between enabling and disabling USO Client, check or un-check the **Globally Disable USO Client** option from the **Settings** menu. Alternatively, if USO Client is minimized, user can right-click the USO Client icon on the icon bar, and select **Enable** or **Disable** to globally enable or disable USO Client respectively. USO Client will indicate that it has been globally disabled from the USO Client icon.



Notes:

- USO Client will remember the previous setting. Thus, logging in and out from USO Client will not alter the setting. Therefore, user needs to change the setting manually.
 - Exiting from USO Client will reset the settings to the default settings.
-

Enable Debug Log

This sub-menu will enable/disable the debug log.

If this sub-menu option is enabled (checked), USO Client will log debugging information to local log file, otherwise, USO Client will not log any information. By default, this option is disabled.

To enable/disable this sub-menu option, follow the steps below:

1. Click on the **Settings** menu option on the menu bar.
2. If the **Enable Debug Log** sub-menu is not currently enabled, i.e. not checked, select the option to enable it.
3. A **confirmation message** dialog will be displayed.
4. Click the **Yes** button to run USO Client in debug mode.



Note: If this option is enabled, the confirmation message dialog will appear each time the USO Client starts up.

5. If the **Enable Debug Log** sub-menu is currently enabled, i.e. checked, deselect the option to disable it.



Note: The change takes effect immediately without the need to restart USO Client.

Language

This sub-menu will display the list of languages supported by USO Client.

By default, the following languages are supported:

- English
- Simplified Chinese
- Traditional Chinese
- Japanese
- Thai

To change the language selection:

1. Click on the **Settings** menu option on the menu bar.
 2. Select the desired language from the **Language** sub-menu.
-

-
3. User will be prompted to restart USO Client for the change of language to take effect.
 4. Click **OK** and restart the USO Client.



Note: The selected language must also be supported by the operating system.

Java Runtime Settings

While performing screen matching for Java-based application screen, USO Client needs to communicate with the Java runtime environment (JRE) used by the application. If the path of the JRE used by the application is not within the JRE list obtained from the Windows registry, USO Client will fail to communicate in matching the application screens. To resolve this problem, user is required to specify the correct path of JRE used by the application to allow USO Client to communicate properly with the application in order to match the application screens.

Thus, user is provided an option to specify the additional path of the JRE used by Java-based application as below:

1. Click on the **Settings** menu option on the menu bar.
2. Select the **Java Runtime Settings** sub-menu. The **Java Runtime Setting** dialog will be displayed.
3. Click **Browse** to display the **Browser for Folder** dialog.
4. Browse the desired JRE directory and click **OK**.

The desired JRE directory will be updated to the JRE list accordingly.

5. Click **Save** and restart USO Client.

USO Client Help Menu

The USO Client **Help** menu provides **Show Log** and **About** sub-menu options.

Show Log

This sub-menu will display the USO Client log file, which stores the process and status log information of the USO Client. It is catered mainly for system administrator to monitor and troubleshoot the USO Client activities.

To view the USO Client log file, proceed with the following steps:

1. Start USO Client if it has not already started.
-

-
2. Click on the **Help** menu option and select **Show Log** sub-menu option, or simply press F10 key anytime whenever USO Client main window is in focus.
 3. The USO Client log file will be opened in the user-preferred text editor.
 4. To close the USO Client log file, simply exit the file.

**Notes:**

- USO Client logs are stored in a text file saved in the user's local workstation.
- The USO Client log file is located here:
C:\<User_Profile_Directory>\AppData\Roaming\USOClient\log\usoclient.log
- Depending on the local computer's settings, user may be prompted to select the desired application for opening the USO Client log file.

About

This sub-menu will display the **About USO Client** screen.

The **About USO Client** screen provides information on USO Client version number and edition, product copyright information, as well as product's company information.

To view the USO Client version information installed on the user's workstation, proceed with the following steps:

1. Start USO Client if it has not already started.
2. Click on the **Help** menu option and select the **About sub-menu** option, or simply press F1 key anytime whenever USO Client main window is in focus.

Toolbar

This section displays the **Login, PSE Login, Logout** and **Exit** toolbar icons.

- **Login** - It is similar to **Login** to Server sub-menu as discussed in [Login to Server](#).
- **PSE Login** - It is similar to **Login** to Local PSE sub-menu as discussed in [Login to Local PSE](#).
- **Logout** - It is similar to **Logout** sub-menu as discussed in [Logout](#).
- **Exit** - It is similar to **Exit** sub-menu as discussed in [Exit](#).

User Information

This section displays the User Name and the login status of the logged-in user.

Login status has 3 possible values:

- **Server + Local**

This value is displayed if user logs in to USO server while **Enable PSE value** is set to "Yes" and **PTD Type** is "Software PTD".

- **Server Only**

This value is displayed if user logs in to USO server while **Enable PSE** is set to "No".

- **Local**

This value is displayed if user logs in to local PSE.

Application List

This section displays the list of applications that the user is authorized to access. The user can perform SSO to the desired application by double-clicking on the application name. Refer to [Manipulating USO Client - Application Settings](#) section for more details.

5.3. Using the USO Client

Performing USO Primary Sign-On

This chapter describes the steps to performing Primary Sign-On to USO system via USO Client.

After starting the USO Client, users are provided with two modes of login to the USO system via the USO Client, i.e.:

- USO Server Login (Online Mode)
- Local PSE Login (Offline Mode)

The USO login modes mentioned above are referred to as USO Primary Sign-On.

Upon successful USO login, the list of applications that the user is authorized to access will be displayed in the USO Client. User will then be able to perform SSO to the desired application. This is referred to as USO Secondary Sign-On.

Performing USO Server Login (Online Mode)

In online mode, users will perform primary sign-on to the USO Server in an online manner. Users may login to the USO Server either via the USO Client or via a web browser.

Login to USO Server via Web Browser

To login to the USO Server via a Web browser, proceed with the following steps:

1. Start a web browser and enter the URL address to access the USO Web UI.
2. Upon launching of the USO URL, the page will be redirected to the Common Login Page (CLP) to handle user login.
 - If the **Default authentication for USO users** attribute is set with a realm, e.g. "40151", then user will not be prompted to select an authentication realm. Instead, the **USO Login** page is automatically displayed and the assigned realm is used for login.
 - If the **Default authentication for USO users** attribute is empty, then user will be prompted to select the authentication realm from the list of realms configured in CLP.



Note: An error message will be displayed if no authentication realm is configured in CLP.

-
3. Enter the valid **USO User ID** and **password**, and click on the **Login** button from CLP's **USO Login** page.
 4. Upon successful login, system will display the list of USO applications that the user is authorized to access.

Depending on the USO Client Configuration settings, user may be prompted to set PSE password and/or password-protected screen saver.

i **Note:** Only USO applications will be displayed on the USO Client list.

Login to USO Server via USO Client

To login to the USO Server online via the USO Client, proceed with the following steps:

1. Start the USO Client if it has not already started.
2. Click the **Login** icon from the toolbar. Alternatively, click **File > Login to Server**. If USO Client is minimized, right-click the USO Client icon on the Windows taskbar, and select **Login**.
3. Upon launching of the USO URL, the page will be redirected to the Common Login Page (CLP) to handle user login, unless it's the first time login to the USO server.

i **Note:** If logging in for the first time (i.e. USO Web UI is not launched yet and USO properties are not loaded yet), launching USO Web UI will fail and display an error stating to enter the USO Web UI URL address. Therefore, it is a must to launch and login USO Web UI first to load the USO properties to registry before using USO Client for login.

- If the **Default authentication for USO users** attribute is set with a realm, e.g. "40151", then the user will not be prompted to select an authentication realm. Instead, the **USO Login** page will be automatically displayed and use the assigned realm for login.
- If the **Default authentication for USO users** attribute is empty, then user will be prompted to select the authentication realm from the list of realms configured in CLP.

i **Note:** An error message will be displayed if no authentication realm is configured in CLP.

4. Enter a valid USO **User ID** and **password**, and click on the **Login** button from CLP's **USO Login** page.
 5. Upon successful login, both the SSO Portal and the USO client will be displayed. A list of USO applications that the user is authorized to access will be shown. Depending on the USO Client Configuration settings, user may be prompted to set PSE password and/or password-protected screen saver.
-

i **Note:** If the **Close SSO Portal When USO Client Active** attribute has been set to "Yes", the SSO Portal will be closed upon successful login and only the USO client will remain displayed. As only USO applications (e.g. ESSO applications and native applications) are displayed in the USO client, federated applications (e.g. SAML and OIDC) will not be displayed. For more details about this attribute, refer to [Administer USO Client Configuration - Close SSO Portal When USO Client Active](#).

Performing Local PSE Login (Offline Mode)

While in offline mode, where user has no access to the network, user can perform a PSE Login locally. Changes administered while offline will take effect in USO server once user is connected to the network.

Creating PSE File

- **Creating Password-Protected PSE File**

In order to enable a password-protected local PSE meant for login under offline mode, the USO Client configuration attributes related to PSE must be configured accordingly:

- **Enable PSE** should be set as "Yes"
 - **PTD Type** is set to "4 (Server PTD)"
 - **PSE File Option** is either set to "2" or "3"
 - **Realm for PTDType** is set with "Server" value
1. After the attributes are configured, upon successful USO server login, if user's PSE file has not been created before and has not previously specified any PSE password, then user will be prompted to set a PSE password.
 2. Enter the **PSE password** and click **OK** to allow the system to create the local PSE file accordingly.
 3. For subsequent USO server logins, user will be logged in to the local PSE simultaneously. Since the PSE contains application login information, user will be able to perform local PSE login, i.e. offline mode.

i **Note:** This setting will allow any user who has their own PSE file on the machine to do PSE login.

- **Creating Non-Password-Protected PSE File**

In order to enable a non-password-protected local PSE meant for speeding-up primary login process, the USO Client Configuration attributes related to PSE must be configured accordingly:

- **Enable PSE** should be set as "NoPwd"
 - **PTD Type** should be set as "Auto PTD"
-

After the attributes are configured, upon successful login to USO server, if user's PSE file has not been created before, then the system will create the local PSE file accordingly.



Note: Since the PSE does not contain any application login information, user will not be able to perform local PSE login, i.e. offline mode.

Performing Local PSE Login

In offline mode where the relevant password-protected PSE file has been created as described in [Creating PSE File](#), users will perform primary sign-on to the local PSE in an offline manner. This scenario is applicable especially when server network connection is not available where user can only login to the local PSE via the USO Client.

To login to the local PSE, follow the steps below:

1. Start the USO Client if it has not already started.
2. Click the **PSE Login** icon from the toolbar. Alternatively, click the **Login to Local PSE** sub-menu option from the **File** menu.
3. Enter the PSE password and click the **Login** button.
4. Upon successful login, system will display the list of USO applications that the user is authorized to access and notice that the login status is only showing **Local**.

User is now ready to perform USO i.e. Secondary Sign-On to the applications as described in [Performing USO Secondary Sign-On](#).

Performing USO Secondary Sign-On

This chapter describes the steps to perform Secondary Sign-On to USO applications.

Performing Secondary Login to USO Application

This section describes the scenario of performing secondary login to an USO application.

To login to the desired application, perform the following steps:

1. Launch the desired application by double-clicking on it from the application list or by launching the application directly, for example, running the application executable for a desktop application or directly accessing to the login page for a Web application, etc.

i **Note:** For this option to function properly, the application should be trained with **Process External Launch** attribute set to "Yes". Refer to [Configure USO Application](#) for detailed explanation.

2. USO Client will launch the application and match the application login screen, and automatically login to the application using the login information stored in the USO system.

i **Note:** If USO Client is unable to locate the application path, it will prompt the user to specify the correct application path. Click on the **Browse** button to specify the correct application path.

3. If there is no login information available (e.g. the user is accessing the application for the first time), USO Client will display the application initialization screen to obtain the login information from the user.
4. Enter a valid application user Id and password, as well as the user-defined attribute for the particular application, if applicable.
5. Click **OK** and the system will update the login information to the USO Server as well as to the local PSE (if PSE is enabled).

i **Note:** Once the login information for the particular application has been updated to the USO Server and local PSE, user will not be prompted to enter the login information again for the subsequent logins to that particular application.

USO Application Login with Re-Authentication

Depending on the screen processing options selected during application training (refer to [Configuring Application for Re-authentication](#) for detailed discussion on screen processing options), for applications trained with re-authentication option, user will be required to provide primary login password for re-authentication upon matching the application login screen as described below.

- **Windows Domain Authentication (DOMAIN)**

For application trained with such re-authentication method, USO Client will prompt for user's Windows domain logon or Windows local logon password for re-authentication.

- **User-Defined**

For application trained with such re-authentication method, USO Client will prompt for user-defined authentication method that could be simply a password-based authentication method. This password-based authentication method must be the same as the desired **Login Module Id** currently assigned to the user in the AccessMatrix system, e.g. "SystemPasswordBasic".

On the USO Client **User Verification** screen, enter the password and click **Login** to proceed with the login process.

USO Application Login with Another/Second User Authentication

Depending on the screen processing options selected during application training (refer to [Configuring Application for Second User Authentication](#) for detailed discussion about screen processing options), for applications trained with another/second user authentication option, USO Client will prompt for the other or the second user authentication to perform primary sign-on to the server upon matching the application screen. This feature is meant for types of applications that requires a second user (e.g. supervisor) to enter the credential.

Perform the following steps to enable this feature:

1. Launch USO Trainer and train an application, e.g. DemoDesktop. Tick **Require Authentication by Another User** under the **Processing Option wizard** screen.
2. Export the ADF and import to Admin Console as **USO Application**.
3. Entitle the new imported application to 2 users, e.g. user1 and user2.
4. Input the user application attributes, e.g. **user Id**, **password** and **domain** to both users. Save the changes.
5. Login to USO Web UI using 1st user, e.g. user1, and launch USO Client or login directly to USO Client.
6. Launch the application. If the application requires a second user at any point during operation, the **Second USO User Sign-On** dialog will be prompted.
7. Enter the valid primary login information of the second user, i.e. user2, and submit to proceed with the authentication process. The application process will continue if the second user is successfully authenticated.



Notes:

- Secondary user login supports a password-based authentication realm.
- The authentication realm used is the same for both primary and secondary user.

USO Application Login Using Multiple Profiles Feature

For applications with multiple login profiles, USO Client may prompt user to select the appropriate application login profile upon matching the login screen.

Check the **Don't remember profile** option to allow USO Client to forget the last chosen profile and prompt for profile selection each time the application is launched. Refer to [Manage Multiple Application Login Profiles](#) for detailed description on managing multiple application login profiles.

Select the desired profile and click **OK** to proceed with the login process.

Changing USO Secondary Password

This chapter describes the steps to change USO Secondary Password, i.e. the application's password.

There are two ways to change application's password via USO Client, that is:

- Through manual application password change
- Through automatic application password change

Manual Application Password Change



Note: It is assumed that:

- The desired application is currently enabled.
- The application utilizes the "Manual" password option for the **Change Password Mode** attribute during training of the application.
- The user has already logged on to the desired application prior to accessing the application password change screen.

To change the desired application's password manually, perform the following steps:

1. Access the application password change screen.
2. USO Client will be able to detect and match the application password change screen, and prompts to collect the new password from the user.



Note: It is assumed that the application password change screen has been previously trained properly using USO Trainer.

3. User needs to enter a valid new application password twice accordingly, and click **OK**.
 4. USO Client will submit the new application password to the application.
-

Automatic Application Password Change



Note: For automatic application password change, the auto-generated password can conform to certain password policy, which is configurable during application training. Refer to [USO Application Training - Password Policy](#) for the detailed explanation of the password policy.

It is assumed that:

- The application password change screen has been previously trained properly using the USO Trainer.
- The desired application is currently enabled.
- The application utilizes the "Auto" password option for the **Change Password Mode** attribute during training of the application.
- The user has already logged on to the desired application as described in previous chapter prior to accessing the application password change screen.

To change the desired application password automatically, perform the following steps:

1. Access the application password change screen.
2. USO Client will be able to detect and match the application password change screen.
3. User will not be prompted to provide the new application password as USO Client will generate the new application password automatically and submit to the application.
4. The whole application password change process is performed seamlessly to the user.

Setting Password-Protected Screen Saver

In order to enable a password-protected screen saver, the USO Client Configuration attributes related to screen saver must be configured accordingly:

- **Force Screen Saver** should be set as "OS"
 - **Screen Saver Timeout** should be set to the desired timeout value
1. After configuring the attributes, upon successful USO server login, if user workstation's screen saver has not been enabled or is without password-protection, USO Client will be able to detect it and force the user to set password-protect screen saver.
 2. Click on the **Set Screen Saver** button to set a password-protected screen saver.
 3. Select the desired screen saver.
-

-
4. Set the **Wait** period on **Screen Saver Settings** dialog.



Note: For the screen saver timeout, USO Client will enforce that the time set on user's workstation is shorter than the timeout set in the policy.

5. Check the **On resume, display logon screen** checkbox.
6. Click **Apply** on **Screen Saver Settings** dialog.
7. Click **OK** on USO Client's **Set Screen Saver** dialog.

Manipulating USO Client - Application Settings

This chapter describes the various application settings provided by USO Client.

Enabling/Disabling Application

User is provided a feature to specify whether to allow USO Client to match and fill in the application's information automatically. If the selected application is enabled, USO Client will fill in and submit the login information automatically for the selected application.

This feature provides the following flexibilities for disabling the application:

- **Disable Application for Current Session**

This allows the user to disable the selected application temporarily, i.e. only for current logged-in session.

- **Disable Application Permanently**

This allows the user to disable the selected application persistently, i.e. for current and subsequent logged-in sessions.

User is able to enable/disable the application anytime as long as the user has been authorized to do so. Refer to [Administer USO Client Configuration](#) for further details.

By default, all the applications are enabled. Administrator is given an option on whether to disable the application for the user when performing USO entitlement during user administration. Refer to [Configure User USO Entitlement](#) for further details.

Assuming that user is allowed to disable applications, and the entitled USO application is enabled by default, to enable/disable the application, perform the following steps:

1. Right-click on the desired application.
-

-
2. To disable application for current session, select the **Disable Application for Current Session** option.
 - If the selected application is currently disabled, a cross sign **[X]** will be displayed next to the application icon and the application will be dimmed (grayed out) to indicate that the application has been disabled.
 - To enable the application, right-click on the application and uncheck **Disable Application for Current Session** option.
 3. To disable application for current, as well as subsequent sessions, select the **Disable Application Permanently** option.
 - Similar to the settings for **Current Session**, if the selected application is disabled permanently for all sessions, a cross sign **[X]** will be displayed next to the application icon and the application will be dimmed (grayed out) to indicate that the application has been disabled.
 - Take note that the **Disable Application for Current Session** option will also be checked and dimmed at the same time.
 - To enable the application, right-click on the application and uncheck **Disable Application Permanently** option.

Sign-Off Application

The feature of **Signoff Application** enables USO Client to automatically sign-off all the applications that it processes when user logs out from USO Client. User is also allowed to manually signoff the application.



Notes:

- The **Enable Single Signoff** attribute under USO Client Configuration via Administration Console must be set as "Yes" for the Single Sign-off function to work properly. Refer to [Administer USO Client Configuration](#) for the detailed discussion about the USO Client Configuration.
- The **Signoff Application** option will only be activated when the application is trained with the **Single Sign-off** option enabled and set to **Graceful**, **Graceful Then Force**, or **Force**. Refer to [USO Application Training - Single Sign-Off](#) for further explanation of the **Sign off** parameter.

To manually signoff the application, proceed with the following steps:

-
1. Right-click on the application to be signed-off and select **Signoff Application**.
 2. All the instances of the application that were processed by USO Client will be signed-off.

Continue Matching Application

USO Client may stop matching the application even though it matches the screens trained for the application. This happens when the application is trained such that USO Client stops matching the application it processed.

The **Continue Matching Application** function allows USO Client to continue on matching the application when the above occurs.

- i** **Note:** The Continue Matching option will only be activated when the application is trained with the following conditions:
 - The **Process the application launched outside of USO Client** option is checked while training the application. Refer to [USO Application Training - Application Launching](#) for further explanation on this parameter. This option is equivalent to the **Process external launch** attribute in USO Application. Refer to [Configure USO Application](#) for more details.
 - The screen's action script contains the system-defined action: `_StopMatching()`. Refer to [USO Application Training - Custom Behavior](#) for the detailed discussion on the action scripts.

User has to manually activate the above-mentioned function in USO Client by following steps below:

1. Right-click on the application to be matched and select the **Continue Matching** option.
2. USO Client will continue matching the application and act accordingly again.

Reset Application Login Information

This feature to reset application login information allows users to reset the selected application login information, if the login information stored in the USO server's security registry is not synchronized accordingly.

To reset application login information, perform the following:

1. Login to the USO Server if not already logged in.
 2. Right-click on the desired application and select the **Reset App Info** option.
 3. The Set Application Information window is displayed.
-

-
4. Make the necessary changes to the login information and click **OK**.

Manage Multiple Application Login Profiles

USO Client provides a feature that allows users to define multiple login profiles for an application. When an application is defined with multiple login profiles, USO Client will indicate it in an expandable tree form with a **[+]** sign before the application.



Note: Application without the prefixing **[+]** sign indicates the application contains only one login profile.

Adding New Application Login Profile

To add a new login profile for an application, perform the following steps:

1. Right-click on the desired application and select the **Add Profile** option. A new login profile will be added to the application.
2. User may either accept the default name or specify a new name for the newly added profile and press Enter.

USO Client will update the changes to both the server and PSE. If the update is successful, "USO server/PSE updated successfully." message will be displayed on USO Client's status bar.

User is allowed to manage the application login profiles themselves.



Note: For applications under USO Application group, creating a profile to one application will automatically add a profile to another application within the same application group.

Viewing Multiple Application Login Profiles

To view the list of login profiles of an application, perform the following steps:

1. Click on the prefixing **[+]** sign of the desired application to view the list of login profiles defined for the application.
2. Click on the prefixing **[-]** sign to hide the list of login profiles.

Setting Active Application Login Profile

User is allowed to change the current active login profile for the desired application.

To set an login profile active, perform the following steps:

-
1. Click on the prefixing **[+]** sign of the desired application to view the list of login profiles.
 2. Right-click on the desired profile and select the **Set Active** option to activate the selected profile. If the activation is successful, "USO server/PSE updated successfully." message will be displayed on USO Client's status bar and the profile will be highlighted.

**Note:**

- The inactive profile(s) will be dimmed (grayed out).
- USO Client will remember the last selected profile if the **Don't Remember Profile** option is unchecked in **Set Default Profile** dialog. Refer to [USO Application Login Using Multiple Profiles Feature](#) for related information.

Renaming Application Login Profile

To rename a login profile, perform the following steps:

1. Click on the prefixing **[+]** sign of the desired application to view the list of login profiles.
2. Select the login profile to be renamed, right-click and select **Rename Profile**. Alternatively, select to highlight the desired profile and click once on the highlighted profile.
3. Type the desired name for the selected profile and press Enter to apply the change. If the update is successful, "USO server/PSE updated successfully." message will be displayed on the USO Client status bar.



Note: For applications under USO Application group, renaming a profile under one application will automatically rename the equivalent profile under another application within the same application group.

Deleting Application Login Profile

To delete a login profile, perform the following steps:

1. Click on the prefixing **[+]** sign of the desired application to view the list of login profiles.
2. Select the login profile to be deleted, right-click and select **Delete Profile**. If it is deleted successfully, "USO server/PSE updated successfully." message will be displayed on the USO Client status bar.



Note: For applications under USO Application group, deleting a profile under one application will automatically delete the equivalent profile under another application within the same application group.

Resetting Application Login Information for a Login Profile

Resetting application login information for a login profile is similar to resetting application login information mentioned in [Reset Application Login Information](#).

To reset application login information for a particular login profile, perform the following steps:

1. Click on the prefixing **[+]** sign of the desired application to view the list of profiles.
2. Right-click on the desired profile to be reset and select **Reset App Info**. The **Set Application Information** window appears.
3. Enter the information accordingly and click **OK** to save the changes. If it is reset successfully, "USO server/PSE updated successfully." message will be displayed on the USO Client status bar.



Note: For applications under USO Application group, resetting application information of a profile under one application will automatically update the same information to the equivalent profile under another application within the same application group.

Reset Login Profile Selection

The feature to reset application login profile selection is only applicable for applications that possess multiple application login profiles as described in [Manage Multiple Application Login Profiles](#).

When the login profile selection is reset, the application will not have any default login profile, in other words, there will be no active login profile selected. Thus, user will be prompted to select the login profile when USO is in process.

To reset login profile selection, perform the following steps:

1. Login to the USO Server if not already logged in.
2. Right-click on the desired application and select the **Reset Profile** option. If it is reset successfully, "USO server/PSE updated successfully." message will be displayed on the USO Client status bar.

Change Application Path

The feature to change application path allows user to modify the application path information if the current settings are no longer valid or there is a change in the application path.

To change the application path information, perform the following steps:

-
1. Login to the USO Server if not already logged in.
 2. Right-click on the desired application and select the **Change App Path** option. The **Set Application Path** window is displayed.
 3. Click the **Browse** button next to the **Application Path** input field to specify the application's executable file.
 4. Click the **Browse** button next to the **Working Directory** input field to specify the working directory of the application.
 5. If applicable, specify the application argument on the **Application Argument** input field.
If necessary, user may check the **Reset back to default application path** option to reset all the application path information according to the path information stored in the server.
 6. Click **OK** to save the changes.

Show Application State History

The feature of Show State History allows user to view various actions that USO Client has performed on an application, by showing the previous as well as the current state of the application, as trained.

To view the state history of a particular application, perform the following steps:

1. Login to the USO Server if not already logged in.
2. Launch an application. USO Client will try to match and fill in the login information of the application, and at the same time record down the change of state information.
3. Right-click on the desired application and select the **Show State History** option. The State History dialog is displayed.
4. Click **OK** to close the window.



Note: An empty state history will be shown when the application has not been launched or is closed.

Clear Application State History

The feature of **Clear Application State History** allows user to clear various actions that USO Client has performed on an application.

To clear the state history of a particular application, perform the following steps:

-
1. Login to the USO Server if not already logged in.
 2. Launch an application. USO Client will try to match and fill in the login information of the application, and at the same time record down the change of state information.
 3. Right-click on the desired application and select the **Clear State History** option.
A warning message will be displayed.

 **Important Note:** As stated in the warning message, this function must be used with caution since it will clear the state history of the application as well as other instances of the same application. Thus, this function should be used if and only if there is no other instance of the same application is running.

4. Click **OK** to clear the state history for the application.

Upon successful clearing of the state history for the application, an empty state history will be shown if user attempts to view the application state history as described in [Show Application State History](#).

View Application Properties

USO Client provides application properties to users as additional information of the application. This information includes the number of running instances, application id, application name, process id of the application, and current state of the application.

Perform the following steps to view the application properties:

1. Login to the USO Server if not already logged in.
2. Launch an application. USO Client will try to match and fill in the login information of the application, and at the same time record down the change of state information.
3. Right-click on the desired application and select the **Properties** option. The **Application Properties** window will be displayed.
4. Click **OK** to close the window.

 **Note:** Application properties is empty if there is no instance running for that particular application.

Allowing External DLL to Switch Profile

USO Client allows the application to switch profiles dynamically without asking the user. The profile to use is dynamically chosen by the third party components via external DLLs.

Perform the following steps to enable this feature:

1. An external DLL should be created to specify which profile to be used, e.g. **DLLTest.dll**. Place this DLL under USO Client installation directory, e.g. *C:\Program Files (x86)\USOClient*.
2. Train an application or load an existing ADF of an application, e.g. DemoDesktop.
3. Navigate to **Custom Behavior** and modify the application login screen custom behavior by adding the `_SwitchProfileByExternalDll` JavaScript function.

e.g.:

JavaScript

```
_SwitchProfileByExternalDll('DLLTest.dll')  
$ProcessLogin(0, '', '')
```

4. Save the changes and export the ADF.
5. In Admin Console, import the ADF file and add "%externDllProfileKey" as a custom attribute in USO Application. The value should be the shared secret which will be validated by the DLL.
6. Entitle the new imported application to the user.
7. Login to USO Web UI and launch USO Client. Create multiple profiles.
8. Launch the application. The profile that will be active is the one specified by the external DLL. It will auto login using the profile set by the external DLL.



Note: Switching of profile will not work for the following cases:

- External DLL profile application attribute, i.e. "%externDllProfileKey", does not exist or is empty.
- External DLL path specified does not exist.
- External DLL shared secret key is null or empty.
- Shared Secret keys between external DLL and USO server does not match.

Automatic Identification of Profile Based On Dynamic User Id

USO Client provides automatic identification of profile based on dynamic user Id. This will automatically create new profile and prompt the user for the new profile information if the user Id is not found under existing profiles.

Perform the following steps to enable this feature:

1. Launch USO Trainer and train a desktop/winapi application, e.g. DemoDesktop.
-

-
2. Train the **User Id** field as **Screen Identifier (Label or Static Text)**.
 3. Under **Trained Screen Objects**, double click **[Screen Identifier]** and input "%uidDynamic". Press the Enter key. The value will become "%uidDynamic [Screen Identifier]".
 4. Continue to train other fields (password, submit button, etc.) as normal.
 5. Export the ADF and import to Admin Console as **USO Application**.
 6. Entitle the new imported application to the user.
 7. Launch the application, e.g. DemoDesktop and input the user Id, e.g. user1.
 8. Login to USO Web UI and launch USO Client or login directly to USO Client.
 9. When USO Client matches the application, the user Id value manually entered to the application will be automatically created as profile and prompts the application info screen if the user Id is not found and is not an existing profile. Changing the user Id entered in the application, e.g. user2, will also auto create the profile and prompts the application info screen.

**Notes:**

- Only applicable for Desktop/WinAPI applications.
- Only works when connecting USO Client directly to AxMx server. This is not applicable for USO Trainer (test mode) due to multiple profiles restriction.

6. USO Client Mobile App User Guide

The USO Client Mobile App (iOS/Android) is a mobile app that supplements the USO Client web application. It is used to auto-fill the user name and password fields of applications. Unlike the web application version of the USO Client, the app version can only store up to one profile (that is, one set of login credentials) per application.

The following sections serve as a guide for the installation and usage of the USO Client Mobile App:

- [Installing the USO Client Mobile App](#)
- [Using the USO Client Mobile App](#)

6.1. Installing the USO Client Mobile App

iOS

The USO Client Mobile App (iOS) can be installed from the iOS App Store. To locate the app, search for "USO Client" using the App Store search function.

Android

The USO Client Mobile App (Android) can be installed from the Google Play Store. To locate the app, search for "USO Client" using the Play Store search function.

6.2. Using the USO Client Mobile App

Options

After starting the USOCClient mobile app, the login screen is displayed:

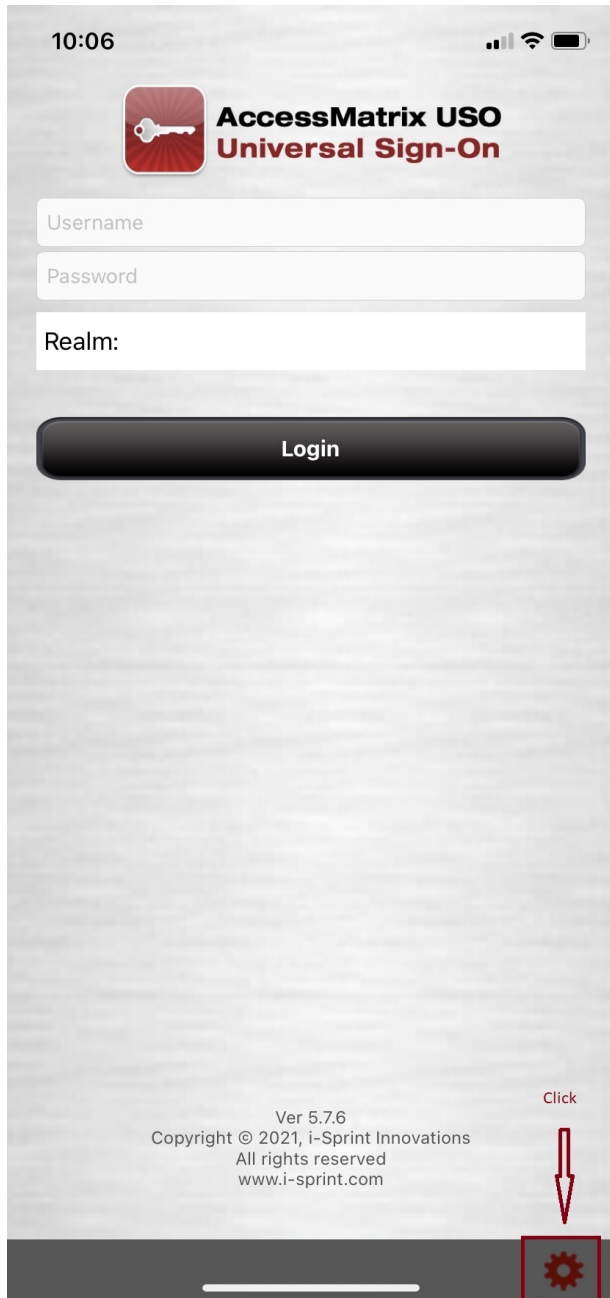


Figure1-1 Login

If you are starting the app for the first time after installing it, you must perform the following steps:

1. Press the **Options** icon in the lower-right corner to enter the **Options** screen.
2. Configure the following fields:

- a. **IP Address / Server Name:** Enter the IP address of the server (e.g. "192.168.40.52").
 - b. **Port** - Enter the port used (e.g. "8080").
 - c. **Enable SSL** - If you are using the https protocol, you must press the **Enable SSL** toggle switch to enable the use of SSL. Otherwise, disable it.
3. Press **Test Connection** to verify that the app can connect to the configured server.
 4. Press **Done** on the top right corner of the screen to save your configurations.

Note: Refer to **Figure 1-2** below for an example of the Options screen.

2:00

Cancel Options Done

SERVER INFORMATION

IP Address / Server Name:

192.168.40.52

Port: 8080

Enable SSL: ☐

Test Connection

Figure1-2 Options

Authentication Realm

After completing the configurations in the **Options** screen, the **Login** screen displays the authentication realm from the target server under the **Realm** field:

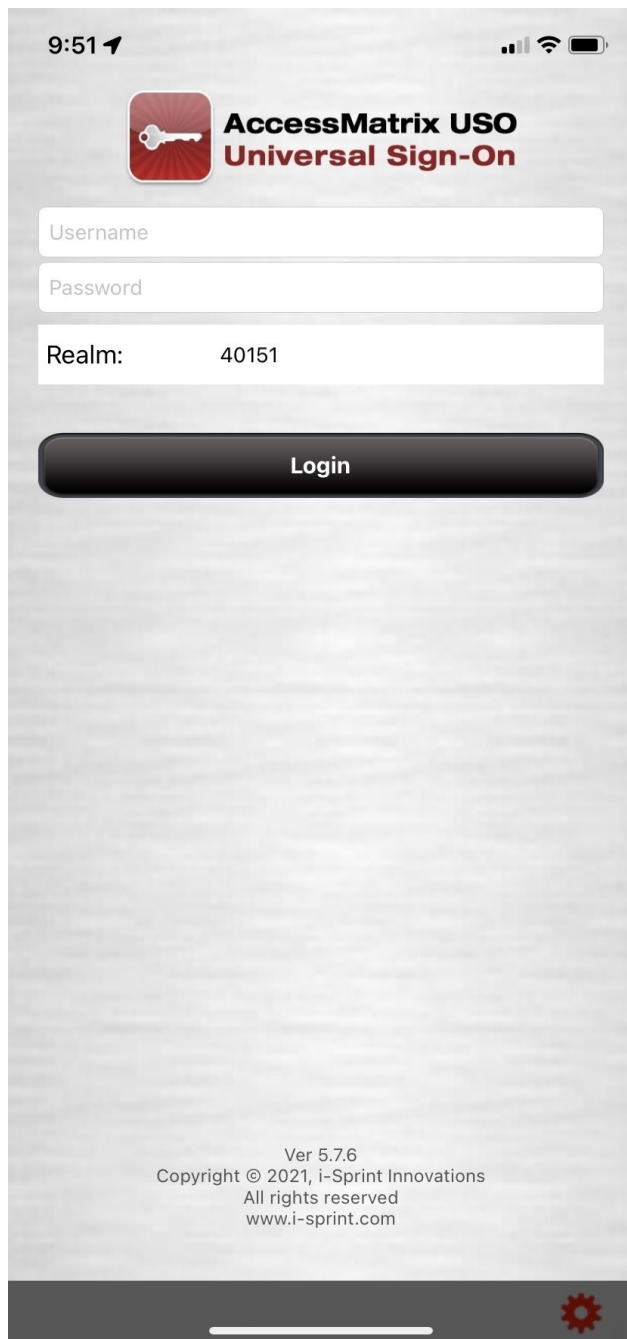
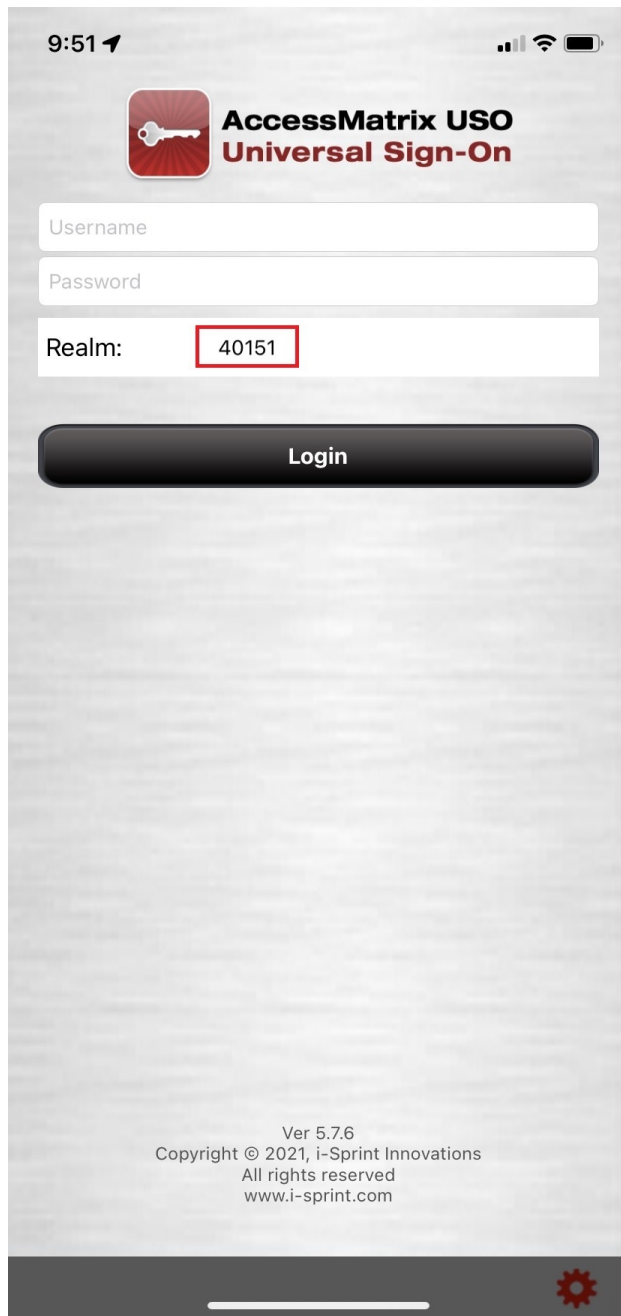


Figure 2-1 Authentication Realm

You can add new authentication realms if needed. To do so:

-
1. Press the realm Id (the area within the red box, pictured in **Figure 2-2**). A dialog box for adding an authentication realm pops up:



**Figure 2-2 Click
Authentication Realm**

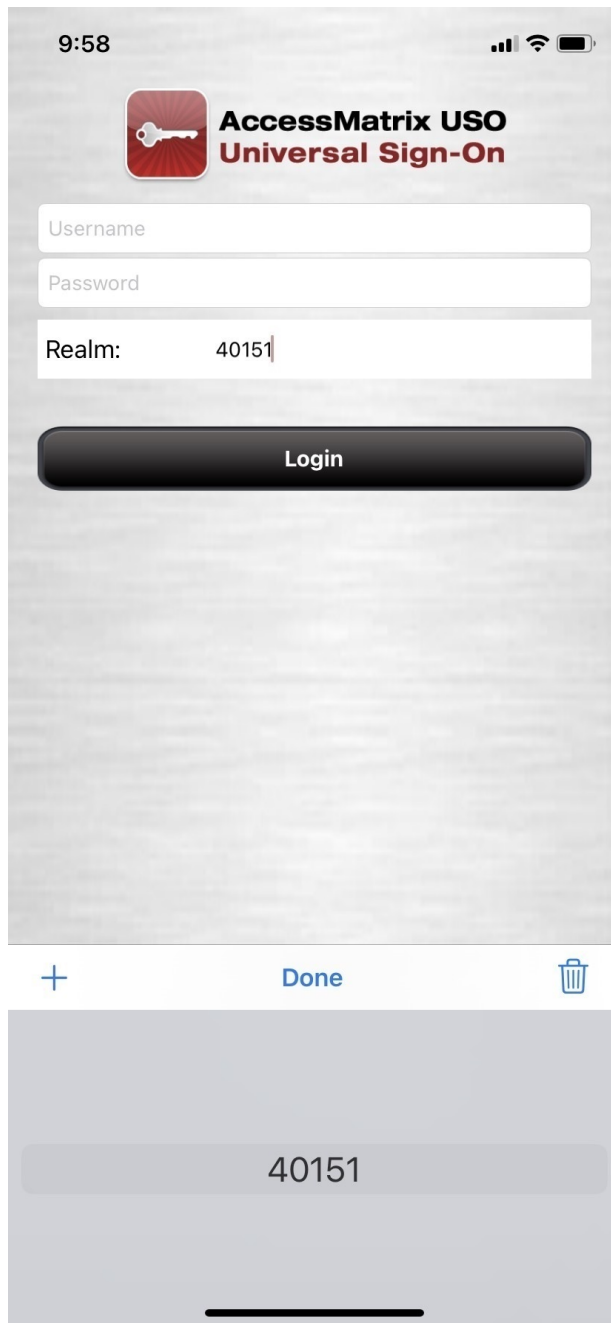


Figure 2-3
Authentication Realm
editor

2. Press the **[+]** icon in the upper left corner of the dialog box, enter the Id of the desired authentication realm and press the **Done** button on the on-screen keyboard:

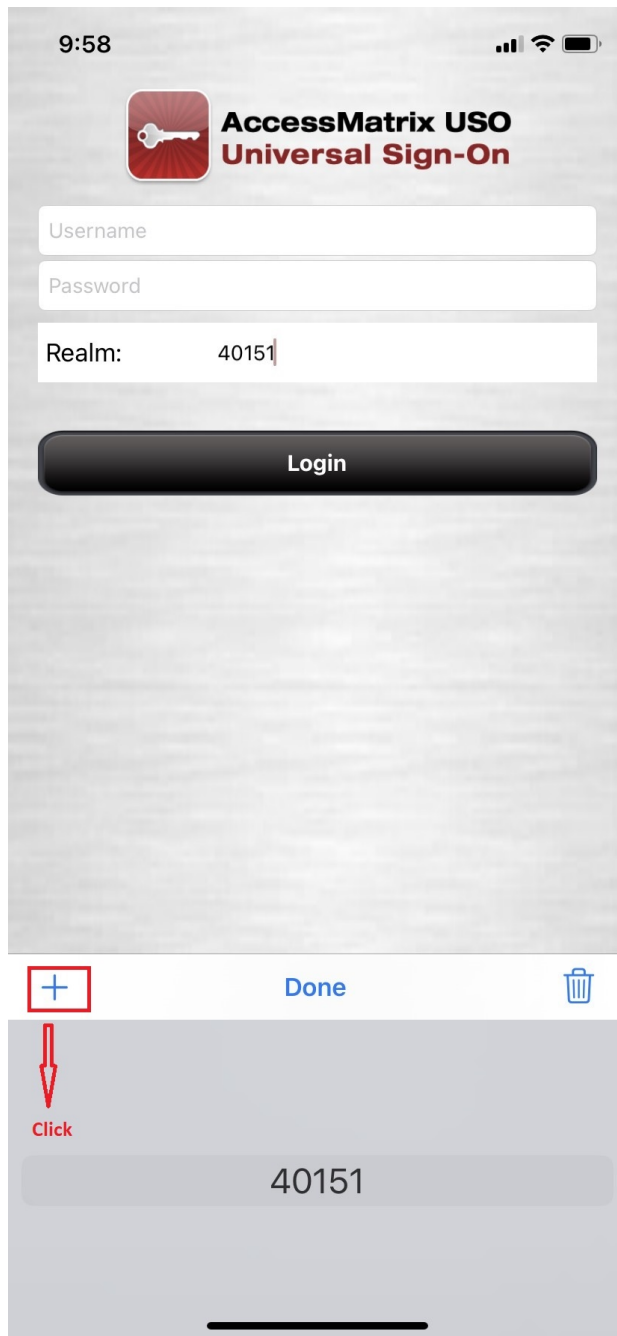


Figure 2-4 Add new Realm

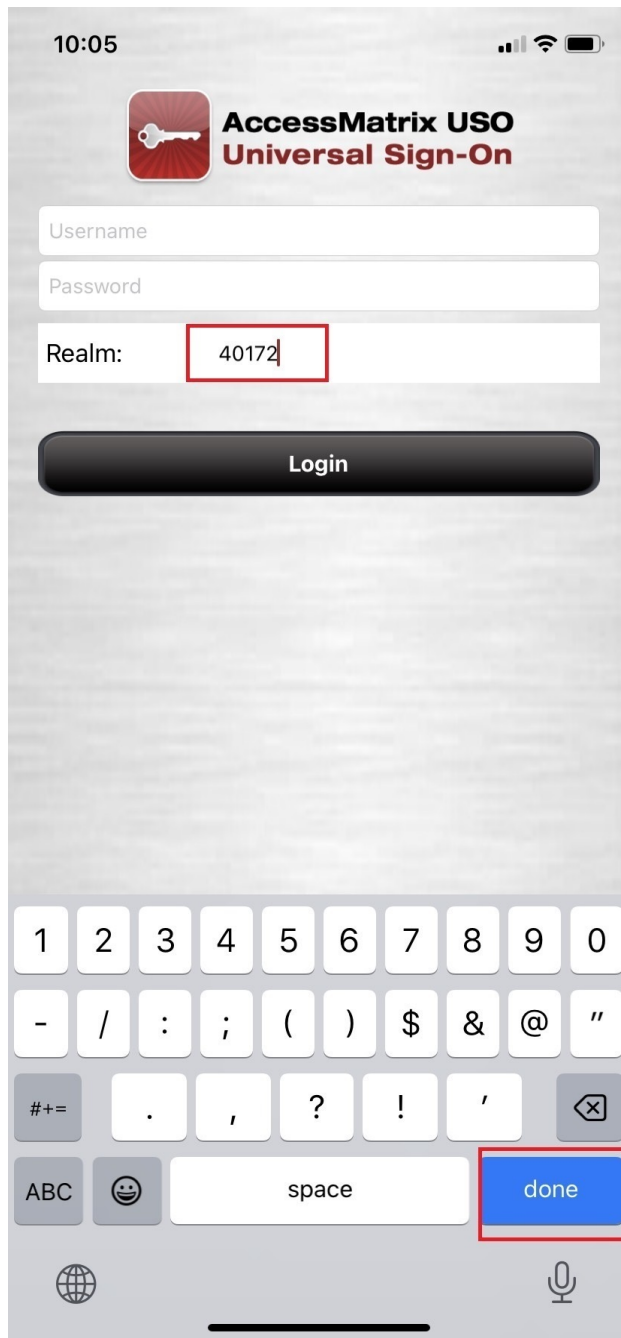


Figure 2-5 Add new Authentication Realm (40172)

After adding a new authentication realm, the newly added realm will be selected by default. To select a different authentication realm:

1. Press the realm Id. A scroll wheel containing the added authentication realms is displayed:

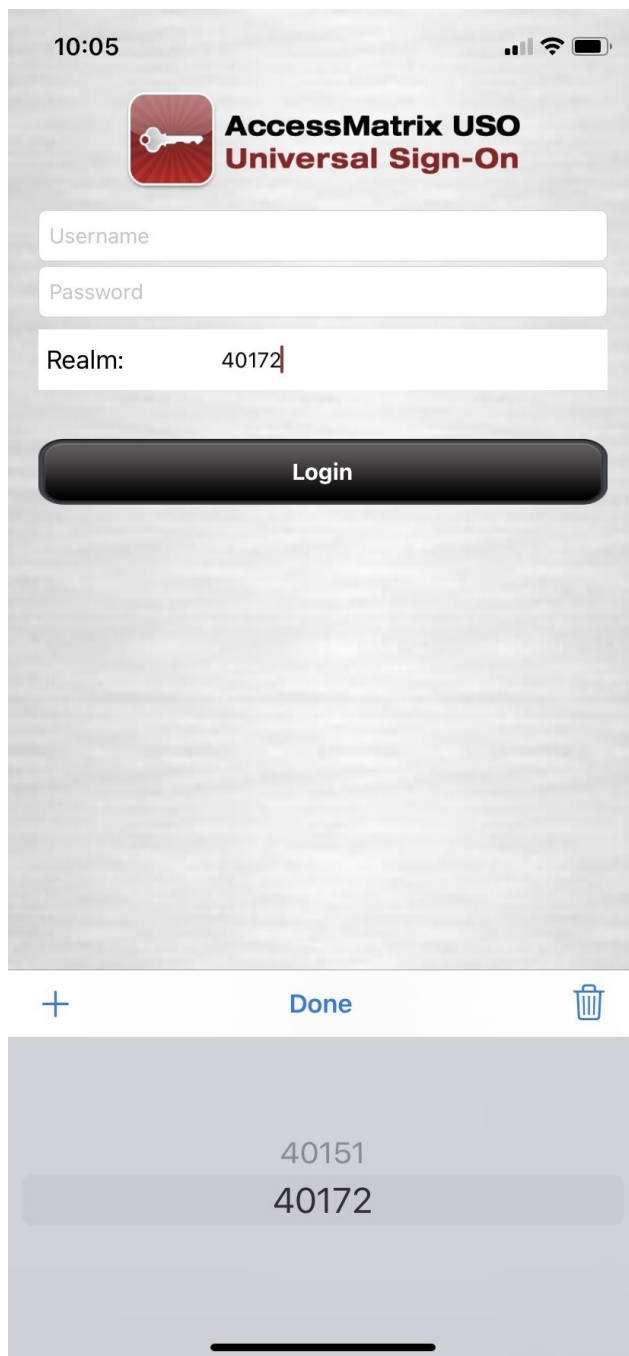


Figure 2-6 New Authentication Realm

-
2. Scroll up/down and press on the desired authentication realm:

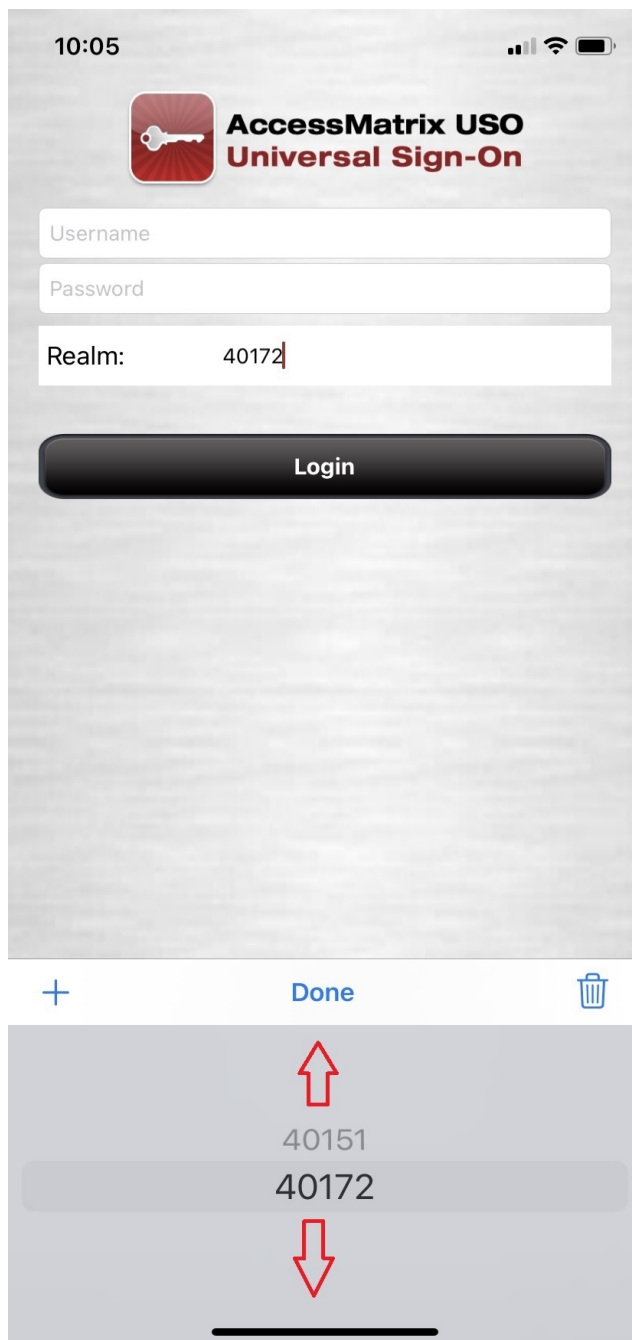


Figure 2-7 Select Authentication Realm

You can also delete unneeded authentication realms. To do so:

1. Press the realm Id. A scroll wheel containing the added authentication realms is displayed.

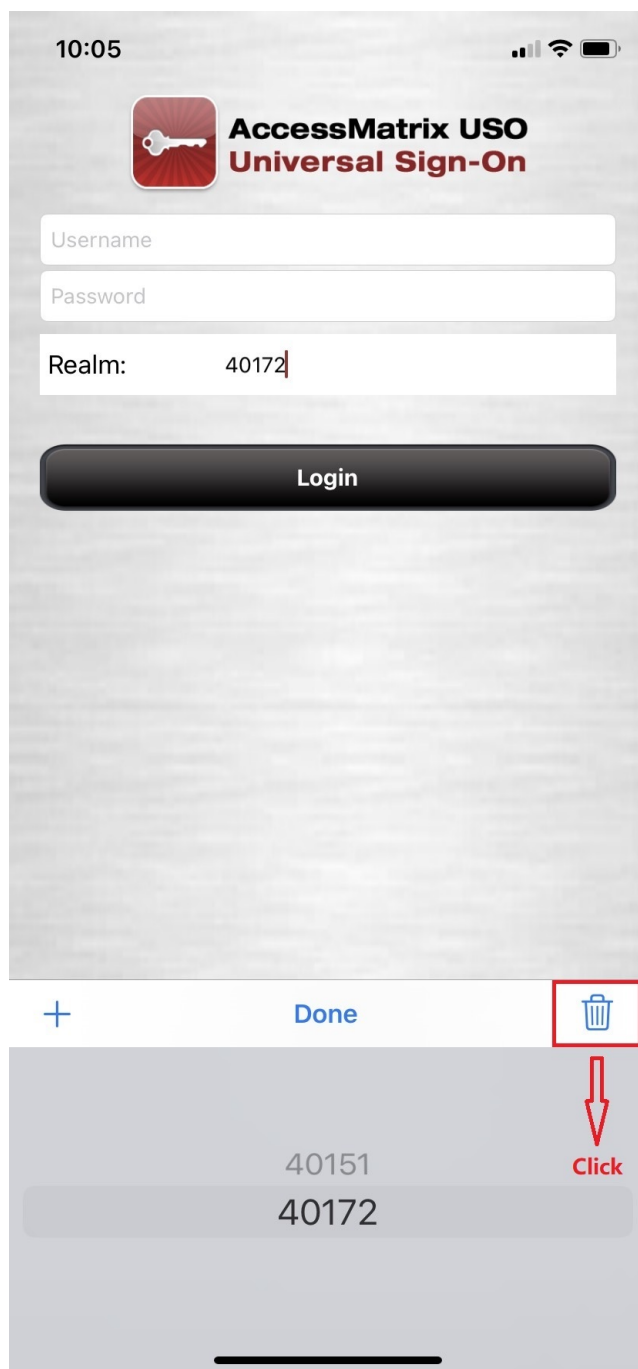
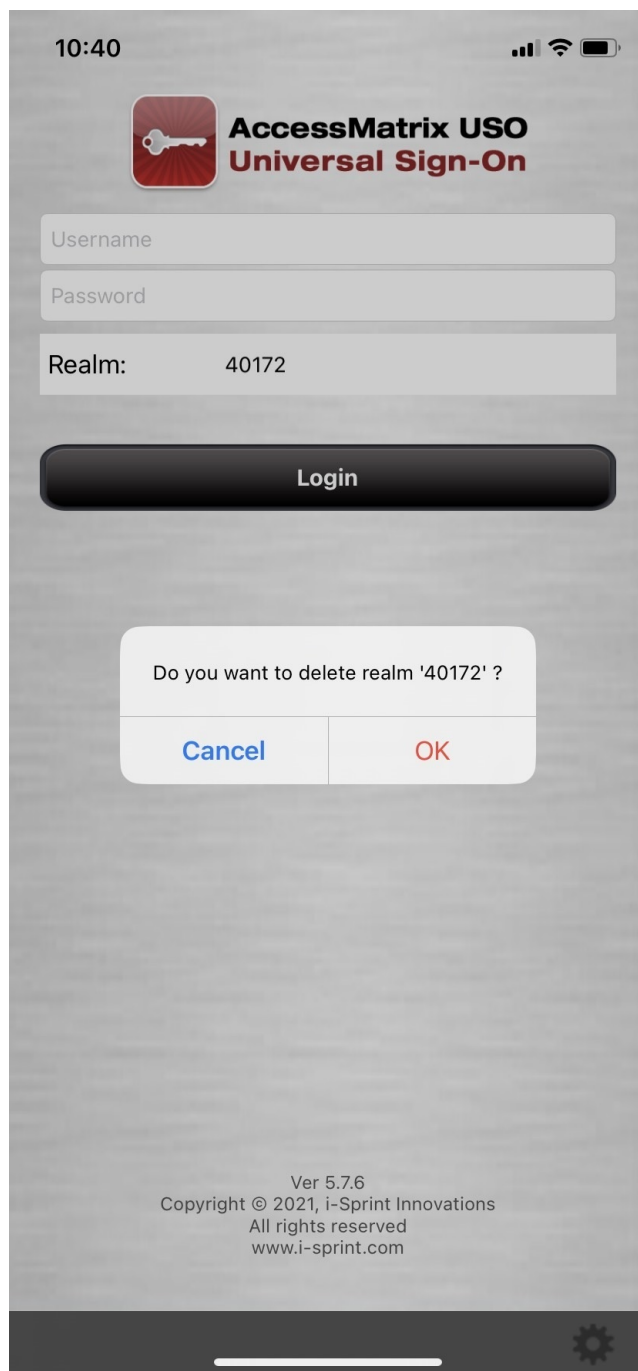


Figure 2-8 Click the delete icon

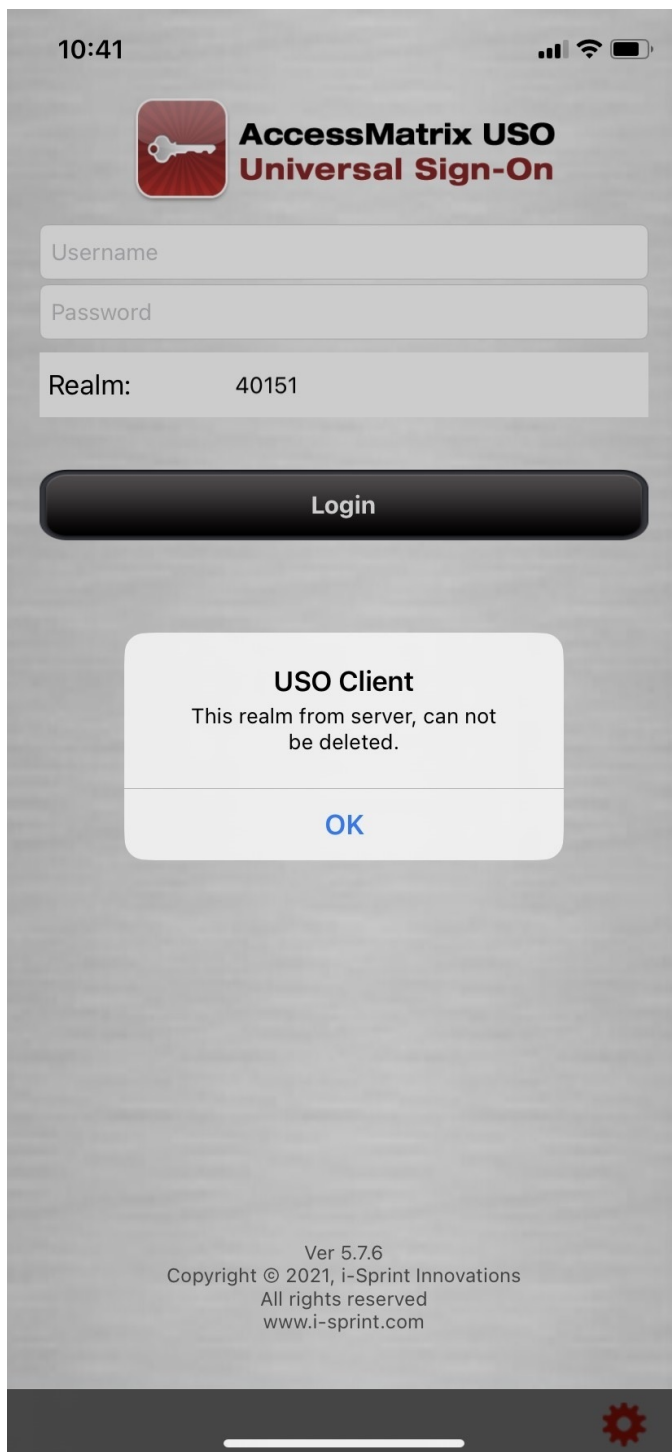
2. Scroll up/down to the authentication realm you want to delete and press the **Delete** icon in the upper right corner. A dialog box asking you to confirm the deletion is displayed:



**Figure 2-9 Delete
[40172] Authentication
Realm**

3. Press **OK** to delete the authentication realm.

i Note: The default authentication realm obtained from the server cannot be deleted (see **Figure 2-10**).



**Figure 2-10 Default Realm
(cannot delete)**

Login

When logging in for the first time, you will be prompted to change your password:

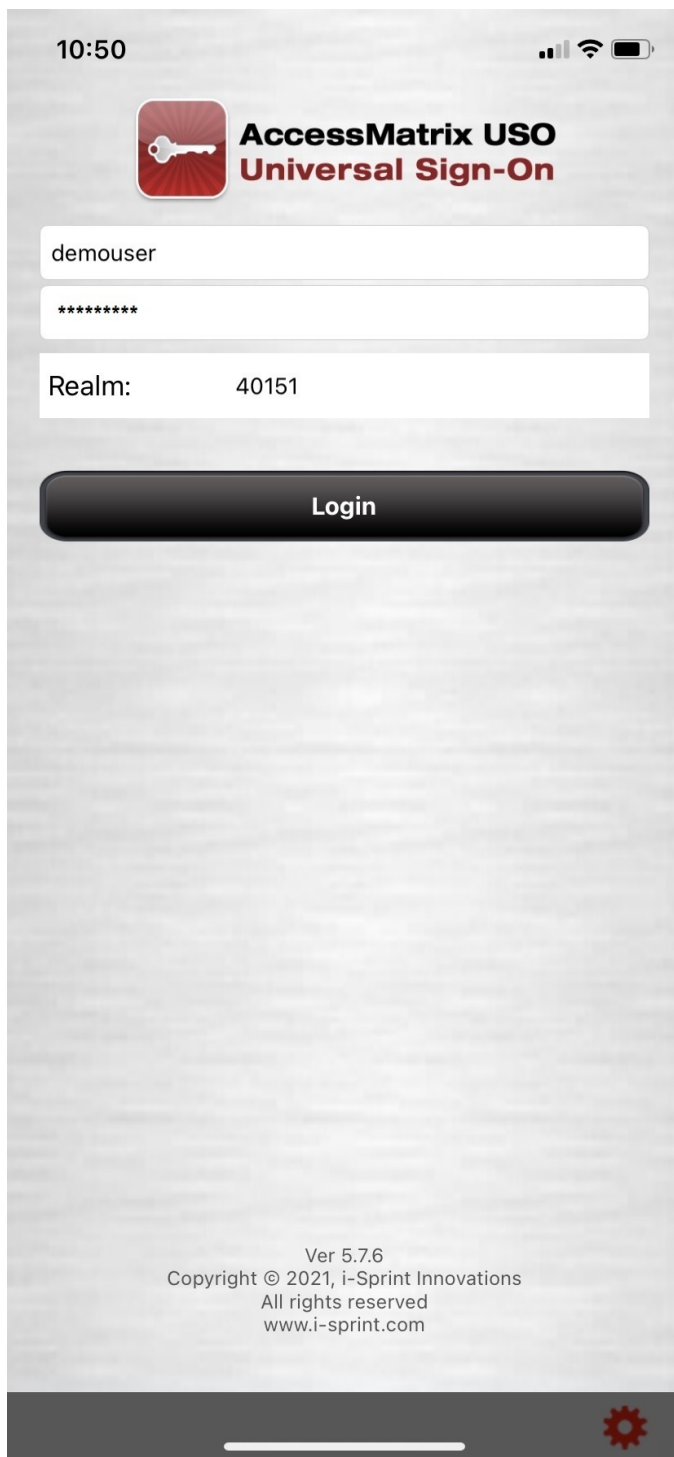


Figure 3-1 Login

4:30

Cancel

Change USO Password

Done

PLEASE SPECIFY YOUR NEW PASSWORD

Old Password

New Password

Confirm New Password

Figure 3-2 Change Password

Once the password has been changed and the next login is successful, you will be asked if you want to enable Touch ID or Face ID verification:

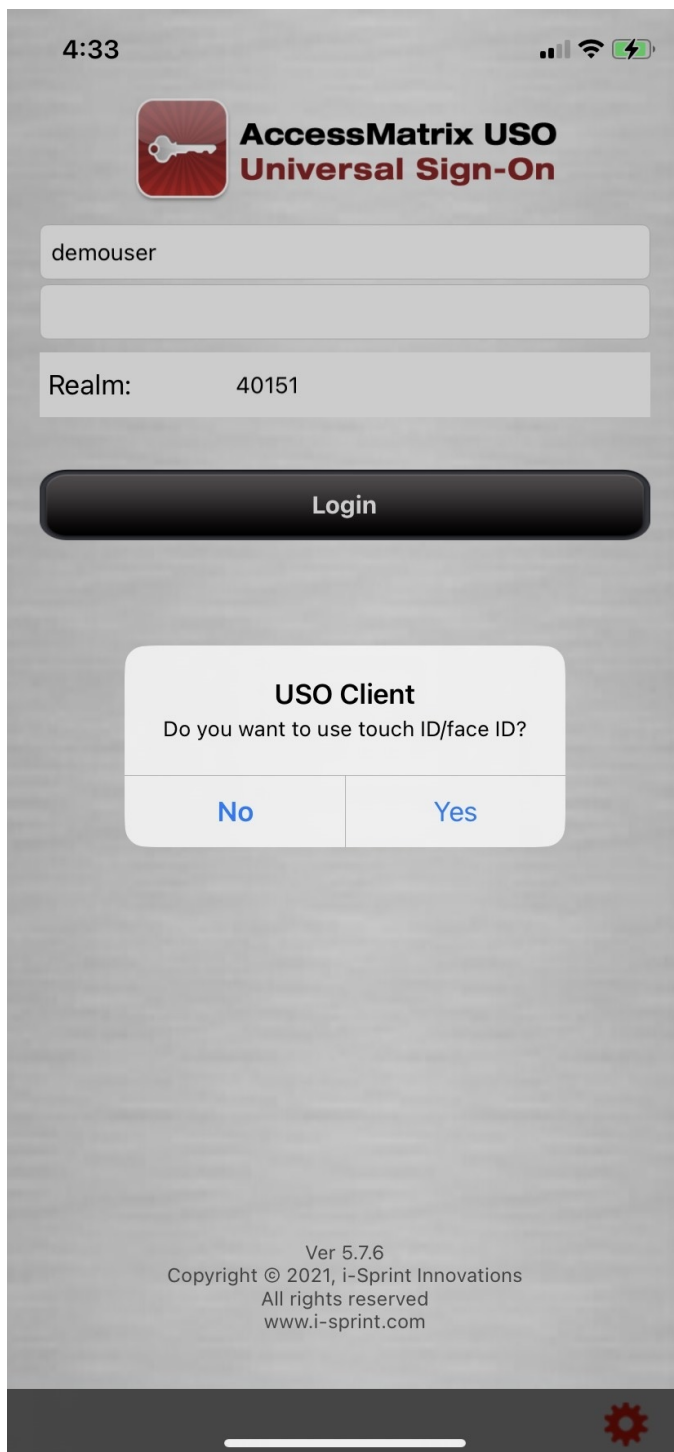


Figure 3-3 Enable touchID?

You may press **No** if you do not wish to enable this feature. However, the prompt to enable Touch ID/Face ID will continue to pop up with each successful login until you agree to enable this feature.

If Touch ID/Face ID verification is enabled, the Touch ID/Face ID verification process will be carried out:

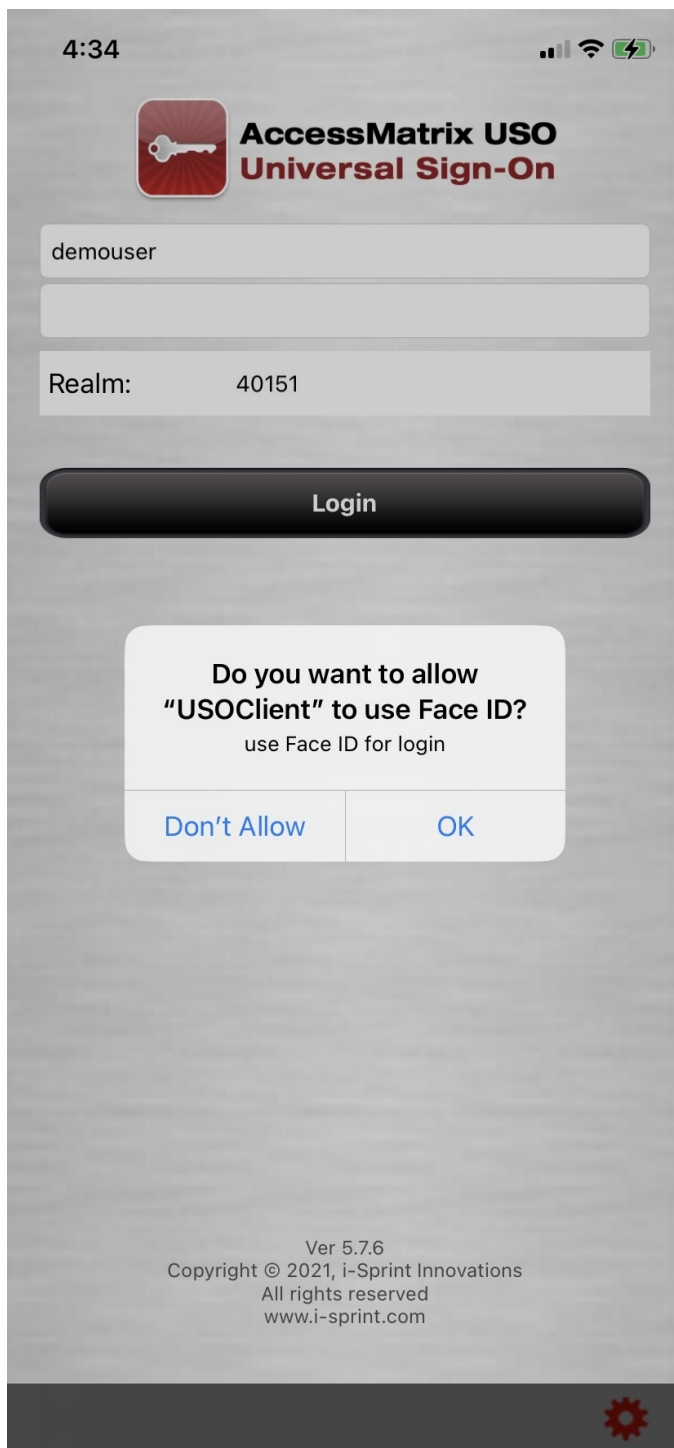


Figure 3-4 Verification touchID/FaceID

After successful verification, Touch ID/FaceID verification will first be used to log in each time the app is started.

User Application

User Application View Modes

After successfully logging in and entering the homepage, the **USO Application** screen is displayed. This screen contains the user's applications and can be displayed in two view modes - **List View** and **Tile View**:



Figure 4-1 List View

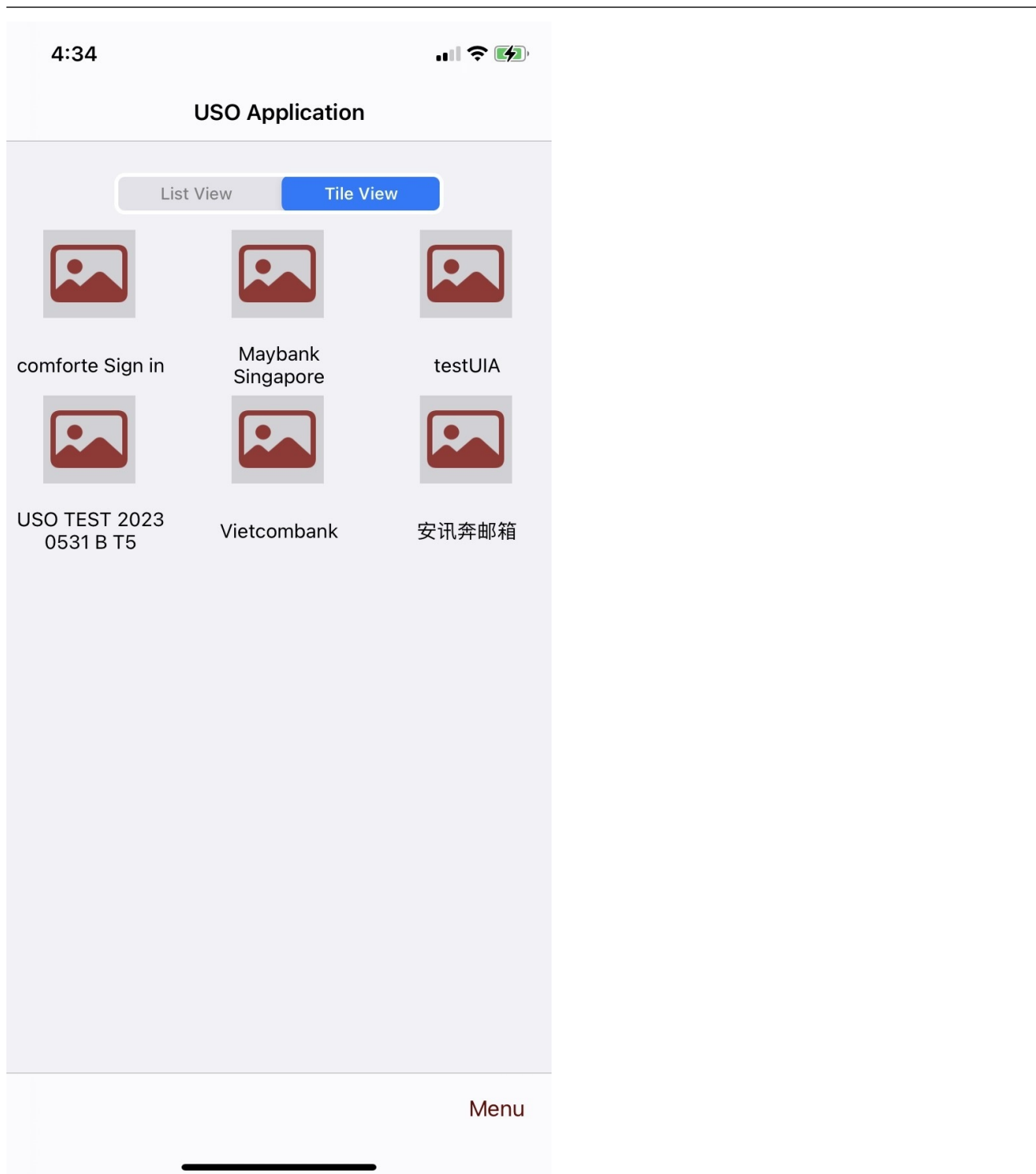


Figure 4-2 Title View

You can sort the applications according to your preferences. To do so, drag and drop an application to the desired location.

**Notes:**

- Each application can only be dragged within the same level, not across levels (e.g. applications in Group1 can only be dragged and sorted within Group1).
- The user's ranking of applications will be recorded and saved locally on the device only; it will not be uploaded to the server.

User Application Login

Pressing an application in the **USO Application** screen will redirect you to the login screen of that application:



Figure 4-3 Login Maybank App

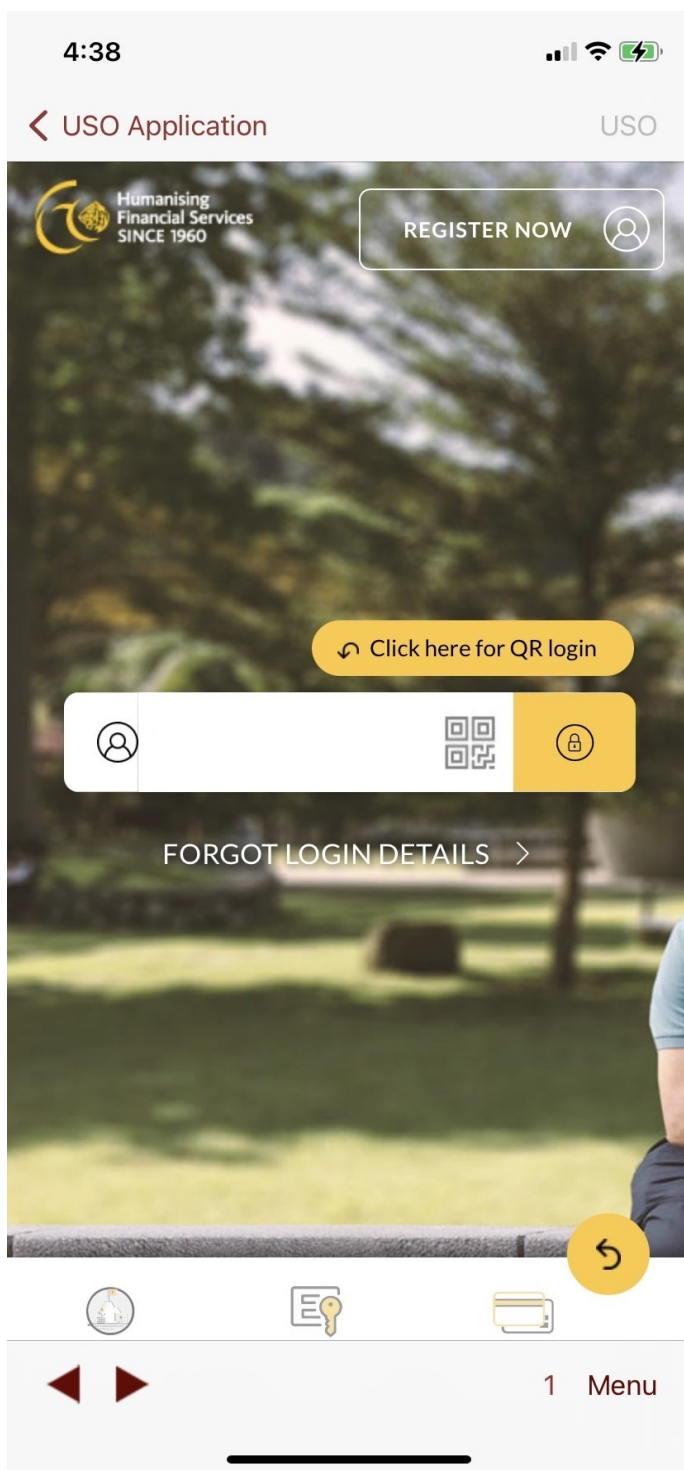


Figure 4-4 Maybank App Login Screen

If the USOCient mobile app does not have the account and password of the user application, it will automatically redirect you to the **Application Info** screen:

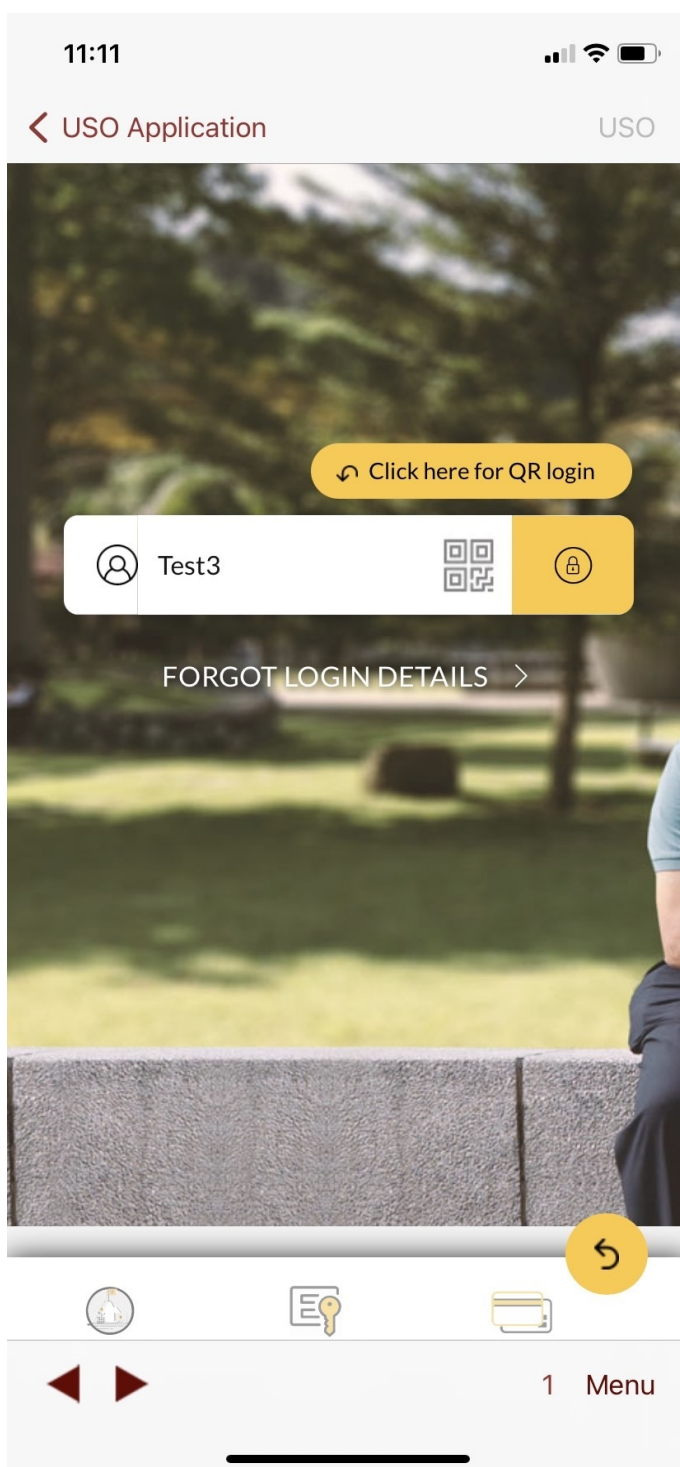


Figure 4-6 Auto Fill in the ID



Notes:

- The USOCClient mobile app only supports one set of account passwords. Adding a new set of account passwords will overwrite the first set of account passwords.

- The AM5 and desktop versions of the USO Client support multiple sets of account passwords.
 - If multiple sets of account passwords are set, the first set of account passwords will be used automatically.

Mobile App Training with USO Trainer on Desktop

1. Navigate to the website of the application to be trained on your desktop using the Chrome web browser.
2. On the browser toolbar, click the **Menu** icon and select **More Tools > Developer Tools** from the drop-down list:

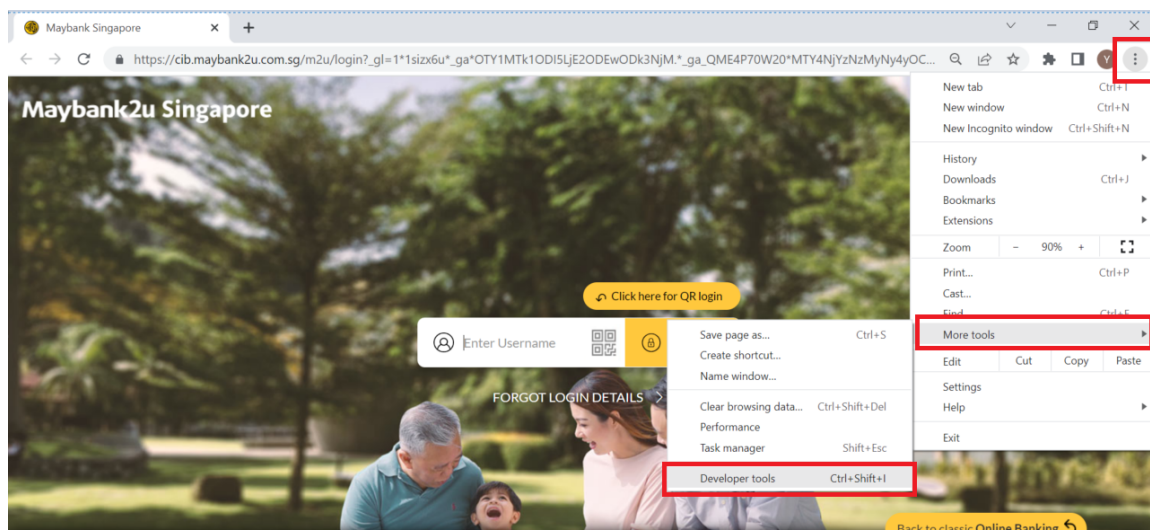


Figure 5-1

The developer tools pane is displayed.

3. Click the **Toggle device toolbar** icon:

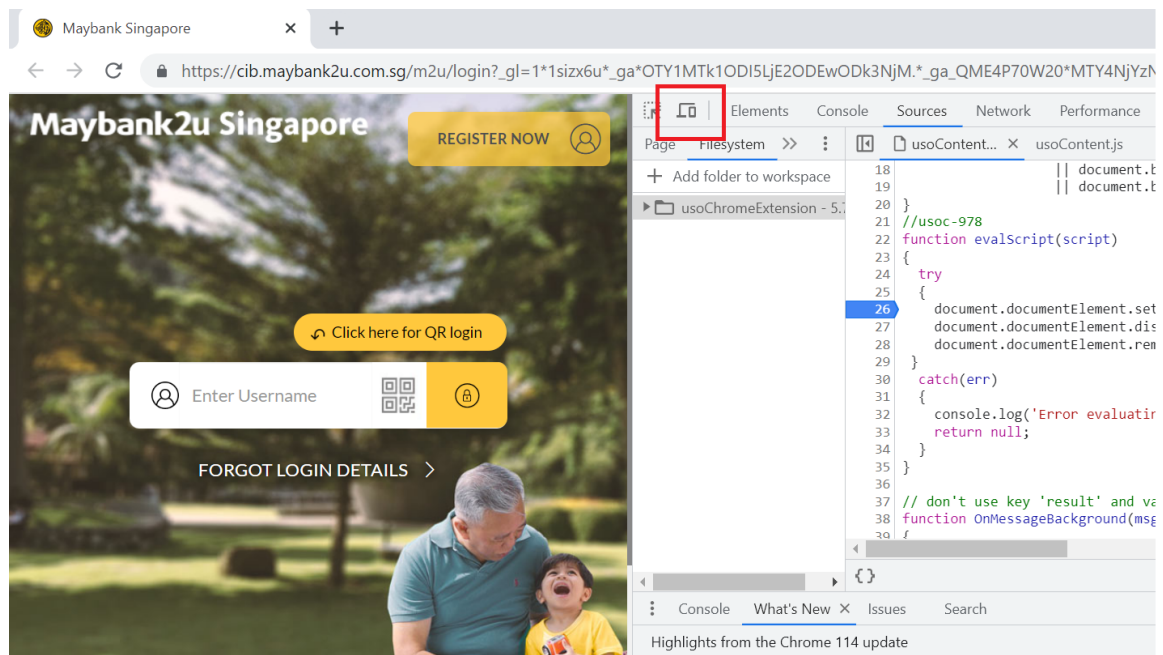


Figure 5-2

The web page is displayed in mobile device dimensions, depending on the mobile device model selected:

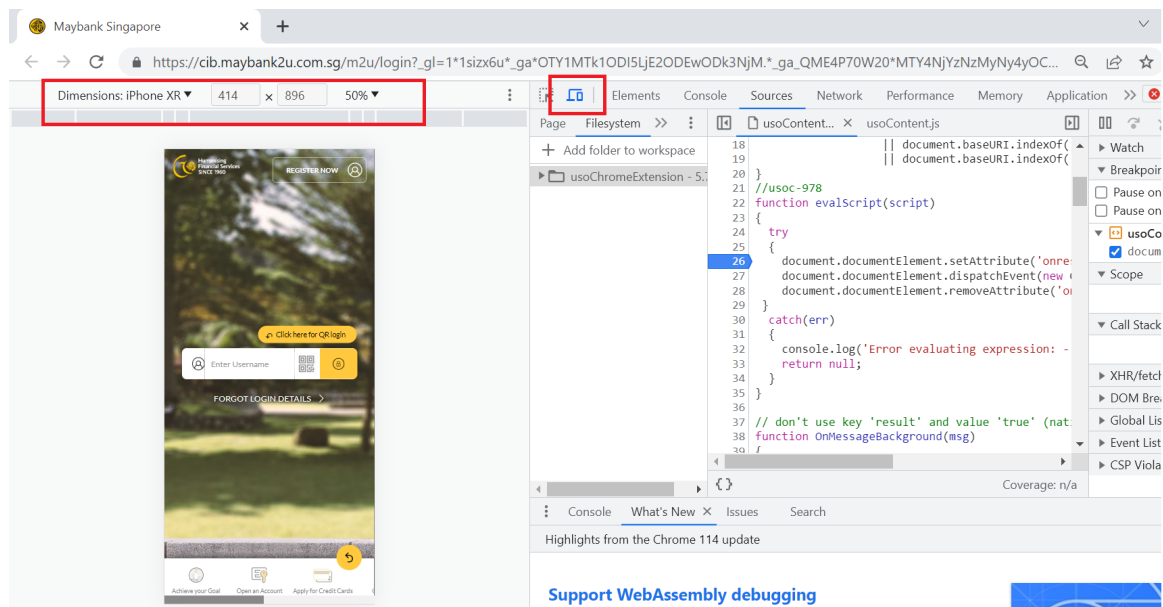


Figure 5-3

4. Modify the display dimensions by clicking the **Dimensions:...** drop-down menu and selecting the relevant device:

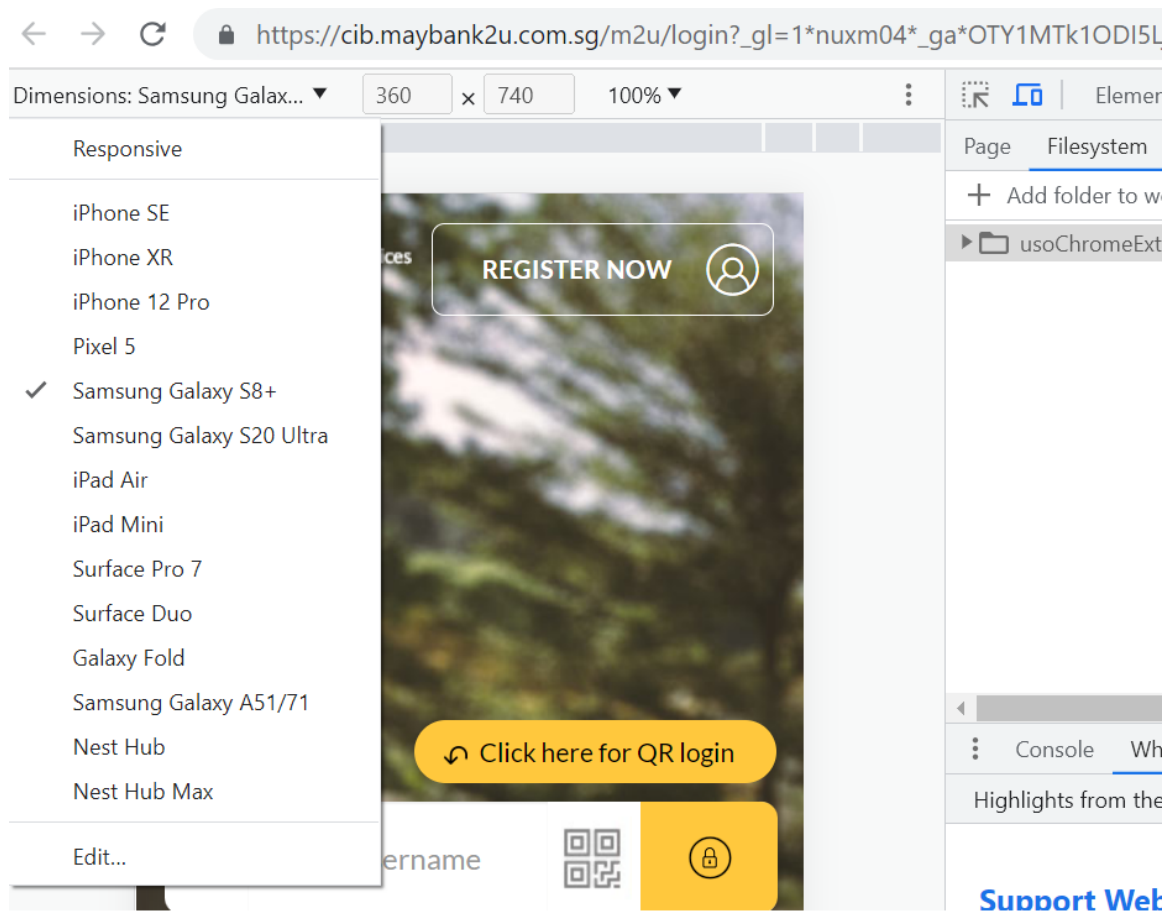


Figure 5-4

5. While in this mobile mode, perform application training as you would on a mobile device.

7. USO Application Training

The USO Trainer allows administrator to train various types of applications. This training task is a prerequisite task for all applications to be effected for USO before user can perform SSO function to the applications. Thus, each application needs to be trained at least once.

USO Trainer Unique Features

USO Trainer 5 provides a number of unique features. Few amongst the notable ones are described below:

- **Wizard-Driven Application Training**

Administrator may use the wizard to guide him in training each desired screen of the target application to be effected for USO.

- **Application Login Information and Attributes**

USO Trainer is used to learn the login and password change sequence of each target application to be effected for USO. USO Trainer also records various attributes such as screen identification and field mappings so that during run time, the appropriate login information will be automatically passed to the application.

- **Default Application Security Policy**

USO Trainer includes the default application level security policy for login behavior and password change.

- **Sophisticated Handling for Screen Matching and Processing**

USO Trainer utilizes State Transition mechanism and JavaScript engine to provide sophisticated handling for screen matching and processing. This state information is extremely useful to allow administrator to determine the exact status of the login or password change process so that the user action can be guided more appropriately to achieve a better user experience.

- **Test Mode Option**

USO Trainer provides a testing option to test the login and password change screens trained and sequence learnt before exporting the trained application.

- **Application Definition File**

The information learnt by USO Trainer is exported to an Application Definition File or ADF (.adf), which will then be imported and used in the Security Server. The ADF is in the form of XML format.

Types of Supported Applications

USO Trainer 5 provides the flexibility to handle the SSO requirements for various types of applications. These applications include:

- **Desktop Application**

Desktop application is an application that uses MS Windows Graphical User Interface as the screen interface.

- **Web Application**

Web application is an application that is accessed through a Web browser (Edge, Chrome or thinforie).

- **Java Application**

Java application is an application written in Java language and runs in a JRE environment.

- **Host Emulator Application**

Standard-based host/terminal emulator using HLLAPI for Mainframe, AS400, and UNIX Host Systems.

- **Console Application**

Console application is an application that is based on command-line input such as DOS Command, Telnet, or FTP.

- **Metro Application**

Metro application is a touch-screen friendly application written especially for Microsoft's WinRT programming interfaces.

For certain complex applications, a single application training process may involve training a combination of different types of application screens/interfaces. USO Trainer allows you to address this requirement with ease. For example, MSN Messenger requires the user login via a desktop application but the password change screen is on a web page.

Another significant benefit of USO is that there is not any programming required or change to be made to the target applications to achieve SSO. USO can address most of the login and password policy scenarios. However, USO Trainer allows the administrator to further customize the login or change password process to handle complex scenarios.

List of Certified Applications

- **Desktop Applications:** Skype v6.18
- **Web Applications:** Gmail, Yahoo Mail
- **Host Emulator Applications:** AS/400
- **Java Applications:** Java Client applications
- **Metro Application:** MetroMail

Known Unsupported Applications

There are cases where USO Trainer cannot match the applications. Below is the list of known unsupported applications:

- SQL Plus (64-bits)
- Firefox

7.1. Installing the USO Trainer

System Requirements of USO Trainer

- USO Trainer must be installed on Windows OS machine, with Windows 10 as the minimum version.
- Application training must be done using the Microsoft Edge (Chromium) browser.

USO Trainer Installation Startup

Installation of USO Trainer

USO Trainer 5 can be installed through AccessMatrix Installer, i.e. it can be installed during the installation of AccessMatrix USO. It can also be installed solely on a computer. Thus, administrator may opt to install the USO Trainer using the standalone USO Trainer Installer.

To see the detailed instructions, refer to this video:

Link: https://www.youtube.com/watch?v=bWM4k_kPA2Y

Starting USO Trainer

Administrator may start USO Trainer 5 by accessing following shortcut:

- **Start > Programs > AccessMatrix > Universal Sign-On > USO Trainer**

Alternatively, you may start the USO Trainer 5 by directly running its executable where the default installation folder is:

- *C:\Program Files\AccessMatrix\USOTrainer5\USOTrainer.exe*

USO Trainer Uninstallation

If USO Trainer 5 is installed together with other AccessMatrix components during AccessMatrix installation, administrator can choose to uninstall USO Trainer 5 either by uninstalling entire AccessMatrix system or by uninstalling USO Trainer 5 solely without uninstalling other AccessMatrix components.

To see the detailed instructions, refer to this video:

Link: <https://www.youtube.com/watch?v=-dfB3Km5Acw>

7.2. USO Trainer User Interface

There are two main user interfaces of USO Trainer 5, namely the Welcome Screen and the Main Menu.

Welcome Screen

The **Welcome Screen** displays options for manipulating USO Trainer features. The explanation of each option is given below:

- **Train an Application by Using Wizard**

The option allows the administrator to train the desired screens of the target application by using the full-fledged wizard.

- **Train an Application**

The option allows the administrator to train the desired screens of the target application without using the full-fledged wizard.

- **Load an ADF**

The option allows the administrator to load the ADF of an application trained previously into the USO Trainer.

- **Do not show this welcome screen**

The option allows the administrator to specify whether to display the **Welcome Screen** upon start-up of USO Trainer. By default, this option is not checked, i.e. to display the **Welcome Screen** upon USO Trainer startup.

To close the **Welcome Screen**, click on the **Close** button.

Main Menu

The main menu consists of following options:

- **USO Trainer Application Menu**

This menu provides further sub-menu options for administrator to test, export the application trained, and so on.

- **USO Trainer Tools Menu**

This menu provides further sub-menu options for administrator to change the view of the **Main Display Panel** and to specify the general options, as well as to manage user-defined action.

- **USO Trainer for Emulator**

This sub-menu option will launch USO Trainer for Emulator that will be used to train **Host Emulator**, **Console** and **Manual** types of applications. Refer to [USO Trainer for Emulator User Interface](#) section for the UI details.

- **USO Trainer Help Menu**

This provides information about the USO Trainer software such as software version and product copyright.

Quick Access Toolbar

The quick access toolbar provides certain command buttons (as shortcuts to main menu options) such as **New**, **Wizard**, **Load**, **Test**, **Export**, etc. that allow the administrator to access to the desired function quickly.

Main Display Panel

The main display panel shows some general information or the list of application screens added for the application, and certain application screen information for the screens added. From the list of application screens added, administrator can select the desired screen to be managed for the screen training.

USO Trainer for Emulator User Interface

Administrator may start USO Trainer for Emulator by selecting **USO Trainer for Emulator** sub-menu under **Tools** menu.

The main user interface of USO Trainer for Emulator displays **Main Menu** and **Screen List**.

- **Main Menu**

The main menu consists of the following options:

- **Application**

This provides further sub-menu options for users to load an ADF file, enable/disable the application integrity checking, test trained application, exported trained application, and exit from USO Trainer.

- **Screen**

This provides further sub-menu options for users to add, edit, and delete screen, move the trained screen up and down, and clear the entire trained screen.

- **User Defined Action**

This menu provides further sub-menu options for users to add, edit, and/or delete the user-defined actions.

- **Help**

This provides information about the USO Trainer software such as software version and product copyright.

- **Screen List**

This section displays a list of the trained screens. From the list, user can select the desired application screen to be managed.

7.3. Using the USO Trainer

This section discusses on the different training methods in using USO Trainer.

Full-Fledged Wizard-Driven Application Training

If not familiar with USO Trainer 5, the administrator may proceed to train the application by using the application training wizard in a full-fledged manner. The full-fledged wizard will guide the administrator through out the entire application training process in a complete manner. The full-fledged wizard allows training of the application for either of following purposes:

- **Application Training for USO Enablement**

This option allows training of the application for USO enablement, i.e. to train the application for single sign-on via USO Client.

- **Application Training for Launching Only**

This option allows training of the application for launching only, i.e. to train the application for launching through the application list but not for any single sign-on purpose.

To start training an application using the full-fledged wizard, perform following steps:

1. Start USO Trainer. Depending on the setting, either of following will be presented:
 - a. If administrator is presented with the **Welcome Screen**, click on the **Train an Application by Using Wizard** button.
 - b. If administrator is presented with the **Main Screen**, click on the **Wizard** quick access button. Alternatively, administrator may click on the **Application** menu and select the **Wizard** sub-menu option.
2. As described earlier, determine the purpose of the application training and then proceed to one of following relevant sections to continue with the application training.

Application Training for USO Enablement

To train an application using the full-fledged wizard for USO enablement, refer to this video:

Link: <https://www.youtube.com/watch?v=Wnhccftk9x8>

Alternatively, follow the steps below:

1. Start Full-Fledged Application Training Wizard

Perform steps described in [Full-Fledged Wizard-Driven Application Training](#) section. The Application Training Wizard will be displayed.

2. Application Training Wizard: Screen Function

Select the **Login and/or Change Password** and/or **Logout Automation** options and select the desired screen function. There are several screen functions to choose from as described below:

- **Login Screen:** This is used to train the login screen of the application.
- **Login with Success Screen:** This is used to train the login success screen of the application, i.e. the screen returned upon successful login.
- **Login with Fail Screen:** This is used to train the login fail screen of the application, i.e. the screen returned upon failed login.
- **Change Password Screen:** This is used to train the change password screen of the application.
- **Change Password with Success Screen:** This is used to train the change password success screen of the application, i.e. the screen returned upon successful password change.
- **Change Password with Fail Screen:** This is used to train the change password fail screen of the application, i.e. the screen returned upon failed password change.
- **Logout Screen:** This is used to train the logout screen of the application.

For example: If the application has a login screen, select the **Login** and/or **Change Password** and/or **Logout Automation** options, and then check the **Login Screen** option, and so on. Click on the **Cancel** button to cancel the application training operation.



Notes:

- At least one screen function must be selected and trained in order to allow for exporting to ADF and the screens selected will be trained according to the sequence shown. It is recommended to select all the desired screen functions and then proceed to train the screens accordingly until all the selected screen functions are trained completely.
- Manage the screens trained after using the wizard to train all the selected screen functions.

3. Application Training Wizard: Start Screen Training

-
- a. Having selected the desired screen functions, click on **Start Screen Training** button to start training the desired screen functions. The **Screen Training Wizard** will be displayed. Perform the screen training as described in [Using Screen Training Wizard](#) section for the basic and advanced option descriptions. Click the **Finish** button to end screen training and the **Application Training Wizard - Next Action** dialog will be displayed. Otherwise, click the **Cancel** button to cancel the training operation.

4. **Application Training Wizard: Next Action**

- a. Having trained all the desired screens, select either to test the existing screens trained or to go to the screen list. If selecting screen list, the application training wizard will be ended.
- b. If none of the screens is trained, select to reset the wizard and start all over again.

5. **Application Training Wizard: Test**

- a. This screen will be displayed if Proceed to test the screens trained option was selected in **Application Training Wizard - Next Action** dialog.
- b. At this stage, proceed to test the screens trained right away or if necessary, configure the application information, e.g. the application path, application launching option, etc. before proceeding to test the screens trained.
- c. In addition, the Administrator is allowed to set the default values for certain system-defined user-application attributes as well as to manage user-defined user-application attributes. Refer to the **Application Training Wizard: User Application Attributes** section below for the details or refer to [Using Application Training Wizard](#) to know about the basic and advanced options of Application Training wizard.
- d. Once the trained application is ready for testing, ensure that USO Client is started, and then click **Test** button to continue. At this point, the USO Client is under **Test** mode. On the other hand, USO Trainer will display the **Application Training Wizard - Next Action** dialog with options to export the application trained or to display the screen list. Refer to [Testing Trained Application](#) section for the detailed instructions on testing a trained application.
- e. If the test has failed, go to the screen list to retrain the screens and test again. Otherwise, export the trained application as described in **Application Training Wizard: Export** section below.

6. **Application Training Wizard: User Application Attributes**

-
- a. To define new user-application attribute, click on the **[+]** icon. Specify the name of the custom user-application attribute and then click **OK** to add it to the existing list.
 - b. To delete any of the existing user-application attribute, select the desired attribute and then click on the **-** icon.

**Notes:**

- The Administrator is provided an option to specify whether to allow user to edit User ID if the User ID has been entered during USO in action. By default, user is allowed to do so. To disallow user to do so, select the **Disallow user to edit user ID** option.
- Refer to [Edit USO Application Attribute Descriptions \(User/Group Application Attributes Tab\)](#) for the detailed discussion about the user attributes.

7. Application Training Wizard: Export

- a. To export the trained application, click on **Export**, and refer to [Exporting Trained Application](#) section for the detailed instructions on exporting a trained application.

8. Application Training Wizard: Completed

- a. Upon successful export of the trained application, system will return to the screen list. At this stage, the entire application training with full wizard-driven has been completed. After closing the application and if necessary, you may load the trained application for retraining the screens manually.

Application Training for Launching Only

To train an application using the full-fledged wizard solely for application launching purpose, refer to this video:

Link: <https://www.youtube.com/watch?v=tQT3AkLqUqA>

Alternatively, follow the steps below:

1. Start Full-Fledged Application Training Wizard

- a. Perform steps described in [Full-Fledged Wizard-Driven Application Training](#) section. The **Application Training Wizard** will be displayed.

2. Application Training Wizard: Screen Function

-
- a. Since the application is to be trained for launching purpose only, select the Application launching only and no screen training is required option.
 - b. The application launching option allows the administrator to specify such that the application contains only the application launching information but not any screen matching information. As such, this will enable user to make use of USO to launch the application only without any screen matching operation since no screen training is required.
 - c. Click on **Cancel** to cancel the application training operation.

3. **Application Training Wizard: Start Screen Training**

- a. Having selected the application launching option, click on the **Start Screen Training** button to start training the desired application. The **Application Training Wizard** will be displayed with the **Application Identity** and **Application Path/URL** options. Refer to [Using Application Training Wizard](#) for more details.

4. **Application Information: Application Identity**

- a. Specify the application identity information, i.e. **Application Id** and **Application Name**.
- b. Click on **Next** button to continue and the **Application Path/URL Screen** will be displayed. Otherwise, click the **Cancel** button to cancel the application training operation. Click the **Finish** button to end the application training.

5. **Application Information: Application Path/URL**

- a. Specify the application path/URL as well as other information, if applicable.
- b. Click the **Cancel** button to cancel the application training operation. Otherwise, click on the **Finish** button to end the application training. The USO Trainer **Main Screen** will display the following information message:
"You have chosen to train "For Launching Only" application. The information saved is intended only for launching the application, not for matching the screens. You can immediately export the application."

6. **Application Training Wizard: Test**

- a. User is advised to test the application information specified although it is not necessary to do so. To test the application information, click on the **Test** quick access button. The **Application Training Wizard - Test** will be displayed.
-

-
- b. Proceed to test the application by referring to [Testing Trained Application](#) section for the detailed instructions on testing the application. Once tested successfully, proceed to the **Application Training Wizard: Export** section below.



Note: If the test has failed, re-specify the application information. To do so, click on **Application** menu option and select the **Details** sub-menu option.

7. **Application Training Wizard: Export**

- a. Export the application information after testing or completely specifying the required application information.
- b. To export the application information, click on the **Export** quick access button, and refer to [Exporting Trained Application](#) for the detailed instructions on exporting an application.

8. **Application Training Wizard: Completed**

- a. Upon successful export of the trained application, system will return to the **Main Screen**. At this stage, the entire application training for launching has been completed. After closing the application and if necessary, load the trained application for retraining by re-specifying the application information.

Standard Wizard-Driven Application Training

In view of the general application training process mentioned in previous chapter, if you are experienced and familiar with USO Trainer 5, you may proceed to train the application directly using the standard application training wizard. Meantime, if the application has been trained previously, administrator may load the relevant ADF for reviewing or retraining of the application.

Using Screen Training Wizard

To start training an application without using the full-fledged wizard, perform following steps:

1. Start USO Trainer. Depending on the setting, either of following will be presented:
 - a. If Administrator is presented with the **Welcome Screen**, click on the **Train an Application** button.
 - b. If Administrator is presented with the **Main Screen**, click on the **New** quick access button. Alternatively, administrator may click on the **Application** menu and select **New** sub-menu option.
-

-
2. Click on the **[+]** button in order to add a new screen. Determine the desired screen function to add, e.g. if the application has a login screen, then select the **Login Screen** option and so on.

The following sections will discuss on the basic and advanced options of the **Screen Training wizard**.

Screen Objects

The options are introduced for screen training as described below:

- **Try to auto recognize**
 - Auto recognize is used to train a screen that contains **User Id** and **Password** fields only, and support <Enter> key as submit method. Train the application screen by dragging the magnifying glass to anywhere within the application screen and then release it.
 - To enable auto recognition, select by checking this option. By default, this checkbox is unchecked when **Screen Training Wizard** is displayed.
 - **Training using native approach**
 - Native approach is the default option used in training an application.
 - By default, this check box is checked. If this option is checked, the **Don't process invisible object** and **Recognize Object by Coordinate** options are disabled. Otherwise, if this option is unchecked, then the said options will be enabled.
 - **Don't process invisible object**
 - By default, USO Trainer recognizes/processes both visible and invisible fields. There is a situation where invisible/hidden fields should not be recognized/processed. For example, both application login and change password screens may have the same object structure but different only in the visibility and enabling/disabling state of the objects.
 - Meantime, USO Trainer has also been enhanced such that besides recognizing hidden fields, it has been parameterized to make it more flexible for training other kinds of applications.
 - To enable screen matching by recognizing/processing visible screen objects/fields only, i.e. not to recognize/process invisible/hidden objects/fields, select by checking this option.
 - **Recognize Object by Coordinate**
 - USO Trainer enhances screen training by providing screen object/field recognition based on coordinates, i.e. to recognize and match screen objects/fields by coordinates. This
-

feature is particularly useful for training Windows dialog-based screens which are not resizable in general. Nevertheless, this option cannot be used for training resizable screen such as web-based application.

- To enable screen matching by recognizing screen objects/fields based on coordinate, select by checking this option.



Notes:

- For training application login success, change password success, and logout screen objects, it is required to train screen identifier and/or submission method. The screen identifier inside <script> tag is not included in identifier matching.
- When using the **Screen Identifier (Label or Static Text)** option, ensure that the screen identifier(s) is/are sufficiently unique so USO can properly identify the screen. Generic identifiers such as "OK" or "Cancel" may result in the inaccurate identification of screens.
- To train an application, use the native approach first. If in case the application can't be trained, then uncheck **Train using native approach** checkbox to use non-native approach. If the application still can't be trained successfully, then train using **Recognize Object by Coordinate** option.
- If using native approach, the filling is done by API. If using non-native approach, the filling is done by simulating keystroke.
- Custom Input can be hidden or changed to non-editable by following the steps described in [Hide or Change Custom Input Field to Non-editable](#) section.
- USO Trainer allows changing of windows class name inside the fields manually for desktop applications. This can be done by filling in the text box that will be displayed after double clicking the field/identifier under Trained Screen Objects. Refer to [Dynamic Windows Class Name for WINAPI/Desktop Applications](#) section for details.

Screen Identity

- **Desktop Application**

For a desktop application, the **Screen Id** and **Screen Title** are mandatory fields that will be automatically filled upon successful screen information training. User may modify the **Screen Id** and **Screen Title** values. There are few options for defining the matching rule for the Screen Title:

- **Equals:** USO Client will match the screen that has exact screen title as specified in the **Screen Title** input field.

-
- **Contains:** USO Client will match the screen with title containing the string specified in the **Screen Title** input field.
 - **Starts with:** USO Client will match the screen with title that starts with the string specified in the **Screen Title** input field.
 - **Ignore:** USO Client will ignore the string specified in the **Screen Title** input field.
 - **Wild card:** USO Client will match the string with the wildcard character specified in the **Screen Title** input field. The valid wildcards are:

- **Asterisk (*):** Represents combination of characters.
- **Question mark (?):** Represents single character.

For example, if the value for the screen title is "Login*", USO Client will match the screen title that starts with "Login", e.g. "Login form" or "Login Screen", etc.

The option **Use application path as part of matching criteria** allows administrator to specify whether to include application path as part of screen matching criteria. By default, this option is not checked, i.e. not to include application path for screen matching.

- **Web Application**

For a web application, the **Screen Id**, **Screen Title**, and **URL** are mandatory fields and will be automatically filled upon successful screen information training. User can modify the **Screen Id**, **Screen Title**, and **URL** values.

Refer to above section related to Desktop Application for detailed explanation about the matching rule options for the **Screen Title** and **URL** fields.

As an example, if the URL of the login page contains parameters that change every time the login page is loaded, select Starts with from the URL matching rule drop-down list and edit the entry in the URL input field to remove the dynamic parameters. For instance, if the value of the URL is

https://login.myapplication.com?parameter1=xxx¶meter2=yyy, where parameters 1 and 2 are dynamic variables, the URL definition should be set to Starts with and the dynamic variables should be removed so it becomes *https://login.myapplication.com*.

- **Java Application**

For Java application, the **Screen Id** and **Screen Title** are mandatory fields and will be automatically filled upon successful screen information capturing. Administrator can specify the desired **Screen Id** and the **Screen Title** is also modifiable.

There are few options for defining the matching rule for the **Screen Title**. Refer to above section related to Desktop Application for detailed explanation about the matching rule options.



Note: For training application change password screen object, administrator must also set correctly the screen order for the change password screen as described below:

- **Case 1 - New Password and Confirm New Password on Same Screen:**

If the new password and confirm new password input fields are on the same change password screen, check both **First Screen** and **Last Screen** options.

- **Case 2 - New Password and Confirm New Password on Different Screens:**

If the new password and confirm new password input fields are on different change password screens, the screen order must be selected accordingly.

For example, the new password input field is on one screen called the First screen, it is required to check the **First Screen** option and uncheck the **Last Screen** option.

Likewise, the confirm new password input field is on another screen called the Last screen, uncheck the **First Screen** option and check the **Last Screen** option.

Processing Option

Administrator is provided with several screen processing options to enhance the screen matching mechanism described below:

- **Require Re-authentication**

This option enables the application to implement re-authentication.

- **Require Authentication by Another User**

This option enables the application to implement multiple user authentications.

- **If Same Screen Reappears**

There are 3 options provided for under this option:

1. **Process And Resubmit:** This tells USO Client to process the trained screen when it reappears by filling in and resubmitting the screen information. This is the default option.
 2. **Process But Do Not Resubmit:** This tells USO Client to process the trained screen when it reappears by filling in but not resubmitting the screen information.
 3. **Do Not Process Again:** This tells USO Client not to process the trained screen when it reappears, i.e. not to fill in any screen information.
-

If possible, Administrator is advised to configure the screen to use state control by means of advanced scripting to substitute this option.

Custom Behavior

Custom behavior defines the state-control conditions for handling custom behavior or flow of application screens.

A pre-state is a conditional expression. Example of pre-state values are:

- true
- control == 'LoginFailed'
- state == 'BeforeLogin' && control=="
- control == 'LoginSubmitted'

An action is a Javascript script. Examples of actions are:

- \$ProcessLogin(0,",")
- If(_FillAllFieldsAndSubmit()) control = 'LoginSubmitted';
- \$HandleNormal(); state = 'AfterLogin'; control = ";

The pre-states are evaluated from top to bottom, and for the first one that evaluates to true, its associated action is executed. If there is at least one pre-state/action configured and none of the pre-states evaluate to true, then nothing is performed for that screen. If there is no pre-state/action configured, then the default action is performed. The default action is filling in and then submitting the screen.

Depending on the application behavior, Administrator may modify the default settings for the parameters by means of state control configuration to cater for user's needs. Refer to [B. Basic, Standard and User Defined Actions](#) for the detailed description of user defined actions.



Note: For training application change password screen object, if the application's change password action contains multiple change password screens, i.e. the new password and confirm new password input fields are on different change password screens, administrator is required to perform following changes:

- **Change Password Screen - First Screen:** For training change password's new password input field screen, i.e. the First Screen, no changes are required in this step.

- **Change Password Screen - Last Screen:** For training change password's confirm new password input field screen, i.e. the Last Screen, some changes are required. From the existing list of rules, select the existing rule: control == "", and click on the JavaScript Editor button to edit the rule. Modify the rule to: control == 'ChangePwdSubmitted', and then click on **OK** to save the changes.

Timeout Actions

Screen Training Wizard

Login (Step 5 of 8): Timeout Actions

It is optional to define the actions to be performed when a screen is matched but has reached its timed-out period, or a screen is matched but has disappeared.

Timeout Actions

Timeout: milliseconds

JavaScript Actions for Timeout:

Disappear Actions

JavaScript Actions for Disappear:

Basic

<< Previous Next >> Finish Cancel

Timeout Actions define the actions to be performed when a screen is matched but has reached its timed-out period (Timeout Actions), or a screen is matched but has disappeared (Disappear Actions).

These actions work in conjunction with a timeout value (by default, 5000 ms). If the timeout value exists for a particular screen, USO Client will typically start a countdown timer after it has successfully processed the screen.

The timer can be stopped when there is an explicit command to stop it. Typically, when another screen (of the same instance) is matched, the timer will be stopped. If the timer counts down to zero, the Timeout Action will be executed. In the example above, the script `$HandleLoginSuccess('Timeout')` will be executed.

Before the timer counts down to zero, if the screen disappears (i.e. the window has closed), the Disappear Action will be executed. In the example above, the script `$HandleLoginSuccess('Screen')` will be executed. This is typical for the case where the pop-up login window closes when authentication is successful.

Generally, there's no need to configure these actions if success and failure screens can be trained. For example, both the success and failure screens for web applications can be usually trained. Due to the inconsistent Internet speed, it may not be advisable to assume successful login after 5 seconds.

Depending on the application behavior, Administrator may modify the default settings for the parameters by means of state control configuration to cater for user's needs. Refer to [B. Basic, Standard and User Defined Actions](#) for the detailed description of user defined actions.

- i** **Note:** For training application change password screen object, if the application's change password action contains multiple change password screens, i.e. the new password and confirm new password input fields are on different change password screens, Administrator is required to perform following changes:
- **Change Password Screen - First Screen:** For training change password's new password input field screen, i.e. the First Screen, and remove the pre-defined value for **Disappear Action**.
 - **Change Password Screen - Last Screen:** For training change password's confirm new password input field screen, i.e. the Last Screen, no changes are required in this step, i.e. do not remove the pre-defined value for **Disappear Action**.

Custom Menu

Custom Menu allows the administrator to customize the menu items to be provided on the USO Client's WinAttach icon that appears whenever a screen is matched. These menu items contain generally the JavaScript functions as the actions to be performed.

Right-clicking on the USO WinAttach icon will bring up its menu. Clicking on a user-defined menu item will cause USO Client to execute its related action for the screen. For example, if the user clicks on the **Fill** menu item, then USO Client will execute `_FillAllFields()` for that screen.

Depending on the application behavior, administrator may modify the default settings for the parameters by means of state control configuration to cater for user's needs.

Certificate Info

This advanced feature is only available when training a SSL-secured Web application where administrator can choose to enable certificate integrity checking on a per-screen level for the Web application. For other types of applications, the integrity checking can only be enabled on a per-application level.

To enable certificate integrity checking, check the **Enable Certificate Integrity Checking** option, and Trainer will automatically retrieve the SSL certificate information.

Screen Object Editor

This advanced feature is normally used when there's a need to adjust the HTML tag of web applications manually, for example to support mobile client. It is intended to change the information of a particular object/field that you have captured during training.

Using Application Training Wizard

Administrator is provided with two modes for training the application, namely Basic and Advanced modes. The difference between these two modes is that under Advanced mode, administrator is provided with more advanced features provided by USO e.g. application password policy management, re-authentication, single sign-off, etc. which will be described in details in following sections. Depending on user's requirements, it is optional whether to implement these advanced features that are generally applied to the whole application.

Having trained all the desired screen functions, you may proceed to train the application information. On the **Main Menu**, click on the **Application** menu and select the **Details** sub-menu option.



Note: The **Details** sub-menu option is only enabled provided you have trained at least one of the screen functions defined.

Application Identity

User may edit the information and then click **Finish** to save the changes and return to the **Main Screen**, or click on **Cancel** to cancel the application information configuration operation.

following are descriptions of the attributes:

- **Application Id**

The identity of the application and is a mandatory field.

- **Application Name**

The name of the application and is a mandatory field.



Note: At this stage, you have completed training the application information under **Basic** mode.

Application Path/URL

following are descriptions of the attributes:

- **Application Path/URL**

The executable path or URL of the application and is a mandatory field.

- **Application Working Directory**

The working directory of the application and is only applicable to desktop application.

- **Application Arguments**

The arguments to be passed to the application during run-time execution of the application.

- **Run as Modern UI (Metro) application**

This option, if checked, will specify that the application can be run as metro application. This option is enabled only if training a web application.

- **USO-Citrix Integration (deprecated, legacy support)**

Currently, there are 2 approaches, namely OCR Approach provided by USO and Virtual Channel Approach provided by Citrix, for integrating USO to provide USO for Citrix-published applications:

- **USO OCR Approach**

If this approach is used, Administrator should specify the full path including the file name of the desired ICA file (.ica) in the **App. Path/URL** field. The path refers to the actual directory where the ICA file resides in the user workstation. Similarly, this actual directory will be the value for the **App Working Directory** field.

For example:

- **App. Path/URL:** *C:\USOICAFiles\USO Desktop Demo.ica*
 - **App. Working Directory:** *C:\USOICAFiles*
 - Leave the **This application is published using Citrix server?** option unchecked as it is not applicable.
-

- **Citrix Virtual Channel Approach**

If this approach is used, i.e. USO Client is running on Citrix mode, the option **This application is published using Citrix server** is applicable and should be checked.

In this case, administrator is required to specify the full path including the file name of the desired ICA file (.ica) in the **Citrix App. Path** field. The path refers to the actual directory where the ICA file resides in the user workstation. Similarly, this actual directory will be the value for the **App. Working Directory** field.

For example:

- **Citrix App. Path:** *C:\USOICAFiles\USO Desktop Demo.ica*
- **App. Working Directory:** *C:\USOICAFiles*
- Leave the **App. Path/URL** field unchanged which should be pointing to the actual directory, including the file name of the application executable.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.
- **Next** button to continue.
- **Finish** button to save the changes and return to the **Main Screen**.
- **Cancel** button to cancel the application information configuration operation.

Application Launching

In this step, edit the application launching information, if necessary, as described below:

- **Allow the application to be launched by USO Client**

Determine whether to allow the application to be launched by USO Client. If the option is checked, the application is allowed to be launched by USO Client. Otherwise, the application is not allowed to be launched by USO Client. The option is checked by default. If the option is unchecked, during SSO, the application will be dimmed (disabled) and user will not be able to click and launch the application from within USO Client.

- **Process the application launched outside of USO Client**

If the option is checked, USO Client will process the application regardless of where the application is launched. Otherwise, USO Client will not process the application if the application is launched externally i.e. outside of USO Client. The option is checked by default.

- **If there is an existing instance of the application running**

This option allows the administrator to determine how USO Client should handle the launching of the application when there is an existing instance of the application running. The launch options available for selection are described below:

- **Ignore the existing instance and launch a new instance:** USO Client will ignore the existing instance and launch a new instance. This is the default setting.
- **Switch the focus to the existing instance:** USO Client will automatically switch the focus back to the existing instance.
- **Terminate the existing instance and launch a new instance:** USO Client will terminate the existing instance and launch a new instance.
- **Prompt user to either switch to or just ignore the existing instance:** USO Client will prompt the user to choose whether to switch to the existing instance, or ignore and launch a new instance.
- **Warn the user that an existing instance is still running:** USO Client will give user a warning message saying that existing instance is still running.

- **Record Last Login Time**

If the option is checked, USO Client will record the last login date/time.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.
- **Next** button to continue.
- **Finish** button to save the changes and return to the **Main Screen**.
- **Cancel** button to cancel the application information configuration operation.

Test Credential

In this step, specify an application's test credential that will be used to test login to the application during application testing. This application login information, if specified, will only be used during application testing and will not be exported into the ADF during application export.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.
 - **Next** button to continue.
-

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- **Finish** button to save the changes and return to the **Main Screen**.
 - **Cancel** button to cancel the application information configuration operation.

User Application Attributes

Administrator is allowed to set default value for certain system-defined user-application attributes as well as to manage user-defined user-application attributes. Under typical application training, administrator is not required to make changes to any of the user-application attributes.

- To define new user-application attribute, click on the **[+]** icon. Specify the name of the custom user-application attribute on **Custom Input Field** dialog. Click **OK** button to add it to existing list, otherwise, click the **Cancel** button to cancel the operation.
- To delete any of the existing user-application attribute, select the desired attribute and then click on the **[-]** icon.

Administrator is provided an option to specify whether to allow user to edit User ID if the User ID has been entered during USO in action. By default, user is allowed to do so. To disallow user to do so, select the **Disallow user to edit user ID** option.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.
- **Next** button to continue.
- **Finish** button to save the changes and return to the **Main Screen**.
- **Cancel** button to cancel the application information configuration operation.

Refer to [Edit USO Application Attribute Descriptions \(User/Group Application Attributes Tab\)](#) for the detailed descriptions of user application attributes.

Password Policy

In this step, edit the password policy information governing application login and password change, if necessary, as described below:

Attribute	Description
Password Policy	

Attribute	Description
Enable Offline mode	If checked, USO Client will process the application when user only logs in to local PSE. Otherwise, USO Client will not process the application when user only logs in to local PSE.
Use SID as Password	If checked, USO Client will send the AccessMatrix's Session Id to the application as password field. Otherwise, USO Client will not send the AccessMatrix's Session Id to the application as password field.
Allow Password Reset	If checked, user is allowed to reset the application information via USO Client. Otherwise, user is not allowed to reset the application information via USO Client.
Set SID to Environment	If checked, USO Client will set 2 variables into its environment. The variables are: WSASID - AccessMatrix Session ID APPID - Application ID
Minimum Age (days)	The minimum password age in days. Currently, USO Client does not use this attribute to control the password age.
Password Expiry (days)	The maximum password expiry in days. Currently, USO Client does not use this attribute to control the password expiry.
Change Password Mode	Either of the following values: "Auto" - USO Client will automatically generate a new password based on the password policy after matching the change password screen for the matched application "Manual" - USO Client will prompt the principal to enter the new password
Password Quality	
Let USO generates strong password	This option will be enabled if Change Password Mode is set to "Auto". If option is checked, then USO Client will generate a strong password during password change.
Minimum Length	The minimum length of the newly entered password. Default value is 10.
Maximum Length	The maximum length of the newly entered password. Default value is 15.
Minimum Uppercase	Minimum number of upper case characters to appear in the new password.
Minimum Lowercase	Minimum number of lower case characters to appear in the new password.

Attribute	Description
Minimum Numeric	Minimum number of numeric characters to appear in the new password.
Minimum Special Characters	The minimum number of special characters to appear in the new password.
Maximum Repeat Characters	The maximum number of times the same character can appear in the new password. This field is only applicable for manual change password mode. The auto-generated password will not check against this value.
Customize Special Characters	Specifies the special characters that can be used in the password. Supported special characters: • "~" • "!" • "@" • "#" • "\$" • "%" • "^" • "&" • "*" • "(" • ")" • "-" • "+" Leaving this attribute blank means that all of the above supported special characters may be used.
Password Mask	A mask to control the password generation. A valid mask constitutes a combination of the following masking characters: • "A" - an uppercase letter • "a" - a lowercase letter • "n" - a numeric digit • "s" - a special character • "x" - a random character (i.e. any of the above types of characters) For example, a mask value of "Asanx" may generate a password value of "G\$m8!" or "M@w9v" and so on. This field is only applicable for auto change password mode. If password is manually changed, USO Client will not check the new password entered by the user against this value.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.

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- **Next** button to continue.
 - **Finish** button to save the changes and return to the **Main Screen**.
 - **Cancel** button to cancel the application information configuration operation.

Re-Authentication

In USO Trainer, it can be specified whether to enable re-authentication for the application. Refer to [Configuring Application for Re-Authentication](#) section to see the detailed instructions on how to use this feature.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.
- **Next** button to continue.
- **Finish** button to save the changes and return to the **Main Screen**.
- **Cancel** button to cancel the application information configuration operation.

Single Sign-Off

In USO Trainer, it can be specified whether to enable single sign-off for the application as described below:

- **Enable Single Sign-Off**

If checked, USO Client would perform single sign-off on all the launched applications that are enabled with such option upon logging out from USO Client.

- **Single Sign-Off Enforcement**

This option tells USO Client on how to enforce the single sign-off action as described below:

1. **Graceful:** USO Client will try to sign-off the launched application. If USO Client unable to sign-off the application after the Signoff Timeout lapse, USO Client will continue to logout without signing off the application.
 2. **Graceful Then Force:** USO Client will sign-off the launched application. If USO Client is unable to sign-off the application after the Signoff Timeout lapse, USO Client will force to sign-off the application.
 3. **Force:** USO Client will directly force to sign-off the application, without waiting for the Signoff Timeout to lapse.
-



Note: This option is only enabled when the **Enable Single Sign-Off** option is checked.

- **Single Sign-Off Timeout**

This option tells USO Client on how long USO Client will wait before the sign-off action takes place.



Notes:

1. The default value is 5 seconds.
2. This option is only applicable when the **Single Sign-Off Enforcement** option is set to **Graceful** or **Graceful Then Force**.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.
- **Next** button to continue.
- **Finish** button to save the changes and return to the **Main Screen**.
- **Cancel** button to cancel the application information configuration operation.

Application Checksum

In this step, you can specify whether to enable application integrity checking as described below:

- **Enable Application Integrity Checking:**

If checked, it will enable the fields to specify the executable file and calculate the checksum.

- **Browse for the executable file:**

Click **Browse** and select the executable file of the desktop application being trained.

- **Enter a description and then calculate the checksum:**

Specify a description and click **Calculate Checksum** to calculate the checksum of the executable file.



Note: Checksum is a calculated value of the executable file that will be verified by USO to ensure originality of the application whenever user runs the application to perform SSO. To recalculate, select the calculated checksum and click the **[-]** button to remove it.

After you have finished making the edits, click the:

- **Previous** button to return to the previous step.
 - **Next** button to continue.
-

-
- **Finish** button to save the changes and return to the **Main Screen**.
 - **Cancel** button to cancel the application information configuration operation.

Complete Training of an Application

After performing all the steps described in previous sections, complete the entire application training process by performing the rest of the steps described below:

1. Test Trained Application

Having trained all the desired screen functions as well as application information, proceed to [Testing Trained Application](#) section for detailed instructions on how to test the trained application accordingly.

2. Export Trained Application

Having tested the trained application, proceed to [Exporting Trained Application](#) section for detailed instructions on how to export the trained application information accordingly.

3. Close Trained Application

After finish testing and exporting the trained application, proceed to close the application in order to complete the entire application training cycle and return to **Main Screen**.

Assuming that there is currently an application being trained or loaded in USO Trainer, to close the current application, perform following steps:

1. Click on **Close** button from the quick access toolbar. Alternatively, administrator may select the **Application** menu and then the **Close** sub-menu option.
2. If there is not any screen added to the screen list as shown in following figure, USO Trainer will just close the application directly.
3. Once the application is closed, administrator will return to the **Main Screen**.



Note: After closing the application and if necessary, load the trained application for retraining the application. To do so, refer to the [Loading ADF](#) section for the detailed instructions.

Training Using USO Trainer for Emulator



Note: To train a host emulator application, it is assumed that USO Trainer for Emulator is already launched and displayed. Refer to [USO Trainer for Emulator](#) section for related details.

To start training an application, perform the following steps:

1. Click **Screen** menu and select **Add** sub-menu. The **Set Application Screen Type** dialog is displayed.
2. Select the desired application screen type. USO Trainer for Emulator is only used to train **Host Emulator**, **Console** and **Manual** types of applications.
3. Define the screen recognition criteria for user login screens and the change password screens.
4. Map the screen fields to the respective system-defined or user-defined variables as screen identifier or input fields. Refer to [System-Defined and User-Defined Attributes](#) section for the list of user-defined attributes.
5. Specify the password management policy and other security settings for the specific application
6. Export the screen definitions and password policy into an Application Definition File (ADF).

System-Defined and User-Defined Attributes

System-Defined Attribute	Description
appprn.%uid	User ID This is the variable for storing the user login information.
appprn.%pwd	Password This is the variable for storing the encrypted user password information.
appprn.%newpwd	New Password This is the variable for storing the encrypted new password information. Upon successful password change, the new password information will be stored in the Password variable.
app.identifier	Screen Identifier This is the variable for storing the additional screen identification information to uniquely identify a user screen.
SUBMITBUTTON	Submit Button This is the variable for storing the submit button information so that the submit action can be triggered by USO.

USO also provides the flexibility for Administrators to define user-defined attributes to cater for application specific requirements.

Testing Trained Application

After finished training all the screens for an application, administrator should proceed to test the trained application prior to generating the ADF file. This is to ensure that the screens are trained correctly.

i **Note:** Prior to testing the trained application, administrator is required to install the USO Client on the testing machine.

To test a trained application, perform following steps:

1. Ensure that the USO Client is currently running and not logged in by any user.
2. Click on the **Test** button from the quick access toolbar. Alternatively, administrator may select the **Application** menu and then the **Test** sub-menu option.
3. Specify the application test credential and then click on the **Test** button to test the trained application.

i **Note:** Sometimes, USO Trainer may not be able to communicate with USO Client successfully, i.e. there is not any response in USO Client after clicking the Test button. In this case, administrator just needs to retry by clicking the Test button until USO Client responses accordingly.

4. If the communication is successful, a **Test Mode** login status will appear in the USO Client.
5. Administrator may proceed to test the trained application to see if USO Client behaves as expected.
6. After finished testing the application trained, logout from the USO Client by clicking on the **Logout** button.

To see the detailed instructions, refer to this video:

Link: <https://www.youtube.com/watch?v=9Av4oXDB9ao>

Testing Host Emulator Application

i **Note:** To test a host emulator application, it is assumed that USO Trainer for Emulator is already launched and displayed. Refer to [USO Trainer for Emulator](#) section for related details.

To test a trained application, perform the following steps:

-
1. Ensure that the USO Client is currently running and not logged in by any user.
 2. Click the **Test** sub-menu from the **Application** menu. The **Application Information** window will appear.
 3. Type in the **Application ID** and **Name**, then select the **Launch** Option.
 4. Click on the **OK** button and the **Application Properties** window is displayed.
 5. Make the necessary changes on the **Principal-Application** and **Application** tabs and click on the **Test** button.
 6. A **Test Mode** login status will appear in the USO Client.

Exporting Trained Application

After finished training as well as testing all the screens for an application, administrator may proceed to generate the relevant ADF (Application Definition File) accordingly. This ADF will be used for importing into the AccessMatrix's security registry through the Administration Console.

To generate the ADF, perform following steps:

1. Click on the **Export** button from the quick access toolbar. Alternatively, administrator may select the **Application** menu and then the **Export** sub-menu option.
2. Click on the **Export** button to export the trained application.
3. Select the desired folder and specify the file name to save the ADF (the file will be saved with ".adf" file extension).



Note: The file is now ready to be imported into the AccessMatrix system.

To see the detailed instructions, refer to this video:

Link: <https://www.youtube.com/watch?v=OEMhU6rTb4s>

Exporting Host Emulator Application



Note: To export an ADF in USO Trainer for Emulator, it is assumed that USO Trainer for Emulator is already launched and displayed. Refer to [USO Trainer for Emulator](#) section for related details.

To generate the ADF, perform following steps:

-
1. Click on the **Export** sub-menu from the **Application** menu. The **Application Information** window will appear.
 2. Fill-in the application information and click the **OK** button to launch **Application Properties** dialog box. Otherwise, click the **Cancel** button to cancel the operation.
 3. Make the necessary changes on the **Principal-Application** and **Application** tabs.
 4. Click on the **Export** button when all required properties have been set.
 5. Select the desired folder and specify the file name to save the ADF (the file will be saved with ".adf" file extension). The export successful message box will appear indicating that the ADF file has been successfully created.

Loading ADF

Assuming that an application has been properly trained, tested and exported to an ADF, administrator may load the ADF for reviewing or retraining, if necessary.



Note: USO Trainer version 5.0 or onwards does not support loading ADF that was exported by earlier version of USO Trainer such as version 4.5 or earlier.

To load an ADF file of a trained application, perform following steps:

- **Case 1: USO Trainer Has Not Been Started**

Start the USO Trainer. Depending on the settings, administrator will be presented with either the **Welcome Screen**, or the **Main Screen**. For the **Welcome Screen**, the administrator may click on the **Load an ADF** button. For the **Main Screen**, the administrator may click on the **Load** quick access button.

Alternatively, the administrator may also click on the **Application** menu and then the **Load** sub-menu option.

- **Case 2: USO Trainer Has Been Started**

If there is currently an application being trained or loaded in USO Trainer, close the current application. On the **Main screen**, Administrator may click on the **Load** quick access button, or the **Application** menu and then the **Load** sub-menu option.

For either of above cases:

-
1. Browse to the folder where the desired ADF file resides, select the file and click on the **Open** button.
 2. The administrator may retrain the screens, test and re-export the screens trained as described in previous chapters.

To see the detailed instructions, refer to this video:

Link: <https://www.youtube.com/watch?v=cjFZfzV1FN4>

Loading ADF in USO Trainer for Emulator



Note: To load ADF in USO Trainer for Emulator, it is assumed that USO Trainer for Emulator is already launched and displayed. Refer to the [USO Trainer for Emulator](#) section for related details.

Following are the steps to load/open an ADF file:

1. Select **Load an ADF** File from **Application** menu.
 2. The **Open File** dialog window is displayed.
 3. Browse to the desired ADF file, select the file and click on the **Open** button. The ADF information will be loaded automatically into USO Trainer.
 4. Administrator may retrain the screens, test and re-export the screens trained.
-

7.4. Managing the USO Trainer

Managing Application Screens

The user can manage the application screens meant for training by means of addition, edition, or deletion of the screens.

If the screen added to the list has been trained, it will be prefixed with the thick icon; otherwise it is an empty screen pending to be trained. To add, view/edit, delete, and changing sequence of application screen, refer to this video:

Link: <https://www.youtube.com/watch?v=gFUZ7Z3UTRY>

The application screen sequence matters when there are two or more screens with almost identical screen information trained. In this case, the screen with more specific screen identification information should come first, i.e. at the upper position in the list. This is because during SSO, the screen matching sequence is done from top screen to bottom screen in the list.



Notes:

- If training a new application, it is assumed that user either clicked on the **New** button to start training the application or the **Wizard** button and has completed the wizard.
- If retraining the application, user have to load the ADF file of the trained application first. Refer to the [Loading ADF](#) section for detailed instructions on loading the ADF file of a trained application.

Managing Host Emulator Application Screens

The users can manage as well as maintain the application screens of USO Trainer for Emulator by means of adding, editing, deleting and clearing the screens trained for both new and trained applications.

To maintain the application screens of the trained applications, Administrator has to load the trained applications first. Refer to [Loading ADF in USO Trainer for Emulator](#) section for detailed instructions on loading the ADF file of a trained application.

Once ADF is loaded, click the **Screen** menu and select the desired sub-menu options to manage application screens.

Configuring Application for Re-Authentication

Re-Authentication is one of the security features provided by USO Client to prevent unauthorized login to an application. This chapter presents the detailed instructions on how to train by configuring an application for implementing re-authentication.

There are two levels of configuration involved namely:

1. Application-Level Configuration
2. Screen-Level Configuration

Perform the application-level configuration followed by screen-level configuration or vice-versa to enable re-authentication for the application.

Depending on the user requirement, application-level configuration is a must while screen-level configuration is optional for implementing re-authentication. These configurations will be discussed in details in following sections respectively.

Refer to [USO Client User Guide - USO Application Login with Re-Authentication](#) for detailed description about signing on to application trained with re-authentication option.

Application Level Configuration

As mentioned earlier, re-authentication can be applied on a per-application as well as per-screen basis. Regardless of whichever approach to apply, following steps are prerequisites steps that must be performed.

Having checked the **Enable Re-Authentication** option, USO Client will prompt to re-authenticate user when application screen is matched. When this option is checked, other related options would be provided for selection.

The following are descriptions about other related options:

- **Re-Authenticate Before Launching Application**

This option enables USO Client to prompt to re-authenticate user right before launching the application.

- **Re-Authentication Method**

This option specifies the method used to re-authenticate the user.

Screen Level Configuration

As mentioned earlier, re-authentication can be applied on a per-application as well as per-screen basis. In order to apply re-authentication on a per-screen basis, the relevant application-level re-authentication options must have been enabled or selected as described in previous section.

If the **Require Re-authentication** screen option is checked, USO Client will prompt user for re-authentication before proceeding to process the matched screen during SSO. The re-authentication method to prompt is based on the one configured at application level described in previous section.

Configuring Application for Second User Authentication

Training an application to require another or second user authentication is one of the security features provided by USO Client to prevent unauthorized login to an application. In order to configure an application to require another or second user authentication, the relevant option must be enabled or selected.

If **Require Authentication by Another User** screen option is checked, USO Client will prompt for another or second user authentication, i.e. requesting another or second user to perform primary sign-on to the server, before proceeding to process the matched screen during SSO.

Refer to [USO Client User Guide - USO Application Login with Another/Second User Authentication](#) for detailed description about signing on to application trained with another or second user authentication option.

Configuring Application for Advanced Scripting

Advanced scripting and state-control configuration are competitive features provided by USO Trainer for enhancing the application screen matching mechanism. These features allow the Administrator to configure the matching as well as login behavior for the application in terms of state and control.

State-control is useful for handling application screen matching under unusual scenarios e.g. when the same screen reappears, timed out, disappears, etc. Under such kind of scenarios, administrator may decide on how USO Client should handle the application by implementing state-control for training the application's screen. As such, USO Client will be able to determine the current state of the application and react according to the action defined by the administrator during application training.

Administrator may implement state-control mechanism for the screen being trained as described in following sections.

Advanced Scripting for Screen Timeout

Advanced scripting for screen timeout allows the administrator to specify how the USO Client should react when the matched screen reaches a certain timeout period. Screen timeout configuration encompasses following configuration:

- **Screen Timeout Period**

Specify the screen timeout period in millisecond where 1 second = 1000 millisecond. Take note that certain events may stop the timeout period before it lapses.

- **Screen Timeout Actions**

When the screen timeout period specified above lapses, USO Client will perform the action specified in the **Timeout Action** pane. The action is expressed using JavaScript functions which can be edited by clicking on the **JavaScript Editor** button. User is allowed to add multiple actions, separated by a semicolon (;). Refer to [B. Basic, Standard and User Defined Actions](#) for the list of actions.

- **Screen Disappears Actions**

If the screen disappears before it is timed-out, USO Client will perform the action specified in the **Disappear Action** pane. The action is also expressed using JavaScript functions which can be edited by clicking on the **JavaScript Editor** button.

State Control Configuration for Custom Behavior

State-control configuration for custom behavior allows the Administrator to specify how the USO Client should react to different application flow, behavior or condition. It allows you to set the automated actions to be executed whenever there is a matched screen. The automated actions are executed based on conditions defined. Different condition will trigger different actions to be executed.

The configuration contains a set of conditions. A condition reflects certain situation in terms of state and control of the application. For each condition to be defined, administrator must specify the associated actions to be performed for that specific condition.

When formulating the condition, the value returned should be a logical value, i.e. either true or false. If true is returned then the corresponding actions will be executed.

There are 2 system variables for defining the condition, namely:

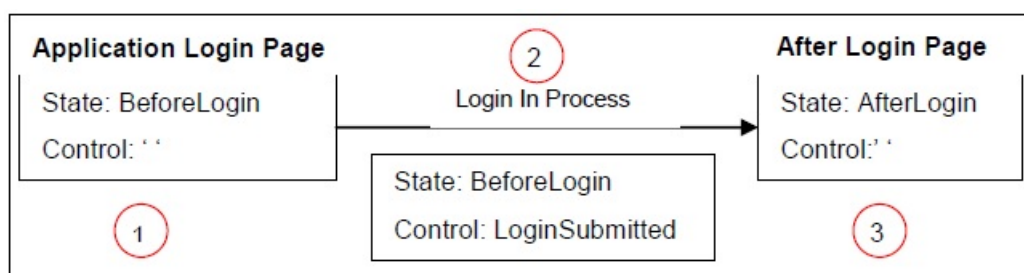
- **State**

State is the variable that reflects a status of the application. USO Trainer has included 5 pre-defined states, namely: NotRunning, BeforeLogin, AfterLogin, AfterLogout, and Disabled. Refer to [A. Application States and Control Variables](#) for additional details.

- **Control**

Control is the variable that contains the interim state of an application. USO Trainer has included 3 pre-defined controls, namely: LoginSubmitted, LoginFailed, and ChangePwdSubmitted. Refer to [A. Application States and Control Variables](#) for additional details.

As an example, following diagram illustrates the transition value of state and control variables for an **Application Login** action:



To translate above diagram into State-Control and Action representation:

Application Login Page

State-Control	Action
control==" && state=='BeforeLogin' (1)	\$ProcessLogin(0,"")

After Login Page

State-Control	Action
control=='LoginSubmitted' && state=='BeforeLogin'(2)	\$HandleLoginSuccess(screen)

Above diagram and table can be further described as below:

-
- When USO Client first matches the application login screen (1), where the condition: `control=="` and `state=='BeforeLogin'` will return true, the action to be carried out is `$ProcessLogin`. The action `$ProcessLogin` changes the control to 'LoginSubmitted' (2).
 - If login is successful, USO Client will match the login successful page, where the condition: `control=='LoginSubmitted'` and `state=='BeforeLogin'` will return true, the action to be carried out is `$HandleLoginSuccess`. The action `$HandleLoginSuccess` changes the control variable to "" and state variable to 'AfterLogin' (3).

Predefined Conditions for Screen Function

For each screen function, there are certain predefined conditions where each condition contains certain control and state variables formed in a logical expression format, e.g.:

```
control == " && (state=='NotRunning' || state == 'BeforeLogin')
```

For each condition, there should be relevant action defined. For example, for the above condition, following action has been defined:

```
$ProcessLogin(0,"")
```

The conditions together with the associated actions vary according to the screen function selected. The condition can be added, deleted, edited, moved up or down.

Advanced Scripting for Custom Menu

Advanced scripting for custom menu allows the administrator to customize the menu items to be provided on the USO Client's **WinAttach** icon that appears whenever a screen is matched. These menu items contain generally the JavaScript functions as the actions to be performed and are to be triggered manually.

For example, if the **Login** function is selected, following menu items are defined:

- **Login Again:** If triggered, this menu item tells USO Client to perform the **LoginAgain** action, i.e. to login to the application again after logout. Normally, this is applicable for application that will return to the login page after logout.
- **Fill:** If triggered, this menu item tells USO Client such that upon matching the screen, USO Client will only fill in the application login information without submitting it.

For each menu item to be defined, there should be relevant action defined for the menu item. For example, for the above menu item **Login Again**, following is the action defined: `$LoginAgain()`.

The menu items together with the associated menu actions vary according to the screen function selected. The menu item can be added, deleted, edited, moved up or down.

Integrating State Control And Scripts

Below are some examples on how the concepts of state control and scripting are integrated.

Default Login Screen Workflow

When the **Login** screen is chosen as the **Screen Type**, USO Trainer will automatically fill the state control parameters and configuration after training of an application.

Custom Behavior

- **State Transition Configuration 1**

Pre-State: control == " && (state == 'NotRunning' || state == 'BeforeLogin')

Action: \$ProcessLogin(0,"")

If the control variable is empty and the state variable is either 'NotRunning' or 'BeforeLogin', then execute the action \$ProcessLogin(0,""). At the beginning of an instance, the state/control is ('BeforeLogin',''), so the above conditional expression will evaluate to true. Therefore the action will be executed. The definition of \$ProcessLogin(not_used, not_used, not_used) is:

Code

```
_ClearTimer()  
_ClearHook()  
if ( _FillAllFieldsAndSubmit() )  
{  
    control = 'LoginSubmitted'  
    _SetTimeout()  
    // _AuditWrite(' ', 'usoAppProcessLogin', 'S', 'USO  
Client successfully processed a login screen')  
}
```

_ClearTimer() will make sure the timer, if any, terminates.

_FillAllFieldsAndSubmit() will perform the playback and submit the screen.

The control variable is set to LoginSubmitted. The state variable remains unchanged.

_SetTimeout() will start the countdown timer, if timeout is configured for the screen.

**Note:**

Change the Custom Behavior for the login screen to allow the user to select the profile to use every time the login screen is displayed. Add `_ResetDefaultProfile()` action before `$ProcessLogin(0,"")` script.

Example value in Script Actions field:

```
_ResetDefaultProfile()  
$ProcessLogin(0,"")
```

- **State Transition Configuration 2**

Pre-State: control == 'LoginFailed'

Action: `$HandleLoginReappearAfterLoginFail()`

If the control variable is 'LoginFailed', this usually means that a "LoginFail" screen has been encountered. In that case, the action `$HandleLoginReappearAfterLoginFail()` will be executed. The definition of `$HandleLoginReappearAfterLoginFail()` is:

Code

```
var result=_PromptCustomButtons(  
_GetMessage('IDS_MSG_CAPTION_LOGIN_TO_APP'),  
_GetMessage('IDS_MSG_PROMPT_RESET_LOGIN_DISABLE'),  
_GetMessage('IDS_MSG_BUTTON_RESET_LOGIN_DISABLE') )  
if (result == 0)  
{  
    $ResetAppLoginInfoAndLoginAgain()  
}  
else if (result == 1)  
{  
    $LoginAgain()  
}  
else if (result == 2)  
{  
    $DisableApplication(false)  
}
```

In short, it will prompt the user to either **Reset Application, Try Again** or **Disable Application**.

-
- If user chooses **Reset**, then the standard action `$ResetAppLoginInfoAndLoginAgain()` will be executed. This prompts the **ResetAppInfo** dialog (like below) and tries to login again.

USO Client - Set Application Information

Login form

USO Client has detected an USO-enabled application screen or you have chosen to reset the application information.

Please provide your login information for Login form:

User ID:

Password:

Confirm Password:

	Attr Name	Attr Value
1	domain	domain

OK Cancel

- If user chooses **Login**, `usoClient` will just try to login again.
- If user chooses **Disable**, `usoClient` will disable the application.

- **State Transition Configuration 3**

Pre-State: `state == 'AfterLogin' && control == "`

Action: `$HandleLogout()`

This means that if the login screen is matched when the state is 'AfterLogin', we should assume that it is a logout. We should not try to fill in and submit the screen. For example, Policy Editor. `$HandleLogout()` will change the state to 'AfterLogout'. Since none of the pre-states will evaluate to true after the change, USO Client will not perform any action on the screen.

Timeout Actions

Timeout: 5000 ms

Timeout Action: `$HandleLoginSuccess('Timeout')`

Disappear Action: `$HandleLoginSuccess('Screen')`

Five seconds after this screen has been processed, if nothing halts the timer, then the script `$HandleLoginSuccess('Timeout')` will be executed. That means, by default, it is assumed that the login is successful.

Before the five seconds are up, and the timer is still active, if the window disappears, then the script `$HandleLoginSuccess('Screen')` will be executed. It is also assumed that the login is successful.

The definition of `$HandleLoginSuccess(reason)` is:

Code

```
_ClearTimer()
_ClearHook()
control = ''
// AuditWrite('', 'usoAppLogin', 'S', 'USO Client
detected application login was successful', '', '',
'reason')
state = 'AfterLogin'
if ( _IsPwdChangeRequired() )
{
    if ( typeof(NavigateToChangePwdScreen) ==
'function')
    {
        NavigateToChangePwdScreen()
    }
}
```

`_ClearTimer()` will make sure the timer terminates, in effect stopping the Timeout Action from occurring.

State/control is set to ('AfterLogin','') to reflect the state after login is successful. The previous state/control was probably ('BeforeLogin','LoginSubmitted').

The last portion is to check if Forced Change Password is required, and if required, it will perform the necessary action.

Custom Menu

Menu Label: **Login Again**

Action: `$LoginAgain()`

In case of `$HandleLogout()`, if the user wants to login again after some time, the user can right-click on the **usoWinAttach** icon and click **Login Again**.

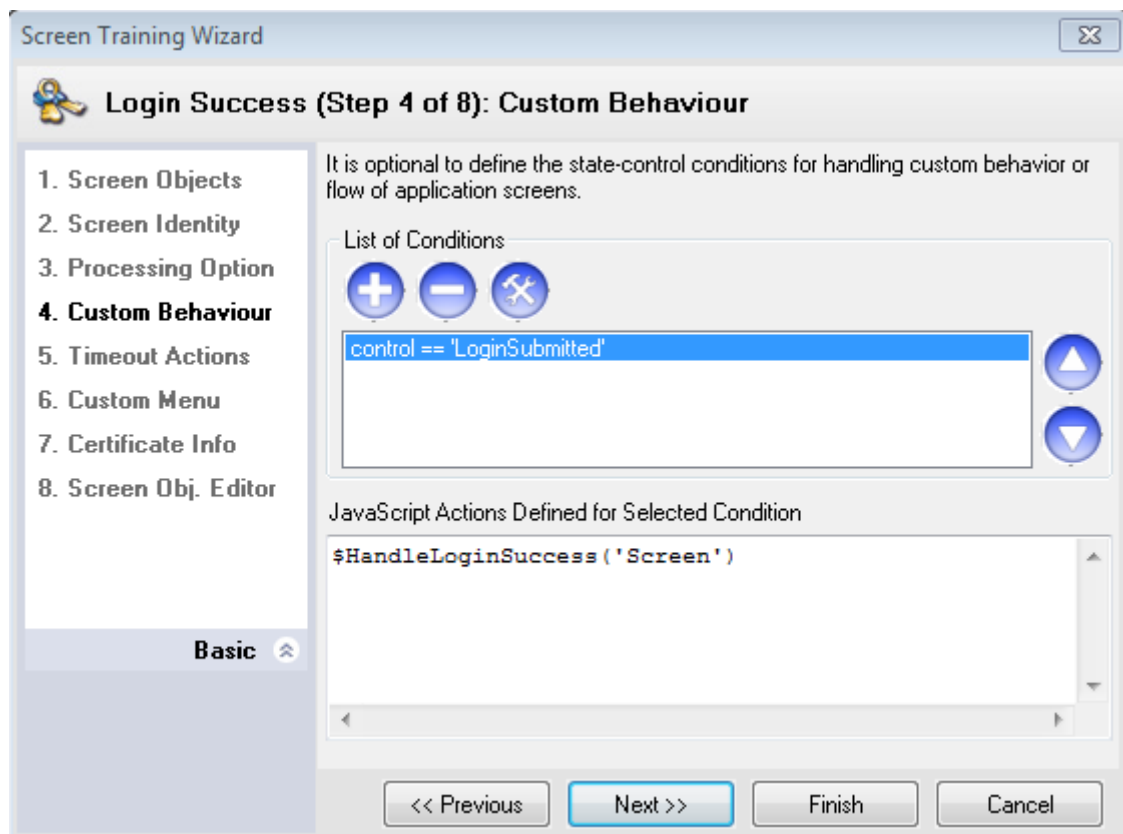
Menu Label: **Fill**

Action: `_FillAllFields()`

If the user clicks **Fill**, then the USO Client will only fill the values without submitting the screen.

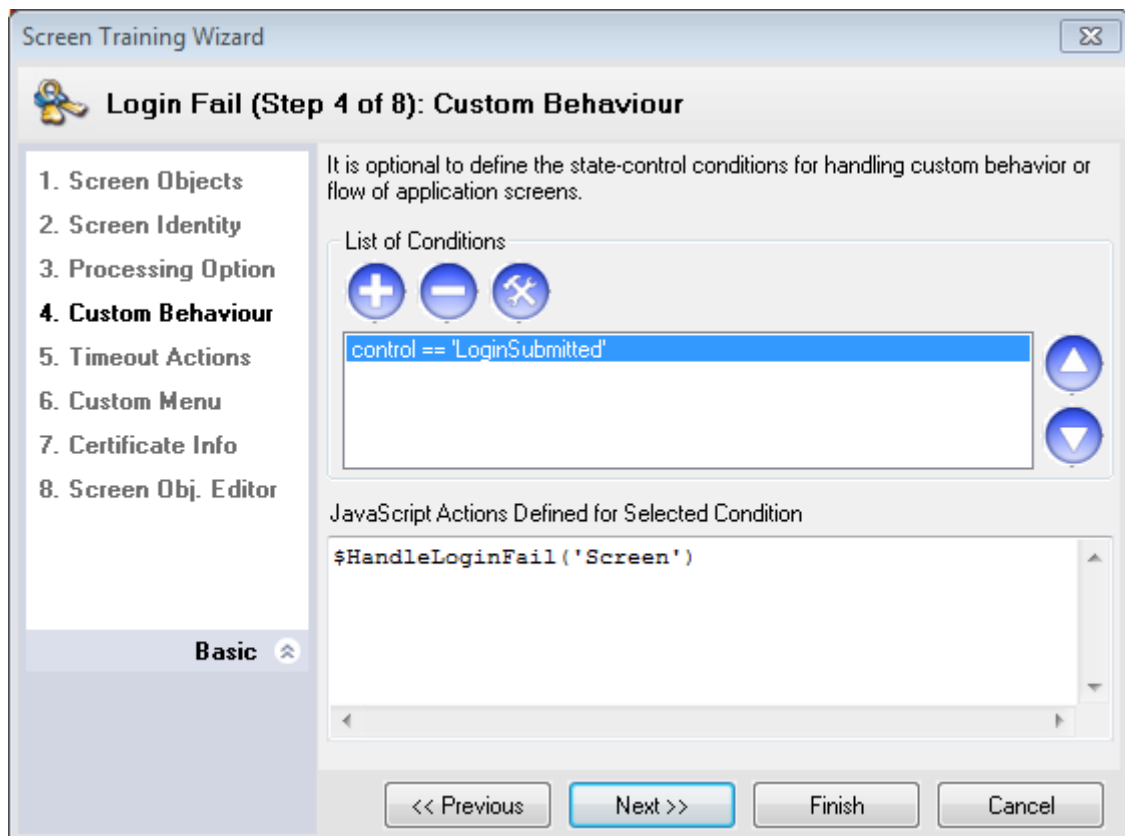
Default Login Success Screen Workflow

When the **Login Success** screen is chosen as the **Screen Type**, the below state-control condition is displayed.



Default Login Fail Workflow

When the **Login Fail** screen is chosen as the Screen Type, the below state-control condition is displayed.



The definition of \$HandleLoginFail(reason) is:

Code

```
_ClearTimer()  
_ClearHook()  
control = 'LoginFailed'
```

Configuring Application Summary

Upon finish training the application screen, administrator may proceed to view or configure the application summary manually, if necessary.

Upon startup, USO Trainer shows the **Welcome Screen** or **Main Screen**. The **Main Screen** is also known as the **List View**. In order to configure the application summary information, administrator must switch the **List View** to **Summary View**. The administrator is allowed to switch freely between the **List** and **Summary Views** throughout the entire application training process.

Basically, the application summary view is divided into following two categories:

-
- **Application Information:** This refers to information pertaining to application itself that also encompasses the screen information trained for the application.
 - **User Profile Information:** This refers to information pertaining to user such as the user profile information for login to application.

Administrator is allowed to configure any of the above-mentioned application summary information as described in details in following sections.



Note: This option should be used only after administrator has finished training all the application screens defined for the application. It acts as a short cut for administrator to configure by editing certain application information directly in a manual manner.

If it is unnecessary, the administrator is not advised to configure the application summary manually especially if he or she is not familiar with the configuration.

Switching Between List and Summary Views

Assuming administrator has finished training all the screens defined for the application, to configure the application summary information, administrator has to switch from **List View** to **Summary View**.

To switch from **List View** to **Summary View**, click on the **Tools** menu and select the **Summary View** sub-menu option. The **Main Screen** will change from **List view** to **Summary View**.

To switch back to **List View**, click on the **Tools** menu and select the **List View** sub-menu option. The **Main Screen** will change from **Summary View** back to **List View**.

Configuring Application Information

To view and edit application information, perform following steps:

1. Switch to **Summary View** as described in the [Switching Between List and Summary Views](#) section.
 2. To configure the application summary information, click on the **[+]** icon next to Application node to expand it.
 3. If necessary, administrator may modify the information carefully. For example, to change the Application ID, select the **ID field** and then right-click on it and select the **Edit** pop-up menu.
-

i **Note:** Some fields may provide drop-down list to choose. In the above case, select the desired option from the drop-down list and then press Enter to accept the change.

Configuring Screen Information

Similar to making changes to application information, administrator may edit the screen information, if and only if necessary.

i **Note:** The administrator is strongly advised not to make changes to the screen information directly through this way, but to train again the screens to avoid any mistakes made to the screens trained.

To view and edit screen information, perform following steps:

1. Switch to **Summary View** as described in [Switching Between List and Summary Views](#) section.
2. From the **Summary View**, click on the **[+]** icon next to **Application** node to expand the application node.
3. Click on the **Screen Info** node under the **Application** node to expand the screen information node. Administrator may further click on **[+]** to expand the desired individual screen node to view or edit the screen details.

Dynamic Windows Class Name for WINAPI/Desktop Applications

Windows class name is different on different platforms (32-bit/64-bit), .NET version and windows version. If a Windows form is trained on Windows 32-bit platform, USO Client will not be able to match for the same Windows form on 64-bit platform as the Windows class name is different. Therefore, it is recommended to change the windows class name into a general name that would work on different windows platforms. The format should be "Classname.FieldType@index", e.g.

Windows class name for a textbox (64-bit platform) :

API_WindowsForms10.EDIT.app.0.2bf8098_r14_ad1@1

Windows class name for the same textbox (32-bit platform) :

API_WindowsForms10.EDIT.app.0.2bf8098_r16_ad1@1

The recommended common general name is "API_WindowsForms10.EDIT@1".

Configuring Miscellaneous Options

Apart from the features described in previous chapters, USO Trainer also provides other miscellaneous configurable options described in details in following sections.

Configuring General Options

Configuring Welcome Screen Option

Upon startup of USO Trainer, Administrator may opt whether to show the **Welcome Screen** which will be shown by default. Administrator may configure the USO Trainer not to show the **Welcome Screen** via either of following means.

- **Through Welcome Screen**

Upon startup of USO Trainer, on the **Welcome Screen**, check the **Do not show this welcome screen** option and then click on **Close**.

- **Through Main Menu Option**

On the **Main Screen**, click on the **Tools** menu and select the **Options** sub-menu. Uncheck the **Show Welcome Screen** option to disable the **Welcome Screen** and then click on the **OK** button to save the change.

Configuring Default Language Option

USO Trainer supports languages other than default English language interfaces. Administrator may choose other supported languages as the default language interface.

To change the interface to another language, perform following steps:

1. On the **Main Screen**, click on the **Tools** menu and select the **Options** sub-menu option.
2. Select the desired language option.
3. Click **OK** and then restart the USO trainer for change of language to take effect.

Configuring Additional Classes Option

USO Trainer provides a competitive feature that allows Administrator to specify additional classes for object recognition purpose during application training. The information of these additional classes

added via USO Trainer will be used by USO Client for matching the application screens trained during SSO in action.

To specify by adding additional classes in particular the class names for object recognition use, perform following steps:

1. On the **Main Screen**, click on the **Tools** menu and select the **Options** sub-menu option.
2. On the **Options** window, click on **Class** tab.
3. Specify the class name accordingly and then click on the **Add** button. The additional class name will be added to the list.

Configuring Java Runtime JRE Option

USO Trainer provides another competitive feature that allows administrator to specify additional JRE used for Java application communication during application training. The information of these additional JRE added via USO Trainer will be used by USO Client to perform screen matching for Java-based applications during SSO in action.

Thus, if the JRE used by the application is not available in the JRE list of USO Trainer, administrator is required to specify the JRE explicitly.

To view the JRE list as well as to specify additional JRE, perform following steps:

1. On the **Main Screen**, click on the **Tools** menu and select the **Options** sub-menu option.
2. On the **Options** window, click on **JRE** tab.
3. Click on **Browse** button to browse through local folders for the additional JRE.
4. Select the desired JRE directory path and then click on the **OK** button. The selected JRE directory patch will be added to the list.
5. Click **OK** button and then restart the USO Trainer for change to take effect.

Configuring User-Defined Actions

USO Client provides a **JavaScript Editor** screen where the user is allowed to select the basic and standard actions and can also create a user-defined actions to be executed based on a specific condition.

The **JavaScript Editor** screen can be launched in USO Trainer 5 in different ways:

-
- By clicking **Tools > User-Defined Actions** menu.
 - By clicking **Edit Selected Condition button** from the **Custom Behaviour** screen.
 - By clicking **JavaScript Editor button** from the **Timeout Actions** screen.
 - By clicking **Edit Selected Menu Item** from the **Custom Menu** screen.

i **Note:** User can click **User Defined Action -> New** to launch **Action Editor** using USO Trainer for Emulator.

JavaScript Editor User Interface

The **JavaScript Editor** has the following fields:

- **Action Name**

Refers to the name of the action to be created. This field is required when adding a user-defined action.

i **Note:** This is only displayed when the **JavaScript Editor** is launched from **User-Defined Actions**.

- **Parameters**

[OPTIONAL] Refers to the parameter(s) of the action to be created.

i **Note:** This is only displayed when the **JavaScript Editor** is launched from **User-Defined Actions**.

- **Description**

Refers to the description or usage of the action to be created.

i **Note:** This is only displayed when the **JavaScript Editor** is launched from **User-Defined Actions**.

- **Condition**

This displays the state-control conditions for the behavior of an application screen.

i **Note:** This is only displayed when the **JavaScript Editor** is launched from the **Custom Behaviour** screen.

- **Menu Item Name**

Displays the name of the selected menu item.

i **Note:** This is only displayed when the **JavaScript Editor** is launched from the **Custom Menu** screen.

- **Script Actions**

Refers to the JavaScript commands that the action will perform.

- **Help Message or Validation Result**

Displays the function name and usage of the selected action in **Defined Actions** field. It also displays the validation result of the JavaScript actions.

- **Defined Actions**

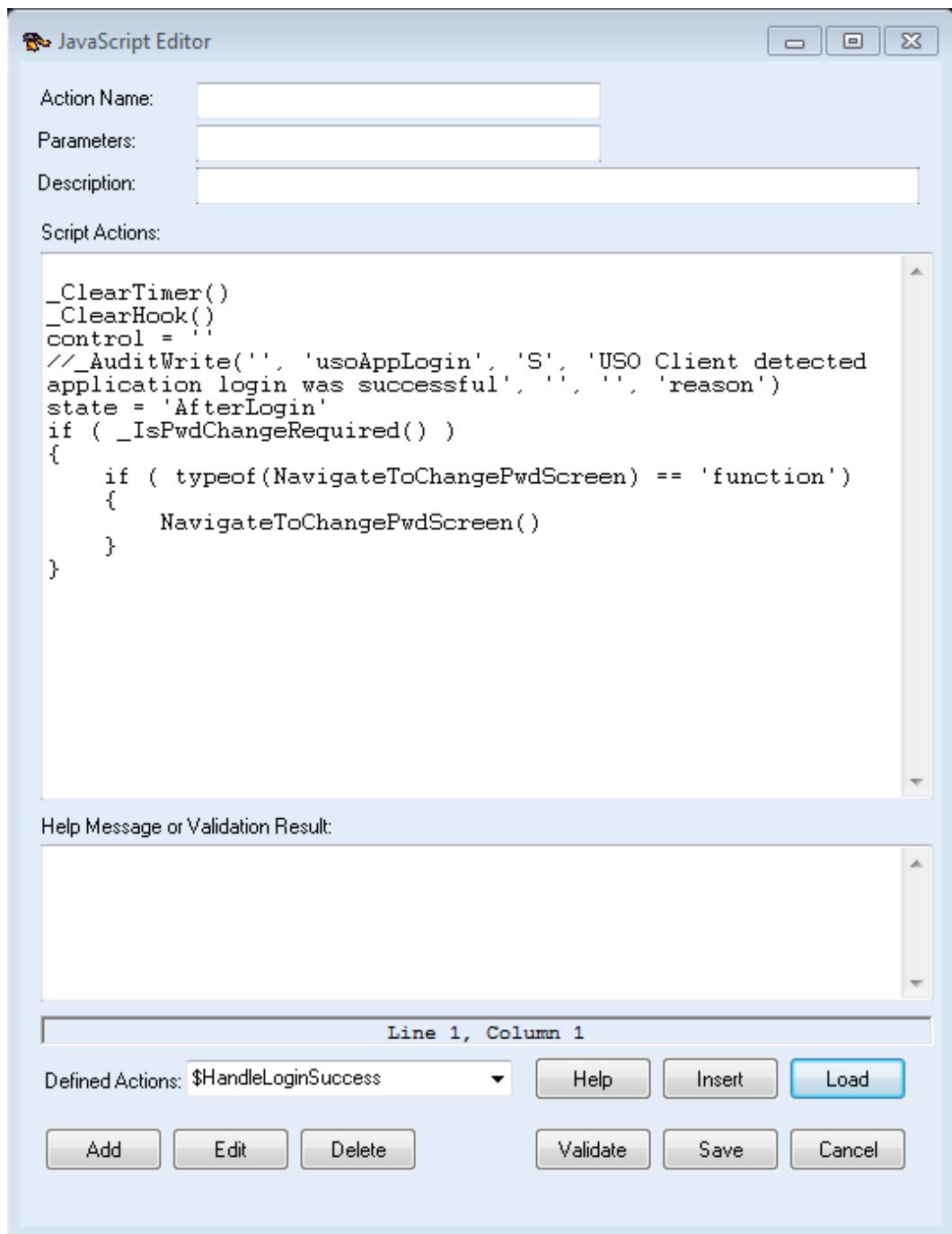
List the basic, standard and user-defined actions.

- **Basic Action**

Basic action is an atomic action. Basic action begins with an underscore "_". For example: `_FillAllFieldsAndSubmit`, `_AuditWrite`, etc.

- **Standard Action**

Standard action is a logic-flow action. Standard action contains JavaScript commands and/or basic actions. The action begins with a dollar sign. For example: `$DisableApplication`, `$HandleChgPwdFail`, etc.



By choosing a standard action and clicking **Load**, the contents of this action can be seen. Some standard actions require parameters. In this case, Administrator must specify accordingly by editing the parameters directly on the action pane.

- **User-Defined Action**

User-defined action is a logic-flow action defined by the user. User-defined action may contain JavaScript commands and/or basic as well as standard actions.

Note: This is only allowed when **JavaScript Editor** is launched from **User-Defined Actions**.

The JavaScript Editor has the following buttons:

- **Help**

This button will display the action's function name and usage description, as well as the validation result to **Help Message or Validation Result** field.

- **Insert**

This button will display the function name & parameter(s), if applicable of the selected action to **Script Actions** field.

- **Load**

This button will load the JavaScript commands of the selected standard or user-defined action to the **Script Actions** field. This button is hidden when basic action is selected.

- **Validate**

This button will verify the action scripts for any error. If there is no error, the Validation Result text area will display the **No Error** message, otherwise it will display the relevant error message.

- **Save**

This button will save the JavaScript commands written in the **Script Actions** field.

- **Cancel**

This button closes the **JavaScript Editor** screen and cancels the changes made.

- **Add**

This button creates a user-defined action. The **Action Name** should be provided for this button to work.

- **Edit**

This button edits the selected user-defined action in **Defined Actions** field.

 **Note:** Only user-defined action can be edited.

- **Delete**

This button deletes the selected user-defined action in **Defined Actions** field.

 **Note:** Only user-defined action can be edited.



Notes:

- USO Trainer for Emulator also offers the same feature and it only differs in user interface display from USO Trainer 5.
- Select **User Define Action->New** to create a user-defined action in USO Trainer for Emulator.
- Select **User Define Action->Edit** to edit a user-defined action in USO Trainer for Emulator.

-
- Select **User Define Action->Delete** to delete a user-defined action in USO Trainer for Emulator.

Enabling Audit In Script

USO Client uses scripts to control how it should react to different application flow, behavior or condition. Please see [State Control Configuration for Custom Behavior](#) and [Configuring User-Defined Actions](#) sections for related information.

There are some standard action scripts that contains the `_AuditWrite` basic action. This basic action script will display the audit events in report. Currently, this action script is disabled/commented in some standard action scripts. To activate it, just enable/comment out the `_AuditWrite` part as shown below:

\$HandleChangePwdFail

Code

```
_ClearTimer()
_ClearHook()
_control = ''
_RollbackAppNewPassword()
_AuditWrite('', 'usoAppChgPwd', 'F', 'USO Client
detected application change password was failure', '',
'', 'reason')
```

\$HandleChangePwdSuccess

Code

```
_ClearTimer()
_ClearHook()
_control = ''
_CommitAppNewPassword()
_AuditWrite('', 'usoAppChgPwd', 'S', 'USO Client
detected application change password was successful',
'', '', 'reason')
state = 'AfterLogin'
```

\$HandleLoginFail

Code

```
_ClearTimer()
_ClearHook()
control = 'LoginFailed'
_AuditWrite('', 'usoAppLogin', 'F', 'USO Client
detected application login was failure', '', '',
'reason')
```

\$HandleLoginSuccess

Code

```
_ClearTimer()
_ClearHook()
control = ''
_AuditWrite('', 'usoAppLogin', 'S', 'USO Client
detected application login was successful', '', '',
'reason')
state = 'AfterLogin'
if ( _IsPwdChangeRequired() )
{
    if ( typeof(NavigateToChangePwdScreen) == 'function')
    {
        NavigateToChangePwdScreen()
    }
}
```

\$HandleLogout

Code

```
state = 'AfterLogout'
control = ''
_AuditWrite('', 'usoAppLogout', 'S', 'USO Client
detected application logout')
```

\$ProcessChangePwd

Code

```
_ClearTimer()
_ClearHook()
if ( _FillAllFieldsAndSubmit() )
{
    control = 'ChangePwdSubmitted'
    _SetTimeout()
    _AuditWrite('', 'usoAppProcessChgPwd', 'S', 'USO
```

```
Client successfully processed a change password
screen')
}
```

\$ProcessLogin

Code

```
_ClearTimer()
_ClearHook()
if ( _FillAllFieldsAndSubmit() )
{
    control = 'LoginSubmitted'
    _SetTimeout()
    _AuditWrite('', 'usoAppProcessLogin', 'S', 'USO
Client successfully processed a login screen')
}
```



Note: Refer to [B. Basic, Standard and User Defined Actions](#) to see the list of defined actions.

Train Web Applications with Multi-level Frame

There are web applications with multi-level frames. An example of this is a text box for user Id that resides in a form and this form is part of an iframe, and the iframe is part of another frame/iframe.

Perform the steps below to train this type of application:

1. Train the application using native approach. The field or identifier will be displayed under **Trained Screen Objects**.
2. Double click the field, e.g. **User Id** or **User Name** field and fill the text box under **Trained Screen Objects** with a similar value to the one shown below:
"window.frames['main'].frames[1].document.getElementsByName('userID')[0]"
3. Press ENTER key to save the changes.
4. Do the same for Password field and fill the text box under **Trained Screen Objects** with similar value as below:
"window.frames['main'].frames[1].document.getElementsByName('password')[0]"
5. Press ENTER key to save the changes.



Notes:

-
- | |
|---|
| <ul style="list-style-type: none">• Only supports Login Screen.• Ensure that JavaScript has been enabled in your browser. |
|---|

Cross Frame Support

USO also supports cross frame scripting (different domain) where contents in different windows and frames can interact in powerful ways by scripting with the object model.

Supported Browsers:

- Google Chrome

Train Application by Coordinate

To train an application by coordinate:

1. Follow the steps described in [Full-Fledged Wizard-Driven Application Training](#) section to display the **Screen Training Wizard**. Alternatively, follow the steps described in [Standard Wizard-Driven Application Training](#) section to display the **Screen Training Wizard**.
2. On the **Screen Training Wizard**, uncheck **Training using native approach** option and check **Recognize Object by Coordinate** option.
3. Proceed the training of application as described in [Full-Fledged Wizard-Driven Application Training](#) section or [Standard Wizard-Driven Application Training](#) section.

Train Mobile Application

To train a mobile application:

1. Follow the steps described in [Full-Fledged Wizard-Driven Application Training](#) section to display **Screen Training Wizard**. Alternatively, follow the steps described in [Standard Wizard-Driven Application Training](#) section to display **Screen Training Wizard**.
 2. On **Screen Training Wizard**, expand to display **Advanced Mode**. Select **Screen Object Editor** and modify the HTML tag values, if necessary. Refer to [Screen Object Editor](#) for the option's description.
 3. Proceed to the training of application as described in [Full-Fledged Wizard-Driven Application Training](#) section or [Standard Wizard-Driven Application Training](#) section.
-

Enabling Re-authentication

To enable re-authentication:

1. If using screen training wizard, then check the **Require Re-authentication** option during screen training as described in [Processing Option](#) section.
2. If using application training wizard, then check the **Enable Re-Authentication** option during application training as described in [Re-Authentication](#) section.
3. Refer to [Full-Fledged Wizard-Driven Application Training](#) section for additional details.

Hide or Change Custom Input Field to Non-editable

There are cases where user is not allowed to edit the custom fields and only constant values are used to fill-in the field. To do this, perform the following steps:

1. In the USO Trainer, select the screen to train. At **Screen Objects**, train the fields or identifiers as usual. To test this use case, make sure you train a field as Custom Input (e.g. Domain field in Demo Desktop).
2. The trained fields will be displayed under **Trained Screen Objects**. Right click on the custom input field (e.g. domain) and click the **Disallow user from resetting the field value** option. The **Non editable** description will be added to the custom input field.
3. Export the trained application to ADF once training is done.
4. Open the ADF and set the default value of the custom input field under **PrincipalApplicationAttribute** and save the changes. Refer below for an example:

Code

```
<PrincipalApplicationAttribute>
  <PrnAttrib name="%profilename.1" Encryption="no"
default_value="Profile 1" />
  <PrnAttrib name="%uid.1" Encryption="no"
default_value="" />
  <PrnAttrib name="%pwd.1" Encryption="yes"
default_value="" />
  <PrnAttrib name="%pwd.lastModified.1"
Encryption="no" default_value="" />
  <PrnAttrib name="%domain.1" Encryption="no"
default_value="myDomain" />
```

-
5. Upload the ADF into AM and entitle the application to users. AM will automatically set the default value for all users of the application.
 6. Login to AccessMatrix SSO Portal and launch the application. The custom input field is hidden and not displayed during screen matching and resetting of application information. The default value set in previous step will be automatically supplied to custom input field.

Submit Method using Keyboard Event

The Submit Method can also be trained using the field finder, if applicable. In cases where there is no submit button, the submit method can be trained as Keyboard Event.

1. Follow the steps described in [Full-Fledged Wizard-Driven Application Training](#) section to display the **Screen Training Wizard**. Alternatively, follow the steps described in [Standard Wizard-Driven Application Training](#) section to display the **Screen Training Wizard**.
2. On the **Screen Training Wizard**, select **Submit Method**.
3. Select **Keyboard Event** option. The **Select the keyboard event** combo box is displayed.
4. Select the keyboard event to use, i.e. {ENTER} or {TAB},{ENTER}. The event will be added under **Trained Screen Objects**.
5. Proceed the training of application as described in [Full-Fledged Wizard-Driven Application Training](#) section or [Standard Wizard-Driven Application Training](#) section.

Train Submit Method Using JavaScript

For Web Application, the submit method can also be trained as Javascript function.

1. Follow the steps described in [Full-Fledged Wizard-Driven Application Training](#) section to display **Screen Training Wizard**. Alternatively, follow the steps described in [Standard Wizard-Driven Application Training](#) section too to display the **Screen Training Wizard**.
2. On the **Screen Training Wizard**, select **Submit Method**.
3. Select the **JavaScript** option. Enter the javascript to use to submit, e.g. javascript:submit(). Press Enter key. The script will be added under **Trained Screen Objects**.
4. Proceed the training of application as described in [Full-Fledged Wizard-Driven Application Training](#) section or [Standard Wizard-Driven Application Training](#) section.



Note: Use any JavaScript that will trigger the submit event.

Enable Support for SAP SSO Using WinAPI

1. To perform SAP SSO, load the SAP ADF template to USO Trainer. The ADF template is provided below:

Code

```
<?xml version="1.0" standalone="yes" ?>
<!USO Application Definition File, created by USOTrainer 5>
<USOADF version="1.1" id="USOADF100XMLISPRINT">
  <Application name="SAP @ ASL">
    <Description value="SAP @ ASL" />
    <Type value="USO" />
    <TabApplication>
      <ApplicationId value="SAPWin" />
    </TabApplication>
    <TabMethod />
    <TabRole>
      <Role name="user" />
    </TabRole>
    <TabApplicationAttribute>
      <Attrib name="%data.version" value="2" Encryption="no"
/>
      <Attrib name="%info" Encryption="no"
value="AsRT+yBNv31VxPty1ZmJN9lvleE6ZoHH+CGjUTB0P7K52sy0idEmBnqCEPSZ
K0AVJIHv8sotwvd1
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```

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8GF/8BusJ
+FEa9wAF3JLxldKNUcBVWlfpNZ0tiEgDagEurQ2qA7JpdfEEpJhp3iv09G3ejd9G/Pf
T8D4D2wkp
AZjEbCMkn4WviJvVZRqIM9wMMjwJ6wkYtbg7Xe8SoIlu2MQdLHrqjceQESiDxgrjcwD
UrhxBVPjr
3q5gt8yAm9r1OWrMbZhM3LmmZ/xbKgLzVu9tB38sIGw76pA5xYx1smL4Ql/WH+eO53f
j6CTqxAQh
kKLvpDnIwAKXDDawrWlVSAvZOUPAOGt/vFL9/15VXAYi33P+v/VUbpFFCmNO4kcEoL/
F5v6NP91k
75Js1valIccia76oIOhj8kF/ZJ52aKK+gE3CBoflsVZsOt2TTHvdnETkeNnNeufGNTE
Z/r7DZeEV
PlZiwAlMSNu5u9sOxnxcQ/DBPPpzjM1TY+mYqxWxgBesLEDeoNR5mMatUe4wFmwWdRM
8DB+AVuQ1
7yeKpenAvMfXtQd7mbFPcdKZJ7AHdhI6elArGt3JoFrT23ftGhey1wBX/kRrw4EL5BE
UC2CD2Mmi
4k346yv9HNH7TsA8WsY7p/cwRhbsVq0bcKSyaJsI8foscKTeS54XiDnYQX2qMqp1GeU
sUpDICGEF

aPC94zMUvwPbgM1M1RPyPuVECoaXEcZiWV5nOugM0gOteSaMh1k3C+HoGXW8cE0IY3Q
0gZ+CNJOw
CIr17TvzDKlFIOIS0/VGc7AA1PBFjDGIKbycSKyt9FLBhM41p7oGWaqcnZWDzuc/QT5
W04fHVZAr
FCwlgezsq7cCvxUfi9GwkKqFtPLE/Ppz6BoI+fEstxezM/oVwFZg+78nyx2fKZgTWvZ
GxwEaHEht
IelyG9x0tYn5uT0jIxJR1fvvgwD4+F/8va392P0+DK4ZtAP9y2i7NpCt30vZetCVNOJK
oNdVdIfuO
RujYtGkCNKhfaTz0BhdHLv+rAqQzqx2sAyvtyH0ipHbgSiI+LvYuUobEv5sL8GvOjS7
2voQF567N
9Kd+ziq6WQJybjfPHVtEpMV8+VICyGUF2IIwNI4WnArgQglm97qo5MOimddpztw8b2f
pADgruh0S
BR49bAd00f978awJiSV6159nwmzbmjcwNqq4UXcdDaoDsml18QTZtG4vQWhnWqQmOKu
lH0m9YhWo
owbziDOJqHWYmd2xltVlGogz3AwylNwcrT18TWDejd9G/PfT8KPPCTKm7ioeGSZzSQu
GUJmF/OVK
7XpNJRobW++4G27SCORaytt3Ah525zN4HhhP7GC3zICb2vU5SrXJkrqdOvYYTPuUvkn
o6PVcoMMQ
vvT/pS6vj7cBQhTZh1IZNizHo2f8WyoC81bvt6itH6xTwvgIVDsI/D50bCTNUj3CrkR
ZAJ2d9Vhh
zhY53Bv2rKmIcqHH/N/J3tMfnJkckvtjatc9uG0YZcwzi8liOsgWxptu12U0ebKZJ2E
eXamMW+WZ
oheLhuV9ktzdw0oVvP0RtTq9AJaJvT5zZsVIGs2cfHA8E8o70N2CSyumWADmuHEF7q
UU2+lIsTj
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eXvKWbgyF
BJfxClF+sdJ5Snpiw93F1Um1DadCBGIOw1U1Gby/+ZbluiegubWu1gJsdLoP8fCeqeB
kDRDuXt1T
P/UxIxGzeQYu/nHQKj5BJoTT9jNpqai6Dy7LC72B0xDMnEtOLeehm5R1Qfw8DVAzupD
o80yN+4OF
3xgXZfE9rdJGr6HH/YDYzfM6hw00oNxNwcd/xVAvORkk4xn8NofTrEjWMtmENHm9Fwz
ojarY1/vD
6C1lPFihzEU2U2/kSHMhXvGwMMoVAuY2qSWP9QQ+Pbeb8MWxiAeOsr45qwsaYaBiZRp
gMw== " />

```
<Attrib name="%apppath" value="c:\program files
(x86)\sap\frontend\sapgui\saplogon.exe" Encryption="no" />
<Attrib name="%workDir" value="c:\program files
(x86)\sap\frontend\sapgui" Encryption="no" />
<Attrib name="%argument" value='"/shortcut=&quot; -
sid=&quot;FYC&quot;&quot;' Encryption="no" />
<Attrib name="%apppath.citrix" value="" Encryption="no"
/>
<Attrib name="%launchOption" value="Ignore"
Encryption="no" />
<Attrib name="%processexternlaunch" value="Yes"
Encryption="no" />
<Attrib name="%offline" value="Yes" Encryption="no" />
<Attrib name="%multiple.profile" value="1"
Encryption="no" />
<Attrib name="%multiple.pwd.prn" value="1"
Encryption="no" />
<Attrib name="%multiple.pwd.concat.uid" value="0"
Encryption="no" />
<Attrib name="%multiple.pwd.concat.pwd" value="0"
Encryption="no" />
<Attrib name="%recordLastLoginTime" value="No"
Encryption="no" />
```

```

        <Attrib name="%allow.launchapp" value="Yes"
Encryption="no" />
        <Attrib name="%allow.draganddrop" value="No"
Encryption="no" />
        <Attrib name="%pwd.reset" value="Yes" Encryption="no"
/>
        <Attrib name="%pwd.isSID" value="No" Encryption="no" />
        <Attrib name="%setenv" value="No" Encryption="no" />
        <Attrib name="%pwd.chg.mode" value="Manual"
Encryption="no" />
        <Attrib name="%pwd.min.length" value="0"
Encryption="no" />
        <Attrib name="%pwd.max.length" value="0"
Encryption="no" />
        <Attrib name="%pwd.gen.minuc" value="0" Encryption="no"
/>
        <Attrib name="%pwd.gen.minlc" value="0" Encryption="no"
/>
        <Attrib name="%pwd.gen.minn" value="0" Encryption="no"
/>
        <Attrib name="%pwd.min.special" value="0"
Encryption="no" />
        <Attrib name="%pwd.maxrepc" value="0" Encryption="no"
/>
        <Attrib name="%pwd.min.age" value="0" Encryption="no"
/>
        <Attrib name="%pwd.max.expiry" value="0"
Encryption="no" />
        <Attrib name="%pwd.gen.mask" value="" Encryption="no"
/>
        <Attrib name="%pwd.randomizedBySystem" value="No"
Encryption="no" />
        <Attrib name="%pwd.includeSpecChars" value="No"
Encryption="no" />
        <Attrib name="%verifyReplay" value="Disable"
Encryption="no" />
        <Attrib name="%verifyReplayMethod" value=""
Encryption="no" />
        <Attrib name="%verifyReplayBeforeLaunch" value="No"
Encryption="no" />
        <Attrib name="%rememberprofile" value="No"
Encryption="no" />
        <Attrib name="%emu.name" value="" Encryption="no" />
        <Attrib name="%emu.version" value="" Encryption="no" />
        <Attrib name="%emu.host" value="" Encryption="no" />
        <Attrib name="%signoff" value="Graceful"
Encryption="no" />
        <Attrib name="%signofftimeout" value="5"
Encryption="no" />
        <Attrib name="%integritycheckingchecksum" value=""
Encryption="no" />
        <Attrib name="%enableintegritychecking" value="No"
Encryption="no" />
        <Attrib name="%debugModeForMobile" value="No"
Encryption="no" />
    </TabApplicationAttribute>
    <PrincipalApplicationAttribute>

```

```

        <PrnAttrib name="%profilename.1" Encryption="no"
default_value="Profile 1" />
        <PrnAttrib name="%uid.1" Encryption="no"
default_value="" />
        <PrnAttrib name="%pwd.1" Encryption="yes"
default_value="" />
        <PrnAttrib name="%pwd.lastModified.1" Encryption="no"
default_value="" />
        <PrnAttrib name="client.1" Encryption="no"
default_value="" />
        <PrnAttrib name="%pse.version" Encryption="no"
default_value="" />
        <PrnAttrib name="%prnapppath" Encryption="no"
default_value="" />
        <PrnAttrib name="%workDir" Encryption="no"
default_value="" />
        <PrnAttrib name="%argument" Encryption="no"
default_value="" />
        <PrnAttrib name="%dllPath" Encryption="no"
default_value="" />
        <PrnAttrib name="%APIName" Encryption="no"
default_value="" />
        <PrnAttrib name="%rememberprofile" Encryption="no"
default_value="" />
        <PrnAttrib name="%lastprofile" Encryption="no"
default_value="" />
        <PrnAttrib name="%lastLoginTime" Encryption="no"
default_value="" />
        <PrnAttrib name="%disableUseridIfNotEmpty"
Encryption="no" default_value="" />
    </PrincipalApplicationAttribute>
</Application>
</USOADF>

```

2. Select **LoginScreen1** and modify the screen identifier's IP value by double clicking the screen identifier under **Trained Screen Objects**.
3. Export the modified SAP ADF. Follow the steps described in [Exporting Trained Application](#) section for exporting steps.
4. Create an USO Application and import the ADF file.
5. Entitle the application to the user and proceed to login to USO Server via AccessMatrix SSO Portal to launch USO Client.
6. USO Client can now support SAP single sign-on.

i Note: To support SAP application with multiple servers, change the IP identifier on ADF template to this format:
IP1:PORT1;IP2:PORT2;IP3:PORT3;...;IPn:PORTn

Enable Support for Web Application in Chrome Browser

Perform the following steps to enable USO Client support for Web Application in Chrome browser:

1. Access the web application in the Chrome browser and train the application with the USO Trainer. Refer to the [Using the USO Trainer](#) section for the training methods and steps.



Note: Ensure that the USO Chrome extension is already installed before testing the application. Refer to [AccessMatrix SSO Portal User Guide - Install USO Chrome Extension](#) section for the installation details.

2. Once the web application is successfully trained, proceed to verify if the USO Client is able to match and fill by using the USO Trainer test mode. Refer to the [Testing Trained Application](#) section for the details.
3. Once verified to be working in Chrome browser, log out from the USO Client to exit the test mode.
4. On the USO Trainer, export the trained application to generate the ADF accordingly. Refer to the [Exporting Trained Application](#) section for the details.
5. Import the ADF into the Admin Console to create the USO application. Refer to the [Configure USO Application](#) section for the details.
6. Navigate to **Operation Dashboard > User & Segment > User** and assign the USO entitlement of this USO Application to the test user and save. Refer to the [Configure User USO Entitlement](#) section for the steps.

Enable Support for Web Application in Edge (Chromium) Using IE Enterprise Mode

Internet Explorer (IE) and Microsoft Edge (Chromium) can work together to support legacy web applications by using Internet Explorer's Enterprise Mode. This means specific websites and apps that are having compatibility problems with Microsoft Edge can be automatically opened using IE 11. Additionally, you can continue using Microsoft Edge as your default browser, while also ensuring that your apps continue working on IE 11.

The following sections describe the steps on how to enable USO Client to support web applications in Microsoft Edge (Chromium). Ensure that the USO Chrome extension is not installed on Microsoft Edge by typing `edge://extensions` in the address bar.



Note: In this guide, <ACMATRIX_HOME> is used to refer to the installation directory of AccessMatrix, e.g. /opt/acmatrix/tomcat.

Configure the Enterprise Mode Site List

The Enterprise Mode Site List is an XML document that specifies a list of sites, their compat mode, and their intended browser. This list is used to determine which sites should open in IE 11 browser.

To configure:

1. Navigate to <ACMATRIX_HOME>/webapps/am5/lib and open the **sitelist.xml** as shown below:

XML

```
<site-list version="2">
  <site url="localhost:8080">
    <compat-mode>Default</compat-mode>
    <open-in>IE11</open-in>
  </site>
  <site url="portal.i-sprint.com:9443">
    <compat-mode>Default</compat-mode>
    <open-in>IE11</open-in>
  </site>
</site-list>
```



Notes:

- **siteurl** - specifies the URL (and port number) of the site you wanted to put on the Enterprise Mode site list. The URL can be a domain, sub-domain, or any path URL.
- **compat-mode** - controls what compatibility setting is used for specific sites or domains. If value is Default or not specified, then it loads the site using the default compatibility mode for the page. Otherwise, you can specify IE[x], where [x] is the document mode number into which the site load, e.g. IE11.
- **open-in** - controls what browser is used for sites. Specifying IE11 will open the site in IE11.

2. Configure and add other USO web applications following the format shown above.
3. Save the changes.

Enable Internet Explorer's Enterprise Mode

Before you can use a site list with Enterprise Mode, you need to turn the functionality on and set up the system for centralized control. There are two ways to enable this feature:

- using Group Policy, and
- using the Windows registry

To turn on IE's Enterprise Mode using Group Policy:

1. Launch the local group policy editor by pressing Windows Key + R, then type "gpedit.msc" into the **Run** dialog, and press Enter.
2. Navigate to **Computer Configuration > Administrative Templates > Windows Components > Internet Explorer** and double-click **Use the Enterprise Mode IE** website list setting.
3. Click **Enabled**, and then in the **Options** area, type the location to the site list mentioned in the previous section. For example:
http://localhost:8080/am5/lib/sitelist.xml
4. Click **Apply** and click **OK** to close the **Use the Enterprise Mode IE website list** dialog.

To turn on IE's Enterprise Mode using the registry:

1. Run the USO Client installer to configure the Windows registry.
 - **Using USO Client Installer (AllUser)**
Running this installer will install the USO Client; however, the registry keys will not be created. The user is required to run the **usoIEModeOnEdgeAllUser.reg** file found in the USO Client installation folder manually due to permission issue: *C:\Program Files (x86)\USOClient\usoIEModeOnEdgeAllUser.reg*. The following registry keys will be created:
 - *HKEY_LOCAL_MACHINE\Software\Policies\Microsoft\Internet Explorer\Main\EnterpriseMode\SiteList*
 - *HKEY_LOCAL_MACHINE\Software\Policies\Microsoft\Edge\InternetExplorerIntegrationLevel*
 - **Using USO Client Installer (PerProfile)**
Running this installer will install the USO Client; however, the registry keys will not be created. The user is required to run the **usoIEModeOnEdge.reg** file found in the USO Client installation folder manually due to permission issue:
C:\Users\[USERNAME]\AppData\Roaming\USOClient\usoIEModeOnEdge.reg. The following
-

registry keys will be created:

- *HKEY_CURRENT_USER\SOFTWARE\Policies\Microsoft\Internet*

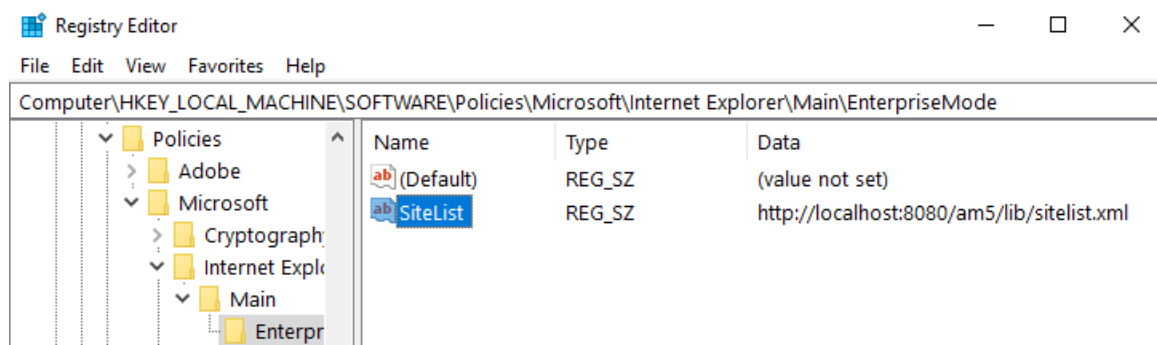
Explorer\Main\EnterpriseMode\SiteList

- *HKEY_CURRENT_USER\SOFTWARE\Policies\Microsoft\Edge\InternetExplorerIntegrationLevel*

2. To verify, launch the registry editor by typing "regedit" into the **Run** dialog and pressing Enter.

Note: If the User Account Control requests for your permission to proceed, agree to it.

3. Edit the **Sitelist** registry key to point to where you want to keep your Enterprise Mode site list file. For example:



4. Save the changes.

To check if Microsoft Edge is recognizing the site list correctly, launch the Microsoft Edge browser and type *edge://compat/enterprise* in the address bar. The enterprise mode site list should be displayed.

Note: Only those sites that you specifically configure in site list will use IE mode.

Train and Import the Web Application into the Admin Console

1. Access the web application in IE browser and train the application with the USO Trainer. Refer to the [Using the USO Trainer](#) section for the training methods and steps.
2. Once the web application is trained successfully, proceed to verify if the USO Client is able to match and fill by using the USO Trainer test mode. Refer to the [Testing Trained Application](#) section for the details.

Note: Clicking the web application from the USO Client (Test Mode) will launch the application using IE browser.

-
3. Once verified to be working in IE browser, right-click on the web application from USO Client and select the **Change App Path** context menu to change the **Application Path** and **Working Directory** information from IE browser to those of Microsoft Edge (Chromium) browser:
 - **Application Path**
For example, change from *C:\Program Files\Internet Explorer\iexplore.exe* to *C:\Program Files (x86)\Google\Chrome\Application\chrome.exe*.
 - **Working Directory**
For example, change from *C:\Program Files\Internet Explorer* to *C:\Program Files (x86)\Google\Chrome\Application*.
 4. Click **OK** button and double-click on the application to launch the web application in Microsoft Edge (Chromium) browser or you can also launch the web application directly by accessing the login page of the web application in Microsoft Edge (Chromium) browser.
 5. Once verified to be working in Microsoft Edge (Chromium) browser, log out from the USO Client to exit the test mode.
 6. Modify the values of **Application Path/URL** by clicking on **Application > Details > Advanced > App. Path/URL** on the USO Trainer. Change the **Application Path** and **Working Directory** information from IE browser to those of Microsoft Edge (Chromium) browser as shown in step 3.

 **Note:** This step is optional. The modification of the **Application Path** and **Working directory** attributes can also be done after the ADF is imported into the Admin Console.

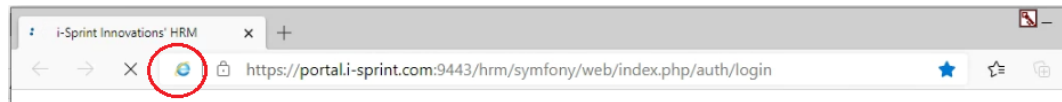
7. Click the **Finish** button and proceed to export the trained application to generate the ADF accordingly.
8. Import the ADF into the Admin Console to create the USO application.
9. Navigate to **Operation Dashboard > User & Segment > User** and assign the USO entitlement of this USO Application to the test user and save. Refer to the [Configure User USO Entitlement](#) section for the steps.

Verify the Web Application in Microsoft Edge (Chromium) Browser

Perform the following steps to verify if the web application is already supported in Microsoft Edge (Chromium) browser:

- Launch Microsoft Edge (Chromium) browser and access the AccessMatrix SSO Portal by entering the SSO Portal URL, e.g. *http://localhost:8080/usoserver/start.uso*.
-

-
- Log in as the test user.
 - After a successful login, the list of USO and non-USO applications entitled to the logged-in user will be displayed and the USO Client will be launched too.
 - Click the web application icon either from the SSO Portal page, or from the USO Client.
 - The USO Client should match and autofill the user Id and password.
 - The IE logo indicator is displayed on the left side of the navigation bar as shown:



7.5. Host Emulators

Generic Emulator

USO Trainer for Emulator has existing emulators that were already created. User can either use this existing emulator or user can also add, modify or delete emulators.

- To use an existing emulator, just choose any emulator from the **Emulator** drop-down list.
- To delete an emulator, just choose the desired emulator from the **Emulator** drop-down list and click **Delete** button.

 **Note:** Take precautions when deleting emulator.

- To modify an emulator, choose the desired emulator from the **Emulator** drop-down list and click **Modify** button. The **USO Trainer - Add A Generic Emulator** screen will be displayed.
- To add an emulator, click **Add** button. The **USO Trainer - Add A Generic Emulator** screen will be displayed. Refer to [Adding New Emulator](#) for details.

Adding New Emulator

USO Trainer for Emulator supports adding of a new emulator as it needs to know how it can get the contents of an emulator screen, so that matching can be done. The idea is to copy the screen into the clipboard, so that USO Client can retrieve the screen from the clipboard.

 **Note:** This is similar to the Copy (Ctrl-C) command found in Notepad. Paste the contents later into another editor.

In **USO Trainer - Add A Generic Emulator** screen, fill in the appropriate **Name**, **Product Name** and **Product Version**. There are no restrictions (except user should not use a Name which has been used before) but the values should be reasonably clear to others who read it.

Executable is referring to the emulator program itself. Choose the correct one so that USO Client is able to launch the emulator. Ensure that if the same for the **Working Directory** field.

For **Copy & Paste Commands**, refer to [Syntax for Copy and Paste Command-Lists](#) for details.

Syntax for Copy and Paste Command-Lists

A copy or paste command-list is a comma-separated list of commands.

A command can be:

Command	Description
{MouseSelectAll}	Use mouse to select the whole screen
{KeyboardEvent}	Send playback value by keyboard event
{CopyFromClipboard}	Get clipboard data and clear clipboard
{CopyToClipboard}	Copy playback value to clipboard
{ClearClipboard}	Clear the clipboard
{SendKeys(keys)}	Send keys by keyboard event
WM_COMMAND:id	Send WM_COMMAND msg with CtrlID id
WM_COMMAND:wParam:lParam	Send WM_COMMAND msg with params
WM_SYSCOMMAND:id	Send WM_SYSCOMMAND msg with CtrlID id
WM_SYSCOMMAND:wParam:lParam	Send WM_SYSCOMMAND msg with params
Message_ID	Send Message_ID msg without params
Message_ID:wParam	Send Message_ID msg with one param
Message_ID:wParam:lParam	Send Message_ID msg with params

SendKeys is a standard API for sending keystroke events. Refer to [C. Keystroke Code for SendKeys](#) for the codes for special characters.

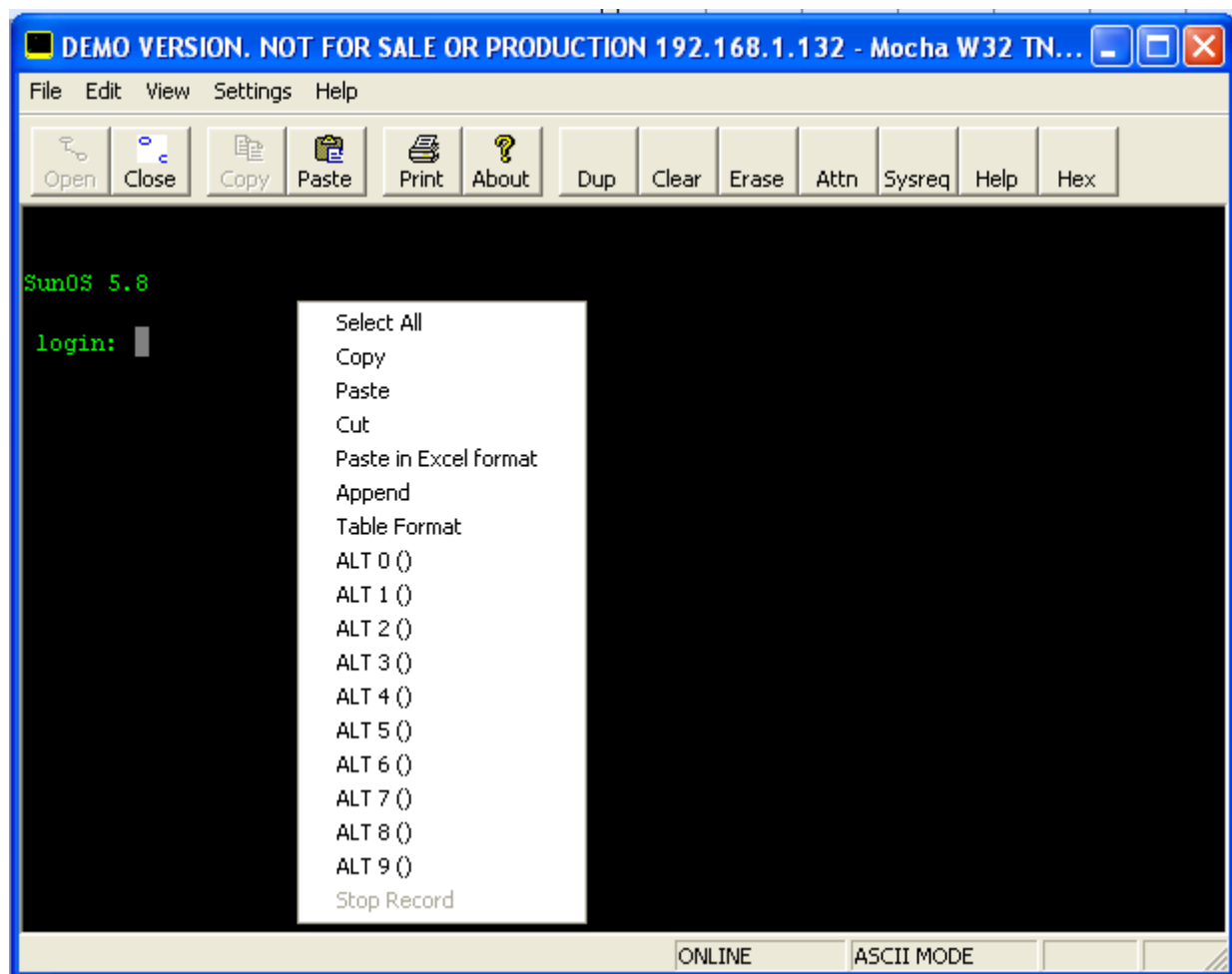
Copy Command-List

The whole idea of this command-list is to copy the emulation screen into the clipboard, where USO Client can then retrieve from.

The method of copying the screen depends on each emulator. The common methods are:

- "Select All" command, followed by "Copy" command
 - "Copy All" command
 - Using mouse to select the screen, followed by "Copy" command
-

The commands are usually found in the menu bar or drop-down menu when you right-click on the screen.

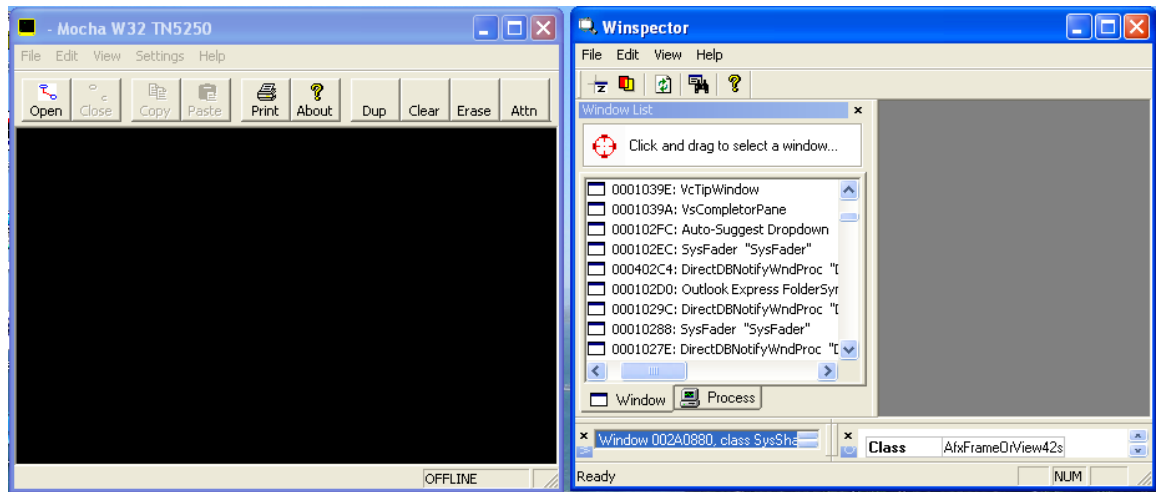


Using the example of the Mocha emulator, notice that the drop-down menu (when right-clicking the screen) includes the "Select All" and "Copy" commands. The commands can also be found in the edit menu.

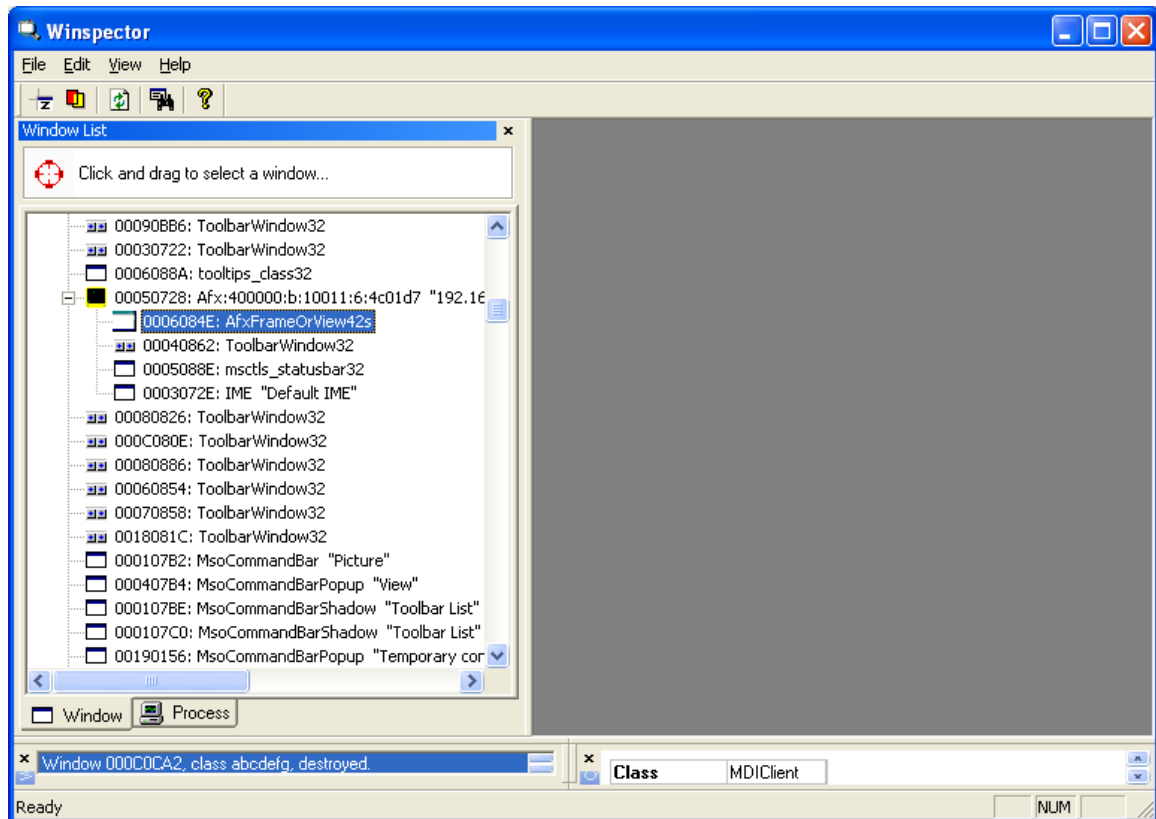
Getting the Message and Control ID of a Command

Perform the below steps to get the message and control Id of a command:

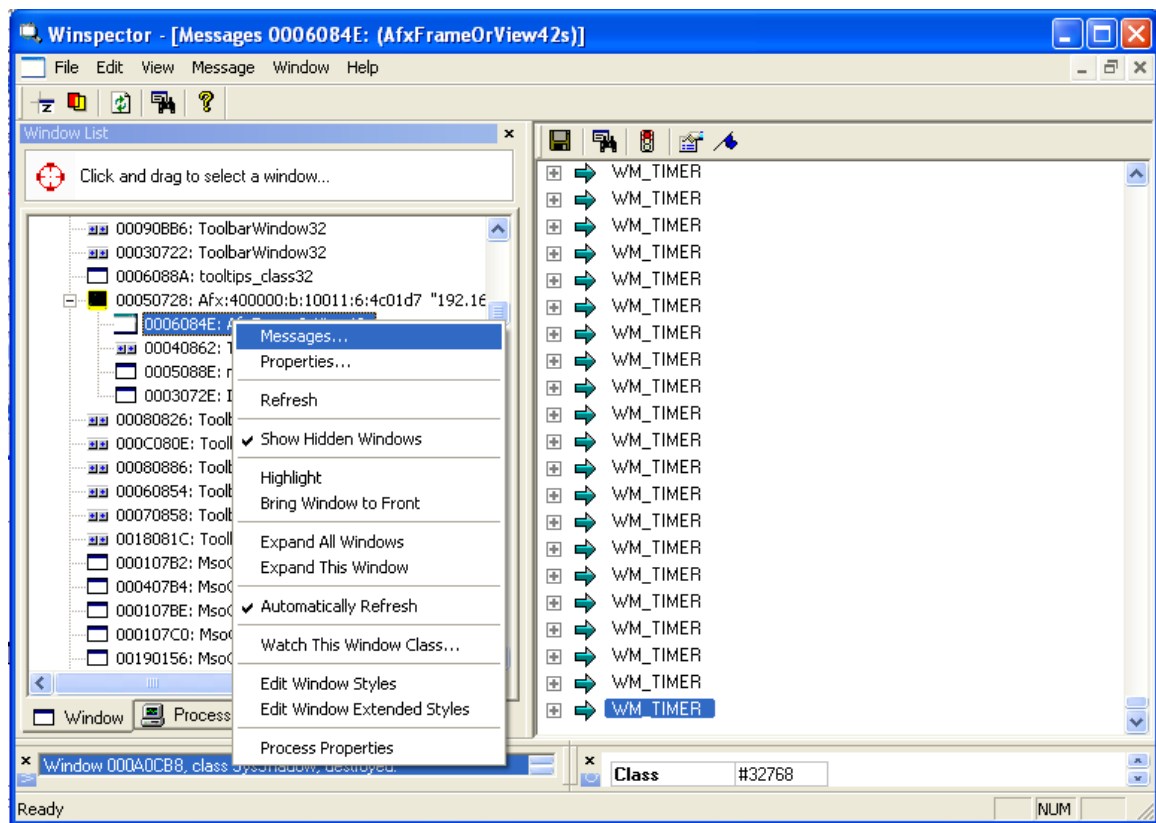
1. With Winspector, find the emulation window by dragging the target icon to the target window.
See below.



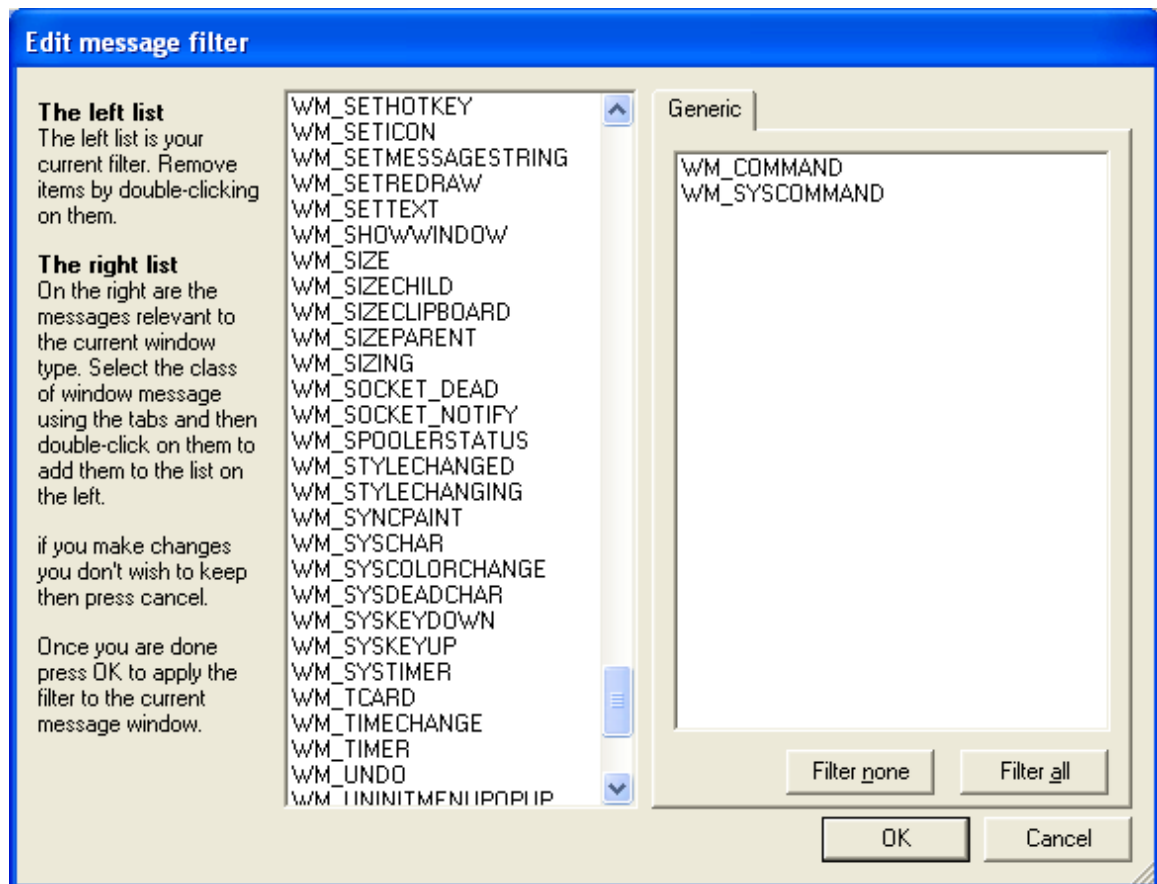
2. The following window will be displayed.



3. View the messages of the window by right-clicking on the item and selecting "Messages...".

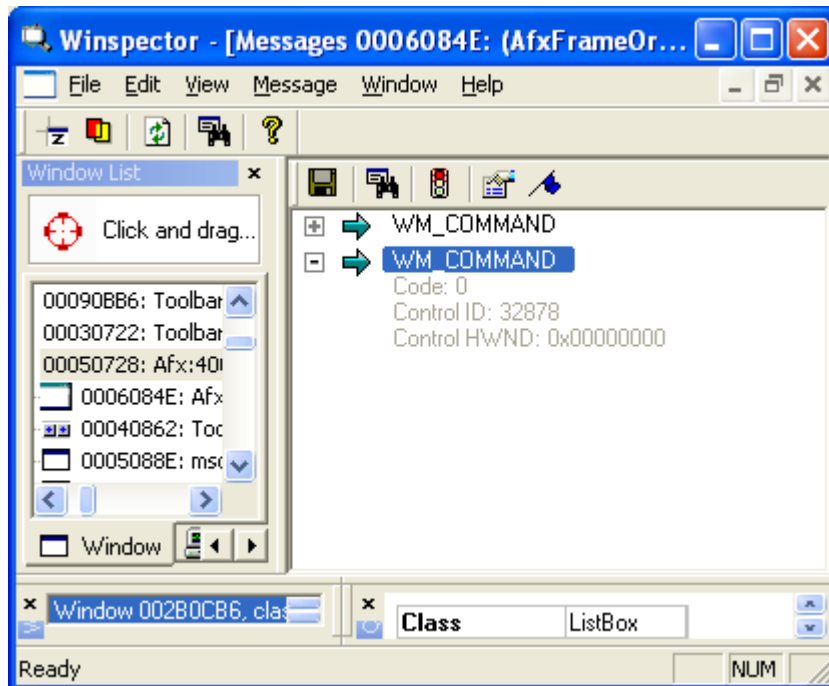


-
4. Usually, there will be a lot of messages. This is typical of most windows. To pinpoint to the relevant message, filter the messages.



5. Normally, the command message is WM_COMMAND. If this fails, try WM_SYSCOMMAND. If still unsuccessful, then do an exhaustive search. There is no knowing how the emulator can behave.
-

-
6. Perform a command on the emulator. For instance, perform "Select All". The following message will appear:



7. This means that for the emulator, the "Select All" command message is WM_COMMAND and its Control ID is 32878.
8. Similarly, find the message and Control ID of the "Copy" command. (For example, WM_COMMAND with 57634.)

Configuring the Copy Command-List

Following the same example, in order to copy the screen to the clipboard, first perform "Select All" followed by "Copy".

Therefore, build the Copy command-list as follows: *WM_COMMAND:32878,WM_COMMAND:57634*.

The final step of the Copy procedure is to tell USO Client to get the contents in the clipboard. Therefore, the final Copy command-list is: *WM_COMMAND:32878,WM_COMMAND:57634,{CopyFromClipboard}*

Variations of the Copy Command-List

If the "Select All" function is not available, then there is no choice but to simulate mouse events selecting the whole screen. In this case, the Copy command-list will look like:

{MouseSelectAll},WM_COMMAND:57634,{CopyFromClipboard}

Sometimes, the emulator joins the "Select All" and "Copy" functions into a single function. In this case, the Copy command-list will look like: `WM_COMMAND:57634,{CopyFromClipboard}`

At rare times, `WM_SYSCOMMAND` is used instead of `WM_COMMAND`.

Paste Command-List

For every value that USO Client needs to playback on the emulator, this set of command-list will be executed.

The common methods are:

- Copy value to clipboard, then emulator "Paste" command, followed by clearing the clipboard
- Send keystrokes to input the value.

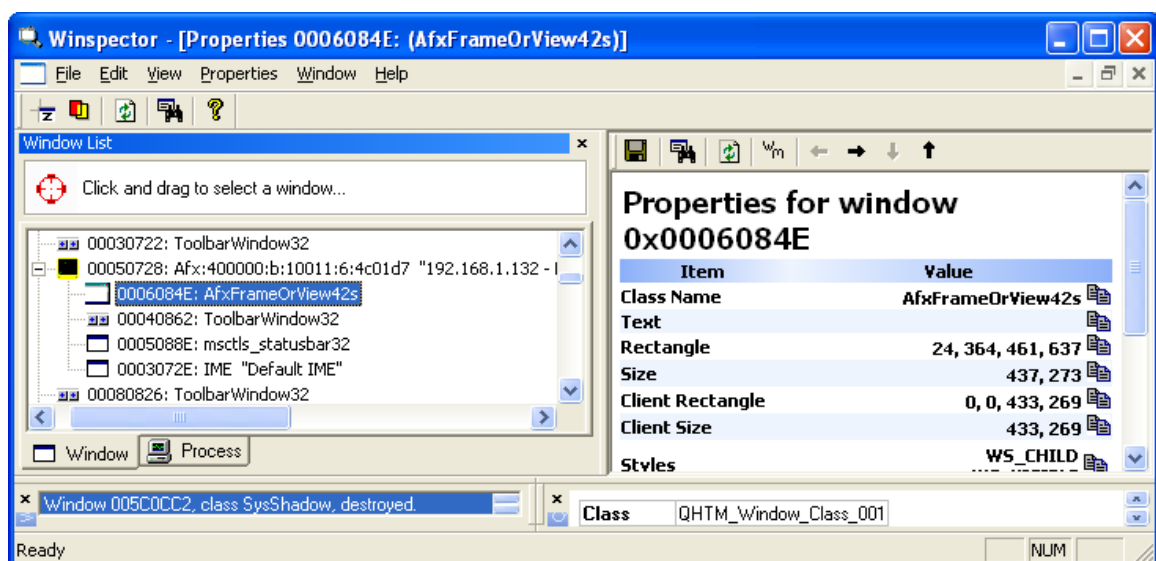
For the first method, the Paste command-list (assuming the control ID for Paste is 1234) will look like: `{CopyToClipboard},WM_COMMAND:1234,{ClearClipboard}`

For the send keystroke method, the Paste command-list will be: `{KeyboardEvent}`

Window Info

Configuration of Window Info is only necessary if:

- The command message must be sent to a child window, or
- The `{MouseSelectAll}` operation is to be performed on a child window



Using Winspector, the class name of the window can be found like above.

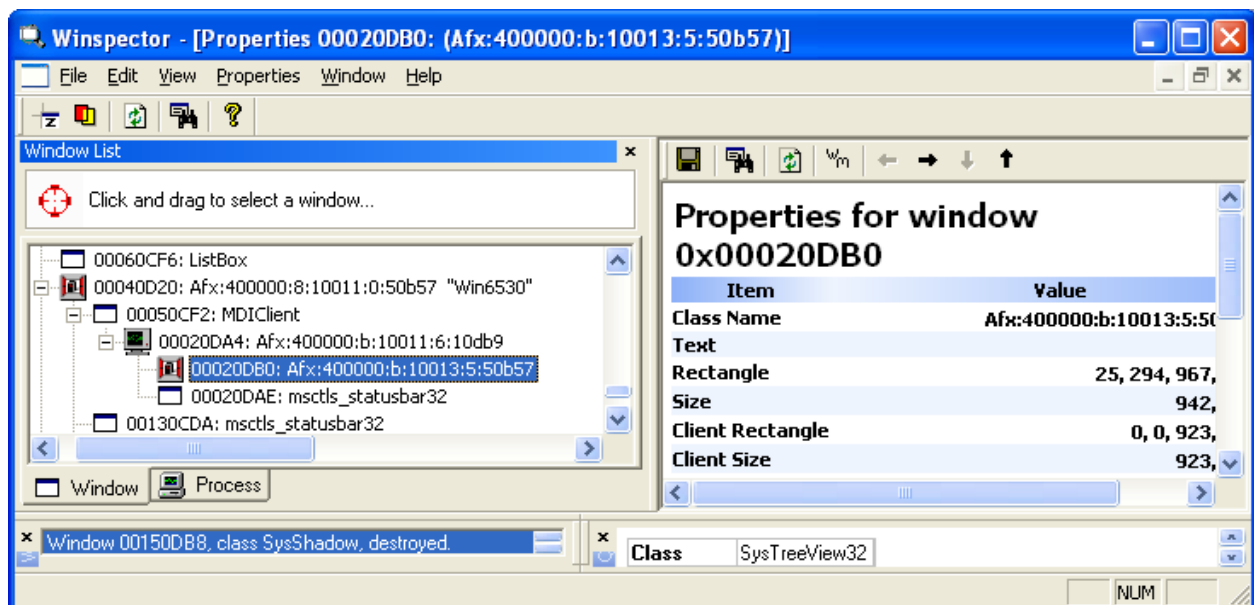
Window Class Hierarchy

The Window Class Hierarchy is a comma-delimited list of class names. The list may contain just one class name. If more than one names, the list must be ordered from child to parent, i.e. the window for the 2nd class name must be the parent of the window of the 1st class name, the 3rd must be the parent of the 2nd, and so on.

Basic regular expression (with symbols * and ?) can also be for the class name. Use regular expressions only if absolutely necessary, i.e. the class name can change. Regular expressions will slow down the searching.

From the example above, the value for the Window Class Hierarchy can be:

- AfxFrameOrView42s
- AfxFrameOrView42s,Afx:400000:b:10011:6:4c01d7
- AfxFrameOrView42s,Afx:400000*



From the example above, if the highlighted window is the target window, the value for the Window Class Hierarchy can be:

- Afx:400000:b:10013:5:50b57
 - Afx:400000:b:10013:5:50b57,Afx:400000:b:10011:6:10db9
 - Afx:400000:b:10013:5:50b57,Afx:400000:b:10011:6:10db9,MDIClient
-

-
- Afx:400000*,Afx:400000*,MDIClient, Afx:400000*

The reason for specifying a hierarchical list of class names is because a class name may be too general. By using a list, there is a better chance to pinpoint the exact window.

Target Window Position For Commands

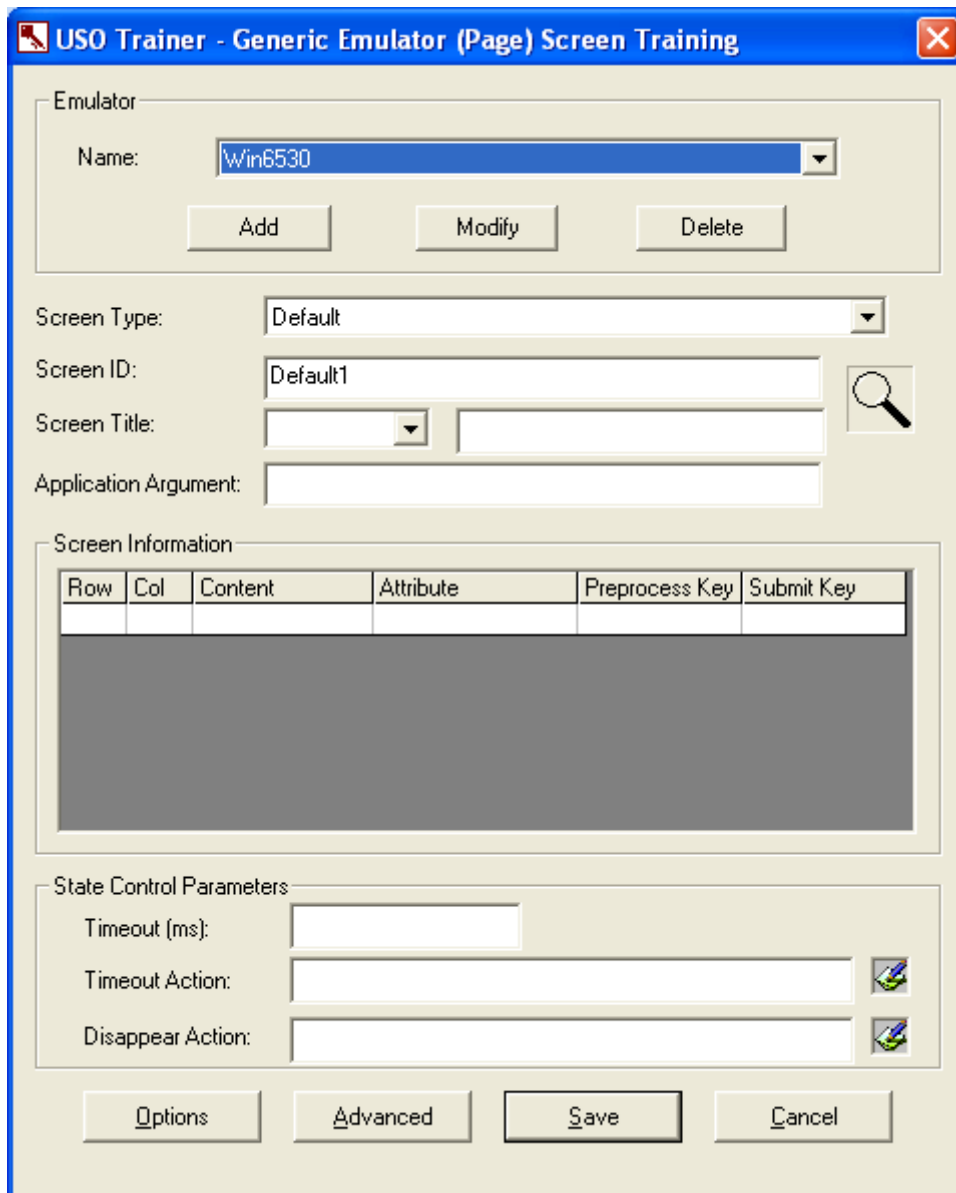
The "Target Window Position For Commands" is used to specify the exact window where command messages, *{MouseSelectAll}* operations and *{KeyboardEvent}* keystrokes will be sent.

If this value is "-1", all operations will be performed on the top parent window. In the example above, that would be the window 00040D20.

If this value is "0", all operations will be performed on the window whose class name is the first one specified in the Window Class Hierarchy. In the example above, that would be the window 00020DB0.

If this value is "1", all operations will be performed on the second window, and so on.

Training a Generic (Page) Screen



The dialog box is titled "USO Trainer - Generic Emulator (Page) Screen Training". It contains several sections for configuring a screen training session.

Emulator Section:

- Name: Win6530 (dropdown menu)
- Buttons: Add, Modify, Delete

Screen Configuration Section:

- Screen Type: Default (dropdown menu)
- Screen ID: Default1 (text field)
- Screen Title: (text field with a dropdown arrow)
- Application Argument: (text field)
- Search icon: A magnifying glass icon.

Screen Information Section:

Row	Col	Content	Attribute	Preprocess Key	Submit Key

State Control Parameters Section:

- Timeout (ms): (text field)
- Timeout Action: (text field with a file icon)
- Disappear Action: (text field with a file icon)

Buttons: Options, Advanced, Save, Cancel

Application Argument

Application argument must be correct in order for USO Client to launch the application properly. USO Client will use the emulator executable (trained when adding a new emulator) and then append the argument behind it.

For instance, if the emulator exe is *C:\Program Files\EmuTest\emutest.exe* and the argument is */ipaddress 192.168.1.132*, then USO Client will launch the application by executing: *C:\Program Files\EmuTest\emutest.exe /ipaddress 192.168.1.132*

The trainer must find out how to use the command line to launch the application.

Screen Information

To train the screen information, firstly copy the whole content of the emulation and paste it into a text editor (for instance, Notepad).

```
1234567890123456789012345678901234567890123456789012345678901234567890
1      >> Welcome to the best known public AS/400! <<
2
3      System name is:      M170PUB1      subsystem : QINTER2      devicename: QPADEV001P
4      Username/Benutzer:      Password      (enter blind)
5
6
7      /*****
8      ] Rechenzentrum Kreuznach      ]
9      ] powered by AS400Profis.de      ]
10     ] -      ]
11     ] Holger Scherer      ]
12     ] Jungstrasse 25-27      ]
13     ] 55543 Bad Kreuznach      ]
14     ] Germany      ] => eMail-Help: support400@rzkh.de
15     ] holger(at)rzkh.de      ] (Please read FAQ before!)
16     ] Tel. +49-89-35319553      ]
17     ] iSeries-Hosting, ERP-Software,      ] *** English help pages ***
18     ] advanced data processing      ] http://m170pub1.rzkh.de/forumeng.html
19     ]      ] *** Deutsche Hilfeseiten ***
20     ]      ] http://m170pub1.rzkh.de/forum.html
21     ] _____/
22
23     Number of currently active Users:
24
25     sin Service von www.AS400Profis.de
```

The example above is used. At least one identifier is needed each screen. To train "Username/Benutzer" and "Password" as identifiers,

- Set Row as 4, Col as 2
- Set Content as Username/Benutzer
- Set Attribute as app.identifier
- Ignore Preprocess Key and Submit Key
- Add new entry
- Set Row as 4, Col as 35
- Set Content as Password
- Set Attribute as app.identifier
- Ignore Preprocess Key and Submit Key

To train the userid field,

- Add new entry
 - Ignore Row, Col and Content
 - Set Attribute as appprn.%uid
-

- Set Submit Key as {TAB}

To train the password field,

- Add new entry
- Ignore Row, Col and Content
- Set Attribute as appprn.%pwd
- Set Submit Key as {ENTER}

The syntax for Preprocess Key and Submit Key is the same for SendKeys.

USO Trainer - Generic Emulator (Page) Screen Training

Emulator

Name:

Screen Type:

Screen ID:

Screen Title:

Application Argument:

Screen Information

Row	Col	Content	Attribute	Preprocess Key	Submit Key
4	2	Username/Benutzer	app.identifier		
4	35	Password	app.identifier		
			appprn.%uid		{TAB}
			appprn.%pwd		{ENTER}

State Control Parameters

Timeout (ms):

Timeout Action:

Disappear Action:

Training an Identifier

USO Client will match the contents from the clipboard with what has been trained here. Make sure the screen identifier(s) is/are sufficiently unique to properly identify the screen.

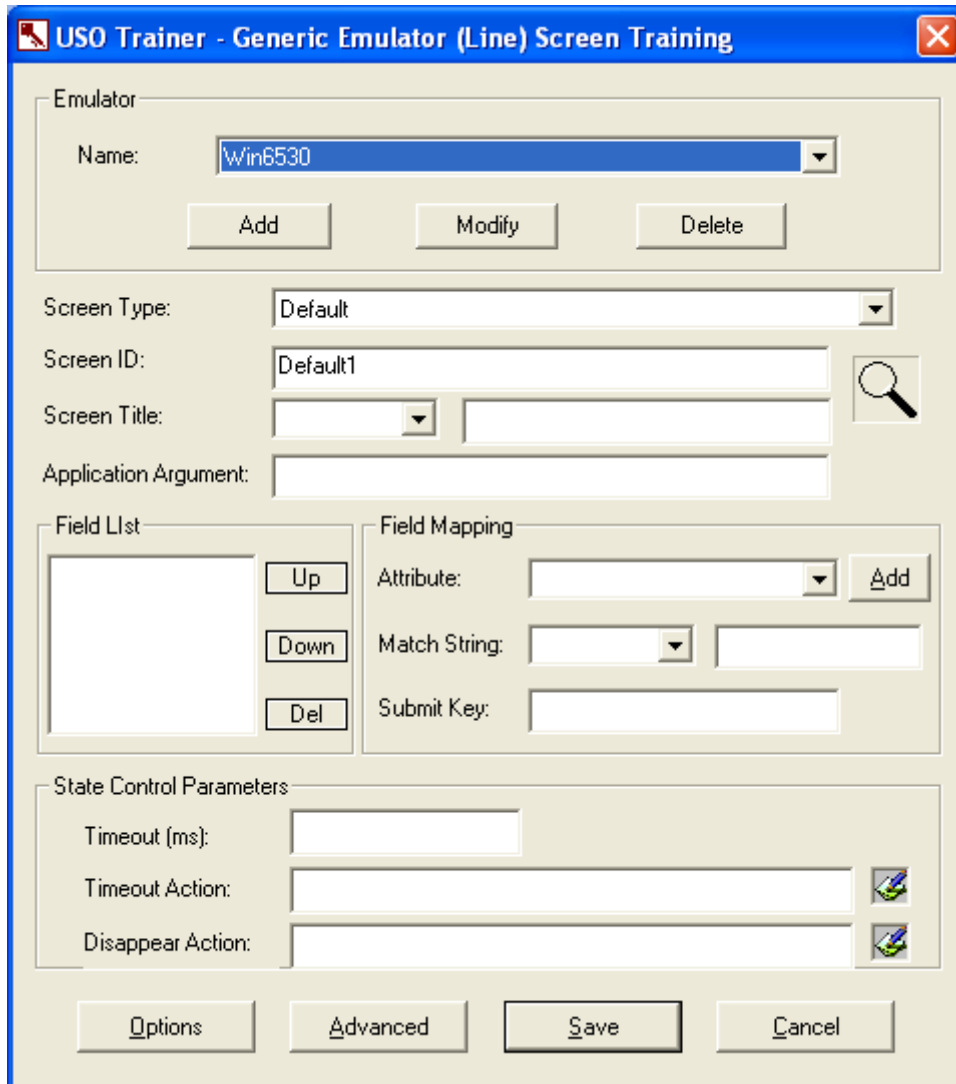
Training a Field

Once a screen is matched, USO Client will fill in the field values one by one starting from the top. In the example above, USO Client will do the following:

- Key in the username
- Key in TAB
- Key in the password
- Key in ENTER

Sometimes, the cursor is not at the required position at the beginning. For example, in order for the cursor to reach the username field, three tabs may first be needed. In that case, train the appprn.%uid Preprocess Key as {TAB}{TAB}{TAB}.

Training a Generic (Line) Screen



The dialog box is titled "USO Trainer - Generic Emulator (Line) Screen Training". It contains several sections for configuring a screen training session.

- Emulator:** A dropdown menu for "Name" is set to "Win6530". Below it are "Add", "Modify", and "Delete" buttons.
- Screen Type:** A dropdown menu set to "Default".
- Screen ID:** A text field containing "Default1".
- Screen Title:** A dropdown menu and a text field.
- Application Argument:** A text field.
- Field List:** An empty list box with "Up", "Down", and "Del" buttons to its right.
- Field Mapping:** Contains "Attribute:" (dropdown), "Match String:" (dropdown and text field), and "Submit Key:" (text field). An "Add" button is next to the Attribute dropdown.
- State Control Parameters:** Contains "Timeout (ms):" (text field), "Timeout Action:" (text field with a small icon), and "Disappear Action:" (text field with a small icon).
- Buttons:** "Options", "Advanced", "Save", and "Cancel" are at the bottom.

Application Argument

This is similar to [Application Argument](#) section.

Screen Information

SunOS 5.8
login:

By nature, each screen should only have one field (even though this is not enforced by the USO Trainer).

For the example above, to train the username,

-
- Set **Attribute** as appprn.%uid
 - Set **Match String** as "Ends with" login:
 - The default **Submit Key** is ENTER so there is no need to train the Submit Key
 - Click **Add**
-

7.6. USO Trainer Advanced Mode

USO Trainer Advanced Mode is a tool that converts ADF files to ADS files and vice versa. ADS files are intermediary text files, in JSON format in this case, for better readability. As the "Screen Info" attribute is encrypted and unreadable in ADF files, USO Trainer Advanced Mode converts them into ADS, producing a decrypted plain text format to enable manual editing. Manual editing is required under circumstances such as when Screen Identifier is unable to detect the intended object on the screen (e.g. a hidden invisible frame object covering the intended textbox object). Even though the object may be invisible to user, USO Trainer will detect that object instead of the intended one.

For applications running on java version below 7u6, it is possible to manually edit the Screen Identifier via USO Trainer (under Trained Screen Objects). However, USO Trainer does not support this function for applications running on java 7u6 or higher. Therefore, USO Trainer Advanced Mode is required. Below table shows applications that could and could not be edited via USO Trainer:

Application Type	Field Editing by USO Trainer UI
Desktop	Yes
Web based	No
Java (>=7u6)	No

Usage of USO Trainer Advanced Mode

USO Trainer Advanced Mode is a command line control tool unlike USO Trainer which has a UI. It should only be used as a last resort when USO Trainer is completely unable to detect the intended object or manual editing via USO Trainer is impossible.

USO Trainer Advanced Mode requires a complete ADF file. Therefore, the standard application training procedure has to be performed using USO Trainer to create an ADF file first, even if the necessary screen objects are unable to be detected or the ADF file does not work.

To understand the usage options of USO Trainer Advanced Mode:

1. Open the command prompt and execute **usota.exe**.

2. The following will be displayed:

```
CA. C:\Windows\system32\cmd.exe

C:\Program Files (x86)\AccessMatrix\usotrainer5>usota
Usage:
usota app.adf [-t] -> result app.ads
usota app.ads [-t] -> result app.adf
param [-t] will send the adf to USO Client for testing mode
-----
usota -[java|uia|dom] HandleWindow [object highlighting]
Sample: usota -java 0xC60796
        usota -java C60796 "USO_JAB_password text@Password,3"
```



Notes:

- USO Trainer Advanced Mode only supports ADF created by USO Trainer 5.
- `usota -java` is the command to retrieve USO fields for Java applications.
- `usota -uia` is the command to retrieve USO fields for Desktop applications.
- `usota -dom` is the command to retrieve USO fields for Web applications.
- `usota -chromium` is the command to train the Web applications on Microsoft Edge (Chromium-based) browser.

Convert ADF File to ADS File

1. Open the command prompt and type: `usota <adf filename>` to convert the file to ADS, e.g.
`usota "C:\demoApp.adf".`
2. If successful, the ADS file will be created in the same directory of the ADF file and the following message will be displayed:
"Non HexDigit encountered...File 'C:\demoApp.ads' created successfully."

To test the ADS file in USO Client, add `-t` behind the command: `"usota <adf filename> -t"`, e.g. `usota "C:\demoApp.adf -t".`

If successful, the application will be displayed under USO Client applications list ready for testing.



Note: Ensure USO Client is running before executing the command in test mode. Otherwise, "Error: USO Client is not running!" will be prompted.

Convert ADS File to ADF File

1. Open the command prompt and type: `usota <ads filename>` to convert the file to ADF, e.g.
`usota "C:\demoApp1.ads"`

-
2. If successful, the ADF file will be created in the same directory of the ADS file and the following message will be displayed:

"File 'C:\demoApp1.adf' created successfully."

To test the ADF file in USO Client, add `-t` behind the command: `usota <ads filename> -t`, e.g. `usota "C:\demoApp1.ads -t"`.

If successful, the application will be displayed under USO Client applications list ready for testing.



Note: Ensure USO Client is running before executing the command in test mode. Otherwise, "Error: USO Client is not running!" will be prompted.

Edit ADS File Content

There are two ways of getting the screen information of an application:

- using Win API on USO Trainer Advanced
- using UI Automation on USO Trainer Advanced

Using Win API

1. Use a preferred text editor to open the ADS file (e.g. Notepad++).
2. Look for Screen Information, identify the correct screen by checking "Info.Title" for screen name.
3. Look for that screen's Field Objects. Each field object possesses five fixed attributes;
 - Info.FieldClass: refers to the name of the field. It could be anything.
 - Info.FieldFlag: (Unused)
 - Info.FieldID: The "Field Id" value depends on the type of application. Please see below table:

Application Type	Info.FieldID
Desktop	API_Classname@Index e.g. API_Edit@0
Web based	DOM_TagName@TagAttribute(id/name/classname)@Index,frameIndex.frameLevel e.g. DOM_INPUT@UID@4,-1,-1
Java (>=7u6)	USO_JAB_FieldType@FieldName,Index e.g. USO_JAB_text@Login Id,2

-
- Info.FieldOption: (Unused)
 - Info.FieldValue: refers to the identifier of the application to recognize what field it is.
4. Look for the Screen Identifier field, by looking for "Info.FieldValue" with value "app.identifier".
 5. Update the "Info.FieldClass" value of the Screen Identifier field to the correct value to be recognized by the application.

i **Note:** Do not change the value of "Info.FieldValue".

6. Keep the default values for all other attributes.
7. Save the ADS file.

i **Note:** Before converting the ADS file back to ADF file, user should check for JSON correctness by using any JSON Formatter/Validator tool to minimize error during conversion. (Recommended to use Notepad++ JSON Plugin or <https://jsonformatter.curiousconcept.com/>)

! **Important Note:** Do not make changes to any other Field Objects.

Using UI Automation

1. Run a third party software, e.g. Spy++ to retrieve the Window handle.
 2. Use the following commands to retrieve automatically the USO Fields and Identifiers of an application:

```
usota -[java|uia|dom] HandleWindow [object highlighting]
```

where:

 - `Handle Window` is the window handle retrieved by using Spy++.
 - `object highlighting` is the USO Field Identifier. This argument is optional for the first usota execution, i.e. before retrieval of the field identifier. The field identifier can then be provided for next usota execution to enable highlighting of the field.
 3. Use a preferred text editor to open the ADS file (e.g. Notepad++).
 4. Look for Screen Information, identify the correct screen by checking "Info.Title" for screen name.
 5. Look for that screen's Field Objects. Each field object possesses five fixed attributes;
 - Info.FieldClass: refers to the name of the field. It could be anything.
 - Info.FieldFlag: (Unused)
 - Info.FieldID: The "Field Id" value depends on the type of application. Please see below table:
-

Application Type	Info.FieldID
Desktop	USO_UIA_Classname@TagAttribute(id/name),Index e.g. USO_UIA_Edit@UID,2
Web based	DOM_TagName@TagAttribute(id/name/classname),Index e.g. USO_DOM_INPUT@login,3
Java (>=7u6)	USO_JAB_FieldType@FieldName,Index e.g. USO_JAB_text@Login Id,2

- Info.FieldOption: (Unused)
- Info.FieldValue: refers to the identifier of the application to recognize what field it is.

6. Look for the Screen Identifier field, by looking for "Info.FieldValue" with value "app.identifier".
7. Update the "Info.FieldClass" value of the Screen Identifier field to the correct value to be recognized by the application.

i Note: Do not change the value of "Info.FieldValue".

8. Keep the default values for all other attributes.
9. Save the ADS file.

i Notes:

- Before converting the ADS file back to ADF file, user should check for JSON correctness by using any JSON Formatter/Validator tool to minimize error during conversion. (Recommended to use Notepad++ JSON Plugin or <https://jsonformatter.curiousconcept.com/>)
- Supports Unicode character (UTF-8) in ADF/ADS format, but it cannot display properly on command prompt in Windows with English code page.

! Important Note: Do not make changes to any other Field Objects.

Train Web Application on Microsoft Edge (Chromium-based) Browser

Take note of the following checklist before training the web application on Edge Chromium:

- Ensure that IE Enterprise Mode is disabled.
- Ensure that there's only one chromium browser running. For instance where the Google Chrome browser is running, then see to it that the USO Chrome Extension is disabled.
- Secure a sample ADS file as a template.

To train the web application:

1. Launch Microsoft Edge (Chromium-based) browser and install the USO Chrome Extension using the Chrome Web Store. Refer to the [AccessMatrix SSO Portal User Guide - Install USO Chrome Extension](#) section for the installation steps.
2. Install the USO Trainer, if not installed yet.
3. Install the USO Client, if not installed yet.
4. Open the Microsoft Edge (Chromium-based) browser and open the Web application to train.



Note: Ensure that the current active tab on Edge Chromium browser is the web application to be trained.

5. Then, open the command prompt and run this command:

```
usota -chromium
```

For example: `C:\>usota -chromium`

6. If successful, the screen information (such as **Title** and **Input** fields) of the Web application will be displayed.
 7. Proceed to edit the ADS file content using the provided screen information.
 8. Save the ADS file.
 9. Convert the ADS file to ADF file. Refer to [Convert ADS File to ADF File](#) section for the conversion and testing steps.
-

8. USO Integration

This user guide presents detailed discussions and steps on how AccessMatrix Universal Sign-On (USO) can be configured to provide support for the following:

- [Active Directory Password Filter](#)
- [SAML Integration](#)
- [UAM Integration for USO](#)
- Integrated Windows Authentication (IWA)

There are two alternatives of doing integrated windows authentication:

- **Integrated Windows Authentication with CLP**

This is the default option of doing IWA. It supports complex domain environment. For example, if you have users registered in different domains or event forests. The integrated windows authentication is done in more reliable way as it relies on Microsoft IIS to handle the integrated windows authentication. Note that this method does not support the "PTDType = Server" USO feature which secures the Personal Security Environment (PSE) file by using the AES key that is derived from the same password being used by the user for authentication to the AccessMatrix server. Other PTDType values are supported.

You may refer to the [AccessMatrix Common Login Page Deployment Guide - Configure CLP with Windows Integrated Authentication](#) section for the integration details.

- **Integrated Windows Authentication with SPNEGO/Kerberos (deprecated)**

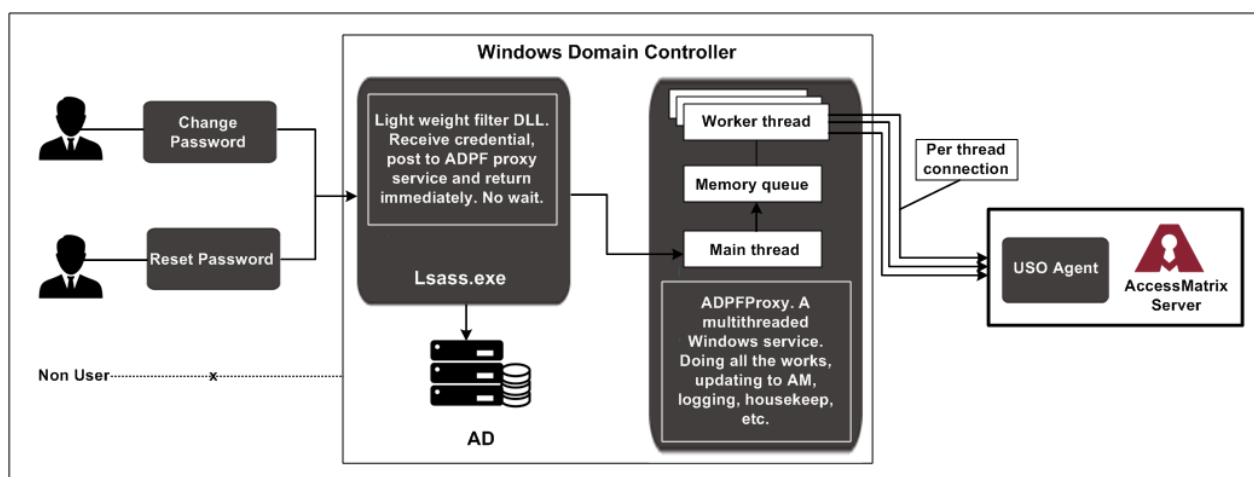
This is the option for legacy deployments that need to support IWA with SPNEGO/Kerberos. It supports the "PTDType = Server" USO feature and instead of using CLP, this method uses the USO Agent (deprecated) login page. This method does not support complex domain environment.

8.1. Active Directory Password Filter

Access Matrix ADPF is meant to synchronize AD password (as the primary password provider) with a secondary password stored in AM database. This synchronization is achieved by using AD Password Filter technology provided by Microsoft. This filter is a DLL module, where we provide a callback function, that will be invoked by Local Security Authority Subsystem Service (**Lsass.exe**). **Lsass.exe** is a process that is responsible to process and enforce security policy in Windows.

Lsass.exe is a very critical process in Windows. Any DLL module that will be loaded by LSASS shall be (ideally) error-free. Like a module that is loaded in Lsass address space, any problem in our DLL module will impact LSASS. The problem in LSASS will usually mean a problem in the whole AD system, including slowness in authentication or hanging/unresponsiveness. Some previous versions of ADPF had shown some problems, e.g. memory leak and slowness due to heavy synchronous process in it.

ADPF Architecture



There are two modules to serve password synchronization:

- Password filter module, a DLL that will be loaded by **Lsass.exe**.

Note: Password filters are an official feature provided by Microsoft to synchronize password changes to foreign account databases. For more information, refer to this [link](#).

- The password filter module is an add-on provided by i-Sprint that performs read-only operations on the AD domain controller. It does not modify or add data to the AD.
-

-
- Upon changing the password, the password filter sends the user id and new password to the ADPFProxy. This will be the only data that leaves the password filter. There is no data traffic entering the password filter.
 - ADPFProxy, a Windows service that will execute the necessary tasks to update passwords to the AccessMatrix database.
 - The ADPFProxy resides on the same server as the password filter.
 - Upon receiving the user id and the new password from the password filter, the ADPFProxy uses it to log into the AccessMatrix server and fetch the ESSO application assigned to the user. If the user does not exist in the AccessMatrix database, or if the user does not have any relevant ESSO application, the ADPFProxy will not send the password out from the AD domain controller.

As shown in the above picture, for any actions of change password (by users) or reset password (by Administrator), Lsass will invoke the ADPF module provided by AccessMatrix. The password filter DLL is a very lightweight module, that's only receiving a credential from **Lsass.exe**, and post it to another module (that we call ADPFProxy) using Windows message communication mechanism, and return immediately without waiting all the tasks finished. It is a very simple DLL to ensure the robustness of Lsass, virtually only about 10 lines of C codes. Very low risk to the production domain controller.

The rest of the tasks will be done by ADPFProxy. It is a Windows service application that is running in the background. ADPFProxy is a multi-threaded process, where the different task is served by a different thread. The most important work is performed by worker threads, i.e. to submit/update password to USO server. The connection to the USO server is also not only one, but as many as the worker threads. Each worker thread will spawn its HTTP connection. Currently, due to the limitation of NTLM authentication, only one worker thread is running.

In the peak hours, the unprocessed credential will be queued to be processed soon. So, it will still give the users a good experience. By using this approach, any problem/delay related to updating the password to Access Matrix database will not affect the performance of LSASS.

Configuring ADPF

System Requirements of ADPF

Prior to ADPF deployment, it is recommended to prepare the environment and configure the required files for ADPF to work.

ADPF supports Windows 2008 R2, Windows 2012 and Windows 2012 R2 servers. There are two sets of ADPF files provided, one for 32-bit and one for 64-bit OS versions. Each set has two binary files:

- **AxMxADPFilter.dll** (the password filter)
- **AxMxADPFProxy.exe** (the proxy service)

In addition, an **AxMxADPF.ini** file is also provided that contains settings to managed the AxMxADPFProxy behavior. Any modification will require ADPFProxy windows service restart.

Deploying ADPF

Perform the following steps to deploy ADPF:

1. Place **AxMxADPFilter.dll** into the *C:\Windows\System32* directory. Deploy the correct DLL depending on the bitness of the operating system, i.e. 32-bit or 64-bit.
2. Register the password filter by navigating to the following registry key:
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa.
3. Under *Notification Packages* subkey, add the name of ADPF filter, i.e. "AxMxADPFilter", to the existing data value.



Notes:

- Do not overwrite the existing values, and do not include the .dll extension.
- Registering a password filter requires computer restart for changes to take effect.
- There is no log provided from the filter. To test whether the filter is working, you must install the AxMxADPFProxy first.

4. Copy **AxMxADPFProxy.exe** and **AxMxADPF.ini** into a temporary folder, e.g. *C:\TEMP\ADPF*.

5. Open the **AxMxADPF.ini** file and specify the correct configurations. Please refer to the sample configuration below:

```
AxMxADPF.ini
1 #Please read this part first!
2 # - Maximum number of characters for each value is 250.
3 # - Any changes in this configuration file requires restarting of AxMxADPFProxy service.
4 # - To add comment, please add the '#' symbol at the beginning of the line.
5 # - Please do not put the '#' after the parameter or value.
6
7 [AxMxADPFProxy]
8
9 #FAMID: the FAM that will be used to connect to USO Server. Currently only 40171 and 40172 that are supported.
10 FAMID = 40171
11
12 #DOMAIN: windows domain. This computer and USO/AxMx server must be in the same domain.
13 DOMAIN = qaad1.com
14
15 #APPID: USO Application ID that authenticate to AD Server.
16 #Separate each application by using comma (,)
17 APPID = DesktopDemo
18
19 #SERVER: IP Address or host name of the USO/AxMx/IIS Proxy Server
20 #The format is SERVER:PORT.
21 #If the servers are more than one (for failover), please separate each entry with comma (,)
22 #e.g. 192.168.1.1:8080, 192.168.1.2:8090, 192.168.1.3:9080
23 #Maximum allowable number of servers is only five.
24 SERVER = WIN-FDFGVEGFDHJ.qaad1.com:8080
25
26 #LOGFILEDIR: the log file folder. Please ensure that the folder exists prior running the service.
27 LOGFILEDIR = C:\Temp\ADPF
28
29 #REQUEST: for composing the complete URL for USO request.
30 #Don't change this, unless you really know what you are doing.
31 REQUEST = /am5/start.uso?reqid=usoclientrequest.uso
32
33 #FUNCTIONID: The request code number. Supports backward compatibility. Supported values: 15 and 19
34 #For AM4 connection, use 15. For AM5, use 19.
35 FUNCTIONID = 19
36
37 #DAYSTOKEEPLOG: The number of days for keeping the log files (housekeeping), Default: 30 days
38 DAYSTOKEEPLOG = 30
39
40 #LOGLEVEL: the correct entries are: INFO or DEBUG
41 LOGLEVEL = DEBUG
```

6. Open the command prompt, and run the following commands to install the service into Windows Service: AxMxADPFProxy I.
7. Upon successful installation, run the service manually from Services.

 **Note:** You do not need to reboot the computer to run the service.

Testing ADPF

1. Before testing ADPF, make sure that you assigned a domain user with either the 40171 or 40172 authentication realm. Ensure that the domain user can log in to AccessMatrix SSO Portal or USO Client.
2. Use the domain user to login to a client workstation and perform password change/reset. Check if a log is created in the log folder defined in **AxMxADPF.ini**.

-
3. The application's password will be updated. Successful change password will display a log similar below:

```
2014-03-05 14:06:07 [Th-0x9bc] INFO - Start updating credential to server.
2014-03-05 14:06:07 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(103) Start performing login.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(142) Login is ok. SID=m0hCg2x50dy94fyrGbzksg
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(145) Finish performing login.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(151) Start performing changePassword.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(238) Start performing changePassword for DesktopDemo.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(273) Finish performing changePassword for DesktopDemo.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(231) Finish performing changePassword.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(279) Start performing logout.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(302) Logout is ok.
2014-03-05 14:06:08 [Th-0x9bc] DEBUG - CredentialSubmitter.cpp(304) Finish performing logout.
2014-03-05 14:06:08 [Th-0x9bc] INFO - Finish updating credential to server.
```

Uninstalling ADPF

To uninstall, open the command prompt and run the following commands to uninstall the service:

AxMxADPFProxy U.



Note: If you want to uninstall the service, please make sure the service is not running. Otherwise, you need to reboot the computer to accomplish service uninstallation.

8.2. SAML Integration for USO

Security Assertion Markup Language (SAML) is an XML standard that allows secure web domains to exchange user authentication and authorization data. Using SAML, an online service provider can contact a separate online identity provider to authenticate users who are trying to access secure content.

AccessMatrix acts as the Identity Provider. It authenticates the user credential and provides the user identity to the Service Provider.

To integrate SAML with USO, perform the steps described in the following sections.

Configuring AccessMatrix Server

1. Open `<AxMx Root>\tomcat\webapps\am5\WEB-INF\web.xml`.
2. Uncomment the SAML servlet and servlet-mapping elements as shown below:

XML

```
<!-- SAML -->
<servlet>
  <servlet-name>SAMLAuthnProcessor</servlet-name>
  <description>
    Handles SAMLRequest and generates SAMLResponse
    accordingly based on the AM5 session token
    Expected input condition:
    SAMLRequest (in request attribute or parameter)
    RelayState (in request attribute or parameter)
    SPId (in request attribute or parameter): to
    generate idp-initiated SAML response
  </description>
</servlet>
<servlet-
class>com.isprint.am.server.servlet.SAMLAuthnProcessor</servlet-
class>
  <init-param>
    <param-name>loginPageUrl</param-name>
    <param-value>/start.uso</param-value>
  </init-param>
  <description>
    Login page to authenticate user identity
    Request attribute names forwarded to this page:
    SAMLRequest: this SAMLRequest needs to be
    sent back to this servlet as request attribute or parameter
    (please do HTML encoding to avoid XSS)
    RelayState: this RelayState needs to be sent
    back to this servlet as request attribute or parameter (please
    do HTML encoding to avoid XSS)
    SPId: ID of the Service Provider (for display
```



```

purpose)
        SPName: name of the Service Provider (for
display purpose)
        ProviderName: provider name if provided in the
request (for display purpose)
        </description>
    </init-param>
    <init-param>
        <param-name>samlResponseHandlerUrl</param-name>
        <param-value>/redir.jsp</param-value>
        <description>
            Page to do HTTP POST of SAMLResponse to ACS URL
            Request attribute names forwarded to this page:
            SAMLResponseBase64Encoded: SAMLResponse to
be sent to ACS URL
            RelayState: RelayState to be sent to ACS
URL (please do HTML encoding to avoid XSS)
            ACSUrl: ACS URL to send the SAML response
to
            SAMLResponseXML: SAMLResponse (in XML form,
only for troubleshooting)
            SPId: ID of the Service Provider (for
display purpose)
            SPName: name of the Service Provider (for
display purpose)
            ProviderName: provider name if provided in
the request (for display purpose)
        </description>
    </init-param>
    <init-param>
        <param-name>errorPageUrl</param-name>
        <param-value>/error.jsp</param-value>
        <description>
            Page to handle exception. If not specified, it
throws the exception
            Request attribute names forwarded to this page:
            exception
        </description>
    </init-param>
    <init-param>
        <param-name>sessionTokenAttrName</param-name>
        <param-value>WSASID</param-value>
        <description>Session attribute name for
sessionToken. If not specified, default to WSASID</description>
    </init-param>
    <load-on-startup>5</load-on-startup>
</servlet>
<! SAML >
    <servlet-mapping>
        <servlet-name>SAMLAuthnProcessor</servlet-name>
        <url-pattern>/samlssso</url-pattern>
    </servlet-mapping>

```



Note: Take note that the loginPageUrl value is changed from login.jsp to start.uso:

XML

```
<param-name>loginPageUrl</param-name>
<param-value>/start.uso</param-value>
```

3. Add your hostname IP to the "hosts" file to map **usoserver.i-sprint.demo** to your server IP.
e.g. 192.168.50.211 usoserver.i-sprint.demo
4. Start the AM server.

Configuring Identity Provider

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Federated SSO > SAML > Identity Provider**.
2. Select the default Identity Provider (i.e. "DefaultIdentityProvider") and edit.
3. Provide the path for the key store file in **Keystore** field, e.g. "C:\saml-idp3.keystore".



Note: Key store file can be generated by the keytool in java.

4. Key in the **Keystore passphrase**, e.g. "passphrase", and **Key alias** with any value, e.g. "idp".
5. Save the changes.

Configuring Service Provider

Prior to Service Provider creation, ensure that the User Response is already created. If not created yet, follow the steps below:

1. Navigate to **Administration Dashboard > Configuration > Messaging & Response > User Response**.
2. Click **Create** button and select "User Information Response Collector". Click **Next**.
3. Provide the desired **User Response Id**, e.g. "gapps", the **User Response Name**, e.g. "Google Apps for Business", and **Description**.
4. Under **Attributes**, type the desired **Name**, e.g. "email", and select **email** as its **Attribute Name**. Click **Add**.
5. Save the changes and proceed to create the service provider.

There may be different SAML configurations for different service providers. For this guide, google apps will be used as a Service Provider.

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Federated SSO > SAML > Service Provider**.
2. Click the **Create** button and select "UAM as Service Provider" for the **Implementation**. The **Create Service Provider** page is displayed.
3. Enter the **Service Provider Id**, e.g. "Gmail", the **Service Provider Name**, e.g. "i-Sprint Google App for Business", the **Description**, and **Id used as issuer**, e.g. "google.com/a/i-sprint.com.sg".
4. Input the **Assertion Consumer Service URLs** value, e.g. "https://www.google.com/a/i-sprint.com.sg/acs".
5. Set the **Format** to "Email" and **Value** to "User Response".
6. Select the user response value created in previous steps, e.g. "gapps".
7. Input value on **URL to initiate USSO**.



Note: If this field is empty, then USO will not be displayed in AccessMatrix SSO Portal.

8. Save the changes.

Testing SAML for Single Sign-On

1. Navigate to **Operation Dashboard > User & Segment > User**.
 2. Create or search the user to use for testing, and assign the **Email** value to **Email** attribute.
 3. Configure the USO entitlements, if necessary and login account for the selected user.
 4. Save the changes.
 5. Open the following URL in a browser: *https://mail.google.com/a/i-sprint.com.sg*. The page will be redirected to AccessMatrix SSO Portal page.
 6. Enter the USO credentials to log in to AccessMatrix SSO Portal. Click **Login** to continue. The SAML application will be displayed in AccessMatrix SSO Portal and USO Client is automatically launched.
 7. Open a new tab and enter the same URL: *https://mail.google.com/a/i-sprint.com.sg*. At this time, the user doesn't need to enter their credentials but it will perform SSO successfully.
-

8.3. UAM Integration for USO

Integration with Application Security Agent (ASA)

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Web SSO > Access Agent**.
2. Click **Create** and select the "Application Security Agent" as the access agent implementation.
3. Under the **Attributes** tab, specify the value for **uso.url** attribute, e.g.
"http://<hostname>:<port>/usoserver/start.uso".
4. Specify the rest of the required ASA details and save.
5. Perform the steps described in [Assign Application Privilege To User](#).
6. Perform the steps described in [Launching UAM Application](#).

Integration with Physical Access Application

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Web SSO > Access Agent**.
2. Click **Create** and select the "Physical Access Application" as the access agent implementation.
3. Under the **Attributes** tab, specify the value for **uso.url** attribute, e.g.
"http://<hostname>:<port>/usoserver/start.uso".
4. Specify the rest of the required application details and save.
5. Perform the steps described in [Assign Application Privilege To User](#).
6. Perform the steps described in [Launching UAM Application](#).

Integration with Web Security Agent (WSA)

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Web SSO > Access Agent**.
 2. Click **Create** and select the "Web Security Agent" as the access agent implementation.
 3. Under the **Attributes** tab, specify the value for **URL/Path to launch application** attribute, e.g.
"http://www.testserver.com:7070/wsatest/wsa_login_form.html".
 4. Specify the rest of the required application details and save.
 5. Perform the steps described in [Assign Application Privilege To User](#).
-

-
6. Perform the steps described in [Launching UAM Application](#).

Assign Application Privilege To User

1. Navigate to **Operation Dashboard > User & Segment > User** and select a user.
2. Click the **Application Privilege** tab and add the application defined in the previous section.
3. Save the changes.

Launching UAM Application

1. Launch the AccessMatrix SSO Portal by accessing *http://<hostname>:<port>/usoserver/start.uso*.
 2. Log in using the user with the assigned application.
 3. UAM application will be displayed in AccessMatrix SSO Portal once USO login is successful.
 4. Launch the UAM application. The UAM Application page will be displayed.
-

8.4. Integrated Windows Authentication for USO

Enable Integrated Windows Authentication with SPNEGO/Kerberos (Deprecated)

Integrated Windows Authentication (IWA) is a Microsoft technology that is used in an intranet environment where users have Windows domain accounts. With IWA, the credentials (user name and password) are hashed before being sent across the network. The client browser proves its knowledge of the password through a cryptographic exchange with your Web application server.

The key components of IWA include an Active Directory Controller machine, Kerberos Key Distribution Center (KDC) in a Domain Controller machine, a machine with a client browser, and a Web application server.

Microsoft Windows when configured to use an Active Directory domain is based upon a security infrastructure which is at its core Kerberos. Kerberos has supported distributed authentication for a long time. Microsoft has extended a protocol known as Simple and Protected GSS-API Negotiation Mechanism (SPNEGO) for sharing Windows credentials with Web servers. This protocol defines a way via negotiation for the client workstation to send authentication data to the Web server. If the Web server understands the SPNEGO tokens, this can then be used to implement secure Single Sign On. Thus, the end user's identity is transferred in a secure manner to the Web server.

As an overall picture, each AccessMatrix USO user must have a Windows login session within the corporate domain to login to AccessMatrix USO server. When the user launches MS Internet Explorer (IE) browser, the browser must authenticate the user - given the windows domain session information - using Kerberos. When the user is successfully authenticated, USO agent will extract user information from the HTTP header and authenticate the user in AccessMatrix.

Configure Tomcat for Windows Integrated Authentication

AccessMatrix's bundled Tomcat Application Server is used as a default application server for AccessMatrix USO server and thus will be used to demonstrate this integration in this document.

This document is adapted from the installation guide for tomcat server specified in the SPNEGO SourceForge project website at <http://spnego.sourceforge.net/index.html>.



Note: The term <USO_SERVER_ROOT> will be used to refer to the root or installation directory of AccessMatrix server on both Windows and Unix platforms.

Prerequisites

A working domain controller and at least one client computer in that domain are required for this solution, as testing windows integrated authentication from the domain controller will not work.

We recommend completing the following checklist before configuring:

- A functioning Microsoft Windows 2008 R2/Windows 2012/Windows 2012 R2 Active Directory Domain including:
 - Domain Controller
 - Users that can log into the client workstation
 - AccessMatrix USO server running Tomcat server that is joined to Active Directory Domain
- A Microsoft Windows XP SP3/Windows 7/Windows 8 Client Workstation that is joined to Active Directory Domain
- Domain controller's DNS server is properly configured to have entries of Client workstation and the AccessMatrix USO server is running on a Windows Server.
- All hosts should be joined into the Active Directory Domain. Confirm this by doing a **PING** to AccessMatrix USO server and Domain Controller using FQDN.
- A functional user Id to be created in Active Directory Domain. This user Id will be used by the SPNEGO HTTP filter to make connection to Windows server.
- JDK 1.6 and above must be installed and used to run AccessMatrix USO server. SPNEGO HTTP filter only supports JDK 1.6 and above.

Pre-configuration Checklist

- Ensure that the client workstation is on the domain by checking the full computer name (FQDN) and domain name from **System Properties (Control Panel > System and Security > System)**.
 - Ensure that the client workstation is logged onto the domain and not locally.
 - Ensure that a domain user is created. Confirm that the username and password created is correct by attempting to login on some workstation.
 - Ensure the correct address of the KDC.
-

-
- If the IP address or host name of Active Directory is known, then the address of the KDC is already known (it's usually on the same host).
 - Another way to determine the address of the KDC is to use the `ping` command, e.g. `ping qauat.com`. Notice that by pinging the domain, we were able to get the IP of the KDC (the organization may be running multiple KDCs so a different IP may be retrieved each time). It is preferred to use host names instead of IP addresses. Run the `ping` command on the IP address and use the `-a` flag to get the host name associated with the IP address, e.g. `ping -a 192.168.1.167`.
 - Ensure that latest version of the JDK is installed.
 - Java 6 is required (1.6.x and above). It is best to have both the Java Runtime Environment (JRE) and the Java Developer Kit (JDK) installed on the workstation. The app server will only need the JRE installed. The workstation will only need the JDK if doing some advanced configurations, or if building the SPNEGO project from source, or if writing a java client that uses SPNEGO.
 - Confirm the version of the JRE and JDK installed by running each program (**java.exe** and **javac.exe**) with the `-version` flag.
 - Ensure that test connection to KDC is successful. Perform the steps described in [Test Connection to KDC](#).

Test Connection to KDC

Perform the steps below to confirm the connection to KDC. Simply replace the values in the files with the values from the environment.

1. Create the **krb5.conf** file. A sample content of the file is shown below.

Replace the references to QAUAT.COM and qadc.qauat.com (Domain Controller) with the values from your environment. Notice below that the name of the Domain and Realm is QAUAT.COM. Replace the Domain and Realm values with the name of your Domain.



Note: Please ensure that all Domain values (e.g. QAUAT.COM) are specified using CAPITAL LETTERS.

Code

```
[libdefaults]
    default_realm = QAUAT.COM
    default_tkt_enctypes = aes128-cts rc4-hmac des3-cbc-sha1 des-cbc-md5 des-cbc-crc
```

```
default_tgs_etypes = aes128-cts rc4-hmac des3-cbc-  
sha1 des-cbc-md5 des-cbc-crc  
permitted_etypes = aes128-cts rc4-hmac des3-cbc-  
sha1 des-cbc-md5 des-cbc-crc  
[realms]  
  QAUAT.COM = {  
    kdc = qadc.qauat.com  
    default_domain = qauat.com  
  }  
  
[domain_realm]  
  .QAUAT.COM = QAUAT.COM
```

2. Create the **login.conf** file. A sample content of the file is shown below:

It is not necessary, but the module names `spnego-client` and `spnego-server` can be changed. Part of the servlet filter installation will require the user to enter these module names in the **web.xml** file.

XML

```
spnego-client {  
    com.sun.security.auth.module.Krb5LoginModule  
required;  
};  
spnego-server {  
    com.sun.security.auth.module.Krb5LoginModule  
required  
    storeKey=true  
    isInitiator=false;  
};
```

3. Create the **HelloKDC.java** file in the source code editor. A sample content of the file is shown below. Replace the string values with the values from the environment.

Java

```
/*  
 * Copyright (C) 2009 "Darwin V. Felix"  
<dfelix@users.sourceforge.net>  
 *  
 * This library is free software; you can redistribute  
it and/or  
 * modify it under the terms of the GNU Lesser General  
Public  
 * License as published by the Free Software  
Foundation; either  
 * version 2.1 of the License, or (at your option) any  
later version.  
 *  
 * This library is distributed in the hope that it will  
be useful,
```

```

    * but WITHOUT ANY WARRANTY; without even the implied
    warranty of
    * MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
    See the GNU
    * Lesser General Public License for more details.
    *
    * You should have received a copy of the GNU Lesser
    General Public
    * License along with this library; if not, write to
    the Free Software
    * Foundation, Inc., 59 Temple Place, Suite 330,
    Boston, MA 02111-1307 USA
    */

```

```

import java.io.File;
import java.io.FileNotFoundException;
import java.security.NoSuchAlgorithmException;

import javax.security.auth.callback.Callback;
import javax.security.auth.callback.CallbackHandler;
import javax.security.auth.callback.NameCallback;
import javax.security.auth.callback.PasswordCallback;
import javax.security.auth.login.Configuration;
import javax.security.auth.login.LoginContext;

public final class HelloKDC {

    private HelloKDC() {
        // default private
    }

    public static void main(final String[] args) throws
    Exception {

        // Domain (pre-authentication) account
        final String username = "testuser";

        // Password for the pre-auth acct.
        final String password = "p@ssw0rd";

        // Name of our krb5 config file
        final String krbfile = "krb5.conf";

        // Name of our login config file
        final String loginfile = "login.conf";

        // Name of our login module
        final String module = "spnego-client";

        // set some system properties
        System.setProperty("java.security.krb5.conf",
        krbfile);

        System.setProperty("java.security.auth.login.config",
        loginfile);
        //System.setProperty("sun.security.krb5.debug",
        true);
    }
}

```

```
// assert
HelloKDC.validate(username, password, krbfile,
loginfile, module);

    final CallbackHandler handler =

HelloKDC.getUsernamePasswordHandler(username,
password);

    final LoginContext loginContext = new
LoginContext(module, handler);

    // attempt to login
loginContext.login();

    // output some info
System.out.println("Subject=" +
loginContext.getSubject());

    // logout
loginContext.logout();

    System.out.println("Connection test
successful.");
}

    private static void validate(final String username,
final String password
    , final String krbfile, final String loginfile,
final String moduleName)
        throws FileNotFoundException,
NoSuchAlgorithmException {

        // confirm username was provided
        if (null == username || username.isEmpty()) {
            throw new IllegalArgumentException("Must
provide a username");
        }

        // confirm password was provided
        if (null == password || password.isEmpty()) {
            throw new IllegalArgumentException("Must
provide a password");
        }

        // confirm krb5.conf file exists
        if (null == krbfile || krbfile.isEmpty()) {
            throw new IllegalArgumentException("Must
provide a krb5 file");
        } else {
            final File file = new File(krbfile);
            if (!file.exists()) {
                throw new
FileNotFoundException(krbfile);
            }
        }
    }
}
```

```

        // confirm loginfile
        if (null == loginfile || loginfile.isEmpty()) {
            throw new IllegalArgumentException("Must
provide a login file");
        } else {
            final File file = new File(loginfile);
            if (!file.exists()) {
                throw new
FileNotFoundException(loginfile);
            }
        }

        // confirm that runtime loaded the login file
        final Configuration config =
Configuration.getConfiguration();

        // confirm that the module name exists in the
file
        if (null ==
config.getAppConfigurationEntry(moduleName)) {
            throw new IllegalArgumentException("The
module name "
                + moduleName + " was not found in
the login file");
        }
    }

    private static CallbackHandler
getUsernamePasswordHandler(
        final String username, final String password) {

        final CallbackHandler handler = new
CallbackHandler() {
            public void handle(final Callback[]
callback) {
                for (int i=0; i<callback.length; i++) {
                    if (callback[i] instanceof
NameCallback) {
                        final NameCallback nameCallback
= (NameCallback) callback[i];
                        nameCallback.setName(username);
                    } else if (callback[i] instanceof
PasswordCallback) {
                        final PasswordCallback
passCallback = (PasswordCallback) callback[i];
                        passCallback.setPassword(password.toCharArray());
                    } else {
                        System.err.println("Unsupported
Callback: "
                            +
callback[i].getClass().getName());
                    }
                }
            }
        };
    }

```

```
        return handler;
    }
}
```

4. Compile **HelloKDC.java** by typing this command: `javac HelloKDC.java`.

5. Run **HelloKDC** from the command prompt: `java HelloKDC`. If things worked as expected, a "Connection test successful." message should be displayed in the output. For trouble-shooting, refer to:

http://spnego.sourceforge.net/troubleshoot_hello_kdc.html

Configure the AccessMatrix USO Server

Perform the following steps to enable SPNEGO/Kerberos configuration in AccessMatrix USO server:

1. Ensure that the latest **spnego.jar** is in USO lib directory as shown in below location. By default, **spnego-r7.jar** is bundled together with AM installer.

`<USOSERVER_ROOT>\tomcat\webapps\am5\WEB-INF\lib\`

2. Open the USO **web.xml** file from following location in a text editor:

`<USOSERVER_ROOT>\tomcat\webapps\am5\WEB-INF\`

3. Find the `<filter>` section and uncomment the following lines:

XML

```
<!-- USO-SPNEGO KERBEROS: Remark the following -->
<filter>
    <filter-name>SpnegoHttpFilter</filter-name>
    <filter-
class>net.sourceforge.spnego.SpnegoHttpFilter</filter-
class>

    <init-param>
        <param-
name>spnego.allow.basic</param-name>
        <param-value>true</param-value>
    </init-param>

    <init-param>
        <param-
name>spnego.allow.localhost</param-name>
        <param-value>true</param-value>
    </init-param>

    <init-param>
        <param-
name>spnego.allow.unsecure.basic</param-name>
```

```

        <param-value>true</param-value>
    </init-param>

    <init-param>
        <param-
name>spnego.login.client.module</param-name>
        <param-value>spnego-client</param-
value>
    </init-param>

    <init-param>
        <param-
name>spnego.krb5.conf</param-name>
        <param-value>conf/krb5.conf</param-
value>
    </init-param>

    <init-param>
        <param-
name>spnego.login.conf</param-name>
        <param-
value>conf/login.conf</param-value>
    </init-param>

    <init-param>
        <param-
name>spnego.preauth.username</param-name>
        <param-value>testuser</param-value>
    </init-param>

    <init-param>
        <param-
name>spnego.preauth.password</param-name>
        <param-value>p@ssw0rd</param-value>
    </init-param>

    <init-param>
        <param-
name>spnego.login.server.module</param-name>
        <param-value>spnego-server</param-
value>
    </init-param>

    <init-param>
        <param-
name>spnego.prompt.ntlm</param-name>
        <param-value>true</param-value>
    </init-param>

    <init-param>
        <param-
name>spnego.logger.level</param-name>
        <param-value>1</param-value>
    </init-param>
</filter>

```

4. Find the <filter-mapping> section in the file and uncomment the following lines:

XML

```
<!-- USO-SPNEGO KERBEROS: Remark the following -->

<filter-mapping>
    <filter-name>SpnegoHttpFilter</filter-name>
    <url-pattern>/html/auth/ntlm/*</url-
pattern>
</filter-mapping>
```

5. Copy the **krb5.conf** and **login.conf** files created in [Test Connection to KDC](#) section to the Tomcat Home directory: `<USOSERVER_ROOT>\tomcat\conf\`. Take note that the JRE must be able to find these files.

Register Service Principal Names (SPN)

Use the **setspn.exe** utility to register all of your SPNs.

Login to any machine within the same domain as "Domain Administrator" (either a domain administrator or a domain user that belongs to the domain administrator group). The following is needed before running the **setspn.exe** utility:

- Pre-authentication (domain) account name (in this example, it would be "spnego")
- List of all SPNs to be registered

The pre-authentication username should be the same username used in the [Test Connection to KDC](#) section when running **HelloKDC.java** program.



Note: The list of SPNs should comprise of ALL possible aliases for user's service as well as the USO server's FQDN as described in the following example.

For example, assume there is a USO server named "PSTEST1" that has an HTTP server running on port 8080. The administrator must register the following two SPNs:

- HTTP/pstest1
- HTTP/pstest1.qauat.com

To register the above SPNs, run the following command:

```
setspn.exe -A HTTP/pstest1.qauat.com spnego
```

The result after running the **setspn.exe** utility should be as follows:

```
C:\Documents and Settings\Administrator.QASERVER>setspn.exe -A HTTP/pstest1.qauat.com
spnego
Checking domain DC=qauat,DC=com

Registering ServicePrincipalNames for CN=spnego,CN=Users,DC=qauat,DC=com
HTTP/pstest1.qauat.com
Updated object
```

Note the following:

- As demonstrated are only required to register service principal names for spnego. The commands above are the ONLY commands that need to be performed.
- No other value should be passed in as the 3rd argument other than the pre-authentication account name.
- Again, when running the **setspn.exe** utility with the **-A** flag, the only value needed for the 3rd/last argument is the pre-authentication account name.
- Furthermore, only register the principal names with the **HTTP/** prefix.
- Finally, ensure that the SPN is not registered to more than one username. A username can have more than one SPN registered but an SPN can only have one username.

For example, the following two commands are illegal:

- `prompt>setspn.exe -A HTTP/badexample User1`
- `prompt>setspn.exe -A HTTP/badexample User2`

The SPN HTTP/badexample cannot be resolved to a unique domain account.

Start AccessMatrix USO Server

Startup AccessMatrix USO server using the service manager or **startup.bat** accordingly.

Test SPNEGO Using JSP File (Optional)

1. Ensure that **hello_spnego.jsp** is found in
`<USOSERVER_ROOT>\tomcat\webapps\am5\html\auth\ntlm\` directory.
2. Launch a browser and enter the following URL:
`http://<hostname>:<port>/am5/html/auth/ntlm/hello_spnego.jsp`.
You should be able to see your username displayed on the page.

Configure Admin Console

This is the section where the required administrative and configuration tasks on AccessMatrix USO Server should be performed through Admin Console.

- **Assign 40171 Realm as Default Authentication for USO Users**

1. Navigate to **Administration Dashboard > Configuration > Unified SSO > Enterprised SSO > USO Server Configuration** and click the **OK** button of the **Information** dialog.
2. Click **Edit** button. The **Edit Configuration** page is displayed.
3. Under the **Authentication** section, look for the **Default authentication for USO users** attribute, and then select "Windows Integrated Authentication (NTLM) (40171)".
4. Click **Save**.

- **Edit User Store**

1. Navigate to **Administration Dashboard > Configuration > Authentication > User Store**.
2. Select the pre-defined AD user store called "ActiveDirectory" and click the **Edit** button.
3. Modify the settings to map to the AD server. Perform Test Connection to ensure the settings are correct.



Note: Ensure that the AD domain name is specified correctly in the NT Domain field.

- **Update the User Lookup Module**

1. Navigate to **Administration Dashboard > Configuration > Authentication > User Lookup Module**.
2. Select the predefined AD Lookup Module "DefaultActiveDirectoryLookup".
3. Specify the User Store in the **User Store** field. Click **Save**.

- **Assign 40171 Realm to Domain User**

1. Navigate to **Operation Dashboard > User & Segment > User**.
 2. Select "ActiveDirectory" from the **User Store** field and click the **Search** button.
 3. Click the domain user (i.e. "spnego") under the **Search User Result** page.
 4. Assign the relevant login account with authentication realm "Windows Integrated Authentication (NTLM) (40171)" to the domain user.
 5. Click **Save**.
-

Finally, log out and restart the USO Server for changes to take effect.

Configure Client Workstation Web Browser

For Internet Explorer, navigate to **Internet Options > Security > Local intranet > Sites** and add the SSO domain, e.g. "*.qauat.com".

Start the browser and attempt to surf to the snoop servlet. With security turned on, the snoop servlet will issue an authentication challenge to the browser, which will initiate the SPNEGO Kerberos exchange.

On the other hand, you may launch the USO Agent by accessing *http://<hostname>:<port>/am5/start.uso*. If configurations are all correct, then the domain user will be automatically logged in.

9. AccessMatrix SSO Portal Implementation

About AccessMatrix SSO Portal

AccessMatrix SSO Portal is a web portal that allows the end users to launch the USO applications and Federated SSO (Single Sign-On) applications they are entitled to. In addition, this component features Self-Service Portal (a.k.a Self-Service Facility (SSF)) that allows users to retrieve the entitled USO application password and enable USO application enrollment.

USO Applications

USO applications are applications that the user can SSO to using the target application's user Id and password. The term USO application covers both Windows native apps and web apps.

This type of SSO requires a non-intrusive integration with the target application. The target application does not need to be changed, and it typically does not support federated SSO. Thus, if this application requires user Id and password, the SSO Portal will SSO to the target application by automatically entering the actual user Id and password for the user.

Depending on the OS that the users are using, the supported USO application types will vary. The table below shows the supported USO application types for each OS:

Application Type \ OS	Windows	Linux	macOS
Native apps	Yes	No	No
Web apps	Yes	Yes	Yes

Based on your OS and USO application type requirements, you should choose **only one** of the following SSO Portal configuration types:

- **SSO Portal with Windows native apps and web apps**

(Windows only) This configuration type supports both Windows native apps and web apps. It involves installing the USO Client, along with the Google Chrome Extension "USO Chrome Extension".

- **SSO Portal with web apps only**

(Windows, Linux, and macOS) This configuration type supports only web apps. It involves installing the Google Chrome Extension "USO Chrome Extension for Web Apps" only.

Each configuration type supports a different set of features. The table below provides a detailed look at which features are supported for each OS and SSO Portal configuration type combination:

Supported Features \ Configuration Type	SSO Portal with Windows native apps and web apps	SSO Portal with web apps only
USO Trainer "Custom Behavior" (using JavaScript)	Yes	No
Change Password	Yes	No
Reauthentication	Yes	No
Windows		
SSO to web apps (w/ Chrome & Edge browsers)	Yes	Yes
SSO to native apps	Yes	No
macOS		
SSO to web apps (w/ Chrome browser)	Configuration type not supported	Yes
SSO to native apps	Configuration type not supported	No
Linux (e.g. Ubuntu)		
SSO to web apps (w/ Chrome browser)	Configuration type not supported	Yes
SSO to native apps	Configuration type not supported	No
i Notes: When using SSO Portal in a Windows environment, you have the option to use either one of the two configuration types. Here are the considerations for using each one: • SSO Portal with Windows native apps and web apps - This configuration is strongly recommended, as it covers the widest range of features and supports both USO application types. This makes it applicable for most use cases. • SSO Portal with web apps only - This configuration can be considered in the specific case where only web apps need to be used with the SSO Portal.		

Chrome Extensions

As noted in the preceding section, each SSO Portal configuration type makes use of a Chrome Extension (a JavaScript-based technology from Google). There is a different Chrome extension used for each configuration type, and they **cannot coexist**. Furthermore, they are only applicable for Chromium browsers (Google Chrome and Microsoft Edge). Refer to [AccessMatrix SSO Portal User Guide](#) for the installation steps of USO Client and Chrome extensions.

- **USO Chrome Extension**

Installing this Chrome extension will support USO and Federated SSO applications with functionalities of USO Client UI on Chromium browsers (i.e. Google Chrome and Microsoft Edge). The USO Client application will be installed.

- **USO Chrome Extension for Web Apps**

Installing this Chrome extension will provide USO Client functionalities and acts as the USO Client. Hence, the USO Client application will not be installed. This extension is applicable when all the applications are web apps.



Note: Prior to the AccessMatrix 5.7.5-GA release, "USO Chrome Extension for Web Apps" was known as "AccessMatrix Chrome Extension".

Federated Applications

SSO Portal supports SSO to federated applications that support SAML 2.0 as a Service Provider role. SSO Portal, together with CLP, acts as the SAML Identity Provider and SSO to target application by providing SAML assertion, containing the user identity, to the target application. Federated applications are supported regardless of the SSO Portal configuration type you use and do not require Chrome extensions or the USO Client.

Access SSO Portal and Self-Service Portal

AccessMatrix SSO Portal and SSF login authentication processes are integrated with Common Login Page.

- To access AccessMatrix SSO Portal:

`http://<AM_Server_IP_or_Name>:<Port_No>/<USO_Server_Context_Root>/start.uso`

Example: *`http://127.0.0.1:8080/usoserver/start.uso`*

-
- To access AccessMatrix Self-Service Portal:

http://<AM_Server_IP_or_Name>:<Port_No>/<USOSSF_Context_Root>/start.ssf

Example: *http://127.0.0.1:8080/usossf/start.ssf*

9.1. Implementation Planning

Implementation Checklist

Below is the checklist for implementing AccessMatrix SSO Portal.

Tasks	Reference
Deploy SSO Portal and Self-Service Portal	SSO Portal and Self-Service Portal Deployment
Configure SSO Portal	SSO Portal with USO Client SSO Portal without USO Client
Configure Self-Service Portal Policy	Self-Service Portal Policy

Implementation Planning

Planning prior to environment setup is critical to the success of AccessMatrix SSO Portal implementation. Understanding the customer's nature of business, studying the customer's requirements and its existing environments will help to identify the actions to make prior to actual implementation.

The following sections discusses on the implementation considerations that should be assessed during planning.

Performance Consideration

- **USO Applications (with USO Client)**

The list of user's entitled applications and its information are normally downloaded from AccessMatrix server via SSO Portal. In deployment where there are complex entitlements or many applications, it is recommended that you enable the PSE feature to cache USO Application information. This helps to reduce the load on the AccessMatrix server and provides a better user experience in general.

9.2. SSO Portal and Self-Service Portal Deployment

Deploy AccessMatrix SSO Portal and Self-Service Portal

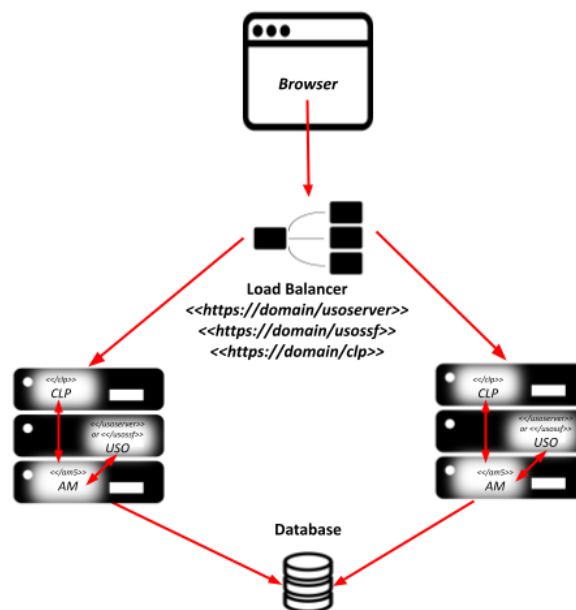
The SSO Portal and Self-Service Portal components can be installed either by running the AccessMatrix installation package (via executable installer or zip archive) or by installing the SSO and Self-Service Portals Web ARchive (WAR) packages. There are two methods of deploying the SSO and Self-Service Portals:

- by bundling SSO and Self-Service Portals with AccessMatrix server; and
- by deploying as a standalone instance from the AccessMatrix server.

i Note: In this guide, `<USOSERVER_HOME>` is used to refer to the installation directory of AccessMatrix SSO Portal components, e.g. `/opt/acmatrix/tomcat/webapps/usoserver`, and `<USOSSF_HOME>` is used to refer to the installation directory of Self-Service Portal components, e.g. `/opt/acmatrix/tomcat/webapps/usossf`.

Deploy AccessMatrix's Bundled SSO Portal and Self-Service Portal

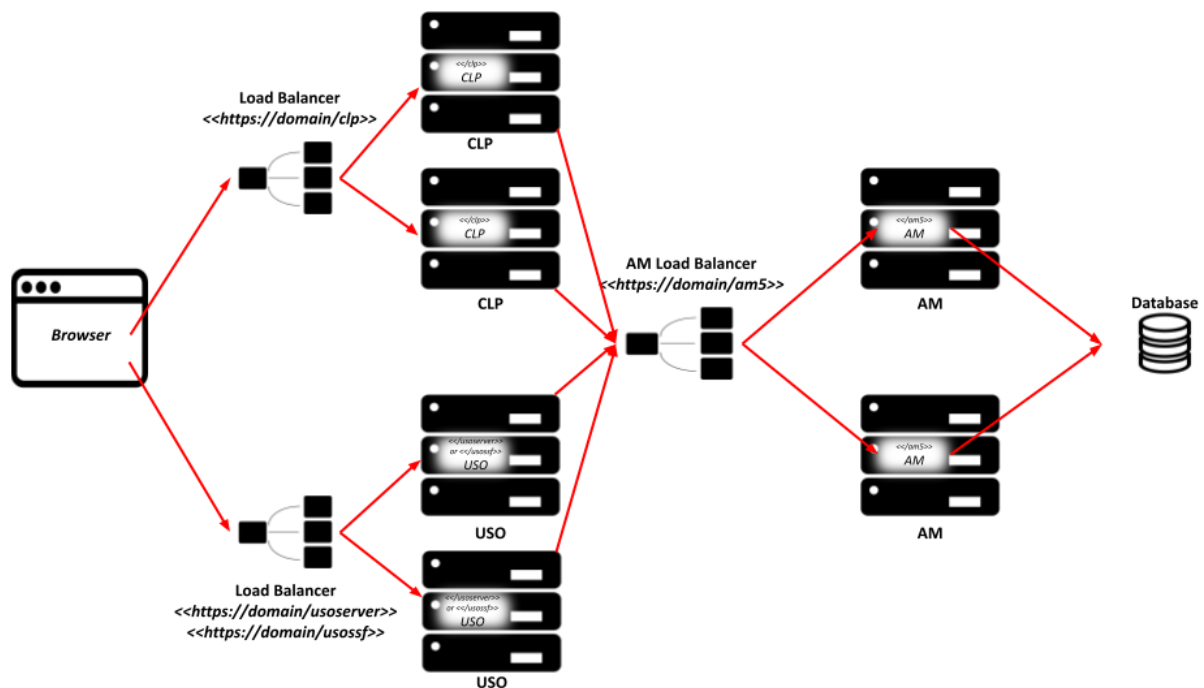
By default, installing the AccessMatrix installation package (via executable installer or zip archive) will deploy SSO Portal and Self-Service Portal together with the Common Login Page, AccessMatrix server and bundled Apache Tomcat server. An example deployment use case is shown below:



As shown in the diagram above, for multiple AccessMatrix instances that consist of the AccessMatrix application servers are deployed with Common Login Page (CLP), SSO Portal and Self-Service Portal running together. A load balancer is needed in front of the CLP and USO instances to distribute the workload among the instances. USO will only communicate to the AccessMatrix Server in the same instance in this type of deployment.

Deploy Standalone SSO Portal and Self-Service Portal

The SSO Portal and Self-Service Portal can be deployed as Web ARchive (WAR) packages onto JAVA-based web or application server (such as Apache Tomcat Server) and deployed as a separate instance from the AccessMatrix server. A load balancer is needed in front of the USO and CLP instances if more than one is deployed for use. Another load balancer is needed between the CLP instances, USO instances and AccessMatrix Servers if more than one AccessMatrix Servers are used. This deployment use case is shown below:



Note: The AccessMatrix installation package comes with an SSO Portal WAR file (e.g. **AccessMatrix-x.x.x.xxxx-GA-usoserver.war**) and a Self-Service Portal WAR file (e.g. **AccessMatrix-x.x.x.xxxx-GA-usosf.war**). Copy the WAR files to the server where the web or application server is installed. The WAR files may be renamed as **usoserver.war** and **usosf.war** for easy reference.

SSO Portal Domain

In this type of deployment, although CLP, SSO Portal and Self-Service Portal can be deployed in different servers, it is recommended that they are using the same domain, *especially when federated SSO is required*. This recommendation is to allow CLP and SSO Portal to access the same AccessMatrix session cookie (WSASID) so that the user experience when doing federated SSO is more seamless. The sharing of cookie can be done by setting the same cookie domain and cookie path (normally root "/") for CLP, SSO Portal and Self-Service Portal.

For CLP, SSO Portal and Self-Service Portal to have the same domain, normally a reverse proxy is deployed to proxy to the respective instances based on the context path. Find a reverse proxy documentation to know on how to configure such deployment. Example of the proxy mapping:

- If CLP is deployed in `clp.internal.example.com` and SSO Portal and Self-Service Portal are deployed in `usoserver.internal.example.com`:
 - `sso.example.com/clp ==> clp.internal.example.com/clp`
 - `sso.example.com/usoserver ==> usoserver.internal.example.com/usoserver`
 - `sso.example.com/usossf ==> usoserver.internal.example.com/usossf`
 - Cookie domain: `sso.example.com`, cookie path: `/`

Deploy SSO Portal and Self-Service Portal on Apache Tomcat Server

To deploy the SSO Portal and Self-Service Portal Web Archive:

1. Navigate to Apache Tomcat's installation directory, e.g. `/ApacheTomcat/apache-tomcat-9.0.35`.
2. Extract the contents of SSO Portal (**usoserver.war**) and Self-Service Portal (**usossf.war**) WAR files to Apache Tomcat's `webapps` folder.
3. Configure SSO and Self-Service Portals to establish connection to the AccessMatrix server as described in [Configure SSO Portal and Self-Service Portal to Connect to the AccessMatrix Server](#) section.
4. Start the Apache Tomcat server.
5. Launch the SSO Portal application.



Note: If changes are introduced to the SSO and Self-Service Portal Web Archives in the future released version, the administrator is required to redeploy the Web Archives.

Configure SSO Portal and Self-Service Portal to Connect to the AccessMatrix Server

SSO Portal

The configuration below establishes the connection between SSO Portal and AccessMatrix server. To configure:

1. Navigate to `<USOSERVER_HOME>/WEB-INF/` and open **web.xml**.
2. Look for the `restAPIURL` element and modify the value to refer to the relevant AccessMatrix server as shown below:

XML
<pre><context-param> <param-name>restAPIURL</param-name> <param- value>http://localhost:8080/am5/rest</param-value> </context-param></pre>

To support client-authenticated SSL (a mandatory configuration for Production environment), the following settings should be configured:

- Modify the `restAPIURL` value to use secure connection to the AccessMatrix server as shown below:

XML
<pre><context-param> <param-name>restAPIURL</param-name> <param- value>https://localhost:8443/am5/rest</param-value> </context-param></pre>

- Configure the following client keystore properties:

XML
<pre><context-param> <param-name>am.hc4.ssl.keystore</param-name> <param-value>/opt/acmatrix/tomcat/conf/ssportal- client.keystore</param-value> </context-param> <context-param> <param-name>am.hc4.ssl.keystore-passphrase</param-name> <param-</pre>

```
value>AesEncryptor:pWtZZ0N6bfIvSlBBF0zXC0ZwSlKny/GexGVFpaqWI1Q
=</param-value>
</context-param>
```

Note: Refer to the [AccessMatrix Common Login Page Deployment Guide - List of Configurable Parameters](#) section for the description of the properties.

3. Save the changes.
4. Restart the USO server for changes to take effect.

Note: Refer to [Connection Pool Settings](#) section if there's a need to fine tune the connection to AccessMatrix server.

Self-Service Portal

The configuration below establishes the connection between Self-Service Portal and AccessMatrix server. To configure:

1. Navigate to `<USOSSF_HOME>/WEB-INF/classes/` and open **web.xml**.
2. Look for the `restAPIURL` element and modify the value to refer to the relevant AccessMatrix server as shown below:

XML

```
<display-name>ssf</display-name>
<context-param>
  <param-name>restAPIURL</param-name>
  <param-value>http://localhost:8080/am5/rest</param-
value>
</context-param>
```

To support client-authenticated SSL (a mandatory configuration for Production environment), the following settings should be configured:

- Modify the `restAPIURL` value to use secure connection to the AccessMatrix server as shown below:

XML

```
<display-name>ssf</display-name>
<context-param>
  <param-name>restAPIURL</param-name>
```

```
<param-  
value>https://localhost:8443/am5/rest</param-value>  
</context-param>
```

- Configure the following client keystore properties:

XML

```
<context-param>  
  <param-name>am.hc4.ssl.keystore</param-name>  
  <param-  
value>/opt/acmatrix/tomcat/conf/ssf_client.keystore</param-  
value>  
</context-param>  
<context-param>  
  <param-name>am.hc4.ssl.keystore-passphrase</param-name>  
  <param-  
value>AesEncryptor:pWtZZON6bfIvSlBBF0zXC0ZwSlKny/GexGVFpaqWI1Q  
=</param-value>  
</context-param>
```



Note: Refer to the [AccessMatrix Common Login Page Deployment Guide - List of Configurable Parameters](#) section for the description of the properties.

3. Save the changes.
4. Restart the USO SSF server for changes to take effect.



Note: Refer to [Connection Pool Settings](#) section if there's a need to fine tune the connection to AccessMatrix server.

Configure Session Token Cookie Domain and Path

Upon SSO Portal login, an AccessMatrix session token cookie (WSASID cookie) is generated. This session cookie is used to verify user identity and the session's validity; and this can be set to a domain and a context path.

To configure:

1. Navigate to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Server Configuration**.
 2. Click **Edit** and specify the values for **Cookie.Domain** and **Cookie.Path** attributes. By default, domain value is empty, thus setting the session cookie in the default domain.
 3. Specify the context path value. By default, path is set to "/", thus allowing the session cookie to be set in all context paths in the current domain.
-

-
4. Save the changes made.
 5. Restart the USO server for changes to take effect.

Configure SSO Portal and Self-Service Portal Log Properties (OPTIONAL)

SSO Portal and Self-Service Portal uses Apache Log4j 2 as the logging utility. It defines the log level and log output file location.

SSO Portal

To configure:

1. Navigate to `<USOSERVER_HOME>/WEB-INF/classes/` and open **usoagentlog4j2.properties**.
2. Modify the log level, if necessary.
3. Save the changes.
4. Restart the USO server for changes to take effect.

Self-Service Portal

To configure:

1. Navigate to `<USOSSF_HOME>/WEB-INF/classes/` and open **usossflog4j2.properties**.
2. Modify the log level, if necessary.
3. Save the changes.
4. Restart the USO SSF server for changes to take effect.

Configure CLP URL

SSO Portal and Self-Service Portal use Common Login Page (CLP) to handle user authentication. Hence, it is required to configure the CLP URL. Refer to [AccessMatrix Common Login Page Deployment Guide - Configure the CLP URL in AccessMatrix Applications](#) for the configuration steps.

Configure Connection Pool Settings to AccessMatrix Server (Optional)

Connection pooling refers to the method of creating a pool of connections and caching those connections so that it can be reused again. It improves the response time of a web-based application. This setting is used to fine tune the connection to AccessMatrix server.

SSO Portal

To configure:

1. Navigate to `<USOSERVER_HOME>/WEB-INF/` and open **web.xml**.
2. Set the connection timeout value of SSO Portal to AccessMatrix server by configuring the following parameter values:

XML

```
<context-param>
  <param-name>am.hc4.timeout.conn</param-name>
  <param-value>20000</param-value>
</context-param>
<context-param>
  <param-name>am.hc4.timeout.socket</param-name>
  <param-value>30000</param-value>
</context-param>
<context-param>
  <param-name>am.hc4.pool.maxconn_per_route</param-
name>
  <param-value>50</param-value>
</context-param>
```

3. You may add more connection pool parameters by referring to [AccessMatrix Common Login Page Deployment Guide - List of Configurable Parameters](#) section.
4. Save the changes.
5. Restart the USO server for changes to take effect.

Self-Service Portal

To configure:

1. Navigate to `<USOSSF_HOME>/WEB-INF/classes/` and open **web.xml**.
 2. Set the connection timeout value of SSF Portal to AccessMatrix server by configuring the following parameter values:
-

XML

```
<context-param>
  <param-name>am.hc4.timeout.conn</param-name>
  <param-value>20000</param-value>
</context-param>
<context-param>
  <param-name>am.hc4.timeout.socket</param-name>
  <param-value>30000</param-value>
</context-param>
<context-param>
  <param-name>am.hc4.pool.maxconn_per_route</param-
name>
  <param-value>50</param-value>
</context-param>
```

3. You may add more connection pool parameters by referring to [AccessMatrix Common Login Page Deployment Guide - List of Configurable Parameters](#) section.
 4. Save the changes.
 5. Restart the USO SSF server for changes to take effect.
-

9.3. SSO Portal for USO Applications

! Important Note: You should have already studied the detailed explanations of each SSO portal configuration and their respective use cases in [AccessMatrix SSO Portal Implementation - USO Applications](#) before proceeding. If you have not, please do so before continuing.

In order to enable the use of SSO Portal with USO applications, you may be required to perform several configurations. The sections below provide instructions on how to do so for each SSO Portal configuration type. You should only refer to **only one** of these sections, based on whichever configuration type is better suited to your needs:

- **SSO Portal with Windows Native Apps and Web Apps**

(Supported on Windows only) This configuration type supports both Windows native apps and web apps. It involves installing the USO Client, along with the Google Chrome Extension "USO Chrome Extension".

If you are using SSO Portal in a Windows environment, it is recommended that you proceed with this configuration.

- **SSO Portal with Web Apps Only**

(Supported on Windows, Linux, and macOS) This configuration type supports only web apps. It involves installing the Google Chrome Extension "USO Chrome Extension for Web Apps" only.

If you are using SSO Portal in a Linux or macOS environment, it is recommended that you proceed with this configuration.

9.3.1. SSO Portal with Windows Native Apps and Web Apps

This is the configuration type for SSO Portal that supports both Windows native apps and web apps. This configuration type is used in Windows environments where the USO applications used are both native applications and web applications, and where Chromium browsers (Google Chrome and Microsoft Edge) are used. It is necessary to install both of the following:

- **USO Client**

- For a manual installation of the USO Client, refer to [USO Client User Guide - Installing the USO Client](#).
- For enterprise setups that require the USO Client to be automatically installed in all of the intended users' computers within an AD domain, it is possible to perform a centralized deployment of the USO Client.

Refer to the [USO Client Centralized Deployment](#) section in this page for the details.

- **USO Chrome Extension**

- For a manual installation of the USO Chrome Extension, refer to [AccessMatrix SSO Portal User Guide - Install USO Chrome Extension](#).
- For enterprise setups that require the USO Chrome Extension to be automatically installed in all of the intended users' computers within an AD domain, it is possible to perform a centralized distribution of the USO Chrome Extension.

Refer to the [USO Chrome Extension Centralized Deployment on Chromium Browsers](#) section in this page for the details.

Once done, you may proceed with the following sections, which detail configurations that may be necessary in order to enable the SSO Portal to work as intended:

- **Set SSO Portal Configuration**

For AccessMatrix versions 5.7.4-GA and earlier, the SSO Portal must be configured to enable the use of the USO Client. Click [here](#) for the configuration details.

- **Enable PSE Login**

For use cases where users need to use the SSO portal when offline/ disconnected from a network. A PSE file must be created to facilitate the local SSO Portal login process. Click [here](#) for the configuration details.

USO Client Centralized Deployment

The USO Client needs to be installed on the user's computer before the user can use it. The SSO Portal provides a link for the user to download the USO Client Windows installer.

Alternatively, for the centralized distribution of the USO Client in an enterprise setup, the administrator may choose to configure the Active Directory Group Policy so that USO Client can automatically be installed on the user's computer without requiring any user action.

The following sections will provide steps on how to automatically distribute USO Client to client computers or users using Group Policy.

Create a Distribution Point

To deploy the USO Client installer, you must create a distribution point on the publishing server (in this example, an Active Directory server):

1. Log on to the server computer as an administrator.
2. Create a shared network folder where you will put the **USOClientInstaller.msi** that you want to distribute, e.g. "sharing".
3. Set permissions on the *sharing* folder to allow access to the distribution package.



Note: If the user does not have access to the MSI file, the user's computer cannot start to install **USOClientInstaller.msi**.

4. Copy or install the **USOClientInstaller.msi** file to the distribution point.

Create a Group Policy Object

This section assumes that you have already installed the Group Policy Management console in your Windows server. The installation and configuration of this console is beyond the scope of this guide.

Perform the steps below to create a Group Policy Object (GPO) to use to distribute the USO Client installer:

1. Start the Group Policy Management console. To do this, in your domain server, open **Server Manager**, then click **Tools** and select **Group Policy Management**.
 2. In the Group Policy Management console tree, right-click your domain, and then click **Create a GPO in this domain, and Link it here**.
-

-
3. Type a name for this new GPO (e.g. "Deploy Software"), and click **OK**.

Assign a Package

Perform the following steps to assign the USO Client installer that are running on Windows 2012 R2, Windows 2016, or Windows 2019, or to users who are logging on to one of these workstations.

1. In the Group Policy Management console tree, right-click **Deploy Software** and click **Edit**.
2. In the **Group Policy Management Editor**, go to **Computer Configuration > Policies > Software Settings**.
3. Right-click **Software installation**, point to **New**, and then click **Package**.
4. In the **Open** dialog box, type the full Universal Naming Convention (UNC) path of the shared USO Client installer. The UNC syntax for Windows systems has the generic form, i.e. *file server\shared folder\file name.msi*. In this example, the UNC path is *myserver\sharing\USOClientInstaller.msi*.



Important Note: Do not use the **Browse** button to access the location. Make sure that you use the UNC path to the shared installer package.

5. Select **Assigned** option from the **Deploy Software** dialog box, and then click **OK**. The package is listed in the right-pane of the **Group Policy Management Editor** window.
6. Close the **Group Policy Management Editor**, and then close the Group Policy Management console.
7. When the client computer starts, the managed software package is automatically installed.

Redeploy a Package

In some cases, you may want to redeploy the USO Client package (for example, if you upgrade or change the installer).

To redeploy a package, perform the following steps:

1. Start the Group Policy Management console. To do this, in your domain server, open **Server Manager**, then click **Tools** and select **Group Policy Management**.
 2. In the Group Policy Management console tree, right-click **Deploy Software** and click **Edit**.
 3. In the **Group Policy Management Editor**, go to **Computer Configuration > Policies > Software Settings**.
-

-
4. Click **Software installation** and in the right-pane of the **Group Policy Management Editor**, right-click the USO Client program, point to **All Tasks***, and then click **Redeploy application**. The following message will be displayed: "Redeploying this application will reinstall the application everywhere it is already installed. Do you want to continue?"
 5. Click **Yes**.
 6. Close the **Group Policy Management Editor**, and then close the Group Policy Management console.

Upgrade the USO Client Application

To upgrade the existing USO Client to the latest version, just replace the copy found in the distribution point with the new one. Refer to [Create a Distribution Point](#) section to know about the distribution point.

Remove a Package

To delete the existing package, perform the following steps:

1. Start the Group Policy Management console. To do this, in your domain server, open Server Manager, then click **Tools** and select **Group Policy Management**.
 2. In the Group Policy Management console tree, right-click **Deploy Software** and click **Edit**.
 3. In the **Group Policy Management Editor**, go to **Computer Configuration > Policies > Software Settings**.
 4. Click **Software installation** and in the right-pane of the **Group Policy Management Editor**, right-click the USO Client program, point to **All Tasks***, and then click **Remove**.
 5. Perform one of the following actions:
 - Click **Immediately uninstall the software from users and computers**, and then click **OK**.
 - Click **Allow users to continue to use the software but prevent new installations**, and then click **OK**.
 6. Close the **Group Policy Management Editor**, and then close the Group Policy Management console.
-

USO Chrome Extension Centralized Deployment on Chromium Browsers

To enable USO Client for SSO Portal using Chromium browsers (i.e. Google Chrome and Microsoft Edge), the USO Chrome Extension must be installed. This extension is available in the Chrome Web Store for users to manually download.

Alternatively, for the centralized distribution of the USO Chrome Extension in an enterprise setup, the Administrator may configure the Active Directory Group Policy so that USO Chrome Extension can automatically be installed on the user's computer without requiring any user action.

The following sections will provide steps on how to automatically distribute USO Chrome Extension on Chromium browsers (i.e. Google Chrome and Microsoft Edge) to client computers using Group Policy.

Prerequisites

- The client computer must be a member of Microsoft Active Directory domain and running on Windows 10 Pro operating system.
- Ensure that the Google Chrome or Microsoft Edge browser is already deployed.

Install Admin (.ADMX) Templates for Browser

Before making any Group Policy Object configurations, install the ADMX template for your respective Chromium browser. ADMX templates add rules and settings for your browser to the Group Policy Central Store in your Active Directory domain.

To install them, do the following:

Google Chrome	1. Retrieve the admin template from https://chromeenterprise.google/browser/download/#manage-policies-tab. ○ To locate the policy files, select "Chrome ADM/ADMX templates" under Policy Templates and click the Download button. 2. Add the admin template to your Active Directory domain. Refer to the Add the Chrome browser policy template section of the Chrome Enterprise documentation for the full steps.
Microsoft Edge	1. Retrieve the admin template from https://aka.ms/EdgeEnterprise. ○ To locate the policy files, click the Download Windows Policy link below the Download button of your respective Windows version. ○ Alternatively, navigate down to the Microsoft Edge version selector and click the Get policy files button after specifying your Edge version.

	2. Add the admin template to your Active Directory domain. Refer to the Add the administrative template to Active Directory section of the Microsoft Edge Enterprise documentation for the full steps.
--	---

Create a Group Policy Object

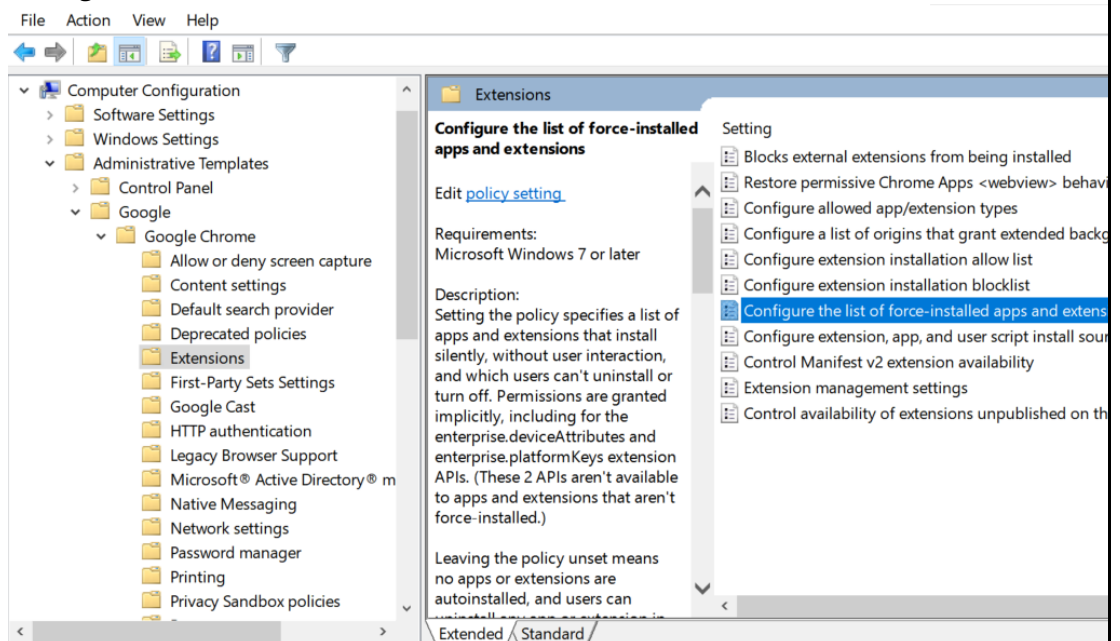
Perform the steps below to create a Group Policy Object (GPO) to use to distribute the USO Chrome Extension file:

1. Start the Group Policy Management console. To do this, in your domain server, open **Server Manager**, then click **Tools** and select **Group Policy Management**.
2. In the Group Policy Management console tree, right-click your domain, and then click **Create a GPO in this domain, and Link it here**.
3. Type a name for this new GPO (e.g. "Browser Policies"), and click **OK**.

Customize Browser Policies

Google Chrome	There are two Google Chrome policies that are required to be configured for USO Chrome Extension deployment: • ExtensionInstallForcelist (Configure the list of force-installed apps and extensions) - Set this policy to enable USO Chrome Extension to be installed silently, without user interaction. Users cannot uninstall or turn off this setting. • ExtensionInstallSources (Configure extension, app, and user script install sources) - Set this policy to define the URL that can install the USO Chrome Extension. Perform the following steps to customize these policies: 1. In the Group Policy Management console tree, right-click Browser Policies and click Edit. 2. In the Group Policy Management Editor, go to Computer Configuration > Administrative Templates > Google > Google Chrome > Extensions.
---------------	--

3. Locate and double-click the **Configure the list of force-installed apps and extensions** setting.



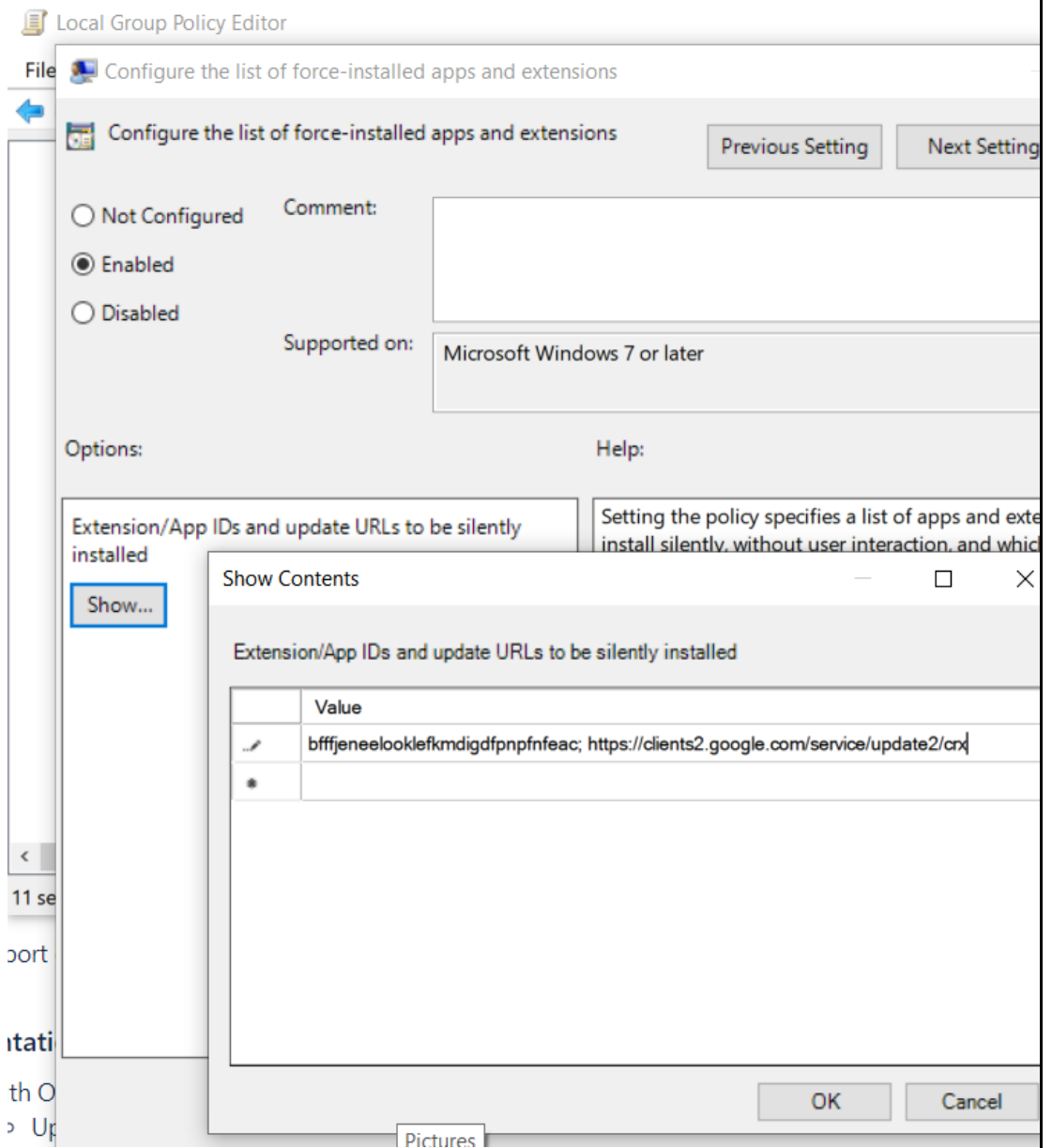
4. Select the **Enabled** radio button and then click the **Show...** button. The **Show Contents** dialog is displayed.
5. Add the following entry: "<USOChromeExtensionID>;
<https://clients2.google.com/service/update2/crx>"



Note: The USOChromeExtensionID value will depend on which USO Chrome Extension Manifest version you are using:

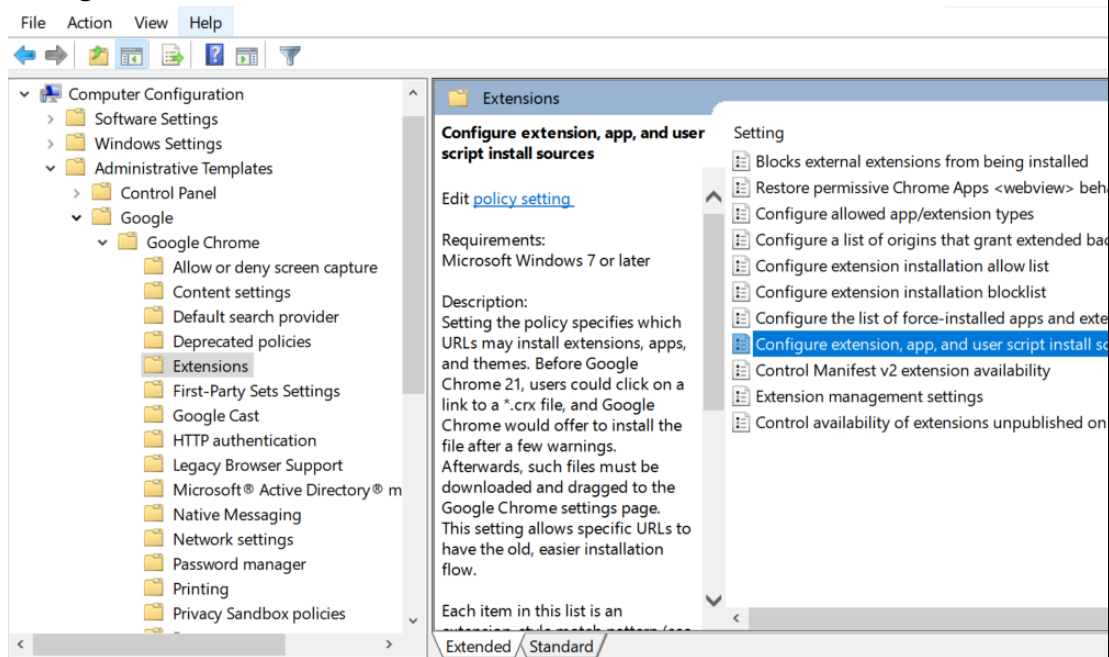
- **USO Chrome Extension (Manifest V3)** -
"bfffjeneelooklefkmdigdfnpfnfeac"
- **USO Chrome Extension** - "cfaiemjbjcbagnibmlflmmfccfdmnbek"

For example: "bfffjeneelooklefkmdigdfnpfnfeac;
https://clients2.google.com/service/update2/crx"



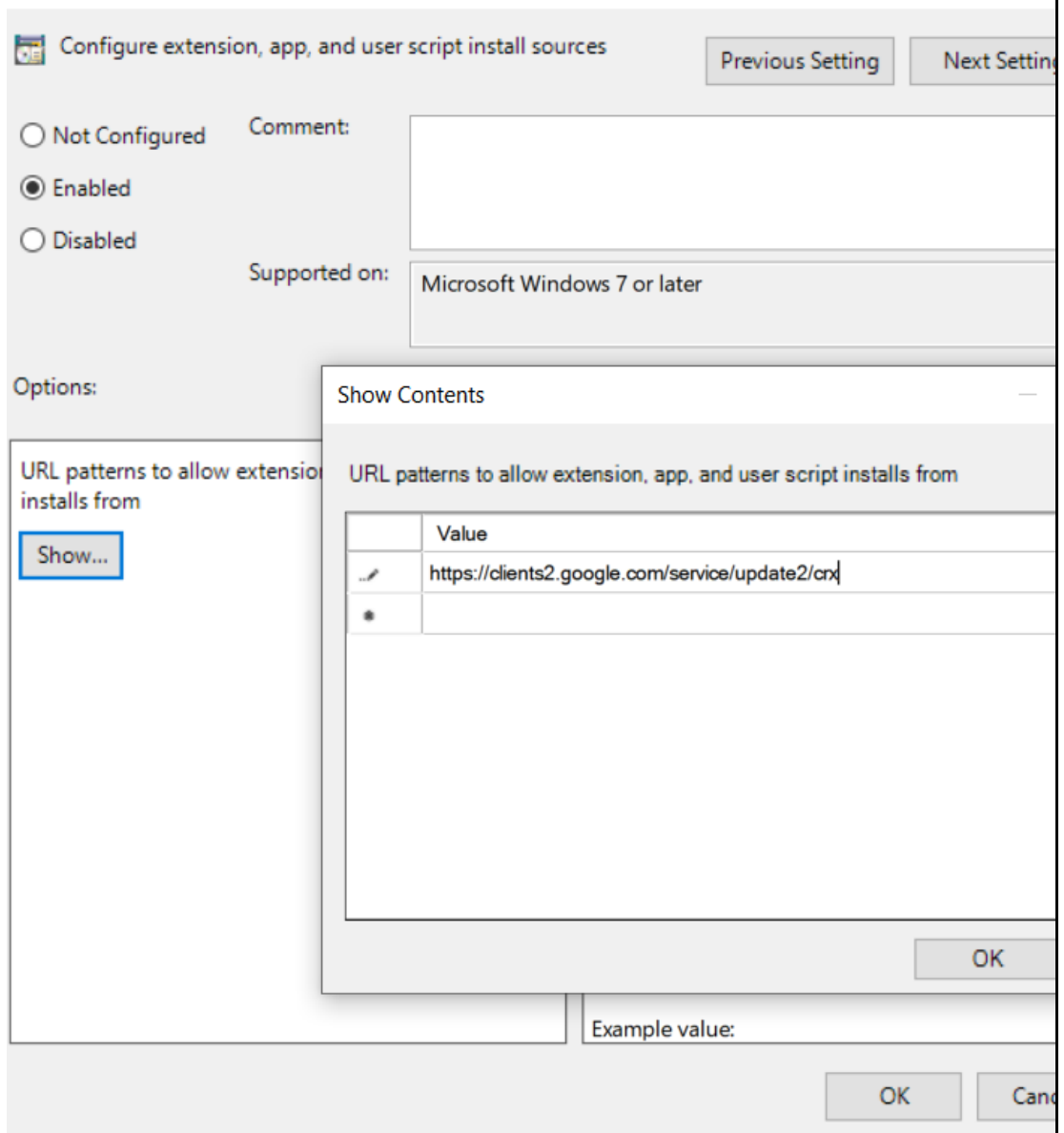
- Click **OK** on the **Show Contents** dialog box. Then, click **Apply** and **OK**.
The **Configure the list of force-installed apps and extensions** setting is now enabled.

7. Locate and double-click **Configure extension, app, and user script install sources** setting.



8. Select **Enabled**, then click **Show** and add the following entry:
"https://clients2.google.com/service/update2/crx"

 Configure extension, app, and user script install sources



Configure extension, app, and user script install sources

Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on: Microsoft Windows 7 or later

Options:

URL patterns to allow extension, app, and user script installs from

Show...

Show Contents

	Value
	https://clients2.google.com/service/update2/crx
	

OK

Example value:

OK Cancel

9. Click **OK** on the **Show Contents** dialog box. Then, click **Apply** and **OK**.
The **Configure extension, app, and user script install sources** setting is now enabled.
10. Close the **Group Policy Management Editor**, and then close the Group Policy Management console.
11. When the client computer starts, the USO Chrome Extension is automatically installed.
12. On the client computer, open Google Chrome and type *chrome://extensions/* to see the installed extension. Type *chrome://policy* to see all policies that are applied.

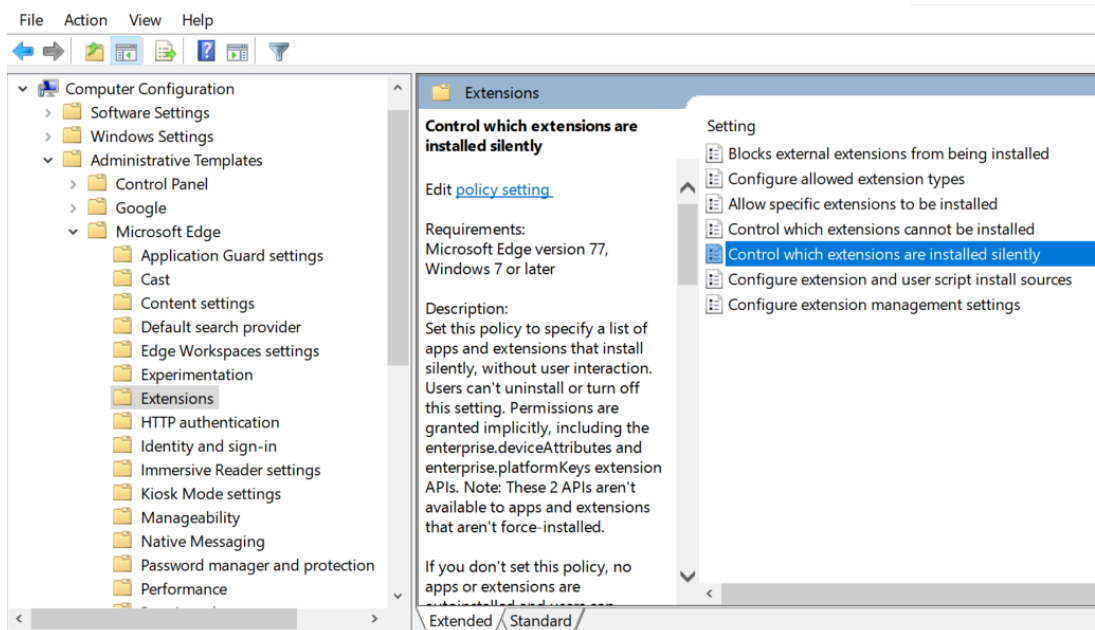
Micros
oft
Edge

There are two Microsoft Edge policies that are required to be configured for USO Chrome Extension deployment:

- **ExtensionInstallForcelist** (Control which extensions are installed silently) - Set this policy to enable USO Chrome Extension to be installed silently, without user interaction. Users cannot uninstall or turn off this setting.
- **ExtensionInstallSources** (Configure extension and user script install sources) - Set this policy to define the URL that can install the USO Chrome Extension.

Perform the following steps to customize these policies:

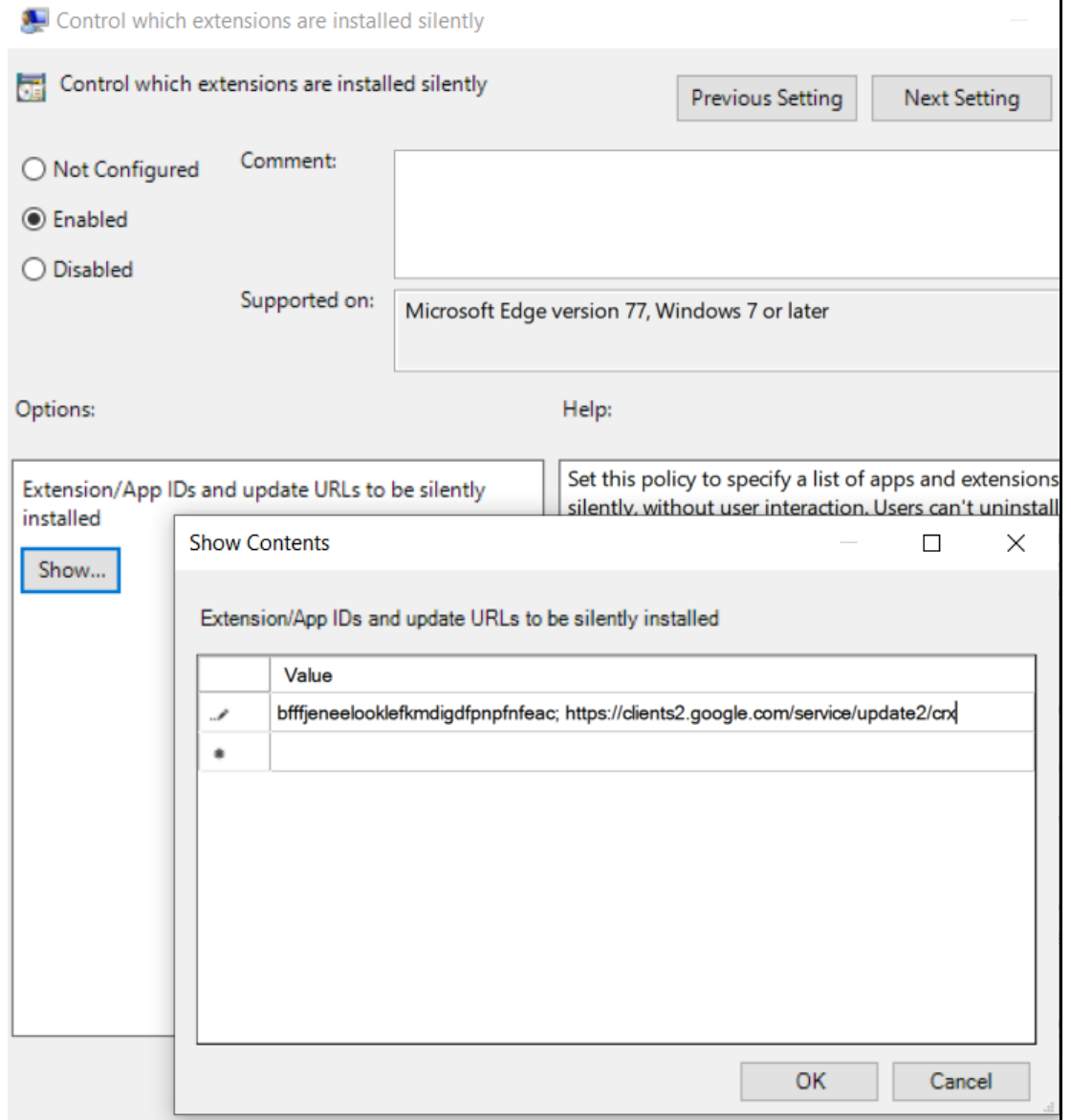
1. In the Group Policy Management console tree, right-click **Browser Policies** and click **Edit**.
2. In the **Group Policy Management Editor**, go to **Computer Configuration > Policies > Administrative Templates > Microsoft Edge > Extensions**.
3. Locate and double-click the **Control which extensions are installed silently** setting.



4. Select the **Enabled** radio button and then click the **Show...** button. The **Show Contents** dialog is displayed.
5. Add the following entry: "<USOChromeExtensionID>; https://clients2.google.com/service/update2/crx"

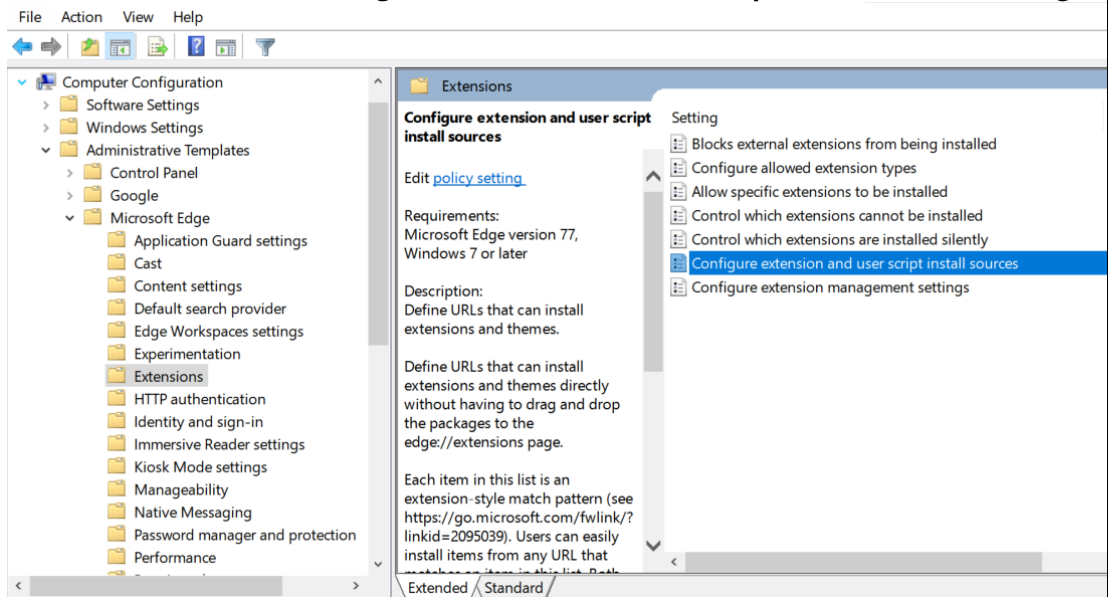
- Note:** The USOChromeExtensionID value will depend on which USO Chrome Extension Manifest version you are using:
- **USO Chrome Extension (Manifest V3)** - "bfffjeneelooklefkmdigdfpnpfnfeac"
 - **USO Chrome Extension** - "cfaiemjbjcbagnibmlflmmfccfdmnbek"

For example: "bfffjeneelooklefkmdigdfpnpfnfeac;
https://clients2.google.com/service/update2/crx"

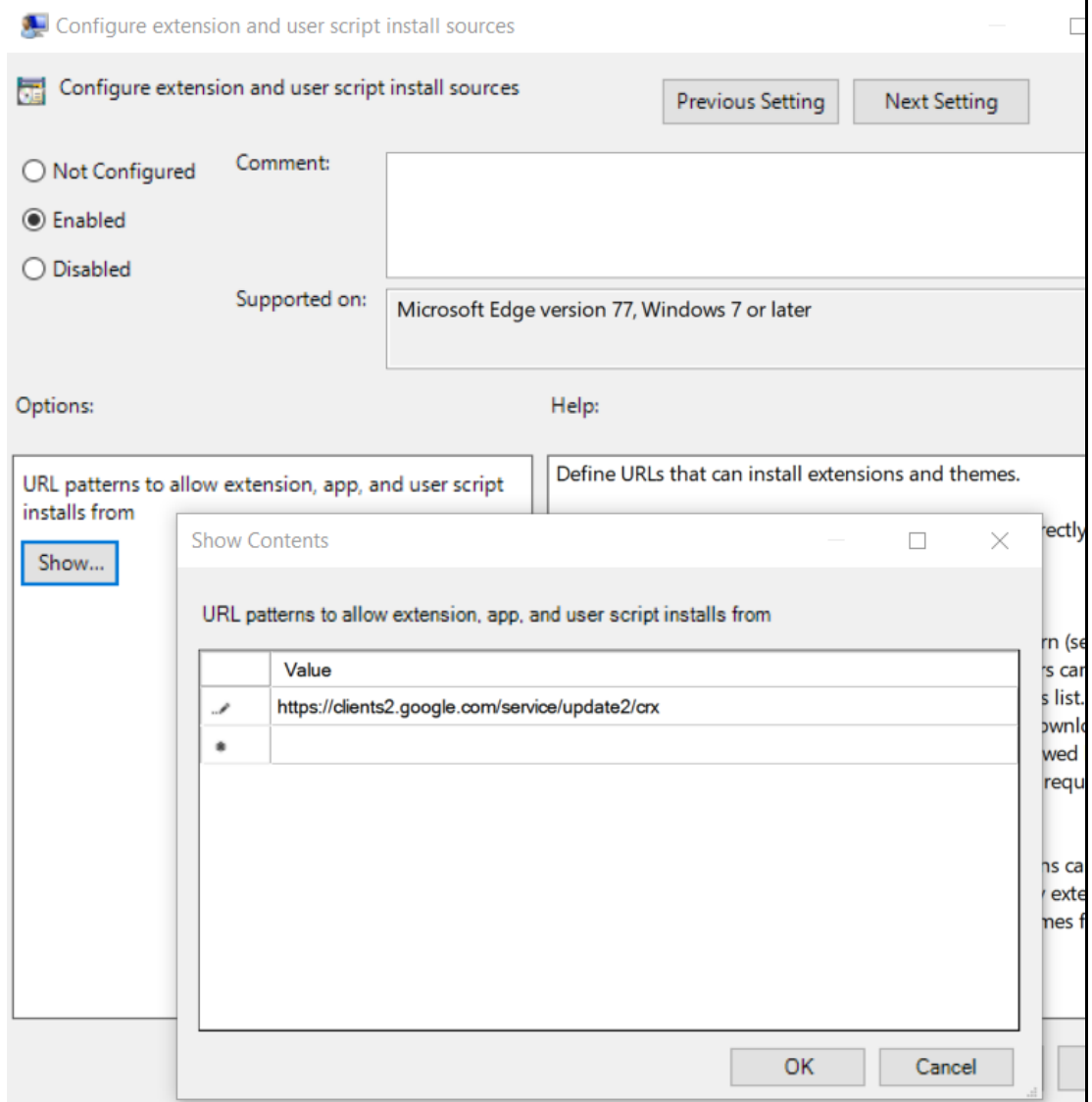


- Click **OK** on the **Show Contents** dialog box. Then, click **Apply** and **OK**. The **Control which extensions are installed silently** setting is now enabled.

7. Locate and double-click **Configure extension and user script install sources** setting.



8. Select **Enabled**, then click **Show** and add the following entry:
"https://clients2.google.com/service/update2/crx"



	- Click OK on the Show Contents dialog box. Then, click Apply and OK. The Configure extension and user script install sources setting is now enabled. - Close the Group Policy Management Editor, and then close the Group Policy Management console. - When the client computer starts, the USO Chrome Extension is automatically installed. - On the client computer, open Microsoft Edge and type <code>edge://extensions</code> to see the installed extension. Type <code>edge://policy</code> to see all policies that are applied.
--	--

Upgrade the USO Chrome Extension

The installed USO Chrome Extension will automatically be upgraded when a new version is available on the Chrome Web Store.

Remove Browser Policies

To remove the customized Google Chrome or Microsoft Edge policies:

Google Chrome	- In the Group Policy Management console tree, right-click Browser Policies and click Edit. - In the Group Policy Management Editor, go to Computer Configuration > Administrative Templates > Google > Google Chrome > Extensions and disable the following policies: Configure the list of force-installed apps and extensions Configure extension, app, and user script install sources
Microsoft Edge	- In the Group Policy Management console tree, right-click Browser Policies and click Edit. - In the Group Policy Management Editor, go to Computer Configuration > Policies > Administrative Templates > Microsoft Edge > Extensions and disable the following policies: Control which extensions are installed silently Configure extension and user script install sources

Set SSO Portal Configuration

	Important Note: - From AccessMatrix 5.7.5-GA and onwards, you no longer need to perform the following configurations in constant.js to enable USO Client for the SSO Portal. If the USO Chrome Extension has been installed, it will be detected and USO Client will automatically be enabled for the SSO Portal. - For AccessMatrix 5.7.4-GA and earlier versions, you are required to proceed with the steps outlined below.
---	--

To enable USO Client for SSO Portal using Chromium browsers (i.e. Google Chrome and Microsoft Edge), ensure that the following configuration is done and the USO Chrome Extension is installed.

1. Navigate to `<USOSERVER_HOME>/scripts/` and open **constants.js**.
2. Look for the `ENABLE_USO_CLIENT` constant and ensure that the value is set to `true` as shown below:

```
JavaScript

angular
  ...
  .constant('ENABLE_USO_CLIENT', true)
  ...
```

3. Save the changes made.
4. Refresh the SSO Portal page for changes to take effect, if necessary.

Enable PSE Login

USO supports offline mode login (a.k.a. PSE login) when user has no access to the network. While in offline mode, the user can perform an SSO portal login locally. A non-password-protected local PSE file is required for this login process.

To create this PSE file:

1. Navigate to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Client Configuration**.
2. Click **Edit** and configure the following attributes accordingly:
 - **Enable PSE** should be set to "Yes"
 - **PTD Type** should be set to "1 (Auto PTD)"
 - **PSE File Option** should be set to "0" or "1"
3. Save the changes.
4. Restart the USO server for changes to take effect.

After the attributes are configured and upon successful SSO portal login, if user's PSE file has not been created before, the system will create the local PSE file accordingly.



Notes:

-
- | | |
|--|---|
| | <ul style="list-style-type: none">• This setting will allow any user who has their own PSE file on the machine to do PSE login.• From USO Client Configuration, set Auto Switch To PSE When Server Unavailable to "Yes" to automatically login to PSE when USO server is down. And set Auto Switch To Server When Available to "Yes" to automatically login to server when USO server becomes available. |
|--|---|

9.3.2. SSO Portal with Web Apps Only

This is the configuration type for SSO Portal that supports only web apps. This configuration type is used in Windows, Linux or macOS environments where the USO applications used are all web applications, and where Chromium browsers (Google Chrome and Microsoft Edge) are used. Installing the USO Client is not needed; only the USO Chrome Extension for Web Apps must be installed. Refer to [AccessMatrix SSO Portal User Guide - Install USO Chrome Extension for Web Apps](#) for the installation steps.

Once done, you may proceed with the following sections, which detail configurations that may be necessary in order to enable the SSO Portal to work as intended:

- **Set SSO Portal Configuration**

For AccessMatrix versions 5.7.4-GA and earlier, the SSO Portal must be configured to disable the use of the USO Client. Click [here](#) for the configuration details.

Set SSO Portal Configuration

 **Important Notes:**

- From AccessMatrix 5.7.5-GA and onwards, you no longer need to perform the following configurations in **constant.js** to disable USO Client for the SSO Portal. If the USO Chrome Extension for Web Apps has been installed, it will be detected and USO Client will automatically be disabled for the SSO Portal.
- For AccessMatrix 5.7.4-GA and earlier versions, you are required to proceed with the steps outlined below.

To avoid initiating the USO Client in SSO Portal, the following configuration must be done to disable the USO Client:

1. Navigate to `<USOSERVER_HOME>/scripts/` and open **constants.js**.
2. Look for the `ENABLE_USO_CLIENT` constant and ensure that the value is set to `false` as shown below:

JavaScript

```
angular
...
.constant('ENABLE_USO_CLIENT', false)
...
```

3. Save the changes made.
 4. Refresh the SSO Portal page for changes to take effect, if necessary.
-

9.4. SSO Portal for Federated Applications

SSO Portal and CLP work together to support Federated Applications. SSO Portal has the ability to list all the federated applications (SAML Service Providers) that the users are entitled to. CLP has the role to handle the user authentication and also the SAML protocol exchange (receiving SAML Request and generating SAML Response). Thus, refer to [AccessMatrix Common Login Page Deployment Guide - CLP for SAML Identity Provider](#) on how to set up CLP as SAML Identity Provider. The sections below discuss the behavior or customization of listing the federated applications (SAML Service Providers).

SAML SP Apps in SSO Portal

SSO Portal supports the display of SAML Service Provider (SP) applications configured in Admin Console. (Refer to [UAM SAML Implementation Guide](#) for the SP app creation steps.)

The display of SAML SP apps in SSO Portal will depend on the access control configuration of the SP:

- If the **Allow all users to access** attribute is set to "true", then all the configured SAML SP applications will be displayed in the SSO Portal even if it's not assigned to the user.
- If the **Allow all users to access** attribute is set to "false", then only the SAML SP applications assigned to the logged in user will be displayed in the SSO Portal. The SAML SP application can be assigned under **SAML Application Privilege** tab in the **User** panel. Refer to [Common Administration Guide - Manage Users](#) section for the tab detail.

Customize SAML SP Application Icon

The tile icon of each SAML SP application can be changed and customized into a desired image.

Following criteria should be met to customize the application icon:

- The image must be a JPG file.
- The image file name and the application ID must be the same. For example, if SAML SP application ID is "Gmail", then image filename should be **Gmail.jpg**
- The dimension of the image must be the same as the size defined in **app.css** file. Default: 300 x 150 pixels.

Place the image file to this location:

- **Tomcat**

`<USOSERVER_HOME>/assets/images/appicon/`

(e.g. `/opt/acmatrix/tomcat/webapps/usoserver/assets/images/appicon/`)



Notes:

- If AccessMatrix server is yet to start, start AccessMatrix and login to SSO Portal page to see the new application icon.
- If AccessMatrix server has been started and has a user already logged in to SSO Portal page, clear the browser cache and re-login to SSO Portal page. The new application icon will be displayed.

9.5. USO Self-Service Portal

About Self-Service Portal

The availability of the self-service functions is governed by USO Self-Service Portal Policy. The Self-Service Portal, a.k.a Self Facility (SSF) is used to perform the functions described below:

- **Retrieve USO Application Password**

This function allows USO users to retrieve and view their entitled USO and non-USO application passwords.

- **Enroll Application**

This function allows USO users to enroll themselves for the USO applications, entitling themselves to perform Single Sign On (SSO) to the enrolled applications.

- **FIDO Registration**

This function allows USO users to register a FIDO2 authentication method. Upon successful registration, users will be able to use this authentication method to log into the AccessMatrix SSO Portal.



Note: The Self-Service Portal timeout is based on the **Session Idle Time (in seconds)** and **Session Lifetime (in seconds)** values configured in Admin Console (navigate to **Administration Dashboard > Configuration > Global Setting > Session**).

Configure USO Self-Service Portal Policy

Before the self-service functions can be used, the administrator is required to do the following configurations:

- Enable the functions in the USO Self-Service Portal Policy.
- Configure the Self-Service Portal attributes in the USO Application.
- Assign **SelfServiceRole** admin role to the user.
- (Optional) Enable the FIDO Registration function in the USO Self-Service Portal Policy.

Enable Self-Service Portal Functions

1. In the Admin Console, navigate to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Self-Service Portal** on the left pane. Then, perform the steps below:
-

-
- Set the **Enable "USO Application Enrollment" function** attribute to "true", if disabled. Default value is true.
 - Set the **Enable "Retrieve USO Application Password" function** attribute to "true", if disabled. Default value is true.
 - Assign a login module to the **Reauthentication Login Module for USO Application Password Retrieval** attribute. Refer to [Common Administration Guide - Configure Login Module](#) for the login module creation steps.

2. Save the changes.

i **Note:** Refer to [USO Admin Console Administration - Configure USO Self-Service Portal](#) for the attribute details.

Configure Self-Service Portal attributes in USO Application

i **Note:** Before performing the steps below, ensure that the USO Applications are already imported.

1. In the Admin Console, navigate to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Application** on the left pane.
2. Select a USO application, e.g. Desktop Demo.
3. Under the **Attributes** tab of the application, set the following Self-Service Portal attributes to "true", if disabled:
 - **Allow user to view password via SSF portal**
 - **Allow user to enroll application via SSF portal**

i **Note:** By default, these attributes are set to "true". If these attributes are disabled, then the USO application will not be displayed on the Self-Service Portal page. Refer to [USO Admin Console Administration - Configure USO Application](#) section for the attribute details.

4. Save the changes.

Assign Admin Role

Performing the steps below will allow the user to do self-service functions.

1. In the Admin Console, navigate to **Operation Dashboard > User & Segment > User** and edit an existing USO user (e.g. demouser).
 2. Select **Admin Privilege** tab.
-

-
3. Then, click **Add** and select "SelfServiceRole" as the admin role.
 4. Save the changes.

i **Note:** Refer to [USO Admin Console Administration - Enable Self-Service Functions for USO User](#) for additional details.

Enable FIDO Registration (Optional)

! **Important Note:** The FIDO registration feature is enabled by default and can be used out-of-the-box with no additional configurations.

If you wish to use a self-created FIDO2 login module for FIDO2 credential registration, or learn more about how FIDO2 credential registration is enabled, you can proceed to view all of the steps below and the linked reference.

Performing the steps below will enable the "FIDO Registration" self-service function.

1. In the Admin Console, navigate to **Administration Dashboard > Configuration > Unified SSO** and click **Enterprised SSO > USO Self-Service Portal** on the left pane. Then, perform the steps below:
2. Ensure the **Enable "FIDO Registration" function** attribute is set to "true". Default value is true.
3. Assign a FIDO2 login module to the **FIDO Registration Login Module Id** attribute. By default, a pre-created FIDO2 token login module ("DefaultFIDO2TokenLogin") is assigned to this attribute. Refer to [Common Login Page Deployment Guide - FIDO2 Authentication](#) for further details on creating the FIDO2 login module and enabling FIDO2 registration.
4. Save the changes.

i **Note:** Refer to [USO Admin Console Administration - Configure USO Self-Service Portal](#) for the attribute details.

9.6. SSO Portal and Self-Service Portal Customizations

SSO Portal Customization Overview

SSO Portal and Self-Service Portal support UI customization which allow users to set their preferences on what they want to see or how they want information to be organized or displayed. The SSO Portal and Self-Service Portal allow the customization of the following components:

- [Application Icon](#)
- [Application Menu List](#)
- [Text and labels](#)
- [Banner Logo and Image](#)

Customize Application Icon

The tile icon of each USO application can be changed and customized into a desired image. Following criteria should be met to customize the application icon:

- The image must be a JPG file.
- The image file name and the application ID must be the same. For example, if USO application ID is "Gmail", then image filename should be **Gmail.jpg**)
- The dimension of the image must be the same as the size defined in **app.css** file. Default: 300 x 150 pixels.

To change the CSS size:

1. Navigate to `<USOSERVER_HOME>/tomcat/webapps/usoserver/styles/`.
2. Open the **app.css** file and edit the `.iconcell` properties, as shown below:

CSS
<pre>.iconcell { background-repeat: no-repeat; height: 150px; width: 300px; margin-bottom:0; }</pre>

3. Save the changes.
-

To change/ add an image, place the image file in this location:

- **Tomcat**

`<USOSERVER_HOME>/assets/images/appicon/`

(e.g. `/opt/acmatrix/tomcat/webapps/usoserver/assets/images/appicon/`)



Notes:

- If AccessMatrix server has not yet been started, start AccessMatrix and login to SSO Portal page to see the new application icon.
- If AccessMatrix server has been started and has a user already logged in to SSO Portal page, clear the browser cache and re-login to SSO Portal page. The new application icon will be displayed.

Customize Application Menu List

Each application on the SSO Portal page has an application menu list (as detailed in the [Application Menu](#) section).

The menu items displayed in this list can be hidden if needed. To do so, perform the following:

1. Navigate to `<USOSERVER_HOME>/scripts/` and open **app.js**.
2. Locate the following lines in the file:

JavaScript

```
function k() {ge.appMenuList=[],
ge.appMenuList.push({item:m["USO.DisableAppCurrentSession"],
getClass:ee,action:H,showCheckIcon:G}),
ge.appMenuList.push({item:m["USO.DisableAppPermanently"],
getClass:te,action:J,showCheckIcon:z}),
ge.appMenuList.push({item:m["USO.SignOffApp"],
getClass:ne,action:q}),
ge.appMenuList.push({item:m["USO.ContinueMatching"],
getClass:X,action:K}),
ge.appMenuList.push({getClass:D}),
ge.appMenuList.push({item:m["USO.ResetAppInfo"],getClass:Y,action:N
}),
ge.appMenuList.push({item:m["USO.AddProfile"],getClass:oe,action:Q}
),
ge.appMenuList.push({item:m["USO.ResetProfile"],getClass:re,action:
ae}),
ge.appMenuList.push({item:m["USO.ChangeAppPath"],getClass:ie,action
:F}),
ge.appMenuList.push({getClass:D}),
ge.appMenuList.push({item:m["USO.ShowStateHistory"],getClass:W,acti
on:x}),
ge.appMenuList.push({item:m["USO.ClearStateHistory"],getClass:W,act
```

```
ion:B}),  
ge.appMenuList.push({getClass:D}),  
ge.appMenuList.push({item:m["USO.Properties"],getClass:W,action:V})  
,
```

3. Delete the value(s) in the list corresponding to the menu item(s) you wish to remove.
For example, delete `ge.appMenuList.push({item:m["USO.ResetAppInfo"],getClass:Y,action:N})` to remove the **Disable Application From Current Session** menu item.



Note: For details on each menu item, refer to [SSO Portal User Interface - Application Menu](#).

4. Save the changes.
5. Restart the USO server for the changes to take effect.

Customize Portal Text and Labels

Customize Labels and Descriptions

The description of the text/labels in the SSO Portal and Self-Service Portal can be customized. For example, the administrator can display contact information for user support on the error page.

SSO Portal

1. Navigate to `<USOSERVER_HOME>/assets/i18n/`.
2. Open and edit **message_en_US.js** file for English labels, **message_ja_JP.js** file for Japanese labels, and **message_zh_CN.js** for Chinese labels.
3. Modify the description of an existing label, e.g. "USO.IdlePrompt".
4. Save the changes.
5. Restart the USO server for the changes to take effect.

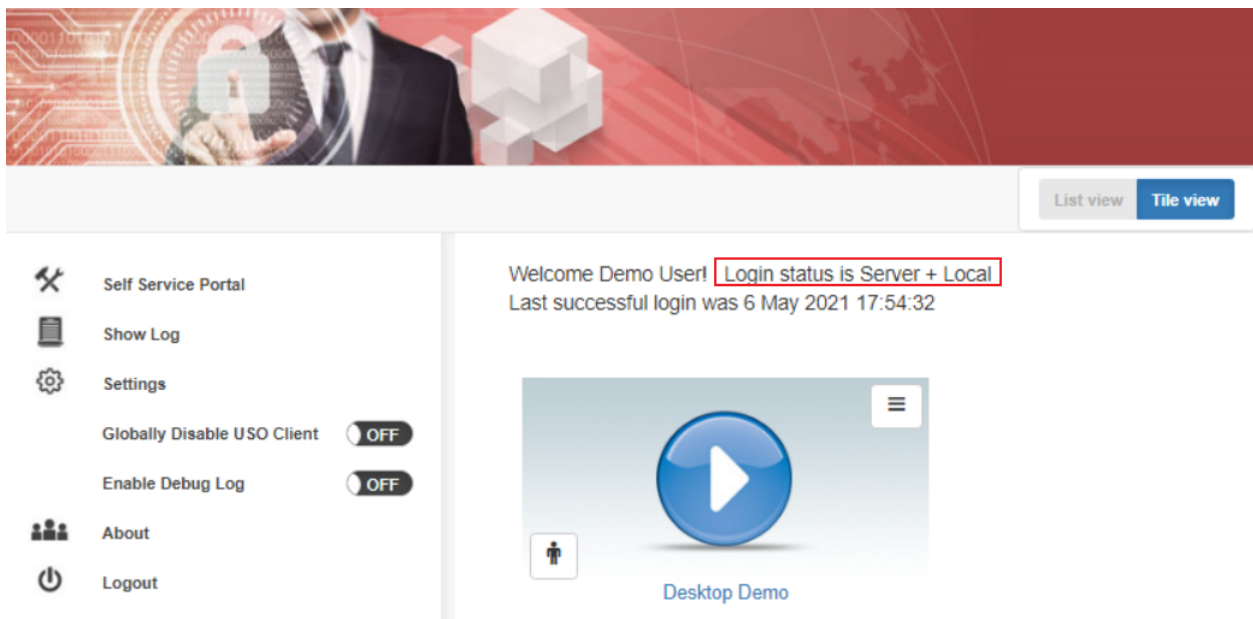
Self-Service Portal

1. Navigate to `<USOSSF_HOME>/assets/i18n/`.
 2. Open and edit **message_en_US.json** file for English labels, **message_ja_JP.json** file for Japanese labels, and **message_zh_CN.json** for Chinese labels.
 3. Modify the description of an existing label, e.g. "Lbl_PasswordDoesNotMeetRequirement".
 4. Save the changes.
-

-
- Restart the USO SSF server for the changes to take effect.

Remove Login Status

Upon logging in to the SSO portal, the user's login status is displayed by default, as shown in the sample image below:



This message can be removed. To do so, perform the following:

- Navigate to `<USO SERVER_HOME>/scripts/` and open **app.js**.
- Locate and comment out/delete the following line:

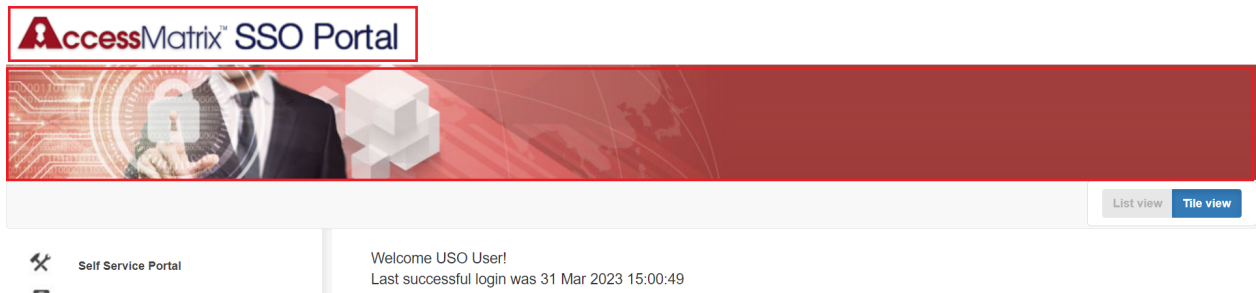
JavaScript

```
{{ 'Brw. LoginStatus' | translate }} {{ vm. usoStatus }}
}}
```

- Save the changes.
- Restart the USO server for the changes to take effect.

Customize Banner

By default, the following banner logo and image are displayed in the SSO Portal upon successfully logging in:



These images can be replaced or removed entirely. To do so, perform the following:

Customize Banner Logo

1. Navigate to `<USOSERVER_HOME>/scripts/` and open **app.js**.
2. Locate the following line:

JavaScript

```
<td colspan="2"></td>
```

- To delete the banner image entirely, comment out/delete this line.
 - To replace the image:
 - a. Save the new image in `<USOServer_HOME>/assets/images/`.
 - b. In **app.js**, change the value of `img src` to the path of the new image (e.g. `"assets/images/new_logo.png"`).
 - c. Update the values of `width` and `height` to the dimensions of the new image.
3. Save the changes.
 4. Restart the USO server for the changes to take effect.

Customize Banner Image

1. Navigate to `<USOSERVER_HOME>/scripts/` and open **app.js**.
2. Locate the following line:

JavaScript

```
<td></td>
```

- To delete the banner image entirely, comment out/delete this line.

-
- To replace the image:
 - a. Save the new image in `<USOServer_HOME>/assets/images/`.
 - b. In **app.js**, change the value of `img src` to the path of the new image (e.g. `"assets/images/new_banner.jpg"`).
 - c. Update the values of `width` and `height` to the dimensions of the new image.
 - 3. Save the changes.
 - 4. Restart the USO server for the changes to take effect.
-

10. AccessMatrix SSO Portal User Guide

This document contains instructions on installing Chrome extensions or USO Client and using SSO Portal and Self-Service Portal interface.

About AccessMatrix SSO Portal

AccessMatrix Single Sign-On (SSO) Portal is a web portal that provides the following SSO functions:

- SSO to USO Applications:
 - SSO to Windows native apps and web apps
 - SSO to web apps only
- SSO to federated applications

It also features the Self-Service Portal that allows users to retrieve the entitled USO application password and enable USO application enrollment.

The login authentication processes of SSO Portal and Self-Service Portal are integrated with Common Login Page (CLP).



Note: Refer to [AccessMatrix SSO Portal Implementation](#) for the concept and implementation steps before proceeding to the next sections of this document.

10.1. Installing USO Chrome Extension for SSO Portal

Before proceeding with the installation steps for the USO Chrome extensions, it is important to understand which SSO Portal configuration type you should use. The following section provides a brief introduction on the different configuration types and their use cases.

Understanding SSO Portal Configuration Types

Which USO Chrome extension you install depends on which AccessMatrix Single Sign-On (SSO) Portal configuration type you intend to use. In turn, the configuration type you choose depends on what Operating System (OS) and USO application types you intend to use. The SSO Portal supports two types of USO applications:

- **Windows native apps** - Applications that must be downloaded and installed on your local machine in a Windows environment.
- **Web apps** - Applications that can be accessed via desktop browsers (in this case, specifically Chromium browsers such as Microsoft Edge and Google Chrome).

Depending on the OS you are using, the supported USO application types will vary. The table below shows the supported USO application types for each OS:

Application Type \ OS	Windows	Linux	macOS
Native apps	Yes	No	No
Web apps	Yes	Yes	Yes

Based on your OS and USO application type requirements, you should choose **only one** of the following SSO Portal configuration types:

- **SSO Portal with Windows native apps and web apps**
(Windows only) This configuration type supports both Windows native apps and web apps. If you are using SSO Portal in a Windows environment, it is recommended that you proceed with this configuration, as it covers the widest range of features and supports both USO application types. This makes it applicable for most use cases.
You must [install both the USO Client and the USO Chrome Extension](#).
-



Note: There are currently 2 coexisting versions of the USO Chrome Extension:

- **USO Chrome Extension** - This version of the extension uses Chrome Manifest V2, which has been deprecated by Google. It should only be used by existing USO Chrome Extension users.
- **USO Chrome Extension (Chrome Manifest V3)** - This version of the extension uses Chrome Manifest V3, the latest version of the Chrome extension platform. New users should use this version of the extension.

In this guide, unless the Manifest version is specified, the term "USO Chrome Extension" is used to refer to whichever version of the extension you are using.

- **SSO Portal with web apps only**

(Windows, Linux, and macOS) This configuration type supports only web apps.

If you are using SSO Portal in a Linux or macOS environment, it is recommended that you proceed with this configuration.

You must [install the USO Chrome Extension for Web Apps](#) only.



Note: Prior to the AccessMatrix 5.7.5-GA release, "USO Chrome Extension for Web Apps" was known as "AccessMatrix Chrome Extension".

The following table provides a brief overview of each of the various Chrome extensions that are available and their differences:

Features \ Name of Extension	USO Chrome Extension	USO Chrome Extension (Chrome Manifest V3)	USO Chrome Extension for Web Apps
Chrome Manifest Version	Version 2	Version 3	Version 3
USO Trainer “Custom Behavior” (using JavaScript)	Yes		No
Windows			
SSO to web apps (Chrome & Edge browsers)	Yes		Yes
SSO to native apps	Yes		No
Required to install USO Client?	Yes		No
macOS			

Features \ Name of Extension	USO Chrome Extension	USO Chrome Extension (Chrome Manifest V3)	USO Chrome Extension for Web Apps
SSO to web apps (Chrome browser)	Use of this extension not supported for macOS		Yes
SSO to native apps	Use of this extension not supported for macOS		No
Required to install USO Client?	USO Client not available for macOS		USO Client not available for macOS
Linux (e.g. Ubuntu)			
SSO to web apps (Chrome browser)	Use of this extension not supported for Linux		Yes
SSO to native apps	Use of this extension not supported for Linux		No
Required to install USO Client?	USO Client not available for Linux		USO Client not available for Linux

Once you have studied the differences between both configuration types and are ready to proceed, refer to **only one** of the following sections based on your required use case:

- [SSO Portal with Windows Native Apps and Web Apps](#)
- [SSO Portal with Web Apps Only](#)

SSO Portal with Windows Native Apps and Web Apps

The USO Chrome Extension supports USO and Federated SSO applications with USO Client functionalities on Google Chrome and Microsoft Edge (Chromium-based) browsers. The user must first download and install the USO Client if there is no previous or existing version of the USO Client installed. Following that, they must install the USO Chrome Extension.

Download and Install USO Client

Follow the installation steps provided below to install the USO Client:

1. Launch the SSO Portal and login. Once authenticated successfully, a page indicating that USO Client has yet to be installed is displayed.

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2. Click on the **USO Client Installer** hyperlink and select **Save Target As** to save the installer to a local folder.
 3. After the installer is successfully downloaded and saved, the download complete window appears.
 4. Click on **Open Folder** and run the USO Client Installer from within the folder to install the USO Client.

! **Important Note:** User must not click on **Run** to run the installer to avoid improper installation!

5. If the security warning window is prompted, click **Run** to continue. USO Client setup wizard will be displayed. Click **Install** button to continue installation.
6. Upon successful installation, click on the **Finish** button to close the USO Client setup wizard.
7. As instructed on the SSO Portal web page, after USO Client installation is completed, click on the **here** hyperlink to continue with the login process.
8. Once user has logged in completely, USO Client user interface will be displayed.
Notice that the USO Client icon will also appear in the Windows task bar.

i **Note:** Detailed information on the installation of the USO Client can be found in [USO Client User Guide - Installing the USO Client](#).

Install USO Chrome Extension

Next, follow the steps provided below to install the USO Chrome Extension. There are two ways to install the USO Chrome Extension:

Installation using Chrome Web Store	1. Launch Google Chrome or Microsoft Edge (Chromium-based) browser. i Note: You cannot add extensions when you browse in Incognito mode or as a guest. 2. Open the Chrome Web Store. 3. Type "USO Chrome Extension (Manifest V3)" in the Search the store textbox and press Enter. a. Alternatively, search for "USO Chrome Extension" if it is necessary to use the Manifest V2 version of the extension. 4. Under the Extensions list, click USO Chrome Extension. The USO Chrome Extension details will be displayed. 5. Click Add to Chrome. 6. To approve, click Add extension. 7. To manage the extension:
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	- For Google Chrome, click the Customize and control Google Chrome icon from the upper-right corner of the screen, then click More tools > Extensions. - For Microsoft Edge, click the Settings and more icon from the upper-right corner of the screen, then click Extensions. 8. On the USO Chrome Extension, click Details and turn on Allow access to file URLs. (This is useful to support PSE.)
Installation using Windows Registry	- Find or create the following key in the registry: <i>HKEY_LOCAL_MACHINE\Software\Wow6432Node\Google\Chrome\Extensions</i> - Create a new registry key (folder) under the Extensions key with the same name as the ID of USO Chrome extension, i.e. "cfaiemjbjcbagnibmflmmfccfdmnbek". - Under this key, create a string registry value "update_url" with this data: "https://clients2.google.com/service/update2/crx" (this points to the USO Chrome Extension crx in the Chrome Web Store). - Launch the browser and go to <i>chrome://extensions</i> to make sure the extension is listed. - On the USO Chrome Extension, click Details and turn on Allow access to file URLs. (This is useful to support PSE.)
(For System Administrators) Centralized Installation	For a centralized distribution of the USO Chrome Extension in an enterprise setup, the system administrator may configure the Active Directory Group Policy so that USO Chrome Extension can be automatically installed on the target user's computer. System administrators seeking to perform a centralized installation of the USO Chrome Extension should refer to the steps outlined in AccessMatrix SSO Portal Implementation - SSO Portal with Windows Native Apps and Web Apps - USO Chrome Extension Centralized Deployment on Chromium Browsers.
 Important Note: - The latest USO Chrome Extension must be added. Remove the existing version if there is any. - Ensure that USO Chrome Extension for Web Apps is not installed.	

SSO Portal with Web Apps Only

 Note: Prior to the AccessMatrix 5.7.5-GA release, "USO Chrome Extension for Web Apps" was known as "AccessMatrix Chrome Extension".
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The USO Chrome Extension for Web Apps acts as the USO Client and it is only applicable when all the applications are web apps. It provides the USO Client functionalities on Google Chrome and Microsoft Edge (Chromium-based) browsers.

Install USO Chrome Extension for Web Apps

There are two ways to install the USO Chrome Extension for Web Apps:

Installation using Chrome Web Store	1. Launch Google Chrome or Microsoft Edge (Chromium-based) browser.  Note: You cannot add extensions when you browse in Incognito mode or as a guest. 2. Open the Chrome Web Store. 3. Type "USO Chrome Extension for Web Apps" in the Search the store textbox and press Enter. 4. Under the Extensions list, click USO Chrome Extension for Web Apps. The USO Chrome Extension for Web Apps details will be displayed. 5. Click Add to Chrome. 6. To approve, click Add extension.
Installation using Windows Registry	1. Find or create the following key in the registry: <i>HKEY_LOCAL_MACHINE\Software\Wow6432Node\Google\Chrome\Extensions</i> 2. Create a new registry key (folder) under the Extensions key with the same name as the ID of USO Chrome Extension for Web Apps, i.e. "khdnocomjejhghmhpmpdoanjdbalbl". 3. Under this key, create a string registry value "update_url" with this data: "https://clients2.google.com/service/update2/crx" (this points to the USO Chrome Extension crx in the Chrome Web Store). 4. Launch the browser and go to <i>chrome://extensions</i> to make sure the extension is listed.
	Important Note: • The latest USO Chrome Extension for Web Apps must be added. Remove the existing version if there is any. • Ensure that USO Chrome Extension is not installed.

10.2. SSO Portal User Interface

This section discusses how to access the AccessMatrix Single Sign-On Portal (SSO Portal) and provides descriptions on the various UI and functionalities available. The SSO Portal allows for users to perform Single Sign On (SSO) to their applications and manage the different user profiles that will be used to log in.

Access SSO Portal and Launch Application

i **Note:** Before accessing the SSO Portal, ensure that Common Login Page (CLP) has been configured correctly, as it is used to handle user authentication. For more details, refer to [AccessMatrix Common Login Page Deployment Guide - Set Up CLP for AccessMatrix Applications](#).

To access the SSO Portal and launch an application, perform the following steps:

1. Open a supported browser and enter the SSO Portal URL (e.g. <https://localhost:8443/usoserver/start.uso>). The CLP will be displayed for user authentication.
2. Enter the login credentials accordingly and click the **Login** button. The SSO Portal will be displayed.
3. Click on the corresponding application name. The SSO Portal will launch the application using the default profile.

i **Note:** If there are multiple profiles where a default profile has not been set, the user will be directed to select a profile before the application is launched.

Session Expiry

If the USO Client is not in use, and the SSO Portal is left idle after login, the user will be automatically logged out once the session expires.

SSO Portal User Interface

The following table provides a description of the menu items available in the user interface:

Menu	Description
Home	Refreshes the SSO Portal.
About	Displays information about the current version of the Admin Console, USO Chrome Extension and USO Client.
Show Log	Opens the usoclient.log in the default text editor, such as Notepad.
Settings	Displays the Settings menu. Allows the configuration of Globally Disable USO Client and Enable Debug Log .
Globally Disable USO Client	If enabled () icon), it deactivates the USO Client from performing SSO for all applications. Note: This configuration is accessed by clicking on Settings.
Enable Debug Log	If enabled () icon), it allows the USO Client to log debugging information to the local log file. Note: This configuration is accessed by clicking on Settings.
Self Service Portal	Launches the Self Service Portal using the current logged in account.

Application Management

The number of profiles and the current default profile will be displayed for each application. Each application allows for several configurations to be made:

- [State History](#)
- [Profile Management](#)
- [App Configuration](#)

State History

Mouse over the state history button (X icon) to access the settings below.

Setting	Description
Show State History	Shows the state history of the application.
Clear State History	Clears the state history of the application.
Properties	Shows the properties of the application such as application name, state, etc.

Profile Management

The **Profile Management** page allows the user manage their profiles in each application. It provides several functionalities:

- [Create New Profile](#)
- [Set Default Profile](#)
- [Rename Profile](#)
- [Update Profile Login Details](#)
- [Delete a Profile](#)

Create New Profile

To create a new profile, follow the instructions below:

1. Click on the **Profiles** button () for the corresponding application. The **Profile Management** page will be displayed.
2. Click on the Add button. A new default profile will be created.



Note: To further configure the profile, you may need to perform further actions. For more details on the configurations that can be made to a profile, refer to [Profile Management](#).

Set Default Profile

To set a default profile, follow the instructions below:

1. Click on the **Profiles** button () for the corresponding application. The **Profile Management** page will be displayed.
 2. Click on the ... button in the row corresponding to the relevant profile. A list of actions available for the profile will be displayed.
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3. Click on the **Set as Default** button ( icon). Upon completion, there will be a **Default** icon beneath the profile name.

Rename Profile

To rename a profile, follow the instructions below:

1. Click on the **Profiles** button ( icon) for the corresponding application. The **Profile Management** page will be displayed.
2. Click on the ... button in the row corresponding to the relevant profile. A list of actions available for the profile will be displayed.
3. Click on the **Rename** button ( icon).
4. Specify a new profile name in the text box provided and click the **Confirm** button ( icon).

Update Profile Login Details

This allows for the user to make changes to the User ID and the password.

To update the profile, follow the instructions below:

1. Click on the **Profiles** button ( icon) for the corresponding application. The **Profile Management** page will be displayed.
2. Click on the ... button in the row corresponding to the relevant profile. A list of actions available for the profile will be displayed.
3. Click on the **Update Login Details** button ( icon). The **Update Profile Login Details** page will be displayed.
4. Specify the relevant fields as necessary and click the **Save** button.

Deleting a Profile

To delete a profile, follow the instructions below:

1. Click on the **Profiles** button ( icon) for the corresponding application. The **Profile Management** page will be displayed.
 2. Click on the ... button in the row corresponding to the relevant profile. A list of actions available for the profile will be displayed.
 3. Click on the **Delete** button ( icon). A confirmation dialog will be displayed.
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- Click on **Yes, Delete Profile** to complete the deletion process.

App Configuration

The settings for each application can be accessed by clicking on the **App Configuration** button ( icon).

Settings	Description
Disable from Current Session	If enabled ( icon), this prevents the user from launching the application for the current session. The application's icon will be grayed out.
Disable Permanently	If enabled ( icon), this prevents the user from launching the application. The application's icon will be grayed out. The application cannot be launched in subsequent sessions.
Signoff Application	Signs the user out of the application, and closes the application.
Continue Matching	Allows the USO Client to continue matching for applications trained with system-defined action <code>_StopMatching()</code> . <code>_StopMatching()</code> is a screen action script that stops the matching of applications it processed. Note: Refer to USO Client User Guide - Continue Matching Application for more information.
Change App Path	Allows users to change the application path and application argument.

10.3. Self Service Portal

The Self Service Portal allows for users to perform several self-service functions:

- [Retrieve USO Application Password](#)
- [Enroll USO Application](#)
- [FIDO2 Credential Management](#)

Accessing the Self Service Portal

The Self Service Portal can be accessed in two different ways.

i **Note:** Depending on how the Self Service Portal has been accessed, the logging out process may differ slightly. Refer to the respective instructions under each method.

Through SSO Portal

To access the Self Service Portal using the SSO Portal, follow the steps below:

1. Access the SSO Portal.

i **Note:** For more details on how to access the SSO Portal, refer to [SSO Portal User Interface](#).

2. Click on **Self Service Portal**. The Self Service Portal will be displayed.

To log out of the Self Service Portal when it has been accessed through the SSO Portal, follow the steps below:

1. Click on the **Back to USO** button. The SSO Portal will be displayed.
2. Click on your profile image ( icon if you have not uploaded one).
3. Click **Logout**.

Through Direct URL

To access the Self Service Portal using a direct URL, follow the steps below:

1. Open a supported browser (e.g. Chrome) and enter the Self Service Portal URL
(e.g. <https://localhost:8443/usossf/start.ssf>)

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2. Select an authentication realm using the drop-down list and click the **Next** button.
 3. Enter the login credentials accordingly and click the **Login** button. The Self Service Portal will be displayed upon successful login.

To log out of the Self Service Portal when it has been accessed through a direct URL, follow the steps below:

1. Click on your profile image ( icon if you have not uploaded one).
2. Click **Logout**.

Retrieve USO Application Password

i **Note:** Ensure that the steps described in the [AccessMatrix SSO Portal Implementation- Configure USO Self-Service Portal Policy](#) section are completed.

The USO Self-Service Portal allows for users to retrieve the password for their enrolled applications. However, they must first be reauthenticated for security purposes. For more details on the reauthentication method and an example of the configurations necessary, refer to [Reauthentication Login Module](#).

To retrieve the USO application password, perform the following steps:

1. Log in and access the Self-Service Portal, as described in [Accessing the Self-Service Portal](#).
2. Click on the **Retrieve Password** tab to display the **Retrieve Password** page. This page shows the list of USO applications the user is entitled to perform SSO with.
3. Tick the checkbox in the row corresponding to your USO application to select it.
4. Click on the **Retrieve Password** button. The user will be prompted to reauthenticate. Enter accordingly and click **Login**.

i **Note:** Depending on the login module used, the prompts for the reauthentication may differ slightly. For example, if FIDO2 authentication has been configured, a **Login With Passkey** will appear instead. In this case, the user will be prompted to use their passkey to complete the reauthentication process.

5. Upon successful reauthentication, a list of USO applications will be displayed. Click on the hide button ( icon) in the row corresponding to your USO application to display the password in
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cleartext. Upon timeout, the application password will be masked and the user will be unable to view the cleartext.

i **Note:** The timeout duration can be configured in the Admin Console using the attribute **Password display time (in seconds)** of the USO Self-Service Portal Policy. For more details, view [Configure USO Self-Service Portal](#).

Reauthentication Login Module

The reauthentication login module specifies the method used for reauthenticating users in Self Service Portal. This login module can be configured using the Admin Console:

i **Note:** For more details, refer to [Configure USO Self-Service Portal](#).

1. Navigate to the **Administration Dashboard > Configuration > Unified SSO > Enterprised SSO > USO Self-Service Portal** and click the **OK** button.
2. Click the **Edit** button.
3. Select the login module that will be used for reauthentication using the attribute ***Reauthentication Login Module for USO Application Password Retrieval**.
4. Click the **Save** button to save the changes.

Depending on the login module used for the reauthentication process, additional configurations may be required for both the login module and the user. By default, "SSFOTPLogin" is used as the login module, which uses SMS/Email OTPs to authenticate the user. As an example, this login module requires configurations such as the setting up of a delivery channel, and for users to have the required attributes (e.g. mobile phone number or email address).

i **Note:** For more details on how to configure the "SSFOTPLogin" login module, refer to [UAS SMS OTP Implementation Guide - Implementation Checklist](#).

Enroll USO Application

i **Note:** Ensure that the steps described in the [AccessMatrix SSO Portal Implementation - Configure USO Self-Service Portal Policy](#) section are completed.

Perform the following steps to enroll or unenroll the selected USO application:

1. Log in and access the Self-Service Portal, as described in [Accessing the Self-Service Portal](#).
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2. Click on the **Enroll Apps** tab to display the **Enroll Application** page.
 3. Click on the **User** tab. The application list displayed shows the USO applications the user is entitled to perform SSO with.

i **Note:** To view the enrolled applications based on your current user group, click on the **Group** tab.

4. To enroll an application, click on the switch ( icon) in the row corresponding to your USO application.
5. To unenroll an application, click on the switch ( icon) in the row corresponding to your USO application.
6. Click the **Save** button to save the changes.
7. If the Self-Service Portal was launched via the SSO Portal, click on the **Back to USO** button to return to the SSO portal. The application list will display all currently enrolled applications.

FIDO2 Credential Management



Important Note: The FIDO2 features described here are enabled by default and can be used out-of-the-box with no additional configurations.

If you still want to learn about the configurations for enabling FIDO2 features in USO SSF, you can refer to the steps described in the [AccessMatrix SSO Portal Implementation - Configure USO Self-Service Portal Policy](#) section.



Note: The following terms are used in FIDO2:

- **Relying Party** - Refers to the online service/application that relies on the FIDO2 authenticator for user authentication. In this scenario, Self Service Portal is the Relying Party service.
- **FIDO2 Credential** - Refers to the cryptographic key pair associated with the user, which is generated during the FIDO2 credential registration process. The public key of the key pair is stored by the Relying Party, while the private key is stored by the FIDO2 authenticator.
- **FIDO2 Authenticator** - Refers to the authenticator device that stores the private key of the cryptographic key pair and facilitates user authentication.

USO SSF provides a range of FIDO2 credential-related features, namely:

- [Register FIDO2 Credential](#)
 - [Edit FIDO2 Credential Name](#)
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- [Delete FIDO2 Credential](#)

These features require the use of a FIDO2-compliant authenticator. The following types of authenticators are supported:

- **Platform authenticators** - Authenticators built into and restricted to a specific device (e.g. fingerprint scanner on a laptop for Windows Hello).
- **Cross-platform authenticators** - External authenticators that can be used with multiple devices (e.g. physical security key).

Register FIDO2 Credential

i Note: Before attempting to register your FIDO2 credential, ensure that the URL has been specified using https. (e.g. `https://localhost:8443/usosf/start.ssf`)

1. Log in and access the Self-Service Portal, as described in [Access Self-Service Portal](#).
2. Click on the **Register FIDO2** tab to display the **FIDO2 Registration** page.
3. Click on the **Register New** button to display a **Create Name** dialog box.
4. Enter a name for the FIDO2 credential and click **Confirm**. The FIDO2 authenticator will prompt the user for authentication.
5. Follow the instructions provided by the authenticator.
6. Upon successful registration, a confirmation message will be displayed.

i Note: If you wish to verify your FIDO2 credential, perform a test login by continuing with the instructions below.

7. Click on your profile image and select **Logout** to return to the Common Login Page (CLP).

i Note: If you have not uploaded a profile image, click on the  icon.

8. Expand the authentication realm drop-down list and select the FIDO2 authentication realm (e.g. "FIDO2 Realm").
9. Perform a test login using the FIDO2 authenticator to verify that it is working as intended.

Edit FIDO2 Credential Name

1. Log in and access the Self-Service Portal, as described in [Access Self-Service Portal](#).
 2. Click on the **Register FIDO2** tab to display the **FIDO2 Registration** page.
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3. Click on the **Edit** button ( icon) in the row corresponding to your registered FIDO2 credential.
 4. Type in the new name for your FIDO2 credential in the text box and click the save button ( icon).

Delete FIDO2 Credential

1. Log in and access the Self-Service Portal, as described in [Access Self-Service Portal](#).
2. Click on the **Register FIDO2** tab to display the **FIDO2 Registration** page.
3. Click on the **Delete** button ( icon) in the row corresponding to your registered FIDO2 credential. A confirmation message will be displayed.
4. Click on the **Yes, Delete Passkey** button.

11. Appendices

USO Administration

A. Admin Role and Admin Right Definitions

Following table provides detailed description for the various predefined admin roles. Administrator is allowed to modify the admin rights assigned to the admin role.

Admin Role	Description
InitialAdminRole	By default, this initial admin role allows administrator to perform all operations in the system.
UserAdminRole	By default, this user admin role allows administrator to perform following operations: - User Administration (except login account time editing); - Group Administration; - USO Resource and Entitlement Administration; - Token Viewing; - Admin Privilege Assignment; - Segment Administration; - UAM Application Administration.
PolicyAdminRole	By default, this policy admin role allows administrator to perform following operations: - System Administration; - Policy Administration; - NAS Client Administration; - Encryption Key Administration (except deletion).
SelfServiceRole	By default, this self-service role allows user to view USO application password.

Apart from the above predefined admin roles, administrator is also able to define other admin roles containing desired admin rights. Following table provides detailed description for the various system-defined admin rights in terms of the menu option accessible and actions performable:

Resource	Operation	Menu Option Accessible	Actions Performable
User Admin Rights			
User	Create, Edit, Delete, View	Operation Dashboard > User & Segment > User > Search, Create	Admin Right Dependency: User - Create, Edit, Delete i) Able to search, view,

Resource	Operation	Menu Option Accessible	Actions Performable
			create, edit, delete user information ii) Only able to view login account
	Assign Group Membership	-	Admin Right Dependency: User - Create, Edit Group - View i) Able to assign/unassign group membership
	Edit Login Account Status, Edit Login Account Time	-	Admin Right Dependency: User - View Policy - View i) Able to edit login account start time and end time
	Reset Password	-	Admin Right Dependency: User - View User - Reset Password i) Able to reset password
	Able to assign/unassign admin privilege	-	Admin Right Dependency: User - View Admin Privilege - Assign i) Able to assign/unassign admin privilege
Group	Create, Edit, Delete, View	Operation Dashboard > User & Segment > Group > Search, Create	Admin Right Dependency: Group - Create, Edit, Delete i) Able to search, view, create, edit, delete group information ii) Only able to view membership
	Able to assign/unassign admin privilege	-	Admin Right Dependency: Group - View Admin Privilege - Assign Able to assign/unassign admin privilege
USO Admin Rights			
USO Resource	Create, Edit, Delete, View	Administration Dashboard > Configuration > UAM & USO > USO Application > Create, Search	Admin Right Dependency: USO Resource - Create, Edit, Delete, View Able to search, view, create,

Resource	Operation	Menu Option Accessible	Actions Performable
		Administration Dashboard > Configuration > UAM & USO > USO Application Group > Create, Search	edit, delete USO application and USO application group information
USO Entitlement	Assign	-	Admin Right Dependency: USO Resource - Create, Edit, Delete, View
USO Application Password	Edit, View	-	Admin Right Dependency: USO Application Password- Edit, View Able to retrieve USO application password (via SSF)
System Configuration Rights			
System	Create, Edit, Delete, View	Administration Dashboard > Configuration > Authentication > User Store > Search, Create; Administration Dashboard > Configuration > Authentication > User Lookup Module > Search, Create; Administration Dashboard > Configuration > Authentication > Login Module > Search, Create; Administration Dashboard > Configuration > Authentication > Authentication Realm -> Search, Create; Administration Dashboard > System > Housekeeping > Housekeeping Task; Administration Dashboard > System > Server & Cache > Server > Search, Create	Admin Right Dependency: System - Create, Edit, Delete, View i) Able to search, view, create, edit, delete user store information ii) Able to search, view, create, edit and delete user lookup module information iii) Able to search, view, create, edit and delete login module information iv) Able to search, view, create, edit and delete authentication realm information v) Able to view, edit housekeeping task information vi) Able to search, view, create, edit and delete server information
Policy	Create, Edit, Delete, View	Administration Dashboard > Security Policy > Admin & Audit > Admin Role > Search, Create Administration Dashboard	Admin Right Dependency: Policy – View, Create, Edit, Delete i) Able to search, view, create, edit and delete

Resource	Operation	Menu Option Accessible	Actions Performable
		> Security Policy > Authentication & USO > Login Policy > Search, Create Administration Dashboard > Security Policy > Authentication & USO > Password Expiry Policy > Search, Create Administration Dashboard > Security Policy > Authentication & USO > Password Quality Policy > Search, Create Administration Dashboard > Security Policy > Authentication & USO > Housekeeping > Housekeeping Policy Administration Dashboard > System > Server Monitoring	Admin Role information ii) Able to Search, View, Create, Edit and Delete Login Policy iii) Able to Search, View, Create, Edit and Delete Password Expiry Policy iv) Able to Search, View, Create, Edit and Delete Password Quality Policy v) Able to View, and Edit Housekeeping Policy Configuration vi) Able to View, and Edit Sever Monitoring Configuration
Report Right			
UAM & USO related reports	View	Real Time Report: Application Summary By Segment Report Dashboard > UAM & USO > Real Time > Application Summary By Segment	Admin Right Dependency: Report - View, Segment - View
		Real Time: USO Entitlements Report Dashboard > UAM & USO > Real Time > USO Entitlements	Admin Right Dependency: Report - View, User - View, USO Resource - View
		Real Time Report: User Application Privilege Report Dashboard > UAM & USO > Real Time > User Application Privilege	Admin Right Dependency: Report - View, UAM Application -View, User - View
		Historical Report: User Application Privilege Report Dashboard > UAM	Admin Right Dependency: Report - View, User - View

Resource	Operation	Menu Option Accessible	Actions Performable
		& USO > Historical > USO Application Login	
		Historical Report: Failed UAM Authorization By Application Resource Report Dashboard > UAM & USO > Historical > Failed UAM Authorization By Application Resource	Admin Right Dependency: Report - View, User - View, UAM Application - View
Others			
Segment	Create, Edit, Delete, View	Segment/OU Store	Admin Right Dependency: Segment - View, Create, Edit, Delete Able to search, view, create, edit and delete segment/sub-segment information

B. Sample USO Reports

Application Summary By Segment Report Sample

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Application Summary By Segment Report

Report Filtering Criteria

Segment Id : All Segment Name : All

Segment Id	Segment Name	Hierarchical Path	Number of Application Security Agent	Number of Web Security Agent	Number of USO Application	Total Applications
RootSegment	RootSegment	RootSegment	0	0	0	0
SampleSegment1	SampleSegment1	RootSegment/ SampleSegment1	0	0	1	1
SampleSegment1.1	SampleSegment1.1	RootSegment/ SampleSegment1/ SampleSegment1.1	0	0	0	0
SampleSegment1.2	SampleSegment1.2	RootSegment/ SampleSegment1/ SampleSegment1.2	0	0	0	0
SampleSegment1.3	SampleSegment1.3	RootSegment/ SampleSegment1/ SampleSegment1.3	0	0	0	0
SampleSegment1.4	SampleSegment1.4	RootSegment/ SampleSegment1/ SampleSegment1.4	0	0	0	0
SampleSegment1.5	SampleSegment1.5	RootSegment/ SampleSegment1/ SampleSegment1.5	0	0	0	0
SampleSegment1.6	SampleSegment1.6	RootSegment/ SampleSegment1/ SampleSegment1.6	0	0	0	0
SampleSegment1.7	SampleSegment1.7	RootSegment/ SampleSegment1/ SampleSegment1.7	0	0	0	0
SampleSegment2	SampleSegment2	RootSegment/ SampleSegment2	0	0	0	0
SampleSegment2.1	SampleSegment2.1	RootSegment/ SampleSegment2/ SampleSegment2.1	0	0	0	0
SampleSegment2.2	SampleSegment2.2	RootSegment/ SampleSegment2/ SampleSegment2.2	0	0	0	0

USO Entitlements Report Sample

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User USO Entitlements Report

Report Filtering Criteria

User Store : Default Store User or Group : User
User Id : All Application Id : All

User Id	: demouser	Segment Id	: -	User Name	: Demo User
Description	: Initial Demo User			Account Status	: Activated
Application Id	Application Name	Entitlement	Password Sharing	Profile	User Id
DesktopDemo	Desktop Demo	User	His/her own	Profile 1	user1
			His/her own	Profile 2	user2
			His/her own	Profile 3	user3

USO Application Login Report Sample

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USO Application Login Report

Report Filtering Criteria

User Store : Default Store
User Id : All
Session End Time - From : 2018-05-02 00:00:00 Session End Time - To : 2018-05-02 15:58:40

No.	Session End Time	User Id@User Store	User Name	User Created Time	Realm Id	Last Login	USO App Login Time
1	2018-05-02 15:42:34	demouser@DefaultStore	Demo User	2018-04-27 10:11:46	40151	2018-05-02 15:58:19	2018-05-02 16:00:50
2	2018-05-02 15:38:10	testuser1@DefaultStore	testuser1	2018-05-02 12:23:36	40151	2018-05-02 12:30:59	

USO Application Training

A. Application States and Control Variables

- **Application States**

USO Trainer provides 5 pre-defined states, namely: NotRunning, BeforeLogin, AfterLogin, AfterLogout, and Disabled, as described in following table:

State	Meaning	Remarks
NotRunning	No application instances are running.	This is the initial state for application when USO Client shows application list after login. After re-enabling an application, the state is also reset to NotRunning.
BeforeLogin	The application instance (indicated by process id) has been detected but the login process has not taken place yet. USO Client will fill in the login information when the login screen is matched.	This state is set when: - After an application is launched from USO Client, USO Client will associate the process id to this application instance and set to this state. - When the application is launched externally and USO Client matches the login screen, USO Client will set to this state. - When the application has just been re-enabled and USO Client matches the login screen.
AfterLogin	USO Client has successfully filled in the login information to the application and login process succeeded.	In case of login failure, the state will remain as BeforeLogin and the control flag will be set to LoginSubmitted.
AfterLogout	USO Client has detected that user has logged out from the application.	Typically this state is valid when Logout screen exists and USO Client has matched the screen, or the login screen re-appears after logout and the previous state was AfterLogin.
Disabled	The application has been disabled in USO Client.	When application is disabled, USO Client will not be able to keep track of the application's state any more. Thus, when the application is re-enabled, the state is changed back to initial state of NotRunning.

- **Control Variables**

USO Trainer provides 3 pre-defined control variables, namely: LoginSubmitted, LoginFailed, and ChangePwdSubmitted as described in following table:

Value	Meaning	Remarks
LoginSubmitted	USO Client submits the application's login information and is waiting for the result screen or timeout.	This control variable is set after USO Client successfully processes the application's login screen.
LoginFailed	USO Client detects a failed login process.	It is set after USO Client successfully matches a login failure screen, or upon login timeout and user confirms that login process has failed.
ChangePwdSubmitted	USO Client submits the change password screen and is waiting for the result screen or timeout.	It is set after USO Client successfully processes a change password screen.

B. Basic, Standard and User Defined Actions

- Basic Actions**

A basic action is an action that is directly linked to a certain execution by the usoClient. A basic action is atomic and cannot be broken down further. Basic action function names are preceded by underscore (_).

The following table shows the basic actions provided by USO:

Action Name	Parameters	Definition
_AuditWrite	app_id, event_type, result, msg, detail1,detail2, detail3	Generates an audit message with the parameter values and write it to the audit table.
_ClearHook	-	Clear existing hook of the target application screen. Note: Unused at the moment.
_ClearTimer	-	Clear existing timeout counter of the current login / change password process.
_CommitAppNewPassword	-	Commit the new password change and update the new password to the server (this action is used when change password is successful). Internally, this will also insert a ChangePwdSuccessful control into the state/control history

Action Name	Parameters	Definition
		and generate an audit message to indicate the success.
_ContinueMatching	-	Re-activate application screen matching after it has been disabled by _StopMatching(). Refer to _StopMatching for further details.
_DisableApplication	prompt	Disable the currently matched application. Set prompt = true to prompt user before disabling the application. Set prompt = false to directly disable the application.
_EvalOnBrowser	evalScript	Evaluates a JavaScript expression on active browsers.
_FillAllFields	-	Fill all field values onto the matched screen. Returned result (a Boolean value): - True if successfully fill in all fields, - False if failed to fill in all fields.
_FillAllFieldsAndSubmit	buttonId	Fill all field values onto the matched screen and submit the filled screen using the buttonId, if specified. Returned result (a Boolean value): - True if the screen is successfully submitted, - False if failed to submit the screen.
_FillFieldByVarName	valueSrc	Fill the field value onto the matched screen for the field information that contains variable name. Returned result (a Boolean value): - True if successfully fill in the field - False if failed to fill in the field
_GetMessage	-	Returns the message based on the message Id.
_GetTextValue	id	Returns the text value from the application screen.
_GetUserId	-	Returns the user Id of the user currently logged in to USO.

Action Name	Parameters	Definition
_GetVar	var_name	Returns the value of the attribute of the current profile. The "var_name" can be app.xxx, pm.xxx, apppm.xxx or just xxx.
_IsPwdChangeRequired	-	Returns true if password change is required, otherwise false. Usually used for checking Forced Change Password..
_LaunchApp	appld	Launches the target application that is specified in the parameter. Note: Unused at the moment.
_MessageBox	msg, title	Shows a message box with specified message and title.
_PromptAppLoginInfo	-	Shows the dialog that prompts for application login information (typically used for first time login, and when login failed). It will also be automatically triggered whenever a USO Client tries to get appprn.%uid or apprn.%pwd and the value is null or ?. Returned result (a Boolean value): - True if application login information is successfully collected. - False if fails to collect the application login information.
_PromptAppNewPassword	-	Shows the dialog that prompts for application new password (typically used when change password screen matched, or when user activate reset application password). It will also be automatically triggered whenever USO Client tries to get appprn.%newpwd and the value is null, or the first screen option of the change password screen is checked. Returned result (a Boolean value): - True if application new password information is successfully collected. - False if fails to collect application new password.
_PromptComboBox	msg, title, choices, def	Shows a message box with a combo box.

Action Name	Parameters	Definition
_PromptCustomButtons	msg, title, buttons	Shows a message box with multiple customized buttons. The buttons are delimited by "\n". For example, to specify 3 buttons: "OK\nCancel\nAbort". The return value starts from 0 (specifying the first button). Maximum number of customized buttons is 10.
_PromptTextBox	msg, title, def	Shows a message box with a text box.
_PromptYesNo	msg	Show a message box that prompts for Yes or No. If the user answers yes, it returns true, otherwise false.
_RecordLoginTime	-	Trigger an event to record the login time.
_ReprocessScreen	-	Trigger an event so that USO Client reprocesses the current screen (typically used when there is a state and control flag change).
_ResetDefaultProfile	-	Resets profile to default.
_RollbackAppNewPassword	-	Rollback the password changes and discards the new password (typically used when change password failed). Internally, this will also insert a ChangePwdFailed control into the state/control history and generate an audit message to indicate the failure.
_SendKeys	keys	Send the keyboard strokes to the window. Refer to C. Keystroke Code for SendKeys for keystroke list.
_SetTimeout	-	Sets timeout of the current login / change password process.
_SetVar	var_name, var_value	Sets the value of an attribute of the current profile. The 'var_name' can be app.xxx, pm.xxx, apppm.xxx or just xxx.
_SetWindowHook	-	Sets windows hook of the target application screen. Note: Unused at the moment.

Action Name	Parameters	Definition
_Sleep	period	Sets USO Client to sleep for a certain period (in milliseconds).
_StopMatching	-	Temporarily turn off application screen matching. Typically used in Generic Emulators where there is a need to stop the flickering screens due to the matching process. Refer to _ContinueMatching to re-enable application screen matching.
_Submit	buttonId	Submit the matched screen by using button Id, if specified. Returns a boolean, true if the screen is successfully submitted, otherwise returns false.
_SwitchApp	appld	Switch the application instance of the current application process to another application id, and from then on the instance will belong to the new application. Typically used in telnet application, where the user telnet from one server (application) to another server (application).
_SwitchProfile	index	Switch the profile to use based on the specified profile in the custom scripts without prompting the user to select manually.
_SwitchProfileByExternalDll	dllName	Returns the value of an attribute of the profile from external DLL
_USO2FillAndSubmit	buttonId	Process the screen by filling in all fields in the screen and submit automatically. It may prompt for application login information or new password if required or it will also try to perform accordingly to the screen option setting. Uses the USO 2.3 screen processing logic for backward compatibility.
_Version	-	Returns the version of the USO Client JavaScript engine.
_Wait	-	Wait for a certain period in seconds

- **Standard Actions**

Following table shows the standard actions provided by USO:

- **Standard Actions for Login Process:**

Action Name	Parameters	Definition
\$DisableApplication	prompt	Disable application for single sign-on. Returned result (a Boolean value): - True if the screen is successfully submitted. - False if fails to submit the screen.
\$HandleLoginFail	reason	Handle login failure.
\$HandleLoginReappearAfterLoginFail	-	Handle the event where login screen is matched again after login submitted or login failure.
\$HandleLoginSuccess	reason	Handle login success.
\$HandleLoginTimeout	-	Handle login timeout.
\$HandleLogout	-	Handle the application logout.
\$HandleNormal	-	Handle the screen without state control, just fill in and submit.
\$LoginAgain	-	Reset the login control variable so that USO Client can login again.
\$ProcessLogin	timeout, timeoutAction, disappearAction	Process login screen.
\$PromptLoginResult	yesAction noAction	Prompt the user for the result of the last login, and execute either yesAction or noAction, based on user's response.
\$ResetAppLoginInfoAndLoginAgain	-	Reset application login information and re-login.

- **Standard Actions for Change Password Process:**

Action Name	Parameters	Definition
\$ProcessChangePwd	timeout, timeoutAction, disappearAction	Process change password screen.
\$HandleChangePwdFail	reason	Handle change password failure.
\$HandleChangePwdSuccess	reason	Handle change password success.
\$HandleChangePwdTimeout	-	Handle change password timeout.
\$PromptChangePwdResult	yesAction, noAction	Prompt the user for the result of the last change password, and execute either yesAction or noAction, based on the user's response.

- **User-Defined Actions**

Administrator is allowed to define custom actions known as user-defined actions to cater for different screen matching behaviors as discussed in following steps:

1. Start USO Trainer if it has not been started.
2. On the **Main Screen**, click on the **Tools** menu and select the **User-Defined Actions** sub-menu option.
3. To add, edit, and delete a new user-defined actions, refer to this video:

Link: https://www.youtube.com/watch?v=3BM_buNCGxl

C. Keystroke Code for SendKeys

Keystroke Event	Special Character Code
BACKSPACE	{BACKSPACE},{BS}, or {BKSP}
BREAK	{BREAK}
CAPS LOCK	{CAPSLOCK}
DEL or DELETE	{DELETE} or {DEL}
DOWN ARROW	{DOWN}

Keystroke Event	Special Character Code
END	{END}
ENTER	{ENTER} or ~
ESC	{ESC}
HELP	{HELP}
HOME	{HOME}
INS or INSERT	{INSERT} or {INS}
LEFT ARROW	{LEFT}
NUM LOCK	{NUMLOCK}
PAGE DOWN	{PGDN}
PAGE UP	{PGUP}
PRINT SCREEN	{PRTSC}
RIGHT ARROW	{RIGHT}
SCROLL LOCK	{SCROLLLOCK}
TAB	{TAB}
UP ARROW	{UP}
F1	{F1}
F2	{F2}
F3	{F3}
F4	{F4}
F5	{F5}
F6	{F6}
F7	{F7}
F8	{F8}
F9	{F9}
F10	{F10}

Keystroke Event	Special Character Code
F11	{F11}
F12	{F12}
F13	{F13}
F14	{F14}
F15	{F15}
F16	{F16}
SHIFT	+
CTRL	^
ALT	%

For example, to send ABC followed by Enter, use `_SendKeys('ABC{ENTER}')`?

USO Integration

A. Troubleshooting

Windows Integrated Authentication

For Windows Integrated Authentication related problems and issues, refer to SPNEGO SourceForge documentation for troubleshooting at the following url: <http://spnego.sourceforge.net/>.

USO Client

A. Troubleshooting

A.1 Log files

Logs are created for troubleshooting purposes:

- USO Client log can be found in this location:
`<Users\AppData\Roaming\USOClient\log\usoclient.log`

-
- USO server log can be found in this location:
e.g. Tomcat
<AM Installation Folder>\tomcat\logs\usoserver.log

A.2 Handling Unsuccessful Application Login

This section explains the unsuccessful application login scenarios. User might fail to login to an application due to various reasons. Two possible reasons are explained below.

- **Fail to Login Due to Incorrect Login Information**

Provided with the wrong login information, USO Client will prompt that an application login failure has been detected. Follow the steps below to correct the login information:

1. Click the **OK** button on the **Login fail** dialog.
2. There are three options to choose from:
 - Select **Reset** to reset the application login information.
 - If **OK** is selected, the new application login information will be saved and used to login to application.
 - If **Cancel** is selected, the **Disable Application** dialog is displayed and prompt the user whether to disable SSO feature for this application.
 - If **Yes** is selected, USO Client will disable the application for current USO session.
 - Select **Login** to re-login to the application, USO Client will then perform the login process using the existing login information.
 - Select **Disable** to disable the application.

- **Fail to Login Due to Session Timeout**

USO Client has a session timeout setting, which can be configured via the Administration Console. For a detailed explanation of this feature, refer to the [AccessMatrix Common Administration Guide](#).

If a user attempts to launch an application after the session expired, an error message will display stating that the session has been invalidated.

A.3 Deleting PSE file

PSE file can be deleted from this location: <Users>\<User Id>\AppData\Roaming\USOClient\uso.pse.
